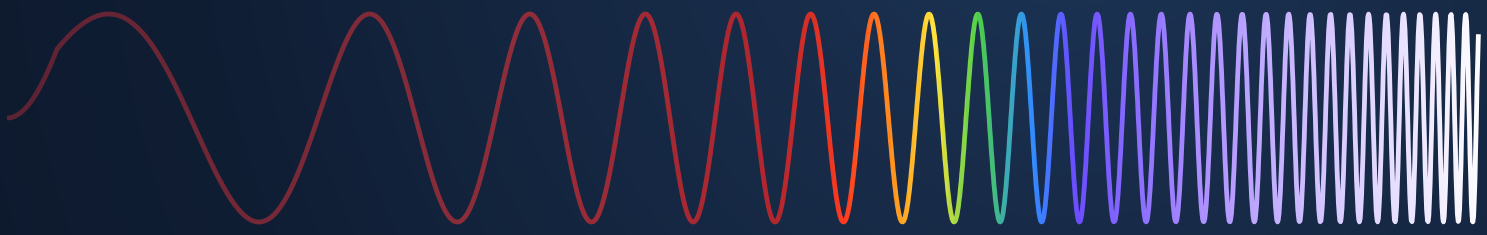
A PERSONAL TEXTBOOK

Light, Waves & Radiation

Electromagnetic waves, photons, atoms and energy levels, reflection and refraction, color, diffraction, infrared and heat, ionizing radiation, and X-ray diffraction — explained from the ground up, with the *why* behind every idea.



RADIO · MICROWAVE · INFRARED · VISIBLE · UV · X-RAY · GAMMA

longer wavelength · lower photon energy

shorter wavelength · higher photon energy

- What a light wave is
- Atoms & energy levels
- Why things have color
- Ionizing radiation
- Photons & photoelectric effect
- Absorption & emission
- Interference & diffraction
- Shielding & space
- The full EM spectrum
- Reflection & refraction
- Infrared & heat
- X-rays & Bragg's law

Written for **Chase Yarnall**

Fall 2026

How to Use This Book

This book is built from the questions you searched while studying: what a photon is made of, why infrared "shows heat," how color works, why a tiny atom can interact with a huge wavelength, what Grace meant about Astrophage. They're all one connected story, told here in order.

How it is built

- **Part I**: what light *is* (a wave in electric and magnetic fields that arrives as photons).
- **Part II**: the matter it talks to (electrons on an energy ladder).
- **Part III**: what happens when they meet (reflection, refraction, color, diffraction, heat).
- **Part IV**: the high-energy end (ionizing radiation, shielding, X-ray diffraction).

Each chapter opens with *The big picture* and ends with a summary and *Check yourself* questions, with full solutions in the Answer Key. Start with Your Questions, Answered, then the ten-minute Prologue. The Contents come next.

SHORT ON TIME? — How to skim

Read the separate **Quick Version** PDF (about 12 pages) first. In this full book, the fast route through any chapter is: *The big picture* → the **Key Idea** and **You Asked** boxes → the *Chapter summary*. That's about a quarter of the text. Skip Rabbit Holes and Worked Examples unless you're curious or studying for an exam.

The boxes

KEY IDEA
What to remember if you forget everything else.

ANALOGY
An everyday picture for intuition, and where it breaks down.

MYTH VS. REALITY
A popular half-truth, corrected.

RABBIT HOLE (optional)
Extra depth for when you're curious. Safe to skip.

YOU ASKED
One of your actual questions, answered in the first sentence.

WORKED EXAMPLE
Given, Find, Think, then the math.

MATERIALS SCIENCE CONNECTION
Where it shows up in ME2340.

PROJECT HAIL MARY CONNECTION
The real physics behind the scene you asked about (spoiler-light).

On rabbit holes: the fine details you love have a home in the Rabbit Hole boxes. Finish the main text of a section first, then dive. If something isn't covered, note it and keep going. It usually clicks a section later.

Reading paths

If you want…	Read
The whole story (recommended)	Prologue, then Chapters 1–11
The core mechanism in one evening	Prologue → Ch 2 → Ch 5 → Ch 6 → Ch 7
ME2340 optical properties & X-rays	Ch 5 (band gaps), 6, 7, 11 (Bragg's law, Miller indices)
Infrared and heat	Ch 3, Ch 5 (molecular vibrations), Ch 9
Your *Project Hail Mary* questions	Ch 8 (Astrophage), Ch 10 (radiation and the tunnel)

Units

Wavelengths: **nm** (10^{-9} m) for visible light, **µm** (10^{-6} m) for infrared, cm or m for microwaves and radio. Energies: **electron-volts**, where 1 eV = 1.602 × 10^{-19} J is the energy an electron gains crossing 1 volt. A visible photon carries 2–3 eV, and ionizing an atom takes about 10 eV. Most useful shortcut in the book: $E\,(\text{eV}) = 1240/\lambda\,(\text{nm})$.

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Your Questions, Answered

This book started from the questions you typed into search bars while studying. Every one of them is answered somewhere in these pages. Here is each question, the one-line answer, and the page where the full explanation lives.

CHAPTER 1 · ELECTROMAGNETIC WAVES: RIPPLES IN FIELDS

"How does EM radiation actually work — what is physically waving?" p. **16**

The electric and magnetic fields themselves are what wave: at each point the light passes through, the strength and direction of the electric field (and the magnetic field with it) swing back and forth, hundreds of trillions of times per second for visible light.

CHAPTER 2 · PHOTONS: LIGHT AS PACKETS OF ENERGY

"What is the photoelectric effect?" p. **23**

The photoelectric effect is the ejection of electrons from a material, usually a metal, when light shines on it.

"Does a bigger amplitude of a wave mean more photons?" p. **28**

Yes: for light of a given color, a bigger amplitude means more photons per second, while each photon keeps exactly the same energy, hf.

"What is a photon made up of?" p. **29**

Nothing smaller: as far as every experiment can tell, a photon is an elementary particle with no internal parts.

CHAPTER 3 · THE ELECTROMAGNETIC SPECTRUM

"What is the EM spectrum — are radio waves, light and X-rays really the same thing?" p. **32**

Yes. Radio waves, visible light and X-rays are exactly the same phenomenon: oscillating electric and magnetic fields traveling at *c*, delivered in photons of energy hf.

CHAPTER 4 · INSIDE THE ATOM: SCALE, ORBITALS, AND QUANTIZED ENERGY

"How big is an atom, from one side to the other?" p. **42**

About 0.1 to 0.5 nanometers, or one to five ten-billionths of a meter.

"If a marble is the nucleus and it's placed at the center of a football field, is the electron probability cloud the size of the football field?" p. **43**

No: the cloud is bigger. With a 1.5 cm marble as the nucleus, a heavy atom such as gold would be about 300 m across, and a hydrogen atom nearly a kilometer.

CHAPTER 5 · ABSORPTION AND EMISSION: ELECTRONS JUMPING UP AND DOWN

"How can an atom that's thousands of times smaller than a light wave absorb it?" p. **52**

Because absorption depends on matching the wave's *frequency*, not its size.

"When an electron jumps down an energy level, how does that make a photon, and what is 're-emission'?" p. **53**

The photon is created during the jump itself; it is not stored inside the atom beforehand.

CHAPTER 6 · REFLECTION, TRANSMISSION, ABSORPTION, AND REFRACTION

“How does light work? What actually reflects the wave when it hits something?” p. **65**

The electrons in the material do: the incoming wave shakes them, and the waves they radiate add up to the reflected wave (and, going forward, to the transmitted wave).

“Is reflection the same thing as the material absorbing and re-emitting the EM wave?” p. **65**

No. Reflection is coherent re-radiation by electrons that are *driven* by the wave; it is not a photon being absorbed into an excited state and re-emitted later.

CHAPTER 7 · COLOR

“How does color work in terms of EM radiation?” p. **75**

Color is how your visual system labels the mix of visible wavelengths (photon energies of about 1.8 to 3.1 eV) entering your eye; an object's color is whatever part of the light falling on it the object sends toward you.

“What are the electron energy gaps that give things their color?” p. **77**

They are the energy differences between states an electron can occupy, and a material is colored when one of its gaps falls between about 1.8 and 3.1 eV, the energies of visible photons.

CHAPTER 8 · INTERFERENCE AND DIFFRACTION (AND WHY SIZE VS. WAVELENGTH MATTERS)

“What is diffraction — and what does it mean in quantum physics?” p. **89**

Diffraction is the spreading of a wave as it passes through an opening or around an edge, by an angle of roughly λ/a; it is strongest when the opening is about a wavelength wide (Section 8.2).

“What is the diffraction limit?” p. **91**

The diffraction limit is the smallest detail any lens-based instrument can resolve, set by the wavelength: about $\lambda/(2\,\mathrm{NA})$, or roughly half a wavelength for the best microscopes.

“Why can big wavelengths interact with small objects?” p. **92**

They can because a wave is an oscillating electric field, and anything containing charges feels it, whatever its size.

“Something so small (an atom) shouldn’t be able to interact with such a long wavelength — the wave should be unnoticeable to the tiny atom. What’s the idea here?” p. **93**

Your intuition is right for *scattering and imaging*, but absorption and emission don’t depend on size at all; they depend on resonance.

CHAPTER 9 · INFRARED, HEAT, AND THERMAL RADIATION

“Is infrared the same thing as heat?” p. **100**

No. Infrared is light, an electromagnetic wave just like visible light but with longer wavelengths (about 0.7 µm to 1 mm).

“Why are infrared waves often called ‘heat waves’?” p. **101**

Because infrared is the radiation warm things give off and the radiation your skin feels as warmth.

“Why does infrared show heat?” p. **101**

Because at everyday temperatures the light an object emits by itself is almost entirely infrared, and the amount rises steeply with temperature ($\propto T^4$).

CHAPTER 10 · IONIZING RADIATION, SHIELDING, AND SPACE

“How does a magnetic field help protect against radiation?” p. **114**

A magnetic field deflects *charged* particles (protons, electrons, alpha particles, cosmic-ray nuclei) so they spiral along field lines instead of reaching the surface, but it does nothing at all to photons (UV, X-rays, gamma rays) or neutrons.

CHAPTER 11 · X-RAYS AND MATERIALS SCIENCE

“What does ‘X-ray vision’ mean — how do X-rays see through things?” p. **119**

X-rays don't "see" anything: an X-ray image is a **shadow**.

“How can parallel planes have different Miller indices — what's the deal with (110) vs (220), or (100) vs (200)?” p. **123**

(220) planes have the same orientation as (110) planes but are packed twice as closely: $d_{220} = d_{110}/2$.

“If the interplanar spacing is higher, does that mean the density is higher?” p. **124**

Within one crystal, yes for the *planar* density: the families of planes with the largest spacing are the most densely packed planes.

PROLOGUE

The Whole Story in Ten Minutes

Radio, light, color, heat, sunburn, X-rays: one story told at different wavelengths. This is the map. When you feel lost in a detail, come back and ask: which step am I on?

1 Light is a ripple in electric and magnetic fields

Charges are surrounded by an **electric field**, and moving charges also make a **magnetic field**. Shake a charge and the change in its field spreads outward at $c = 3.00 \times 10^8$ m/s. A changing electric field makes a magnetic field and vice versa, so the ripple sustains itself and needs no medium. That ripple is an **electromagnetic (EM) wave**, with $c = \lambda f$.

→ CHAPTER 1

2 The wave's energy arrives in packets called photons

Light delivers energy in lumps called **photons**, each with $E = hf = hc/\lambda$. Two separate dials: *frequency* (color) sets the energy per photon; *amplitude* (brightness) sets how many photons arrive. Proof: dim UV knocks electrons out of metal, but blinding red light can't, because no single red photon has enough energy. A 500 nm photon carries $1240/500 \approx 2.5$ eV.

→ CHAPTER 2

3 Radio waves, light and X-rays are the same thing

From radio waves longer than buildings to gamma rays smaller than nuclei, it's all the same thing traveling at *c*. The *only* difference is wavelength, which sets photon energy, which decides what the light does to matter (Figure P.1). Visible light (400–700 nm, 1.8–3.1 eV) is a tiny slice.

→ CHAPTER 3

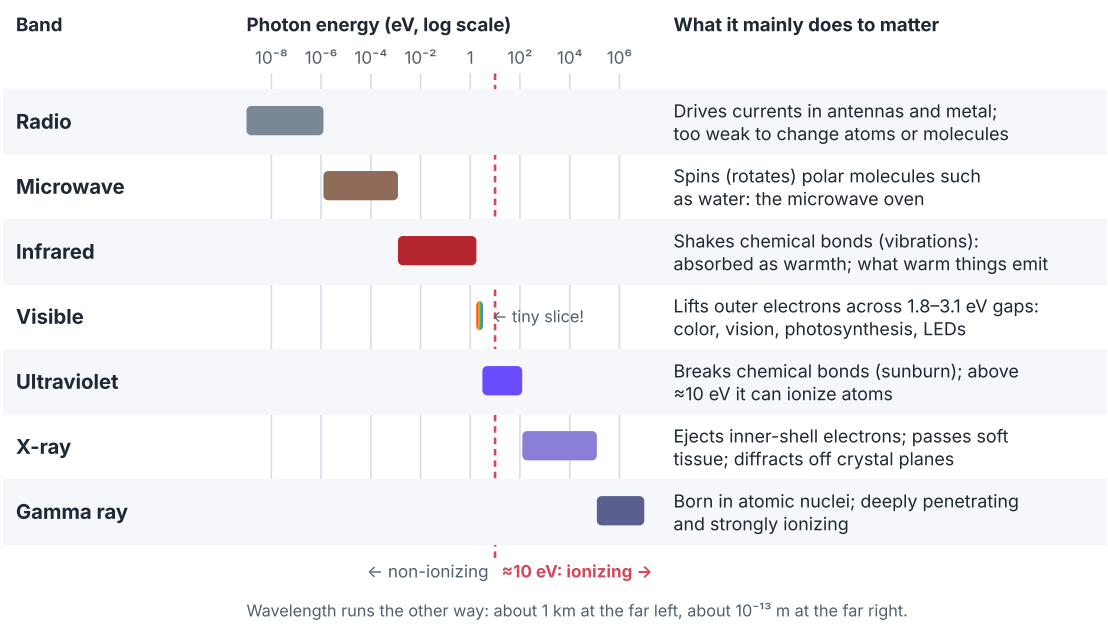


Figure P.1 The same phenomenon over 16 powers of ten. Each band's photon energy matches a different kind of "jump" in matter, which is why each band does something different. The red dashed line marks the rough ionizing threshold (about 10 eV).

4 Matter is electrons on an energy ladder

An atom is 0.1–0.3 nm across; its nucleus is tens of thousands of times smaller. Electrons live in probability clouds and can only have certain energies, like rungs on a ladder. Molecules add smaller rungs for *vibration* (infrared-sized) and *rotation* (microwave-sized). In solids the rungs merge into **energy bands** separated by **band gaps**.

→ CHAPTERS 4 AND 5

5 One question decides what light does to matter

Ask: *is there an allowed jump whose energy matches the photon's hf?* If yes, the photon is **absorbed**, and its energy is re-emitted, turned into heat, used for chemistry, or spent knocking out an electron. If no, the wave just shakes the electrons, and they re-radiate in step with it, which produces **reflection**, **transmission**, **refraction** and **scattering** (Figure P.2). Notice that the atom's *size* isn't in the question: an atom 5,000× smaller than a light wave absorbs it fine, because absorption is about matching *energy* (resonance).

→ CHAPTERS 5, 6 AND 8

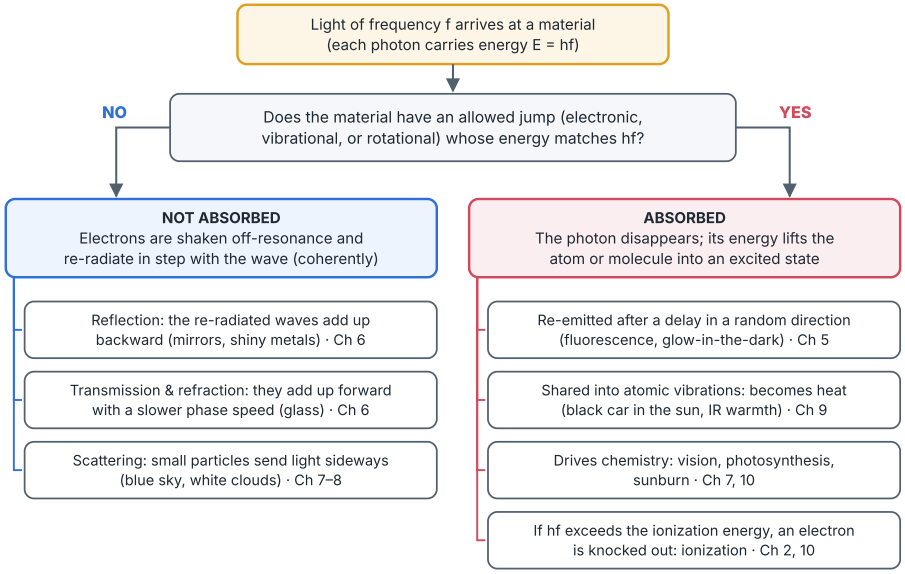


Figure P.2 The master flowchart for light meeting matter. Almost every phenomenon in Part III is one of these boxes.

6 Color is what survives

A material absorbs the wavelengths that match its jumps; your eye gets what's left (a leaf absorbs red and blue, returns green). For solids: band gap above 3.1 eV, no visible absorption (pure diamond is clear); below 1.8 eV, all of it absorbed (silicon looks gray and opaque); in between, colored. The blue sky is different: air molecules scatter short wavelengths far more than long ones.

→ CHAPTER 7

7 Waves spread out and interfere

Light bends around obstacles and spreads through openings (**diffraction**), and overlapping waves reinforce or cancel (**interference**). So you can't image details smaller than about half a wavelength, which is why no light microscope shows atoms. Objects much smaller than the wavelength barely scatter it, which is why a microwave oven's door mesh blocks 12 cm microwaves but passes visible light.

→ CHAPTER 8

8 Everything glows, and warm things glow in infrared

Everything above absolute zero emits **thermal radiation** set by its temperature: you (310 K) peak near 9–10 µm in the infrared, a stove burner (~1000 K) glows dull red, the Sun (5800 K) peaks in the visible. IR is called "heat radiation" because warm everyday objects emit mostly IR and molecules absorb it straight into vibration. But IR is light, not heat, and visible light heats things too.

→ CHAPTER 9

9 Above about 10 eV, light becomes ionizing

A photon with more energy than an electron's binding energy (roughly 10 eV, far UV) can knock it off its atom, breaking bonds and damaging DNA. At everyday intensities, more brightness can't substitute for photon energy. "Radiation" also means fast *particles* (alpha, beta, neutrons, cosmic rays). Magnetic fields steer charged particles away, as Earth's magnetosphere does, but do nothing to photons; only matter stops those.

→ CHAPTER 10

10 X-rays fit between the atoms

X-ray wavelengths (~0.1 nm) match the spacing of atomic planes, so a crystal acts as a 3-D diffraction grating, reflecting only at angles that satisfy **Bragg's law**, $n\lambda = 2d\sin\theta$. That's how we know copper is FCC and iron is BCC. Your ME2340 Miller-index planes are literally the mirrors that reflect X-rays.

→ CHAPTER 11

KEY IDEA — Three equations carry most of this book

$c = \lambda f$ (wavelength and frequency are locked together) · $E = hf = \dfrac{hc}{\lambda}$ (photon energy grows with frequency) · $E_{\text{photon}} = \Delta E_{\text{jump}}$ (light is absorbed or emitted only when its photon energy matches an allowed jump). Nearly everything else is these three applied to a particular wavelength and material.

CHAPTER 1

Electromagnetic Waves: Ripples in Fields

Your Wi-Fi signal, the sunlight on your desk and the glow of a red-hot stove burner are the same kind of thing: ripples in invisible fields that fill all of space. This chapter is about what is actually doing the rippling.

THE BIG PICTURE

- Every charge is surrounded by an electric field; moving charges also make a magnetic field. Fields are real physical things that store energy.
- Shake a charge and kinks in its field race outward at the speed of light. That traveling ripple is an electromagnetic (EM) wave.
- In the wave, a changing electric field makes a magnetic field and vice versa, so the wave carries itself. No medium is needed.
- A wave is described by wavelength, frequency ($c = \lambda f$), amplitude (intensity $\propto$ amplitude²) and polarization.

AFTER THIS CHAPTER YOU CAN

- Explain in plain words what is “waving” in light.
- Sketch the E and B fields of a light wave and state how they are oriented.
- Convert between wavelength and frequency with $c = \lambda f$.
- Explain why light crosses empty space while sound cannot.
- Predict how intensity changes when the field amplitude changes.
- Explain how polarized sunglasses cut glare.

1.1 Charges and fields: the invisible stage

Everything in this book starts with **electric charge**, a property of particles like electrons (negative) and protons (positive). Like charges repel and opposite charges attract, even across empty space.

Physicists describe that reach-across-space push with a **field**: a quantity that has a value and a direction at every point in space. The **electric field** $\vec{E}$ at a point is the force a small test charge q would feel there, per unit charge: $\vec{E} = \vec{F}/q$. Its units are newtons per coulomb, which is the same as volts per meter (V/m).

Moving charges (currents) create a second field, the **magnetic field** $\vec{B}$, measured in teslas (T). It pushes on other moving charges. Figure 1.1 shows both.

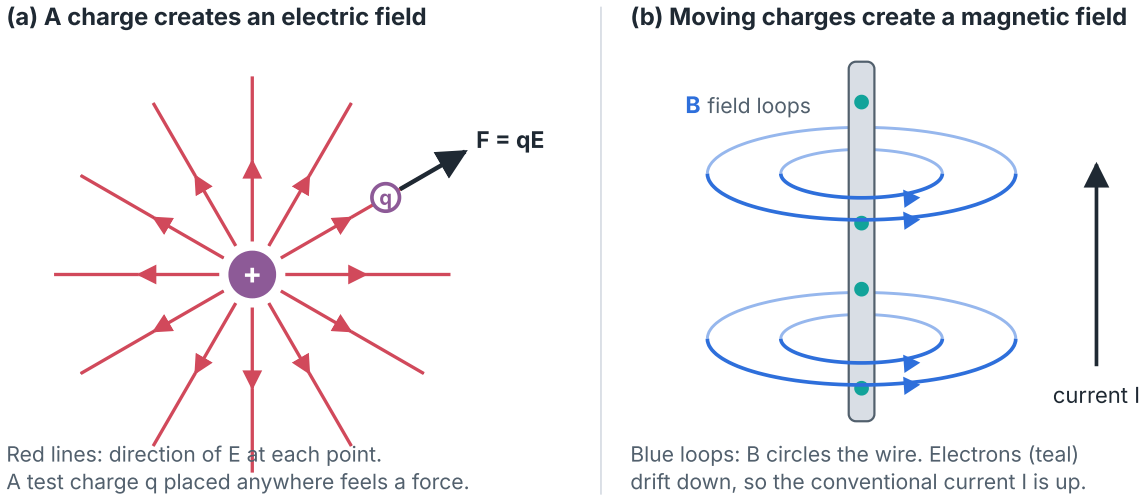


Figure 1.1 (a) The electric field of a positive charge points away from it; a test charge placed anywhere feels a force along the field. (b) A current is wrapped in loops of magnetic field. Field lines are a map: they show the direction of the field at each point, and their crowding shows its strength.

> **ANALOGY**
>
> A field is like a weather map of the wind: at every point there is an arrow with a size and a direction. You can't see the air, but you can see what it does to a leaf. You can't see the field either, but you can see what it does to a charge.

Fields are not just bookkeeping. They store energy, carry momentum and, crucially, they take time to change. If a charge moves, the field far away does not update instantly. The "news" spreads outward at a finite speed, the speed of light. That delay is what makes waves possible.

1.2 Shake a charge, make a wave

Suppose a charge sits still, then is jerked upward a short distance. Close to the charge, the field already points from the new position. Far away, the news has not arrived, so the field still points from the old position. In between, the field lines must connect, and they do it with a **kink** (Figure 1.2a). That kink travels outward at speed c, and it is a pulse of electromagnetic radiation.

Now keep wiggling the charge up and down. Each wiggle launches another kink, and the field out in space flips up, down, up, down in step with the source (Figure 1.2b). That repeating pattern, racing outward, is an **electromagnetic wave**.

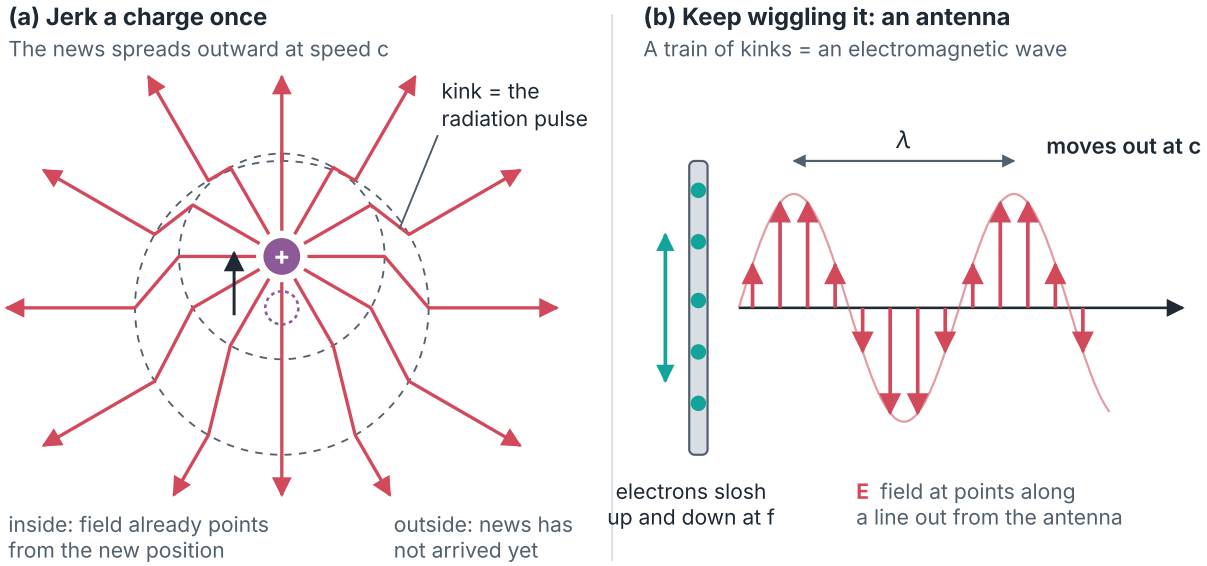


Figure 1.2 (a) Jerk a charge once and a kink in its field lines spreads outward at speed c. (b) Wiggle charges up and down continuously, as an antenna does, and the kinks form a wave train. The arrows show the electric field at points along a line out from the antenna; the magnetic field (not drawn) points into and out of the page.

Two rules follow, and they run the whole book:

- **Only accelerating charges radiate.** A charge sitting still has a static field. A charge coasting at constant velocity carries its field along with it. Only speeding up, slowing down or changing direction makes kinks.
- **Waves push charges.** When the wave reaches another charge, its electric field shoves that charge back and forth at the wave's frequency.

A Wi-Fi router uses both rules. An alternating current drives electrons up and down its antenna about 2.4 billion times per second, which launches the wave. In your laptop's antenna, the arriving electric field drives electrons back and forth, producing a tiny alternating current that the receiver decodes. Many antennas are about a half or a quarter of a wavelength long, because charges slosh most strongly when the length matches the wave.

> **KEY IDEA**
>
> Accelerating charges make electromagnetic waves, and electromagnetic waves accelerate charges. Antennas, light bulbs, atoms and the Sun all emit by shaking charges; eyes, cameras and antennas all detect by being shaken. Light from atoms (Chapter 5) and from hot objects (Chapter 9) works the same way, with electrons as the wiggling charges.

1.3 The self-sustaining wave

Once a wave leaves the antenna, it no longer needs the antenna. Two laws of electromagnetism keep it going on its own:

- **Faraday's law:** a changing magnetic field creates an electric field. This is how generators and induction cooktops work.
- **Maxwell's addition:** a changing electric field creates a magnetic field.

In a wave, the oscillating E field creates an oscillating B field, which creates an oscillating E field a little farther along, and so on (Figure 1.3). The two fields regenerate each other as they travel, with no charges or material needed.

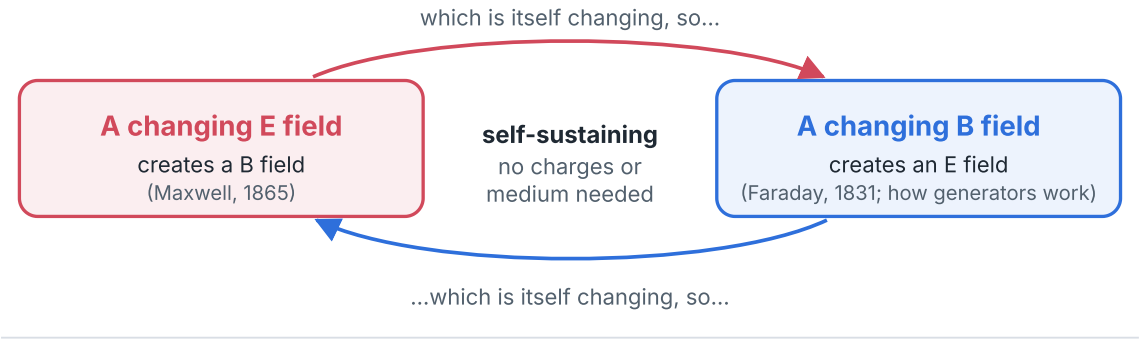


Figure 1.3 The leapfrog that makes light self-sustaining. Each changing field creates the other, slightly farther along, so the pair propagates through empty space.

In 1865 James Clerk Maxwell worked out how fast this leapfrog travels. The speed depends only on two constants that describe electricity and magnetism, ε_0 and μ_0:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{1}{\sqrt{(1.257 \times 10^{-6})(8.854 \times 10^{-12})}}\ \mathrm{m/s} = 3.00 \times 10^{8}\ \mathrm{m/s} \tag{1.1}$$

That number matched the measured speed of light. Maxwell concluded that light is an electromagnetic wave, and in 1887–88 Heinrich Hertz confirmed it by making and detecting radio waves with sparks.

Figure 1.4 is the standard picture of a light wave far from its source, called a **plane wave**. Notice three things:

- $\vec{E}$, $\vec{B}$ and the direction of travel are all perpendicular to each other, so EM waves are **transverse**.
- E and B rise and fall together (they are in phase).
- Their sizes are locked together: $B = E/c$. Because c is huge, B is numerically tiny in teslas, but it carries exactly as much energy as E.

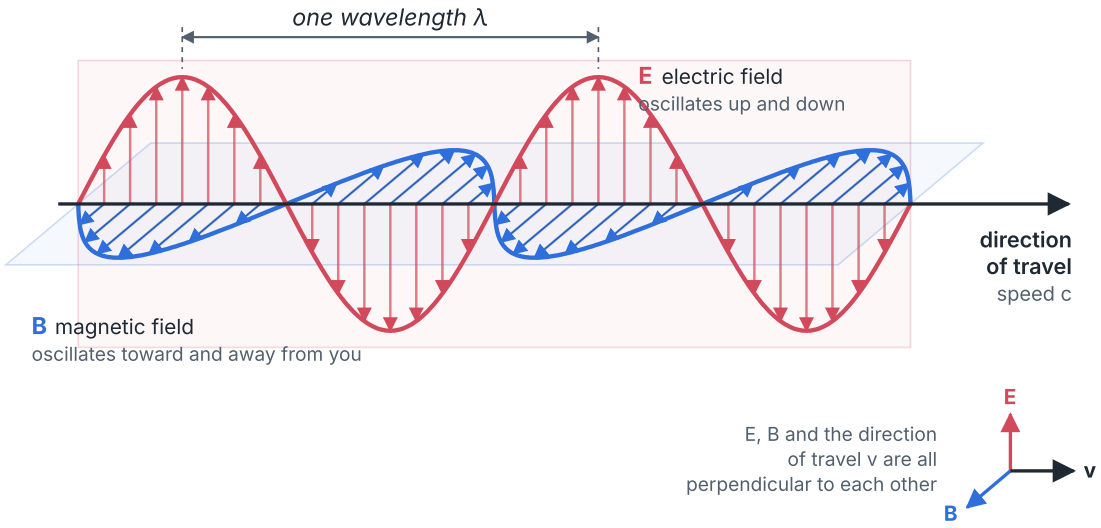


Figure 1.4 A snapshot of a plane EM wave. The red arrows show the electric field and the blue arrows the magnetic field at points along the beam. The whole pattern slides to the right at speed c. E and B are drawn the same size for clarity; in SI units $B_0 = E_0/c$.

YOU ASKED

"How does EM radiation actually work — what is physically waving?"

The electric and magnetic fields themselves are what wave: at each point the light passes through, the strength and direction of the electric field (and the magnetic field with it) swing back and forth, hundreds of trillions of times per second for visible light. Nothing material moves along the beam.

The field is a property of space itself, the "push a charge would feel here." So the most concrete way to picture light is to imagine a single electron sitting in the beam. As the wave passes, the electron is shoved up, then down, then up, at the wave's frequency. That shove is the light.

This is also the root of everything in your color rabbit hole. All interactions of light with matter start with the wave's electric field shaking electrons. What happens next depends on the frequency. Electrons that can resonate at that frequency absorb the energy, and electrons that can't simply re-radiate it or let it pass. That sorting by frequency is what makes objects colored (Chapters 5–7). One twist is coming: the energy turns out to be delivered in lumps called photons (Chapter 2).

MYTH VS. REALITY

Myth: The wavy line in drawings of light is the path the light travels, snaking up and down through space.

Reality: The wavy line is a graph. It plots the field's strength and direction at points along a perfectly straight beam. The light goes straight; only the field value at each point oscillates.

1.4 No medium needed

Sound is a pressure wave: air molecules bump into their neighbors. Remove the air and sound stops, which is why space is silent. Water waves need water, and waves on a guitar string need the string.

Light is different because the thing that waves, the field, exists in empty space. Sunlight crosses 150 million kilometers of near-vacuum and reaches you in about 8.3 minutes. No sound from the Sun ever arrives.

ANALOGY

In a stadium "wave," the pattern sweeps around the stadium but each fan just stands up and sits down in place. In light, the pattern sweeps forward at c, while the field at each point just swings back and forth. The difference: light needs no fans.

MYTH VS. REALITY

Myth: Space must be filled with some invisible substance that carries light, the way air carries sound.

Reality: Nineteenth-century physicists believed exactly this and called the substance the "luminiferous aether." In 1887 Michelson and Morley tried to detect Earth moving through it and found nothing. Light needs no medium; it is a self-sustaining pattern in the fields.

RABBIT HOLE **(optional) — Why is light's speed the same for everyone?**

Safe to skip. Maxwell's formula gives c from ε_0 and μ_0 alone, with no mention of who is measuring. In 1905 Einstein took this seriously: every observer, however they move, measures light in vacuum at the same speed. Keeping that true forces time and length themselves to stretch for fast-moving observers, which is special relativity. For this book, the takeaway is simpler: $c = 299{,}792{,}458$ m/s exactly in vacuum, and we round it to 3.00×10^8 m/s.

1.5 Describing a wave: wavelength, frequency, amplitude

Four numbers describe a simple EM wave (Figure 1.5):

- **Wavelength** λ: the distance from one crest to the next, in meters.
- **Frequency** f: how many crests pass a point each second, in hertz (Hz = 1/s).
- **Period** $T = 1/f$: the time for one full cycle at a fixed point.
- **Amplitude** E_0: the peak electric field strength, in V/m.

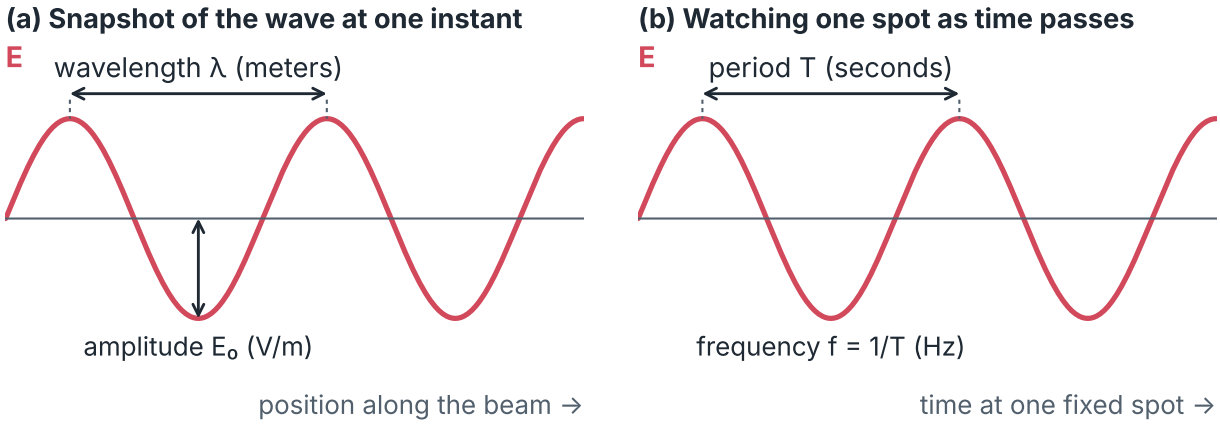


Figure 1.5 The same wave viewed two ways. (a) Freeze time and look along the beam: the repeat distance is the wavelength. (b) Stand at one spot and watch: the repeat time is the period, and the frequency is how many repeats happen per second.

In one period, a crest moves forward exactly one wavelength. Speed is distance over time, so $c = \lambda/T$, or:

$$c = \lambda f \tag{1.2}$$

where $c = 3.00 \times 10^8$ m/s in vacuum, λ is the wavelength (m) and f is the frequency (Hz).

Because c is fixed, wavelength and frequency are locked together: long wavelength means low frequency, and short wavelength means high frequency. In a material such as glass or water, light travels slower than c. The frequency stays the same and the wavelength shrinks (Chapter 6).

WORKED EXAMPLE 1.1 — Wi-Fi, FM radio and green light

Given: Wi-Fi at $f = 2.4$ GHz; an FM station at 100 MHz; green light with $\lambda = 550$ nm.

Find: the wavelength of the first two and the frequency and period of the third.

Think: All three are EM waves in air (essentially vacuum speed), so $c = \lambda f$ applies to each. Convert prefixes first: G = 10^9, M = 10^6, n = 10^{-9}.

$$\lambda_{\text{Wi-Fi}} = \frac{c}{f} = \frac{3.00 \times 10^8\ \text{m/s}}{2.4 \times 10^9\ \text{Hz}} = 0.125\ \text{m} = 12.5\ \text{cm}$$

$$\lambda_{\text{FM}} = \frac{3.00 \times 10^8\ \text{m/s}}{1.00 \times 10^8\ \text{Hz}} = 3.00\ \text{m}$$

$$f_{\text{green}} = \frac{c}{\lambda} = \frac{3.00 \times 10^8\ \text{m/s}}{550 \times 10^{-9}\ \text{m}} = 5.45 \times 10^{14}\ \text{Hz}, \qquad T = \frac{1}{f} = 1.8 \times 10^{-15}\ \text{s}$$

Answer: Wi-Fi waves are about 12.5 cm long (hand-sized), FM waves about 3 m (room-sized), and green light oscillates 5.45×10^{14} times per second, completing a cycle every 1.8 femtoseconds.

Those sizes explain hardware you already own. A quarter-wavelength FM antenna is about 75 cm long, the length of a classic car whip antenna. A quarter-wavelength Wi-Fi antenna is about 3 cm, small enough to hide inside a laptop lid.

1.6 Energy carried by the wave

Fields store energy. The energy per cubic meter in an electric field is proportional to E^2, much as a spring's energy is proportional to its stretch squared. In an EM wave, the energy is split equally between the electric and magnetic fields, and the whole package flows along at c.

The rate at which a wave delivers energy to a surface is its **intensity** I: power per unit area, in W/m². Averaged over a cycle:

$$I = \tfrac{1}{2}\, c\, \varepsilon_0\, E_0^2 \qquad \Longrightarrow \qquad I \propto E_0^2 \tag{1.3}$$

where E_0 is the electric field amplitude (V/m) and $\varepsilon_0 = 8.85 \times 10^{-12}\ \text{C}^2/(\text{N}\cdot\text{m}^2)$.

The squared dependence matters: double the amplitude and you get four times the intensity (Figure 1.6). In everyday terms, amplitude is **brightness**, while frequency is **color**. The two are independent.

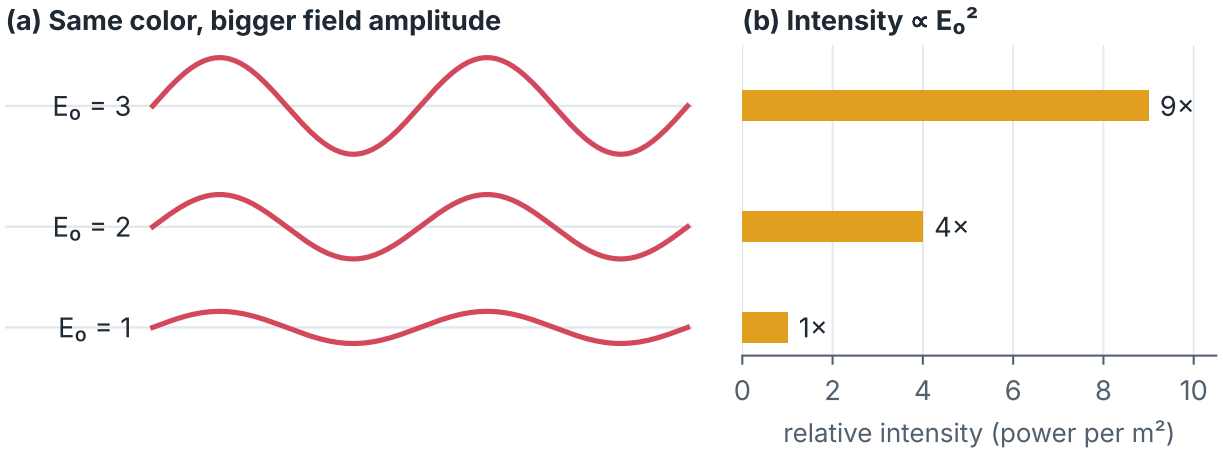


Figure 1.6 Same wavelength (same color), different amplitudes. Tripling the field amplitude makes the wave nine times as intense.

WORKED EXAMPLE 1.2 — How strong are the fields in sunlight?

Given: Bright noon sunlight at the ground delivers about $I = 1000\ \mathrm{W/m^2}$.

Find: the amplitudes E_0 and B_0, and the intensity if E_0 were doubled.

Think: Solve Eq. 1.3 for E_0, then use $B_0 = E_0/c$. Treating sunlight as a single wave is a simplification (it is a mix of many frequencies), but it gives the right size of field.

$$E_0 = \sqrt{\frac{2I}{c\,\varepsilon_0}} = \sqrt{\frac{2(1000)}{(3.00\times10^{8})(8.85\times10^{-12})}}\ \mathrm{V/m} = 868\ \mathrm{V/m}$$

$$B_0 = \frac{E_0}{c} = \frac{868}{3.00\times10^{8}}\ \mathrm{T} = 2.9\times10^{-6}\ \mathrm{T}$$

Doubling E_0 multiplies I by $2^2 = 4$, giving 4000 W/m².

Answer: $E_0 \approx 870$ V/m and $B_0 \approx 2.9\ \mu$T. A wave with twice the amplitude would deliver 4000 W/m².

Two more facts about energy, both developed later:

- **Spreading.** Light from a small source spreads over a sphere, so intensity falls as $1/r^2$ (the inverse-square law, Chapter 10).
- **Momentum.** EM waves also push on what they hit. Sunlight's push is tiny, about $I/c \approx 3\ \mu$N per square meter when absorbed, but it is enough to steer solar sails in space (Chapter 2).

MYTH VS. REALITY

Myth: Brighter light travels faster, or carries "stronger" color.

Reality: All EM waves in vacuum travel at *c*, dim or bright. Brightness is amplitude; color is frequency. Turning up a red laser makes more red, never blue.

1.7 Polarization: which way E points

The **polarization** of a wave is the direction its electric field oscillates. In Figure 1.4 the wave is **linearly polarized** vertically: E only swings up and down.

Sunlight and light bulbs are **unpolarized**. They are made by countless atoms radiating independently, so the E direction is a random, rapidly changing mix of all directions perpendicular to the beam.

A **polarizer** passes only one direction of E, called its transmission axis (Figure 1.7a). Common polarizing film is a plastic sheet whose long molecules have been stretched into parallel lines. Electrons can move easily along those chains, so the E component along the chains drives currents and is absorbed. The perpendicular component passes. An ideal polarizer passes half the intensity of unpolarized light. For light that is already polarized at angle θ to the axis, **Malus's law** gives:

$$I = I_0 \cos^2 \theta \tag{1.4}$$

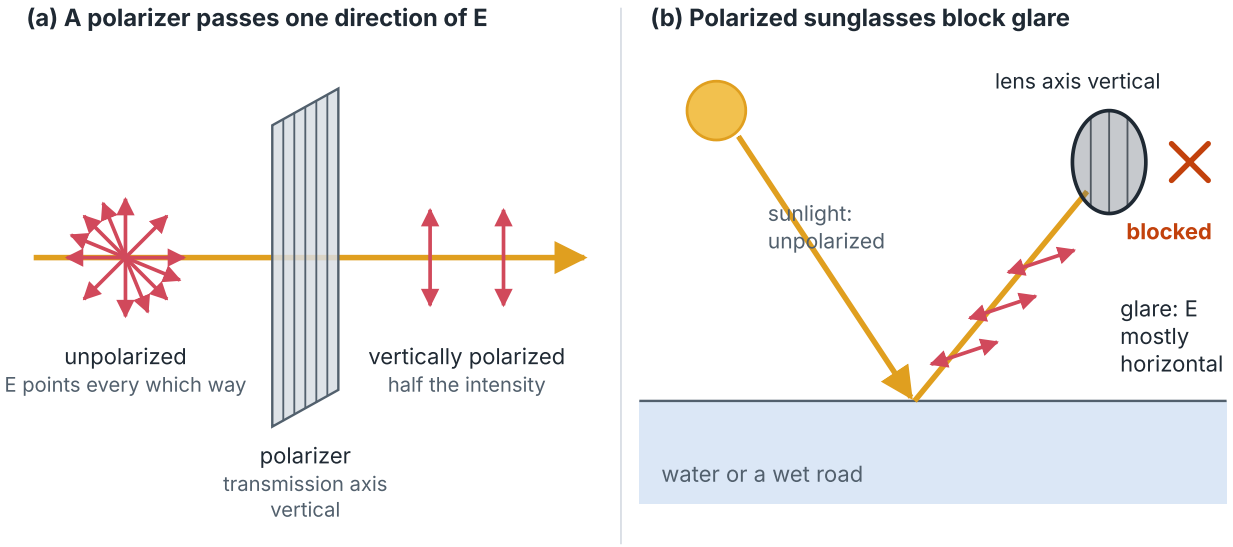


Figure 1.7 (a) A polarizer transmits only the E component along its axis. (b) Light reflected from water or a road is mostly polarized parallel to the surface (horizontal). Polarized sunglasses have a vertical axis, so they block much of that glare while passing other light.

That is exactly how polarized sunglasses work (Figure 1.7b). Light reflecting off a flat surface like water or a wet road comes off mostly polarized parallel to the surface. The reason is the shaken electrons again: electrons driven parallel to the surface radiate strongly into the reflected direction, while electrons driven the other way radiate only weakly in that direction. For water, reflected light is completely horizontally polarized when it arrives about 53° from the vertical (Brewster's angle). Lenses with a vertical transmission axis block that glare. Tilt your head 90° while wearing them and the glare comes back.

LCD screens use the same trick. Each pixel sits between two polarizers, and a layer of liquid crystal rotates the light's polarization by an amount set by a voltage. More rotation lets more light through the second polarizer. This is why an LCD monitor can look dark through polarized sunglasses when you tilt your head.

WORKED EXAMPLE 1.3 — Sunglasses and a monitor

Given: An LCD monitor emits linearly polarized light. You view it through polarized sunglasses and tilt your head so the lens axis makes angles of 30°, 60° and 90° with the screen's polarization.

Find: the fraction of the screen's light that reaches your eyes at each angle.

Think: The light is already polarized, so Malus's law applies directly (ignore the lens's tint).

$$\cos^2 30^\circ = 0.75, \qquad \cos^2 60^\circ = 0.25, \qquad \cos^2 90^\circ = 0$$

Answer: 75%, 25% and 0%. At 90° the screen goes black.

MATERIALS SCIENCE CONNECTION

A polarizer is an engineered **anisotropic** material: its optical properties depend on direction because its polymer chains are aligned. This is the same idea as a crystal whose properties differ along different directions, which you describe with Miller indices in ME2340. Many crystals, such as calcite and quartz, are naturally optically anisotropic, and geologists identify minerals by viewing thin slices between crossed polarizers.

Chapter summary

- An electric field is the force per unit charge at each point in space ($\vec{E} = \vec{F}/q$); moving charges also create a magnetic field $\vec{B}$.
- Accelerating charges launch kinks in their fields that travel at c. A steadily oscillating charge launches an EM wave; arriving waves shake charges in turn.
- Changing E makes B and changing B makes E, so the wave sustains itself and needs no medium. Its speed in vacuum is $c = 1/\sqrt{\mu_0 \varepsilon_0} = 3.00 \times 10^8$ m/s.
- What waves is the field itself. E, B and the direction of travel are mutually perpendicular, and $B = E/c$.
- $c = \lambda f$: Wi-Fi at 2.4 GHz is 12.5 cm long; green light at 550 nm oscillates 5.45×10^{14} times per second.
- Intensity $I = \frac{1}{2} c \varepsilon_0 E_0^2$, so it grows as amplitude squared. Amplitude sets brightness; frequency sets color.
- Polarization is the direction of E. Polarizers pass one direction ($I = I_0 \cos^2 \theta$); polarized sunglasses block horizontally polarized glare.

Check yourself

1. CONCEPT What physically oscillates in a light wave? Does anything material travel along the beam?
2. CONCEPT A charge moving at constant velocity does not radiate, but one moving back and forth does. Explain why, using the idea of field "news" spreading at speed c.
3. CONCEPT Explain why light from the Sun reaches Earth but sound from the Sun does not.
4. CALC Newer Wi-Fi also uses a 5.0 GHz band. What is its wavelength? How does it compare with 2.4 GHz Wi-Fi?
5. CALC Red light has a wavelength of 650 nm. Find its frequency and its period.
6. CALC The Sun is 1.50×10^{11} m away. How long does its light take to reach you?
7. CALC (a) If the electric field amplitude of a laser beam is tripled, by what factor does its intensity change? (b) If the intensity drops to one quarter, what happens to the amplitude?
8. CALC Above the atmosphere, sunlight has an intensity of about 1361 W/m². Find E_0 and B_0.
9. APPLY A car's FM antenna is designed to be a quarter wavelength long for 100 MHz. How long is it?
10. CALC Unpolarized light of intensity 800 W/m² passes through a polarizer and then through a second polarizer whose axis is at 45° to the first. What intensity emerges?
11. APPLY You are fishing on a sunny day and the lake surface is full of glare. Explain why polarized sunglasses help, and predict what happens if you tilt your head sideways by 90°.

Full solutions are in the Answer Key at the back of the book.

CHAPTER 2

Photons: Light as Packets of Energy

Shine the brightest red laser you can find on a sodium plate and nothing comes out. Shine a weak violet light on it and electrons fly off instantly. That stubborn fact forced physicists to admit that light, for all its wave behavior, delivers its energy in lumps.

THE BIG PICTURE

- Light's energy arrives in indivisible packets called photons, each carrying $E = hf$.
- Frequency (color) sets the energy of each photon; brightness sets how many arrive per second.
- The photoelectric effect proves it: below a threshold frequency, no electrons are ejected, however bright the light.
- A photon is an elementary quantum of the electromagnetic field: no mass, no parts, but real energy and momentum.
- Light travels like a wave and is absorbed like a particle. Both descriptions are true at once.

AFTER THIS CHAPTER YOU CAN

- Explain why bright red light cannot eject electrons that dim violet light can.
- Compute photon energies with $E = 1240\ \text{eV} \cdot \text{nm}/\lambda$.
- Use $KE_{\max} = hf - \phi$ to predict photoelectron energies and thresholds.
- Count photons per second from a source of known power.
- Say what a photon is, and what it is not.
- Choose between the wave picture and the photon picture for a given situation.

2.1 What the wave picture predicts

Chapter 1 built a picture of light as a smooth wave whose intensity grows with amplitude squared. That picture explains a great deal, but take it literally and it makes three predictions about light hitting a metal:

- **Any color should work if it is bright enough.** A stronger field shakes electrons harder, so even red light, turned up enough, should eventually tear electrons loose.
- **Brighter light should eject faster electrons.** More energy per second should mean more energy per electron.
- **Dim light should take time.** Energy spread smoothly over the surface should need a while to accumulate in any one electron.

All three predictions are wrong. The experiment that shows it is the photoelectric effect. (Two other failures of the pure-wave picture, the color of glowing objects and the sharp lines in atomic spectra, appear in Chapters 9 and 5.)

2.2 The photoelectric effect

YOU ASKED

“What is the photoelectric effect?”

The photoelectric effect is the ejection of electrons from a material, usually a metal, when light shines on it. Its importance is not that it happens, but how: whether electrons come out depends on the light’s color, not its brightness.

Below a certain threshold frequency, no electrons emerge at all, no matter how intense the light. Above it, electrons emerge immediately, even in very dim light. The only sensible explanation is that light delivers energy in packets (photons), and each electron is freed by absorbing a single packet. Einstein proposed this in 1905, and it won him the Nobel Prize. The rest of this section shows the experiment and the reasoning.

Heinrich Hertz noticed the effect in 1887. Figure 2.1 shows the modern version of the experiment. Light enters a vacuum tube and strikes a clean metal plate. Ejected electrons, called **photoelectrons**, cross to a collector, and an ammeter counts how many arrive per second as a current.

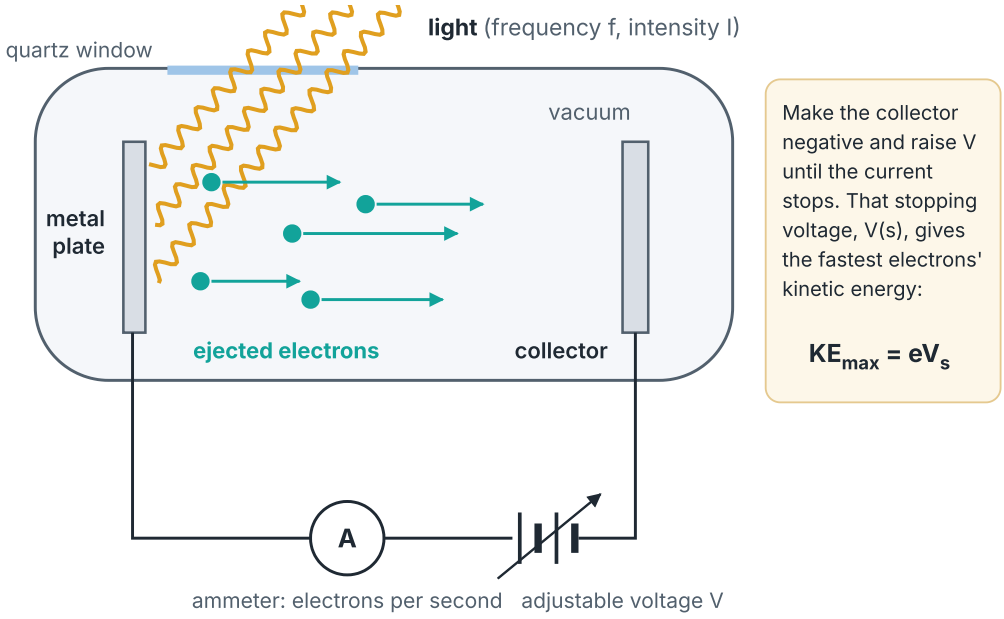


Figure 2.1 The photoelectric experiment. Light ejects electrons from the plate; the ammeter measures how many reach the collector each second. Reversing the voltage and raising it until the current stops gives the stopping voltage V_s, which measures the kinetic energy of the fastest electrons: $KE_{\max} = eV_s$.

To measure electron energies, make the collector negative so it repels the electrons. Raise the voltage until even the fastest electrons are turned back and the current drops to zero. That **stopping voltage** V_s gives the maximum kinetic energy directly: an electron crossing a voltage V_s loses energy eV_s, so $KE_{\max} = eV_s$. In electron volts, the numbers are the same: a 0.82 V stopping voltage means 0.82 eV electrons.

Table 2.1 compares what the wave picture predicts with what is actually observed.

Table 2.1 The photoelectric effect: wave prediction vs. observation.

Question	Pure-wave prediction	What actually happens
Can any color eject electrons?	Yes, if bright enough	No. Below a threshold frequency f_0, none come out at any brightness.
What sets the electrons' maximum energy?	Brightness	Frequency only. Brighter light gives the same $KE_{\max}$.
What does brighter light change?	Electron energy	The number of electrons per second (the current).
Is there a delay in dim light?	Yes, while energy builds up	No. Emission begins essentially instantly.

2.3 Einstein's answer: light comes in packets

Einstein's idea was simple: the energy in light of frequency f comes in indivisible packets, now called **photons**, each carrying

$$E_{\text{photon}} = hf = \frac{hc}{\lambda} \tag{2.1}$$

where $h = 6.626 \times 10^{-34}\ \text{J}\cdot\text{s}$ (Planck's constant), f is the frequency and λ the wavelength.

An electron absorbs one whole photon or nothing. To escape, it must pay an energy toll to leave the metal, called the **work function** ϕ. Whatever is left becomes kinetic energy (Figure 2.2). The least tightly held electrons keep the most:

$$KE_{\max} = hf - \phi \tag{2.2}$$

If $hf < \phi$, the electron cannot escape. The **threshold frequency** is $f_0 = \phi/h$.

This one idea explains every row of Table 2.1:

- **Threshold.** A red photon has too little energy, and electrons cannot save up energy from several photons, because an excited electron loses its extra energy to the metal almost immediately. More red photons just mean more failed attempts. (Only extremely intense pulsed lasers can make one electron absorb two photons at once, a rare exception.)
- **Energy set by frequency.** Each electron gets one photon's worth, hf, regardless of how many other photons arrive.
- **Brightness sets the count.** Brighter light means more photons, so more electrons, each with the same $KE_{\max}$.
- **No delay.** The first photon absorbed can free an electron at once.

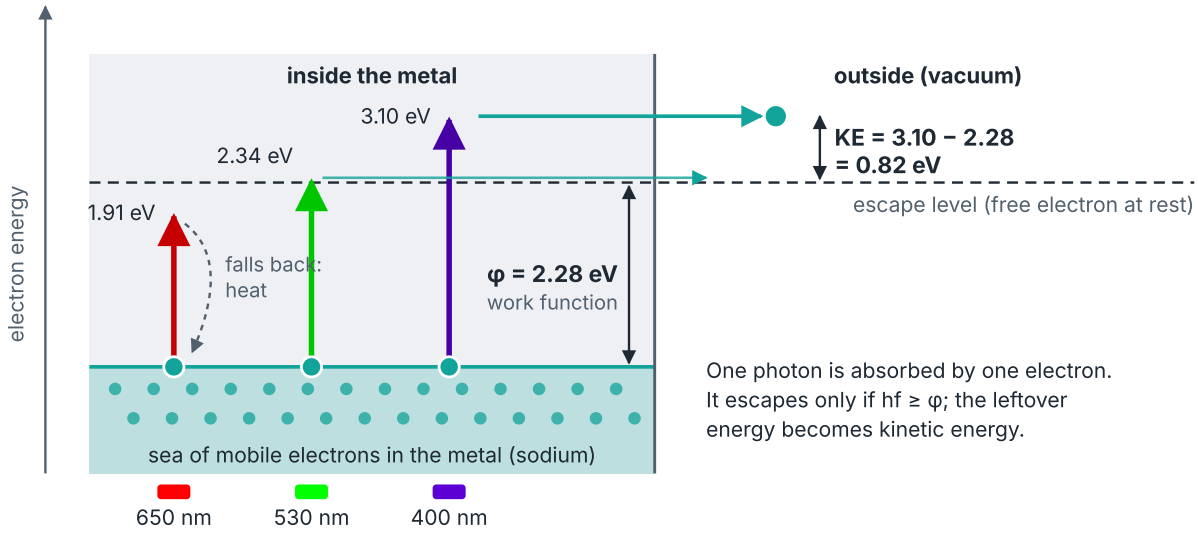


Figure 2.2 The photoelectric effect as an energy budget, for sodium ($\phi = 2.28$ eV). Each arrow is the energy one photon gives to one electron. Red photons fall short of the escape level, so their energy ends up as heat; violet photons clear it with 0.82 eV to spare.

ANALOGY

Each electron is a vending machine that accepts only single coins worth at least 25 cents. A million pennies (red photons) buy nothing, because the machine takes one coin at a time. One quarter (a violet photon) works instantly, and the change is the electron's kinetic energy.

Figure 2.3 plots Eq. 2.2 for two metals. Each is a straight line with slope h; only the starting point differs, set by the work function. Robert Millikan measured exactly these lines in 1916, confirming Einstein's equation (even though he doubted the photon idea). Einstein received the 1921 Nobel Prize in Physics for the law of the photoelectric effect, not for relativity.

Table 2.2 Work functions of some metals (clean surfaces; values vary with surface condition and crystal face).

Metal	Work function ϕ (eV)	Threshold wavelength $\lambda_0 = 1240/\phi$ (nm)	Visible light ejects electrons?
Cesium	2.1	590	Yes, except orange and red
Sodium	2.28	544	Yes, below 544 nm (green to violet)
Potassium	2.3	540	Yes, below 540 nm (green to violet)
Silver	4.26–4.74	291–262	No (needs UV)
Copper	4.7	264	No (needs UV)
Gold	5.1	243	No (needs UV)
Platinum	6.35	195	No (needs far UV)

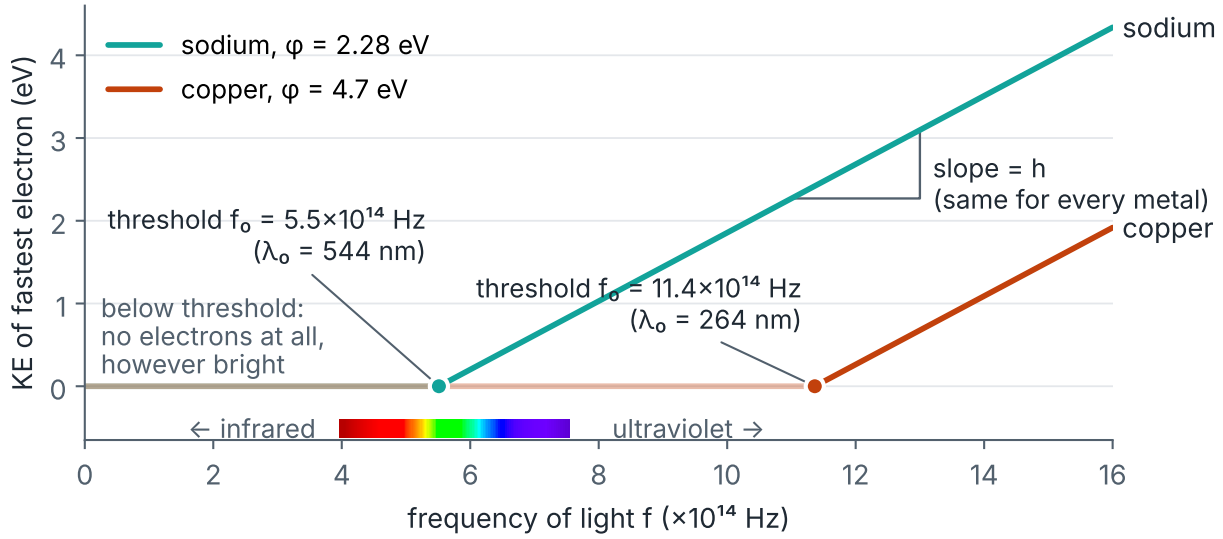


Figure 2.3 Maximum photoelectron energy vs. light frequency, from $KE_{\max} = hf - \phi$. Below the threshold f_0 nothing is ejected; above it the line rises with slope h for every metal. Sodium responds to green, blue and violet light; copper needs ultraviolet. Work functions from HyperPhysics.

MYTH VS. REALITY

Myth: Brighter light knocks electrons out harder, so they come out faster.

Reality: Brighter light of the same color knocks out more electrons, each with the same maximum energy. Only a higher frequency makes faster electrons.

MATERIALS SCIENCE CONNECTION

The work function is a surface property of a material, like hardness, and it depends on which crystal plane is exposed. For silver, it ranges from about 4.26 eV for a polycrystalline surface to about 4.74 eV for the (111) face, the closely packed plane in FCC silver. Engineers exploit low work functions: the light-sensitive surfaces in photomultiplier tubes and night-vision devices use cesium and alkali-metal compounds so that visible photons can free electrons. Solar cells and camera sensors use a close cousin of the effect inside semiconductors, where the photon must exceed a band gap instead of a work function (Chapter 5).

2.4 Photon energy in practice: the 1240 shortcut

Joules are awkward for single photons, so we use the **electron volt** (eV): the energy an electron gains crossing 1 volt, $1\ \text{eV} = 1.602 \times 10^{-19}$ J. Combining constants gives $hc = 1240\ \text{eV} \cdot \text{nm}$, and Eq. 2.1 becomes a shortcut worth memorizing:

$$E\ (\text{eV}) = \frac{1240}{\lambda\ (\text{nm})} \tag{2.3}$$

Visible photons run from about 1.8 eV (red, 700 nm) to 3.1 eV (violet, 400 nm). That is the same range as the energies that hold electrons in molecules and solids, which is why visible light can drive chemistry in your retina (Chapter 3) and give materials their colors (Chapter 7).

WORKED EXAMPLE 2.1 — Photon energies across the spectrum

Given: a green laser (532 nm), Wi-Fi (2.4 GHz) and an X-ray of wavelength 0.10 nm.

Find: the energy of one photon of each, in eV.

Think: Use the 1240 shortcut when you know λ in nm. For Wi-Fi you know f, so use $E = hf$ with $h = 4.136 \times 10^{-15}$ eV · s.

$$E_{\text{green}} = \frac{1240 \text{ eV} \cdot \text{nm}}{532 \text{ nm}} = 2.33 \text{ eV} \; (= 3.73 \times 10^{-19} \text{ J})$$

$$E_{\text{Wi-Fi}} = (4.136 \times 10^{-15} \text{ eV} \cdot \text{s})(2.4 \times 10^{9} \text{ Hz}) = 9.9 \times 10^{-6} \text{ eV}$$

$$E_{\text{X-ray}} = \frac{1240 \text{ eV} \cdot \text{nm}}{0.10 \text{ nm}} = 12{,}400 \text{ eV} = 12.4 \text{ keV}$$

Answer: 2.33 eV, about 10^{-5} eV and 12.4 keV. One X-ray photon carries over a billion times the energy of one Wi-Fi photon.

WORKED EXAMPLE 2.2 — Violet vs. red light on sodium

Given: sodium, $\phi = 2.28$ eV. Violet light at 400 nm; red light at 650 nm.

Find: $KE_{\max}$ and the stopping voltage for each, and sodium's threshold wavelength.

Think: Find each photon energy, then subtract the work function. A negative result means no emission.

$$E_{\text{violet}} = \frac{1240}{400} = 3.10 \text{ eV} \;\Rightarrow\; KE_{\max} = 3.10 - 2.28 = 0.82 \text{ eV}, \quad V_s = 0.82 \text{ V}$$

$$E_{\text{red}} = \frac{1240}{650} = 1.91 \text{ eV} < 2.28 \text{ eV} \;\Rightarrow\; \text{no electrons}$$

$$\lambda_0 = \frac{1240 \text{ eV} \cdot \text{nm}}{2.28 \text{ eV}} = 544 \text{ nm}$$

Answer: violet light ejects electrons with up to 0.82 eV (stopped by 0.82 V); red light ejects none at any brightness. Sodium's threshold is 544 nm, in the green.

2.5 Brightness is photon count

If each photon carries hf, then a beam of power P must deliver photons at a rate $\dot{N}$ (photons per second) such that

$$P = \dot{N}\, hf \qquad \Longrightarrow \qquad \dot{N} = \frac{P}{hf} = \frac{P\lambda}{hc} \tag{2.4}$$

So two beams of equal power but different color contain different numbers of photons (Figure 2.4). The red beam carries more photons, each with less energy. On sodium, only the violet beam's photons can free electrons.

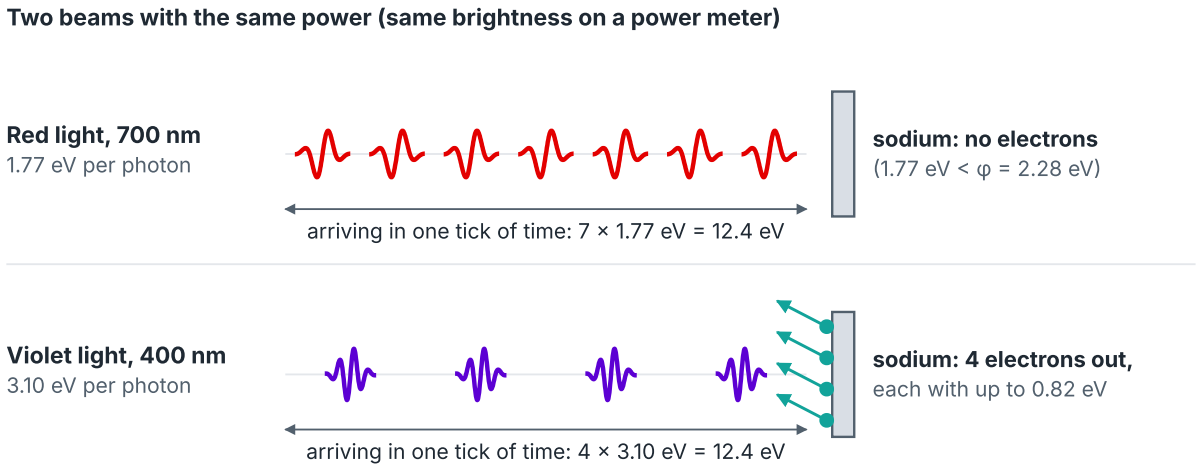


Figure 2.4 Same brightness, different color. Both beams deliver 12.4 eV per tick of time, but red does it with seven low-energy photons and violet with four high-energy ones. Photon energy, not total energy, decides whether an electron can escape.

YOU ASKED

"Does a bigger amplitude of a wave mean more photons?"

Yes: for light of a given color, a bigger amplitude means more photons per second, while each photon keeps exactly the same energy, hf.

The precise link is with amplitude squared. Intensity is proportional to E_0^2 (Chapter 1), and intensity equals photon arrival rate times hf. So at fixed frequency, the photon count is proportional to E_0^2: double the amplitude and you get four times as many photons per second.

Two cautions. First, comparing different colors at equal amplitude, the lower-frequency beam has more photons. Second, for a single photon, the wave's amplitude squared at a spot tells you the probability of detecting that photon there (Section 2.7). Either way, amplitude never changes the size of a photon. Only frequency does.

KEY IDEA

Frequency sets the energy of each photon. Amplitude (brightness) sets how many photons arrive per second. Whether light can do something to a single atom or electron, such as eject it, break a bond or trigger your retina, depends on the energy per photon, not on the brightness.

WORKED EXAMPLE 2.3 — Photons from a laser pointer

Given: a 5.0 mW green laser pointer (the US limit for laser pointers) at 532 nm.

Find: the number of photons it emits per second.

Think: Power is photons per second times energy per photon (Eq. 2.4). Work in joules: $E = 2.33\ \text{eV} \times 1.602 \times 10^{-19}\ \text{J/eV}$.

$$E_{\text{photon}} = 3.73 \times 10^{-19}\ \text{J}, \qquad \dot{N} = \frac{5.0 \times 10^{-3}\ \text{J/s}}{3.73 \times 10^{-19}\ \text{J}} = 1.3 \times 10^{16}\ \text{photons/s}$$

Answer: about 1.3×10^{16} photons per second. A red 650 nm pointer of the same power would emit about 20% more, since each red photon carries less energy.

Table 2.3 shows why you never notice the graininess of light: everyday sources emit staggering numbers of photons. In a very dark room your eye comes close to the limit, since a dark-adapted rod cell in your retina can respond to a single photon.

Table 2.3 Photon rates from everyday sources ($\dot{N} = P/hf$; powers are typical round numbers, and sunlight assumes an average photon energy of about 1.5 eV).

Source	Power	Photon energy	Photons per second
FM radio station (100 MHz)	50 kW	4×10^{-7} eV	$\sim 8 \times 10^{29}$
Wi-Fi router (2.4 GHz)	0.1 W	1×10^{-5} eV	$\sim 6 \times 10^{22}$
Sunlight on 1 cm² (ground, noon)	0.1 W	~ 1.5 eV (average)	$\sim 4 \times 10^{17}$
Green laser pointer (532 nm)	5 mW	2.33 eV	1.3×10^{16}

2.6 What is a photon, really?

YOU ASKED

"What is a photon made up of?"

Nothing smaller: as far as every experiment can tell, a photon is an elementary particle with no internal parts. It is the smallest possible unit, or **quantum**, of energy in the electromagnetic field at a given frequency.

The best way to think of it: a photon is "made of" the electromagnetic field in the way a ripple is made of the pond. The field can only gain or lose energy at frequency f in steps of exactly hf, and one such step is one photon. That is why photons can be created and destroyed. An atom does not store photons inside it; it creates one when an electron drops to a lower energy (Chapter 5), and a photon that is absorbed simply ceases to exist, its energy handed to the absorber.

A photon's properties are few and exact:

- **No mass and no charge.** It can never be at rest; in vacuum it always moves at c.
- **Energy** $E = hf$, set entirely by its frequency.
- **Momentum** $p = h/\lambda = E/c$. Light pushes on what it hits; solar sails use this push for propulsion in space.
- **Spin**, a built-in angular momentum, connected to the light's polarization (Chapter 1).

$$p = \frac{h}{\lambda} = \frac{E}{c} \tag{2.5}$$

MYTH VS. REALITY

Myth: A photon is a tiny ball of light that flies along a definite path, like a microscopic marble.

Reality: A photon has no size or surface, and between emission and detection it does not follow a single trajectory. It spreads out like a wave, can pass through two slits at once (Section 2.7), and then deposits all of its energy at one point when absorbed. "Particle" here means "indivisible lump of energy and momentum," not "little marble."

2.7 Wave and particle: both true

So is light a wave or a stream of particles? It is one thing, a quantum object, that shows wave behavior in how it travels and particle behavior in how it delivers energy. Physicists call this **wave–particle duality**. Electrons and atoms behave the same way (Chapters 4 and 8).

KEY IDEA

Light travels like a wave and arrives like a particle. Use the wave picture for where light goes; use the photon picture for what light does when it is absorbed or emitted.

Table 2.4 Which picture to use.

Use the wave picture for...	Use the photon picture for...
Interference and diffraction (Chapter 8)	The photoelectric effect and solar cells
Reflection, refraction and lenses (Chapter 6)	Absorption and emission by atoms (Chapter 5)
Polarization (Chapter 1)	Whether light can ionize atoms or break bonds (Chapter 10)
Antennas, radio and intensity patterns	Counting detections: cameras, eyes, Geiger counters

RABBIT HOLE **(optional) — Sending photons through two slits one at a time**

Safe to skip. Light passing through two narrow slits makes a pattern of bright and dark stripes on a screen, a classic sign of waves interfering (Chapter 8). Now dim the source until photons arrive one at a time, and replace the screen with a sensitive camera.

Each photon lands at a single point, a particle-like event. Yet the points are not random. As thousands accumulate, the stripes appear (Figure 2.5), even though no two photons were ever in the apparatus together. Each photon's wave passes through both slits and interferes with itself; the wave's intensity at each spot is the probability that the photon is detected there. If you set up detectors to find out which slit each photon used, the stripes disappear.

G. I. Taylor did an early version in 1909 with light so dim that one exposure took months. Modern experiments with true single-photon sources show exactly the buildup in Figure 2.5.

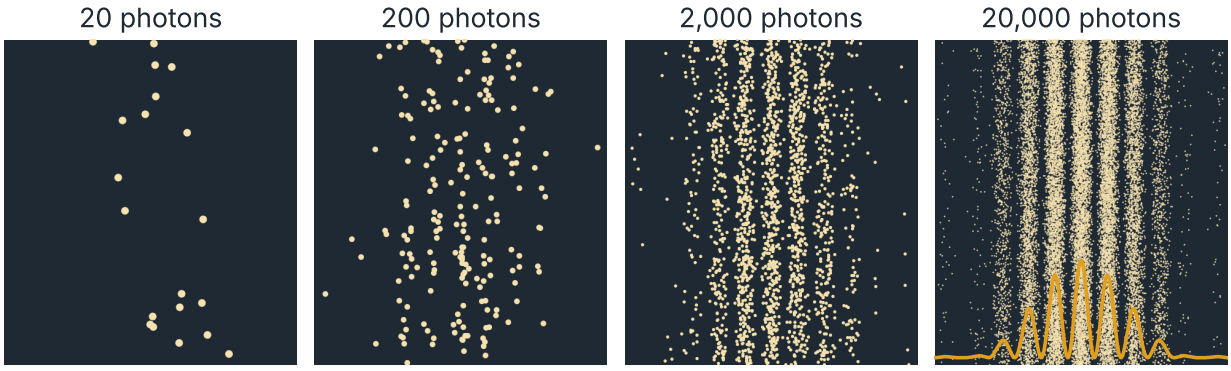


Figure 2.5 Simulated arrivals of single photons behind two slits. Each dot is one photon detected at one point (particle behavior). The striped pattern emerges only from many photons and matches the wave prediction (amber curve). The simulation draws each landing spot at random with probability given by the wave intensity, which reproduces what real single-photon experiments record.

MYTH VS. REALITY

Myth: Light switches between being a wave and being a particle, depending on whether someone is watching.

Reality: Light is always the same quantum object. "Wave" and "particle" are two simplified models, each matching part of its behavior. What a measurement does is force the photon to deposit its energy somewhere definite.

Chapter summary

- Light delivers energy in photons of $E = hf = hc/\lambda$. In practical units, $E\,(\text{eV}) = 1240/\lambda\,(\text{nm})$; visible photons carry 1.8–3.1 eV.
- Photoelectric effect: one photon frees one electron if $hf \geq \phi$, giving $KE_{\max} = hf - \phi$. Below the threshold $f_0 = \phi/h$, nothing happens at any brightness.
- The stopping voltage measures $KE_{\max} = eV_s$. Plots of $KE_{\max}$ vs. f are straight lines with slope h for every metal.

- Brightness is photon rate: $P = \dot{N} h f$. At fixed color, photon count $\propto$ amplitude²; a 5 mW green laser emits 1.3×10^{16} photons/s.
- A photon is an elementary quantum of the EM field: no mass, no charge, no parts; it has energy hf, momentum h/λ and spin.
- Light travels like a wave and is absorbed like a particle. Single photons build up interference patterns one dot at a time.

Check yourself

1. CONCEPT Explain why a very bright red light cannot eject electrons from sodium, while a dim violet light can.
2. CALC Find the energy of one photon of red light (650 nm) and of blue light (450 nm), in eV and in joules.
3. CALC Cesium has $\phi = 2.1$ eV. (a) What is its threshold wavelength? (b) Will 650 nm red light eject electrons? (c) Will 532 nm green light? If so, with what $KE_{\max}$?
4. CALC A germicidal UV lamp emits at 254 nm. It shines on copper ($\phi = 4.7$ eV). Find $KE_{\max}$ and the stopping voltage.
5. CONCEPT You double the intensity of light shining on a metal without changing its frequency (which is above threshold). What happens to (a) the number of electrons ejected per second and (b) their maximum kinetic energy?
6. CALC A 60 W-equivalent LED bulb gives off, say, about 2.5 W of visible light. Treating it all as 550 nm light, roughly how many visible photons does it emit per second?
7. CALC How many 2.4 GHz Wi-Fi photons have the same total energy as one 532 nm green photon? Why can't that many Wi-Fi photons do what one green photon does to a molecule?
8. CONCEPT Does a photon have mass? Can it be at rest? What is it made of?
9. CALC (a) Find the momentum of one 532 nm photon. (b) What force does a 5.0 mW laser beam exert on a surface that absorbs it completely? (Hint: force = momentum delivered per second = P/c.)
10. APPLY Silicon solar cells need a photon energy of at least about 1.1 eV (silicon's band gap) to free an electron. What is the longest wavelength they can use? Which part of sunlight is wasted?
11. CONCEPT In the one-photon-at-a-time double-slit experiment, what tells you where a photon is likely to land? Why can't you see the stripes after only 20 photons?

Full solutions are in the Answer Key at the back of the book.

CHAPTER 3

The Electromagnetic Spectrum

Radio, microwaves, the warmth of a campfire, the color red, sunburn and a dental X-ray feel like completely different things. They are one thing: the wave from Chapter 1, made of the photons from Chapter 2, at wavelengths that span more than eighteen powers of ten.

THE BIG PICTURE

- Every band is the same phenomenon: EM waves traveling at c, delivered as photons. Only the wavelength differs, and with it the frequency and photon energy.
- Each band is made by a different kind of accelerating charge, from electrons sloshing in antennas to nuclei rearranging.
- Photon energy decides what a band does to matter: low energies move charges and make heat, a few eV rearranges electrons (color and chemistry), and above about 10 eV photons ionize.
- The atmosphere is transparent only in a few windows, mainly visible light and radio.
- Visible light is a sliver less than one octave wide, and there are good physical reasons our eyes use it.

AFTER THIS CHAPTER YOU CAN

- Put the bands in order and give rough wavelengths, frequencies and photon energies for each.
- Convert between λ, f and E for any band.
- Name typical sources and uses of each band.
- Predict from photon energy what a band does to matter.
- Explain which bands reach the ground and why.
- Explain why human eyes see 400–700 nm.

3.1 One phenomenon, many names

YOU ASKED

"What is the EM spectrum — are radio waves, light and X-rays really the same thing?"

Yes. Radio waves, visible light and X-rays are exactly the same phenomenon: oscillating electric and magnetic fields traveling at c, delivered in photons of energy hf. The electromagnetic spectrum is simply the full range of these waves, arranged by wavelength.

What differs is only the wavelength, and therefore the frequency and the energy per photon. Those numbers are not minor details, though. They decide how a wave is made, what can detect it, and what it does when it hits matter. A radio wave is meters long and gently drives electrons in an antenna. An X-ray is smaller than an atom and carries enough energy per photon to knock electrons out of it. Same physics, very different behavior, the way a bass note and a dog whistle are both sound.

Figure 3.1 lays out the whole spectrum. Each band can be labeled three equivalent ways, linked by the two equations you already know:

$$c = \lambda f \qquad E = hf = \frac{1240\ \text{eV} \cdot \text{nm}}{\lambda} \tag{3.1}$$

Radio engineers usually quote frequency, optics people quote wavelength, and X-ray and nuclear physicists quote photon energy. They are all describing the same axis.

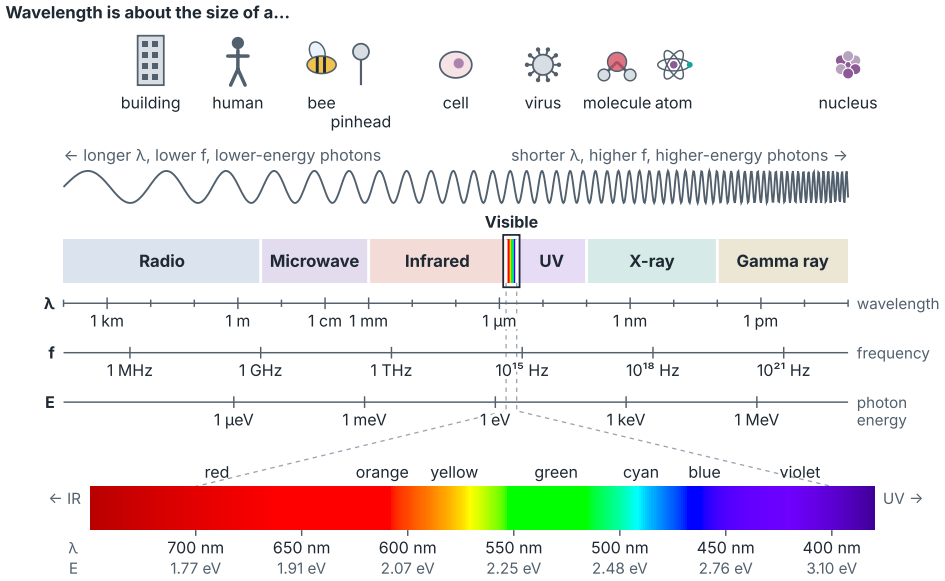


Figure 3.1 The electromagnetic spectrum. Each step to the right is a factor of 10 shorter in wavelength, 10 higher in frequency and 10 higher in photon energy. Icons show objects about one wavelength in size. The visible band, barely a sliver on this scale, is expanded at the bottom. Band boundaries are conventions, not sharp physical edges.

ANALOGY

Think of a piano keyboard stretched for 60 octaves instead of 7. Every key plays the same kind of thing, a vibration, just faster or slower. Visible light is less than one octave of that keyboard: 700 nm to 400 nm is a frequency ratio of only 1.75.

WORKED EXAMPLE 3.1 — How big is the visible slice?

Given: visible light spans 400–700 nm. Figure 3.1 spans wavelengths from 10^{4} m to 10^{-14} m.

Find: what fraction of Figure 3.1's width is visible, and how many octaves (factors of 2) the figure covers.

Think: On a log scale, width is measured in powers of ten (decades), so use logarithms.

$$\log_{10}\left(\frac{700}{400}\right) = 0.24 \text{ decade}, \qquad \frac{0.24}{18 \text{ decades}} \approx 1.3\%$$

$$\text{octaves} = \log_2\left(10^{18}\right) = 18 \times 3.32 \approx 60$$

Answer: visible light is about 1% of the figure's width, under one octave out of roughly 60.

3.2 Where each band comes from

Chapter 1's rule still holds everywhere: accelerating charges make EM waves. What changes across the spectrum is which charge is accelerating, and how fast it oscillates (Figure 3.2). Bigger, slower systems make long waves; smaller, tighter systems make short, high-energy ones.

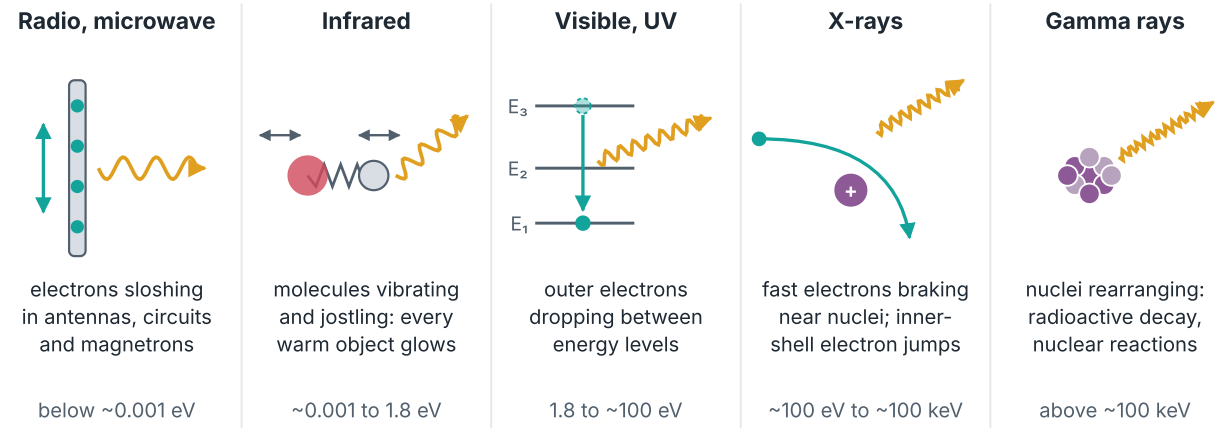


Figure 3.2 Every band is made by accelerating charge, at a different scale. Typical photon energies are listed at the bottom of each panel. The details of each mechanism come later: atoms in Chapter 5, thermal glow in Chapter 9, X-ray tubes in Chapter 11, nuclei in Chapter 10.

MYTH VS. REALITY

Myth: X-rays and gamma rays are different kinds of radiation, and gamma rays always have more energy.

Reality: Both are high-energy photons, and their energy ranges overlap. The names usually describe the source: X-rays come from electrons (braking or jumping between inner shells), gamma rays from atomic nuclei. A 150 keV medical X-ray photon has more energy than some gamma rays from nuclear decay.

3.3 A tour of the spectrum

Table 3.1 is the one-page summary of the spectrum, from longest to shortest wavelength. A few highlights:

- **Radio and microwaves** carry nearly all wireless communication. Wi-Fi (2.4 and 5 GHz), Bluetooth, phones and GPS are microwaves; a microwave oven runs at 2.45 GHz, in the same band as Wi-Fi.
- **Infrared** is emitted by everything warm. Your body glows most strongly near 10 µm (Chapter 9), and modeling the Sun as a 5800 K glowing body, about half of its power is infrared.
- **Visible light** is where the Sun is brightest and where molecules have their electronic transitions, which is why it carries color (Chapter 7).
- **Ultraviolet** photons carry enough energy to break chemical bonds, which causes sunburn and is why UV-C lamps sterilize surfaces.
- **X-rays and gamma rays** ionize atoms and penetrate deeply (Chapters 10 and 11).

Table 3.1 The electromagnetic spectrum at a glance. Boundaries are approximate conventions; photon energy $E = 1240\ \text{eV}\cdot\text{nm}/\lambda$.

Band	Wavelength	Frequency	Photon energy	Sources	Uses	What it does to matter
Radio	> 30 cm	< 1 GHz	< 4 μeV	Lightning, stars, pulsars; antennas and AC circuits	AM (535–1705 kHz) and FM (88–108 MHz) radio, TV, MRI, radio astronomy	Drives currents in conductors; passes through walls and bodies; long waves reflect off the ionosphere
Microwave	1 mm – 30 cm	1–300 GHz	4 μeV – 1.2 meV	Cosmic microwave background; magnetrons, radar, phone and Wi-Fi chips	Ovens (2.45 GHz), Wi-Fi, Bluetooth, cell phones, GPS, radar, satellite links	Twists and jostles polar molecules like water, making heat; reflected by metal
Infrared	700 nm – 1 mm	0.3–430 THz	1.2 meV – 1.8 eV	Every warm object (you peak near 10 μm); the Sun; heaters; IR LEDs	Remote controls (~940 nm), thermal cameras (8–14 μm), fiber-optic links, heat lamps	Vibrates molecules and bonds, making heat; near-IR can still drive some electronic processes (solar cells)
Visible	400–700 nm	430–750 THz	1.8–3.1 eV	Sun, flames, hot filaments, LEDs, lasers	Vision, photography, displays, photosynthesis	Lifts outer electrons; selective absorption gives color; drives vision and photosynthesis
Ultraviolet	10–400 nm	750 THz – 30 PHz	3.1–124 eV	Sun, welding arcs, mercury and black-light lamps, UV LEDs	Sterilizing (254 nm), curing resins, fluorescence, chip lithography	Breaks chemical bonds (sunburn, DNA damage); triggers fluorescence; above ~10 eV ionizes
X-ray	0.01–10 nm	3×10^{16}–3×10^{19} Hz	124 eV – 124 keV	Million-degree gas, the Sun's corona; X-ray tubes, synchrotrons	Medical and dental imaging, CT, airport scanners, crystallography	Ionizes; knocks out inner-shell electrons; passes soft tissue, stopped by dense high-Z matter
Gamma ray	< 0.01 nm	> 3×10^{19} Hz	> ~100 keV	Radioactive nuclei (Co-60: 1.17 and 1.33 MeV), nuclear reactions, supernovae	Cancer radiotherapy, sterilizing food and equipment, PET scans	Ionizes; very penetrating; needs thick, dense shielding (Chapter 10)

WORKED EXAMPLE 3.2 — Name that band

Given: (a) a microwave oven at 2.45 GHz; (b) the infrared glow of your skin at about 10 μm; (c) a dental X-ray photon of 60 keV.

Find: the missing wavelength or frequency, and the photon energy, for each; check the band in Table 3.1.

Think: Use $c = \lambda f$, then $E = hf$ (with $h = 4.136\times10^{-15}$ eV·s) or $E = 1240/\lambda$.

(a) $\lambda = \dfrac{3.00\times10^{8}}{2.45\times10^{9}} = 12.2\ \text{cm}$, $E = (4.136\times10^{-15})(2.45\times10^{9}) = 1.0\times10^{-5}\ \text{eV}$: microwave.

(b) $f = \dfrac{3.00\times10^{8}}{10\times10^{-6}} = 3.0\times10^{13}\ \text{Hz}$, $E = \dfrac{1240}{10{,}000\ \text{nm}} = 0.124\ \text{eV}$: infrared.

(c) $\lambda = \dfrac{1240\ \text{eV}\cdot\text{nm}}{60{,}000\ \text{eV}} = 0.021\ \text{nm}$, $f = \dfrac{c}{\lambda} = 1.5 \times 10^{19}$ Hz: X-ray.

Answer: 12.2 cm and 10^{-5} eV (microwave); 3.0×10^{13} Hz and 0.12 eV (infrared); 0.021 nm and 1.5×10^{19} Hz (X-ray, close to the gamma boundary, but called an X-ray because electrons in a tube make it).

3.4 Photon energy decides what light does to matter

Matter absorbs a photon efficiently when the photon's energy matches a step that something in the material can take. That step might be a molecule spinning faster, a bond vibrating harder, an electron jumping to a higher level or an electron leaving altogether. If nothing matches, the light passes through, reflects or scatters instead (Chapter 6). Figure 3.3 lines up those steps against the spectrum.

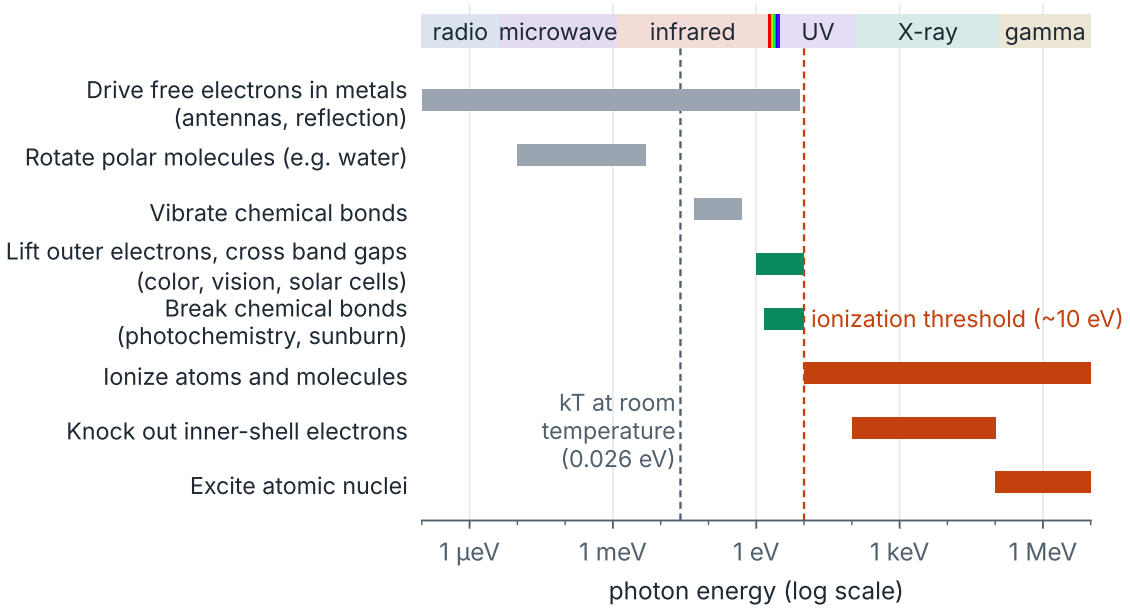


Figure 3.3 What photons of each energy typically do to matter (order-of-magnitude ranges). Gray bars move charges, molecules and bonds and end up as heat; green bars rearrange outer electrons (color, vision, chemistry); orange bars are ionizing. Chapters 5–11 explore each row.

KEY IDEA

Photon energy, not brightness, decides what kind of interaction is possible. Below about 1 eV, light mostly makes things jiggle, which is heat. From about 1.8 to 10 eV it can rearrange outer electrons, which gives color and drives chemistry. Above about 10 eV it can rip electrons off atoms. Because absorption happens one photon at a time (Chapter 2), no number of low-energy photons adds up to one high-energy photon.

WORKED EXAMPLE 3.3 — Which light can break a carbon–carbon bond?

Given: a C–C single bond takes about 3.6 eV to break. Candidates: a germicidal UV-C lamp (254 nm), a green laser (532 nm) and Wi-Fi (2.4 GHz, 9.9×10^{-6} eV per photon from Worked Example 2.1).

Find: which photons can break the bond in one absorption.

Think: One photon, one event. Compare each photon's energy with 3.6 eV.

$$E_{\text{UV-C}} = \frac{1240}{254} = 4.88\ \text{eV} > 3.6\ \text{eV}, \qquad E_{\text{green}} = \frac{1240}{532} = 2.33\ \text{eV} < 3.6\ \text{eV}$$

A Wi-Fi photon falls short by a factor of $3.6/(9.9 \times 10^{-6}) \approx 360{,}000$. It is even about 2600 times smaller than the random thermal energy every molecule already has at room temperature, $k_B T \approx 0.026$ eV.

Answer: only the UV-C photon can break the bond, which is why UV-C damages DNA and kills microbes. Green light and Wi-Fi cannot, at any brightness; their absorbed energy just becomes heat.

MYTH VS. REALITY

Myth: Wi-Fi and microwaves are "radiation," so they are dangerous in the same way as X-rays.

Reality: All light is electromagnetic radiation, including the light you read by. What matters is photon energy and intensity. Radio, microwave, infrared and visible photons are **non-ionizing**: they cannot break molecules apart one photon at a time, and their only effect at high intensity is heating. Ionizing radiation, photons above about 10 eV, is a separate category (Chapter 10).

Wavelength also matters through **size**. Waves bend around objects smaller than or similar to their wavelength and are blocked by openings much smaller than it. That is why AM radio, hundreds of meters long, wraps around buildings, and why the mesh in a microwave oven door blocks 12 cm microwaves but passes 500 nm light (Chapter 8).

MATERIALS SCIENCE CONNECTION

Look at Figure 3.1 again: X-ray wavelengths are about the size of an atom. Copper Kα X-rays have $\lambda = 0.1542$ nm, while the spacing between planes of atoms in metals is a few tenths of a nanometer (BCC iron's lattice parameter is 0.2866 nm). That size match is what makes X-ray diffraction work: crystal planes scatter X-rays like a grating, and the angles reveal the crystal structure and Miller indices (Chapter 11). Visible light, thousands of times longer, simply cannot resolve atomic planes.

3.5 What reaches the ground

Earth's atmosphere absorbs most of the spectrum. Only a few **atmospheric windows** let radiation reach the surface (Figure 3.4):

- **Gamma rays, X-rays and short UV** are absorbed high in the atmosphere. Ozone and oxygen remove UV shorter than about 290 nm (all UV-C and most UV-B); longer UV-A mostly gets through.
- **Visible and near-infrared light** pass through, apart from scattering (Chapter 7) and some water-vapor absorption bands.
- **Most infrared** is absorbed by water vapor and CO_2, except windows near 3–5 µm and 8–14 µm. That absorption is the greenhouse effect at work (Chapter 9).
- **Radio waves** from about 1 cm to roughly 30 m pass freely, even through clouds. Longer waves (below about 10 MHz) are reflected by the ionosphere, which is how AM broadcasts can skip far over the horizon at night.

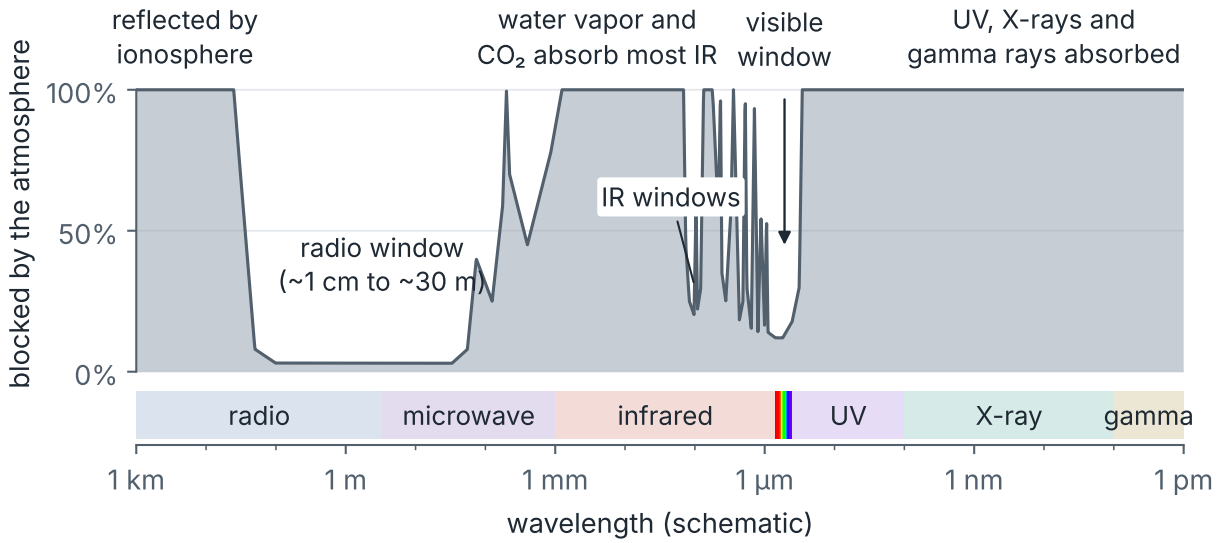


Figure 3.4 How much of each wavelength the atmosphere blocks before reaching the ground (schematic; real curves depend on humidity, altitude and weather). The two big windows are visible/near-IR and radio. Everything else is best observed from space.

This is why astronomy splits in two. Optical and radio telescopes work from the ground. X-ray, gamma-ray and most ultraviolet and far-infrared observatories must fly above the atmosphere, which is one reason the James Webb Space Telescope observes infrared from space.

3.6 Why your eyes see 400–700 nm

Out of 60 octaves, why did eyes settle on less than one? No single reason explains it, but several physical facts line up (Figure 3.5):

- **The light is there.** Sunlight is intense in this band, and the atmosphere is transparent to it.
- **Water is clear there.** Water absorbs strongly in the UV and infrared but very little between about 400 and 700 nm. Eyes are mostly water, and the first eyes evolved in water.
- **The photon energy is just right.** Vision starts when a single photon flips the shape of a retinal molecule in your retina, a switch that takes about 200 femtoseconds. Photons of 1.8–3.1 eV can do this reliably without breaking other bonds.
- **Too much energy is harmful.** UV photons damage molecules, so your cornea and lens absorb almost all UV before it reaches the retina.
- **Too little energy is unreliable.** An infrared-tuned pigment would need such a small trigger that the random thermal jostling of a 37°C eye would set it off, and your own body glows in the infrared.

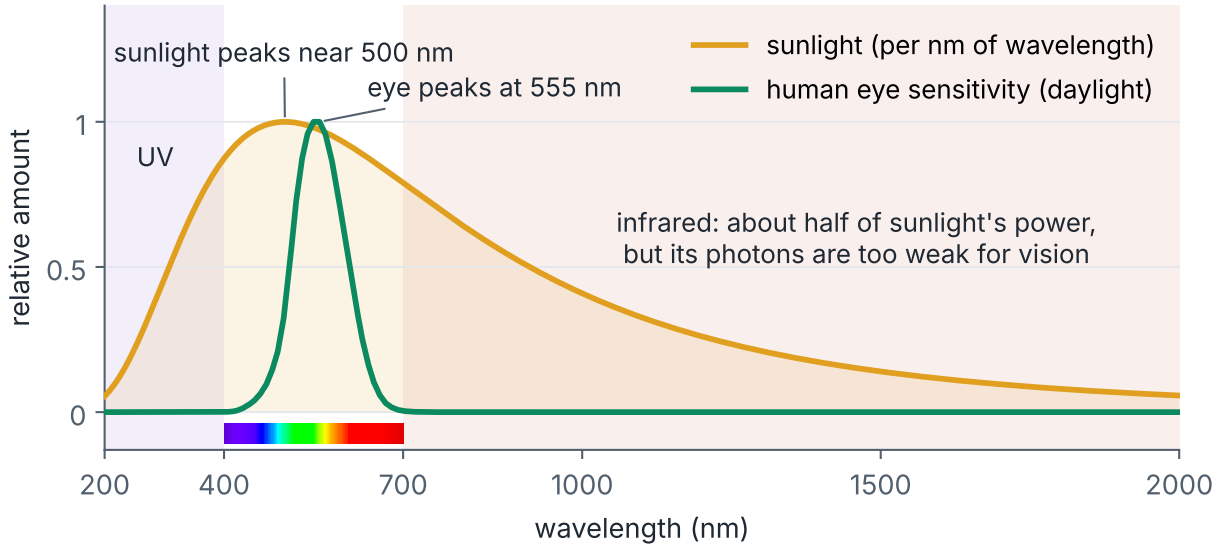


Figure 3.5 Sunlight (modeled as a 5772 K blackbody outside the atmosphere, per nanometer of wavelength) and the eye's daytime sensitivity (the standard CIE curve), each scaled to a peak of 1. The eye responds only to a narrow band near the Sun's brightest wavelengths.

Other animals stretch the window a little. Many insects, including bees, see into the near ultraviolet, which reveals patterns on flowers that are invisible to you.

RABBIT HOLE **(optional) — Where does sunlight really peak?**

Safe to skip. You will often read that "the eye peaks where the Sun peaks." It is half true. Plotted per nanometer of wavelength, as in Figure 3.5, sunlight peaks near 500 nm. Plotted per hertz of frequency, the same sunlight peaks in the near infrared, around 880 nm. Neither plot is wrong: they divide the spectrum into different-sized bins. The robust statement is the one in the bullets above: the visible band sits where sunlight is plentiful, water and air are clear, and photon energies suit molecular switches.

MYTH VS. REALITY

Myth: Visible light is a special kind of radiation, different from the invisible kinds.

Reality: Visible light differs from infrared or ultraviolet only in wavelength. It is "special" only because it is the band your retina's molecules respond to. A thermal camera, a radio and a dental X-ray sensor are simply detectors for other parts of the same spectrum.

Chapter summary

- The EM spectrum is one phenomenon spread over more than 18 powers of ten in wavelength. Each band can be labeled by λ, $f = c/\lambda$ or $E = 1240\ \text{eV} \cdot \text{nm}/\lambda$.
- Order, long to short: radio (> 30 cm), microwave, infrared (700 nm–1 mm), visible (400–700 nm, 1.8–3.1 eV), ultraviolet (10–400 nm), X-ray (0.01–10 nm), gamma ray. Boundaries are conventions; X-ray and gamma overlap and are named by source.
- Each band is made by accelerating charge at a different scale: antennas, vibrating molecules, outer electrons, fast or inner-shell electrons, nuclei.
- Photon energy sets the interaction: heating below ~1 eV, electronic and chemical changes from ~1.8 to 10 eV, ionization above ~10 eV.
- The atmosphere passes visible/near-IR light and radio from ~1 cm to ~30 m; it blocks gamma, X-rays, most UV and most IR.

- Eyes use 400–700 nm because sunlight is plentiful there, air and water are clear, and those photon energies can flip retinal molecules without damaging them.

Check yourself

1. CONCEPT What do radio waves and gamma rays have in common? What is different about them?
2. CALC An AM station broadcasts at 1000 kHz. Find the wavelength and the energy of one photon.
3. CALC A TV remote's LED emits at 940 nm. Find the frequency and photon energy. Which band is it in?
4. CALC Modern computer chips are patterned with "extreme ultraviolet" light at 13.5 nm. What is the photon energy? Is it ionizing?
5. CALC Cobalt-60 emits gamma rays of 1.17 MeV. Find the wavelength and compare it with the size of an atom (about 0.1 nm).
6. CONCEPT A powerful infrared heat lamp can make your skin hot but does not cause sunburn, while a few minutes of UV can. Explain using photon energies.
7. CONCEPT Why can't a telescope on the ground study X-rays from the Sun? Which two broad bands can ground-based telescopes use?
8. APPLY Copper Kα X-rays ($\lambda = 0.1542$ nm) are used to study crystal structures. Find their photon energy, and explain why green light (530 nm) can't do the same job for BCC iron ($a = 0.2866$ nm).
9. CONCEPT Give three physical reasons human vision uses 400–700 nm rather than ultraviolet or infrared.
10. CALC How many octaves (factors of 2 in frequency) lie between a 30 m radio wave and a 0.03 nm X-ray? How many octaves wide is visible light?
11. APPLY Point a TV remote at your phone camera and press a button: the camera shows a flashing light your eye cannot see. Silicon camera sensors respond to photons above about 1.1 eV. Explain what you see.

Full solutions are in the Answer Key at the back of the book.

CHAPTER 4

Inside the Atom: Scale, Orbitals, and Quantized Energy

Almost all of an atom's mass sits in a speck that fills less than a trillionth of its volume, yet your chair still holds you up. And because the electron around that speck behaves like a trapped wave, an atom can hold energy only in exact, fixed amounts. That one rule is behind every color you will ever see.

THE BIG PICTURE

- Atoms are about 0.1–0.5 nm across; the nucleus is 20,000 to 60,000 times smaller still.
- The electron is not a tiny planet. It is a standing wave spread into a probability cloud (an orbital) with no hard edge.
- A wave trapped in a small space can vibrate only in certain patterns, so a bound electron can have only certain energies.
- Hydrogen's allowed energies are $E_n = -13.6\ \text{eV}/n^2$; the energy needed to free the electron completely is the ionization energy.
- Matter feels solid because electron clouds strongly resist overlapping, not because atoms are full.

AFTER THIS CHAPTER YOU CAN

- State the sizes of atoms and nuclei and build an accurate scale model.
- Describe an orbital as a probability cloud and explain why it has no surface.
- Explain, with the standing-wave picture, why a bound electron's energy is quantized.
- Compute hydrogen's energy levels, excitation energies, and ionization energies.
- Explain why mostly-empty atoms make solid, incompressible matter.

4.1 How big is an atom?

No light microscope can show you an atom (Chapter 8 explains why), but atoms can still be measured precisely: by weighing a known volume of solid (Worked Example 4.1) or by diffracting X-rays from a crystal (Chapter 11). Figure 4.1 places the result on a ladder of sizes.

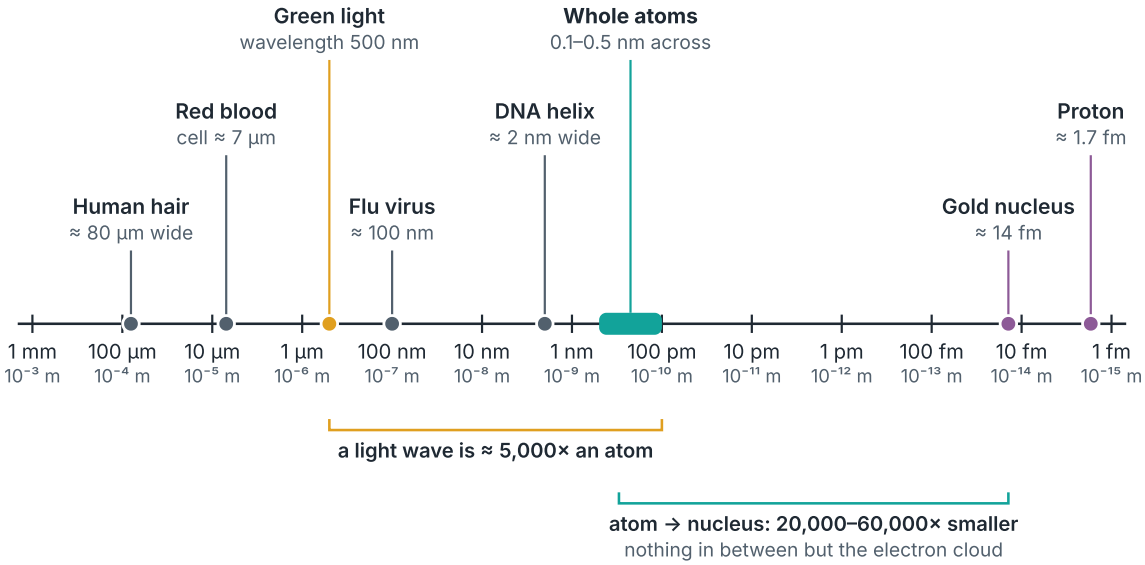


Figure 4.1 Sizes on a powers-of-ten scale: each tick is 10× smaller than the one to its left. Notice the long stretch between "whole atoms" and "gold nucleus": four to five powers of ten in which there is nothing but electron cloud.

YOU ASKED

"How big is an atom, from one side to the other?"

About 0.1 to 0.5 nanometers, or one to five ten-billionths of a meter. The smallest atoms (hydrogen and helium) are roughly 0.1 nm across, the atoms in most metals and ceramics are 0.25–0.3 nm, and the largest common one, cesium, is about 0.5 nm.

For a feel of the scale: about 350,000 copper atoms lined up would span the width of a human hair, and one wavelength of green light (500 nm) is a few thousand atoms long. One caveat: an atom has no hard surface (Section 4.3), so its "diameter" depends on where you draw the edge. That is why tables list slightly different radii for an atom in a metal, in a covalent bond, or as an ion.

WORKED EXAMPLE 4.1 — How far apart are the atoms in copper?

Given: copper's density $\rho = 8.94\ \text{g/cm}^3$ and molar mass $M = 63.55\ \text{g/mol}$; Avogadro's number $N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$.

Find: the volume each atom occupies and the typical atom-to-atom spacing.

Think: one mole of copper has a mass of 63.55 g and contains N_A atoms. Divide the mole's volume by the number of atoms, then take a cube root to get the side of the little cube each atom "owns."

$$V_{\text{atom}} = \frac{M}{\rho N_A} = \frac{63.55\ \text{g/mol}}{(8.94\ \text{g/cm}^3)(6.022 \times 10^{23}\ \text{mol}^{-1})} = 1.18 \times 10^{-23}\ \text{cm}^3 = 1.18 \times 10^{-29}\ \text{m}^3$$

$$d \approx V_{\text{atom}}^{1/3} = (1.18 \times 10^{-29}\ \text{m}^3)^{1/3} = 2.28 \times 10^{-10}\ \text{m} = 0.228\ \text{nm}$$

A hair 80 µm wide is therefore $80 \times 10^{-6} / 0.228 \times 10^{-9} \approx 350{,}000$ atoms across.

Answer: each copper atom gets a cube about 0.23 nm on a side, consistent with Callister's copper radius of 0.128 nm (diameter 0.256 nm), since real atoms pack as spheres, not cubes.

MATERIALS SCIENCE CONNECTION

In ME2340 you treat atoms as touching hard spheres: in an FCC metal the lattice parameter is $a = 2R\sqrt{2}$, so copper's $R = 0.128$ nm gives $a = 0.362$ nm, matching the measured 0.3615 nm. The model works, but "touching" needs care. Callister's R is half the equilibrium distance between neighboring nuclei, the spacing where attraction and repulsion balance (Section 4.7). It is not a physical surface, which is why the same element has different radii as a neutral atom and as an ion in a ceramic.

4.2 A tiny, heavy nucleus

At the center of every atom is the **nucleus**: protons and neutrons (together called nucleons) bound by the strong nuclear force. Rutherford's gold-foil experiments (1909–1911) first revealed it: most alpha particles sailed straight through thin gold foil, but a rare few bounced almost straight back, as if they had hit something tiny and massive. Nuclei are measured in **femtometers** (1 fm = 10^{-15} m), and their size follows a simple rule:

$$R \approx (1.2\ \text{fm})\, A^{1/3} \tag{4.1}$$

where R is the nuclear radius and A the number of nucleons. The cube root appears because nucleons pack like marbles in a bag: volume grows in proportion to the count.

- **Hydrogen** (a single proton): charge radius 0.84 fm, about 1.7 fm across.
- **Gold** ($A = 197$): $R \approx 1.2 \times 5.82 = 7.0$ fm, about 14 fm across.
- **Mass:** a proton outweighs an electron 1836 to 1, so the nucleus holds more than 99.9% of an atom's mass.
- **Density:** nuclear matter is about 2×10^{17} kg/m^3. A teaspoon of it would weigh roughly a billion tonnes.

YOU ASKED

"If a marble is the nucleus and it's placed at the center of a football field, is the electron probability cloud the size of the football field?"

No: the cloud is bigger. With a 1.5 cm marble as the nucleus, a heavy atom such as gold would be about 300 m across, and a hydrogen atom nearly a kilometer. Both are far larger than a 110 m football field (Figure 4.2).

The arithmetic is just the ratio of diameters. For hydrogen, $2(0.0529\ \text{nm})/2(0.84\ \text{fm}) \approx 63{,}000$, so the cloud is $1.5\ \text{cm} \times 63{,}000 \approx 940\ \text{m}$ across. Heavier atoms have bigger nuclei, so their ratio drops to about 20,000–30,000 (Table 4.1), but even gold's cloud would be about three field-lengths wide, spilling well past the stands.

So the analogy is right in spirit: an atom is overwhelmingly empty of mass. It just underestimates the emptiness by a factor of 3 to 9. Two refinements: the cloud has no edge (it thins out smoothly, so "940 m" marks a typical distance, not a wall), and the nucleus is not a hard ball either, so treat these ratios as good to about ±20%.

Table 4.1 Atom vs. nucleus. Atomic diameters are twice the Bohr radius (H) or twice Callister's atomic radius; nuclear diameters use $2(0.84\ \text{fm})$ for the proton and Equation 4.1 otherwise.

Atom	Atom diameter	Nucleus diameter	Ratio	Cloud size if nucleus = 1.5 cm marble
Hydrogen	0.106 nm	1.7 fm	≈ 63,000	≈ 940 m
Carbon	0.142 nm	5.5 fm	≈ 26,000	≈ 390 m
Iron	0.248 nm	9.2 fm	≈ 27,000	≈ 400 m
Gold	0.288 nm	14.0 fm	≈ 21,000	≈ 310 m

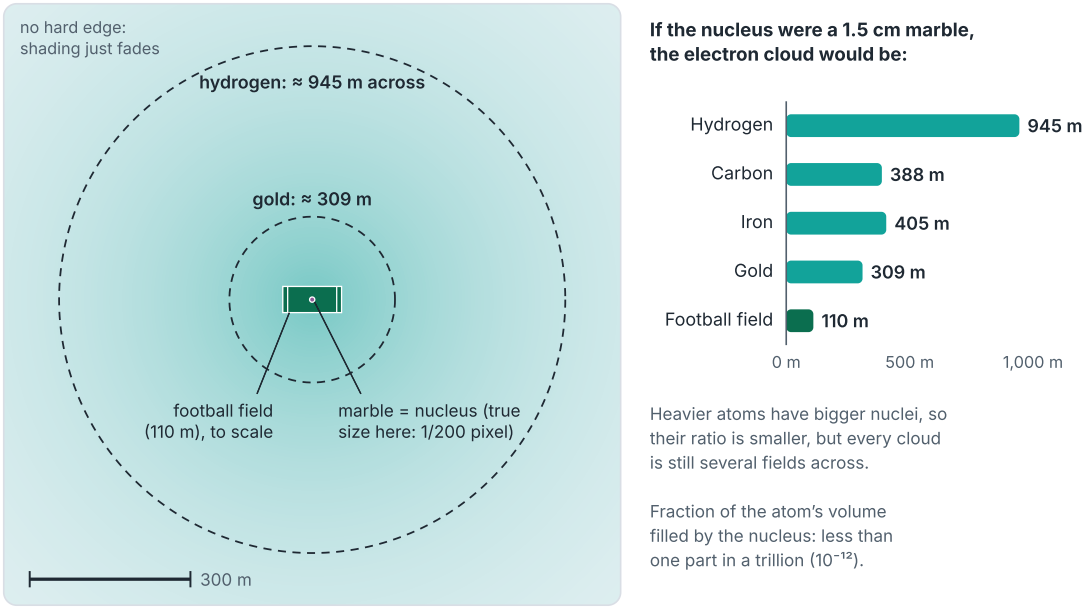


Figure 4.2 The marble-in-a-stadium model drawn to scale. The football field (green) sits deep inside even gold's cloud, and the marble itself would be 1/200 of a pixel. The shading fades smoothly outward because the cloud has no surface.

KEY IDEA

An atom is nearly all empty of *mass*, but it is not empty of *electron*: the electron's cloud fills the whole volume. That cloud, not the nucleus, sets an atom's size, its chemistry, and how it interacts with light.

4.3 Where is the electron? Orbitals and probability clouds

Quantum mechanics gives a strange but well-tested answer: a bound electron has no definite position until something measures it. What the atom does have is an **orbital**, a three-dimensional standing-wave pattern of the electron (its **wavefunction**). Square the wave at any point and you get the probability of finding the electron there. Drawn as shading, that probability is the familiar **electron cloud** (Figure 4.3).

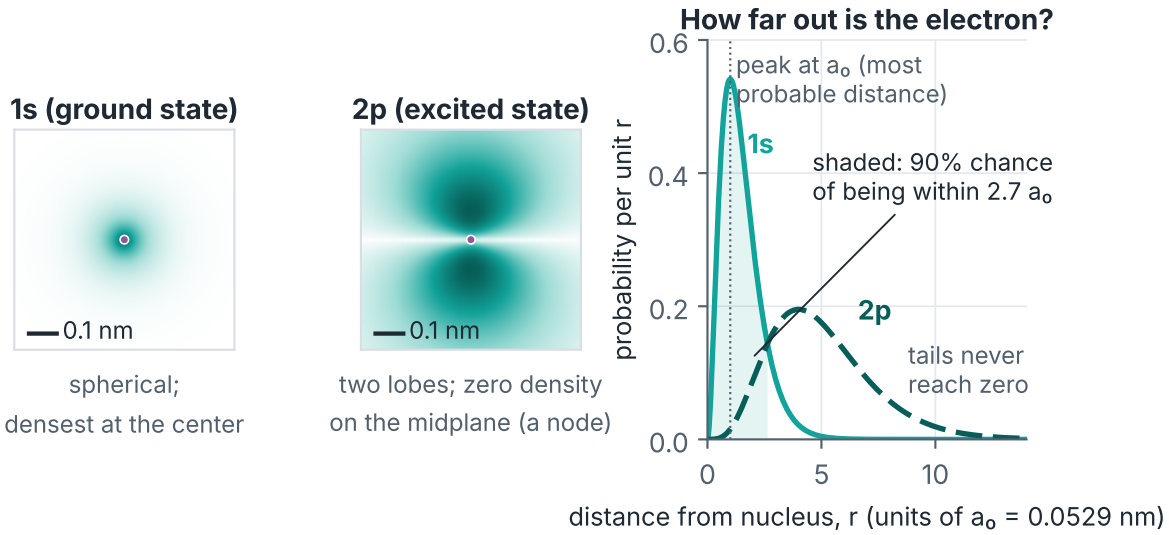


Figure 4.3 Hydrogen's electron density, computed from the exact wavefunctions. The 1s ground state is a fuzzy sphere; the excited 2p state has two lobes and a node (zero density) between them. At right: the electron is most likely found about one Bohr radius (a_0 = 0.0529 nm) from the nucleus, but the curves never reach zero. (Shading is boosted so the faint outer parts show.)

- **Shapes:** s orbitals are spheres; p orbitals have two lobes; d and f orbitals are more elaborate. Each holds at most two electrons.
- **No hard edge:** the density falls off smoothly forever. Chemistry books draw the surface enclosing 90% of the probability, which for hydrogen's 1s is a sphere about 0.28 nm across. That boundary is a convention, not a skin.
- **Excited states are bigger:** higher-energy orbitals spread farther out (compare 1s and 2p).
- **Steady, not swirling:** an orbital is a *stationary* pattern that does not change in time, like a steadily ringing bell. That is why an atom in one orbital does not radiate light (Chapter 5).

MYTH VS. REALITY

Myth: Electrons orbit the nucleus like planets around the Sun, and the cloud is a blur of that fast-moving dot.

Reality: No electron follows a track. A circling charge would radiate light continuously (Chapter 1), lose energy, and spiral into the nucleus almost instantly. Real atoms are stable, the ground state has no orbital motion at all, and the 1s cloud is a sphere, not a flat ring. The cloud is the shape of the electron's wave itself, not a long-exposure photo of a moving particle.

4.4 Why energy comes in steps: the standing-wave picture

ANALOGY

Pluck a guitar string and it rings only at certain pitches, because a whole number of half-waves must fit between the two fixed ends. A trapped electron wave works the same way: only the patterns that "fit" the atom are allowed, and each has its own energy.

In 1924 Louis de Broglie proposed, and experiments in 1927 confirmed, that electrons travel as waves, with a **de Broglie wavelength** $\lambda = h/(mv)$, where m and v are the electron's mass and speed (Chapter 8 shows the experimental proof: electrons diffract). An electron bound to a nucleus is a trapped wave. De Broglie's simple model pictures that wave wrapped around a circle: it must join smoothly onto itself, or successive laps land out of step and cancel (Figure 4.4).

$$n\lambda = 2\pi r_n, \qquad n = 1, 2, 3, \ldots \tag{4.2}$$

where r_n is the radius of the nth allowed orbit and n is the number of wavelengths around it.

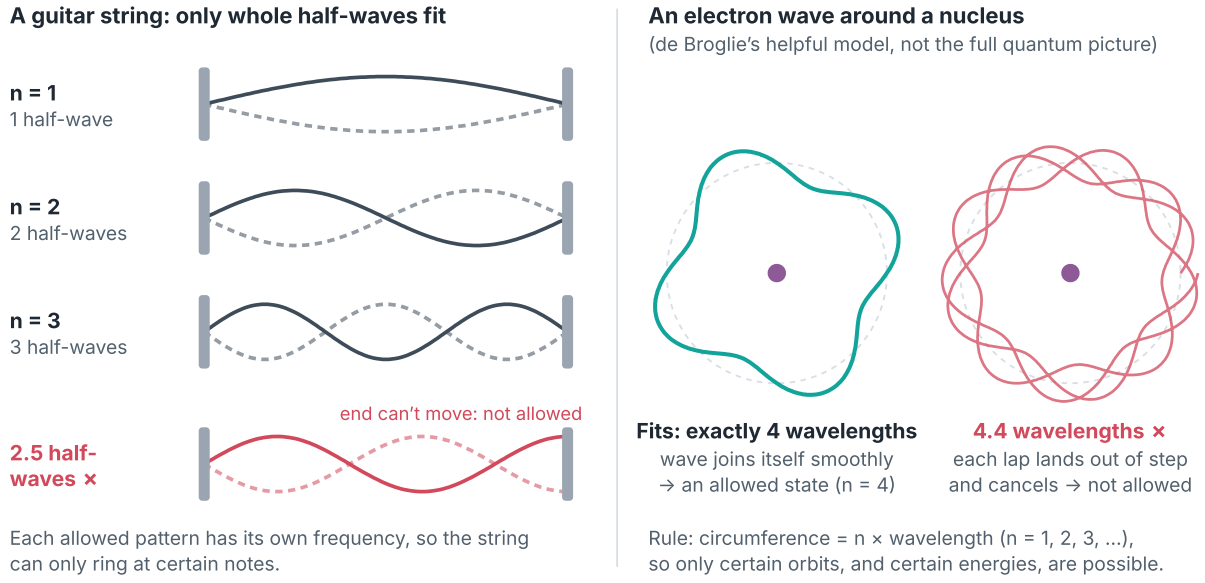


Figure 4.4 Standing waves on a string (left) and de Broglie's electron wave around a nucleus (right). Only whole-number patterns survive, so only certain orbits, and therefore certain energies, are allowed. This is a helpful model, not the full quantum picture.

Only certain wavelengths fit, and wavelength fixes the electron's speed and hence its energy, so only certain energies are possible. Niels Bohr had simply assumed this rule in 1913; de Broglie's wave picture explained it.

KEY IDEA

Confinement causes quantization. Any wave trapped in a limited space (a string, an organ pipe, an electron in an atom) can vibrate only in certain patterns, each with a definite energy. Free electrons, which are not trapped, can have any energy at all.

How far to trust the model. The circle picture gives exactly the right energies for hydrogen, but the real orbitals are 3D standing waves (Figure 4.3), not waves running on circular tracks. Use it for intuition about *why* energy is quantized, not as a picture of the atom. The full theory, Schrödinger's wave equation, keeps the central lesson: allowed patterns, allowed energies.

WORKED EXAMPLE 4.2 — Does the ground-state wave really fit?

Given: in the Bohr model the ground-state electron moves at $v = 2.19 \times 10^{6}\ \mathrm{m/s}$ on a circle of radius $a_0 = 0.0529\ \mathrm{nm}$; electron mass $m = 9.11 \times 10^{-31}\ \mathrm{kg}$.

Find: the de Broglie wavelength, and compare it with the circumference.

Think: if $n = 1$ in Equation 4.2, exactly one wavelength should fit around the circle.

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}\ \mathrm{J \cdot s}}{(9.11 \times 10^{-31}\ \mathrm{kg})(2.19 \times 10^{6}\ \mathrm{m/s})} = 3.32 \times 10^{-10}\ \mathrm{m} = 0.332\ \mathrm{nm}$$

$$2\pi a_0 = 2\pi(0.0529\ \mathrm{nm}) = 0.332\ \mathrm{nm}$$

Answer: λ = 0.332 nm, exactly the circumference: one wavelength fits, so this is the $n = 1$ state.

RABBIT HOLE (optional) — Why doesn't the electron fall into the nucleus?

Safe to skip. The electron is attracted to the proton, so why doesn't it collapse onto it? Squeezing a wave into a smaller space shortens its wavelength, and a shorter wavelength means more momentum and more kinetic energy ($\lambda = h/mv$). Shrinking the cloud lowers the electric energy but raises the kinetic energy even faster. The ground state is the best compromise, and its size is the Bohr radius. There is simply no lower-energy pattern to fall into.

4.5 Hydrogen's energy ladder

For hydrogen, the allowed energies come out remarkably simple:

$$E_n = -\frac{13.6\ \mathrm{eV}}{n^2}, \qquad n = 1, 2, 3, \ldots \tag{4.3}$$

where n is the **principal quantum number**. Energies are in electron-volts (1 eV = 1.602×10^{-19} J).

- **Why negative?** Zero is defined as an electron at rest far from the proton. A bound electron has less energy than that, so you must *add* energy to free it.
- **Ground state** ($n = 1$, −13.6 eV): the lowest level, where an undisturbed atom sits. **Excited states** ($n \geq 2$) are higher rungs: −3.40, −1.51, −0.85 eV, …
- **Rungs crowd together** just below 0 eV. Above 0 the electron is free and can have any energy: a continuum, not a ladder.
- **Higher rungs are bigger:** in the Bohr model the orbit radius grows as $r_n = n^2 a_0$, so $n = 2$ is four times wider than $n = 1$.

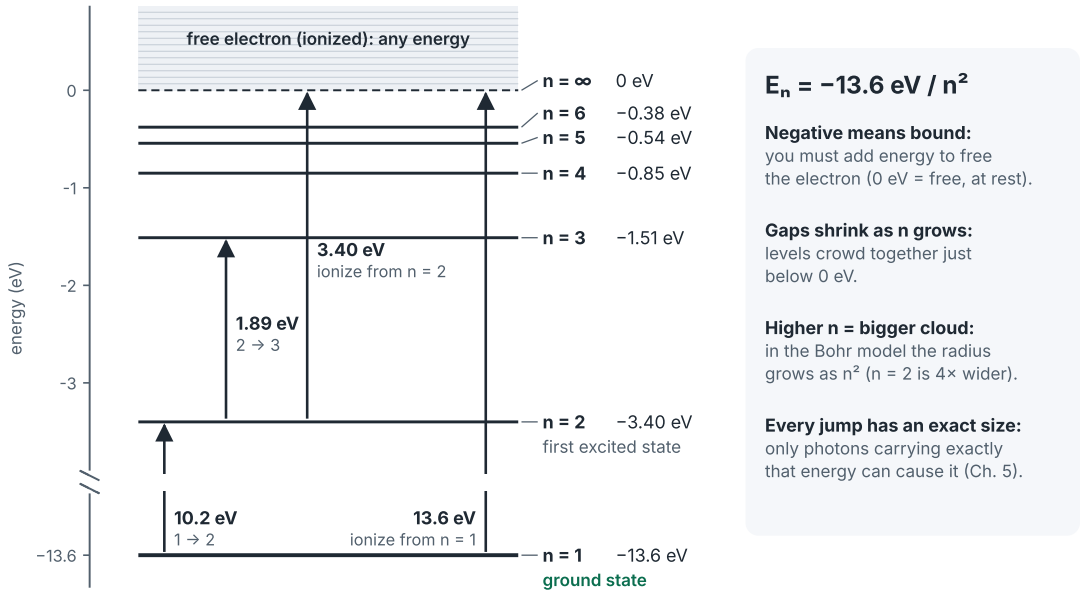


Figure 4.5 Hydrogen's energy ladder, $E_n = -13.6\ \text{eV}/n^2$. Each arrow is a jump of one exact size. The axis is broken between $n = 1$ and $n = 2$; that first gap (10.2 eV) is by far the largest.

Every other element has its own ladder with its own spacings, set by its nuclear charge and its other electrons. No two ladders are alike, which is why each element absorbs and emits its own unique set of colors (Chapter 5).

WORKED EXAMPLE 4.3 — Climbing hydrogen's ladder

Given: $E_n = -13.6\ \text{eV}/n^2$.

Find: (a) the energy to lift the electron from $n = 1$ to $n = 2$; (b) from $n = 2$ to $n = 3$; (c) the energy to free it from $n = 2$.

Think: the energy needed for any jump is the difference between the final and initial rungs.

$E_1 = -13.6$ eV, $E_2 = -13.6/4 = -3.40$ eV, $E_3 = -13.6/9 = -1.51$ eV.

(a) $\Delta E = E_2 - E_1 = -3.40 - (-13.6) = 10.2\ \text{eV}$
(b) $\Delta E = E_3 - E_2 = -1.51 - (-3.40) = 1.89\ \text{eV}$
(c) $\Delta E = 0 - E_2 = 3.40\ \text{eV}$

Answer: 10.2 eV, 1.89 eV, and 3.40 eV. The first step is by far the hardest; an already-excited atom is much easier to excite further or to ionize.

4.6 Ionization energy

The **ionization energy** is the minimum energy needed to remove the most loosely held electron from an isolated atom in its ground state, leaving a positive **ion**. For hydrogen it is 13.6 eV: the full height of the ladder from $n = 1$ to 0. Table 4.2 and Figure 4.6 show measured values.

Table 4.2 First ionization energies (NIST Atomic Spectra Database, rounded).

Element	Ionization energy (eV)	Element	Ionization energy (eV)
Hydrogen (H)	13.6	Sodium (Na)	5.14
Helium (He)	24.6 (highest of any element)	Argon (Ar)	15.8
Lithium (Li)	5.39	Potassium (K)	4.34
Carbon (C)	11.3	Iron (Fe)	7.90
Nitrogen (N)	14.5	Copper (Cu)	7.73
Oxygen (O)	13.6	Gold (Au)	9.23
Neon (Ne)	21.6	Cesium (Cs)	3.89 (lowest of any element)

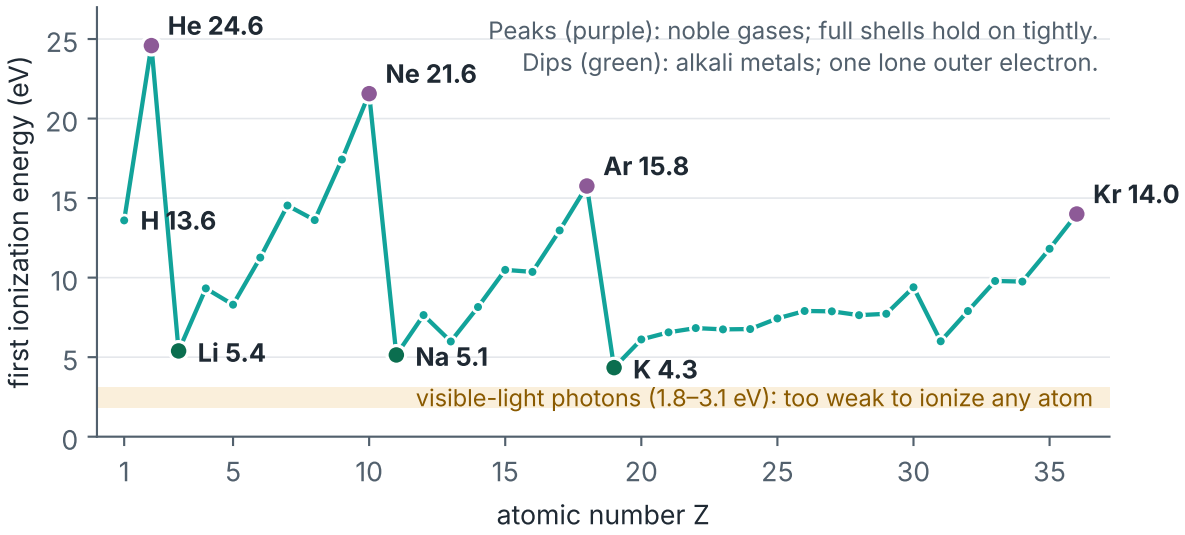


Figure 4.6 First ionization energy versus atomic number. Noble gases (full shells) hold their electrons tightly; alkali metals, with a single electron alone in a new outer shell, give one up easily. Every value lies above the energy of any visible photon (shaded band).

The pattern follows the shells. A noble gas has a completely filled outer shell held close to the nucleus. An alkali metal's outer electron sits alone in a new, larger shell, screened from the nucleus by all the electrons beneath it. Two consequences for this book:

- **Visible light cannot ionize ground-state atoms.** Visible photons carry only 1.8–3.1 eV, less than even cesium's 3.89 eV. Ionizing atoms takes ultraviolet photons of roughly 4–25 eV; the usual threshold for "ionizing radiation" is about 10 eV (Chapter 10).
- **Excited atoms are easier.** From $n = 3$, hydrogen needs only 1.51 eV, which a red photon can supply.

MATERIALS SCIENCE CONNECTION

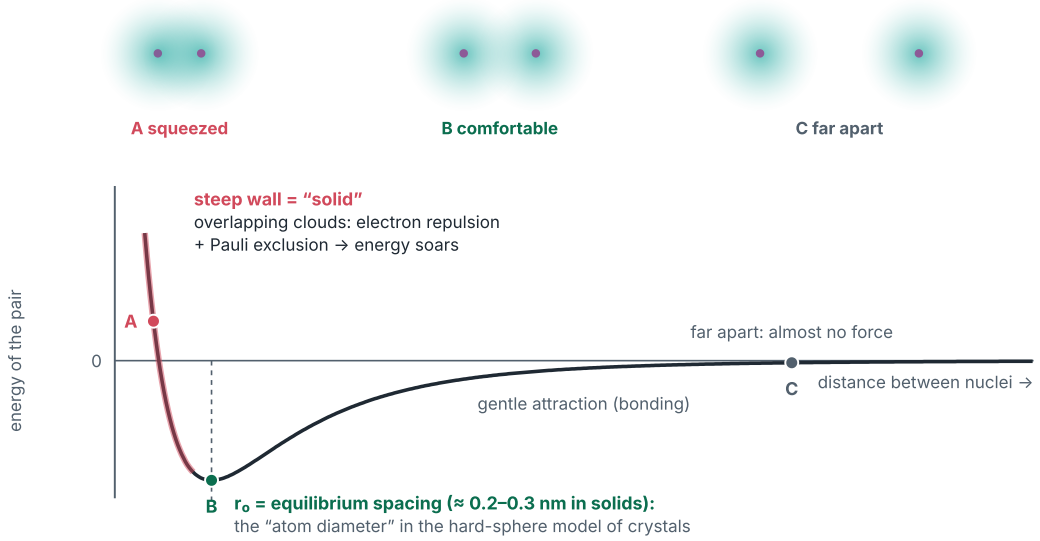
Low ionization energies explain ionic bonding in ceramics (Callister, Chapter 2). Sodium gives up its lone outer electron for just 5.14 eV, and chlorine releases energy when it captures one; the resulting Na^+ and Cl^- ions attract strongly. The same logic produces $Mg^{2+}O^{2-}$ in MgO and Al^{3+} with O^{2-} in alumina. Once the ions form, every ion has a filled, noble-gas-like shell held very tightly. That is why ionic ceramics are electrical insulators and, as pore-free single crystals, transparent (Chapters 5 and 6).

4.7 If atoms are empty, why is matter solid?

When you press on a table, your atoms' electron clouds meet the table's electron clouds, and two effects make overlapping them very costly:

- **Electric repulsion.** The clouds are negatively charged. As they overlap, the electrons repel each other, and each nucleus is less shielded from the other nucleus.
- **The Pauli exclusion principle.** No two electrons can occupy the same quantum state. Forcing two filled clouds into the same space pushes electrons into higher-energy states, and the energy cost soars.

Figure 4.7 shows the result as an energy curve, the same shape as Callister's bonding-energy curve. Far apart, two atoms barely notice each other. At the equilibrium spacing r_0 they are "comfortable." Push closer, and the energy climbs a steep wall. That wall *is* solidity: it is why solids and liquids are nearly incompressible, and why your hand stops at the tabletop.



Schematic, not to scale for a specific material. Cloud shading fades smoothly: there is no hard edge to "touch."

Figure 4.7 Why empty atoms feel solid. Squeezing two atoms closer than r_0 forces their clouds to overlap, which costs energy very steeply (red). The minimum at r_0, typically 0.2–0.3 nm, sets the spacing of atoms in solids.

MYTH VS. REALITY

Myth: Since atoms never really touch, you have never actually touched anything.

Reality: At the atomic scale, electron clouds pushing on each other at about 0.1–0.3 nm *is* what touching means. There is no deeper contact to miss: clouds have no surfaces to meet. The push you feel is electromagnetic repulsion plus Pauli exclusion, and it is completely real.

Chapter summary

- Atoms are about 0.1–0.5 nm across. Nuclei are femtometers across ($R \approx 1.2\ \text{fm}\ A^{1/3}$) and hold more than 99.9% of the mass.
- Scaled so the nucleus is a 1.5 cm marble, the electron cloud would be roughly 300 m (gold) to 940 m (hydrogen) across: bigger than a football field.
- An electron in an atom is a 3D standing wave (orbital). Its square is a probability cloud with no hard edge.
- Only whole-number standing-wave patterns fit ($n\lambda = 2\pi r_n$ in de Broglie's model), so bound electrons have only certain energies.
- Hydrogen: $E_n = -13.6\ \text{eV}/n^2$. The ground state is $n = 1$; 13.6 eV ionizes it.
- Ionization energies run from about 4 to 25 eV, all above visible photon energies.
- Matter is solid because overlapping electron clouds repel (electric repulsion plus Pauli exclusion).

Check yourself

1. CONCEPT Why is it misleading to quote a single, exact diameter for an atom? Why do tables list different radii for the same element?
2. CALC Callister gives aluminum's atomic radius as 0.143 nm. How many aluminum atoms placed side by side would span 1 mm?
3. CALC Use Equation 4.1 to find the diameter of an iron nucleus ($A = 56$). Iron atoms are 0.248 nm across. If the nucleus were a 1.5 cm marble, how wide would the atom be?
4. CONCEPT Give two observations that the "electrons orbit like planets" picture cannot explain.
5. CONCEPT Using the guitar-string analogy, explain why a bound electron has only certain energies while a free electron can have any energy.
6. CALC In the Bohr model the $n = 2$ electron moves at 1.09×10^6 m/s on an orbit of radius $4a_0 = 0.212$ nm. Find its de Broglie wavelength and show that exactly two wavelengths fit around the orbit.
7. CALC A hydrogen atom is in the $n = 3$ state. How much energy ionizes it? Could a 2.0 eV (red-orange) photon do it? Could the same photon ionize a ground-state atom?
8. APPLY Rank sodium, magnesium and neon by ionization energy and explain the order in terms of electron shells.
9. CONCEPT If atoms are mostly empty, why can't you push your hand through a table? Name both effects involved.
10. APPLY Copper is FCC with $R = 0.128$ nm. Compute the lattice parameter $a = 2R\sqrt{2}$ and compare it with the measured 0.3615 nm. In what sense do the "hard spheres" actually touch?

Full solutions are in the Answer Key at the back of the book.

CHAPTER 5

Absorption and Emission: Electrons Jumping Up and Down

The red of a neon sign, the glow of a highlighter under a black light, the heat of sunlight on your arm, the beam of a laser pointer: each is the same tiny event repeated trillions of times a second. An electron jumps between two allowed energies, and a photon carries the difference.

THE BIG PICTURE

- An atom absorbs a photon only if the photon's energy matches a gap between two of its levels: $hf = E_{\text{upper}} - E_{\text{lower}}$.
- When an electron drops, the atom creates a new photon carrying exactly the energy difference.
- Every element has its own ladder of levels, so every element has its own spectral fingerprint.
- Absorbed energy is re-emitted, turned into heat, or used for chemistry. Reflection is a different process.
- Molecules add infrared (vibration) and microwave (rotation) ladders; in solids, the band gap decides metal vs. insulator and an LED's color.

AFTER THIS CHAPTER YOU CAN

- Decide from an energy-level diagram which photons an atom can absorb.
- Compute emitted wavelengths, such as hydrogen's 656 nm red line.
- Explain continuous, emission-line and absorption-line spectra.
- Describe the fates of absorbed energy and distinguish re-emission from reflection.
- Explain fluorescence, phosphorescence and the Stokes shift.
- Use band gaps to classify solids and predict LED colors; explain how a laser works.

5.1 The one rule: the photon must match a gap

An atom's electron can sit only on certain rungs of an energy ladder (Chapter 4). Light deals with that ladder through one rule:

$$E_{\text{photon}} = hf = \frac{hc}{\lambda} = E_{\text{upper}} - E_{\text{lower}} \tag{5.1}$$

where h is Planck's constant, f the light's frequency and λ its wavelength (Chapter 2).

A photon whose energy matches a gap can be **absorbed**: it vanishes completely and the electron jumps up (Figure 5.1A). A photon with the wrong energy, too little *or* too much, cannot drive that jump; it passes by or is weakly scattered (Chapter 6). With $hc = 1240\ \text{eV} \cdot \text{nm}$:

$$\lambda\,(\text{nm}) = \frac{1240\ \text{eV} \cdot \text{nm}}{\Delta E\,(\text{eV})} \tag{5.2}$$

Why so picky? Resonance. The wave's electric field pushes the electron cloud back and forth at the light's frequency. The atom has natural frequencies $f = \Delta E/h$, one per pair of levels (Section 5.2 shows what oscillates). At a match, every push lands in step and drives the atom up; off resonance, the pushes cancel out over many cycles.

KEY IDEA

Light is absorbed by **resonance**: the photon's energy (its frequency) must match an energy gap. Brighter light delivers more photons, but never changes which gaps a photon can bridge.

ANALOGY

A swing builds up only if you push at its own rhythm; push at the wrong rhythm, however hard, and it barely moves. An atom is a swing with several fixed rhythms, one per energy gap.

Two refinements. Above the ionization energy the electron is free, so *any* photon above that energy can be absorbed, the excess becoming kinetic energy. And real lines have a small width, which broadens into wide bands in liquids and solids (Sections 5.7–5.8).

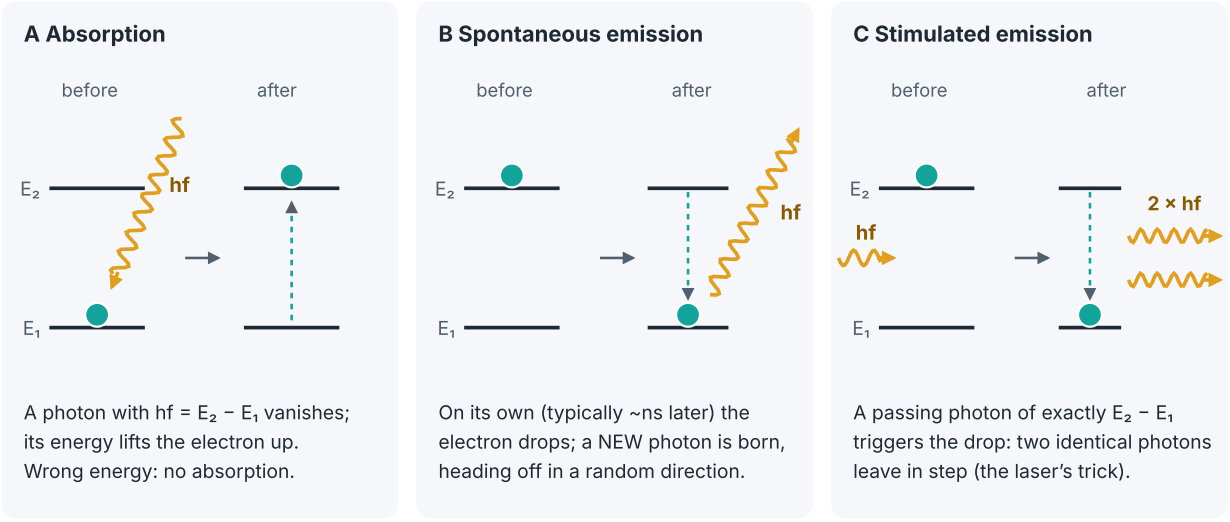


Figure 5.1 The three basic ways light and atoms trade energy. In every case the photon energy equals the gap $E_2 - E_1$. Stimulated emission (C) is the principle of the laser (Section 5.9).

YOU ASKED

"How can an atom that's thousands of times smaller than a light wave absorb it?"

Because absorption depends on matching the wave's *frequency*, not its size. The atom doesn't need to "catch" the whole wave; it needs the wave to shake it at one of its natural rhythms, $f = \Delta E/h$.

A 500 nm wave is about 5,000 times longer than an atom, so the whole atom sits in an almost uniform electric field that reverses direction about 6×10^{14} times per second, shaking the entire cloud in unison, like a cork riding an ocean swell far longer than itself. At resonance the energy builds up and the atom absorbs a photon. In fact, an isolated atom at exact resonance intercepts light from an area roughly the wavelength squared, about a million times its own cross-section. Chapter 8 goes deeper into size versus wavelength.

WORKED EXAMPLE 5.1 — Which photons can ground-state hydrogen absorb?

Given: hydrogen in its ground state ($E_1 = -13.6$ eV). Photons arrive with energies 10.2, 11.0, 12.1 and 15.0 eV.

Find: which are absorbed, and what happens.

Think: gaps from $n = 1$ (from $E_n = -13.6\ \text{eV}/n^2$): 10.20 eV to $n = 2$, 12.09 eV to $n = 3$, 12.75 eV to $n = 4$, and 13.6 eV or more to free the electron.

- **10.2 eV:** matches $1 \to 2$. Absorbed.
- **11.0 eV:** between 10.20 and 12.09 eV; no level fits. Passes through.

- **12.1 eV:** matches $1 \rightarrow 3$. Absorbed.
- **15.0 eV:** above 13.6 eV. Absorbed, ionizing the atom; the electron leaves with 1.4 eV of kinetic energy.

Answer: 10.2, 12.1 and 15.0 eV are absorbed; 11.0 eV is not. Every gap from the ground state is at least 10.2 eV (122 nm, ultraviolet), so ground-state hydrogen ignores all visible light. That is why hydrogen gas is colorless.

5.2 Emission: how a jump makes a photon

An excited atom doesn't stay excited. Typically within nanoseconds (1.6 ns for hydrogen's 2p state), the electron drops on its own and a photon carries away the difference: **spontaneous emission** (Figure 5.1B), again obeying Equation 5.1. An electron high on the ladder may drop in one step or several (say $4 \rightarrow 2 \rightarrow 1$), one photon per step.

YOU ASKED

"When an electron jumps down an energy level, how does that make a photon, and what is 're-emission'?"

The photon is created during the jump itself; it is not stored inside the atom beforehand. While changing levels, the electron is briefly in a mixture of the upper and lower orbitals, and that mixture's charge cloud sloshes back and forth at exactly $f = (E_{\text{upper}} - E_{\text{lower}})/h$ (Figure 5.2). A sloshing charge is a tiny antenna, and oscillating charges radiate electromagnetic waves (Chapter 1). Over a few nanoseconds the mixture shifts entirely into the lower orbital, and the radiated wave carries away exactly ΔE: one photon.

Re-emission means an atom that has just absorbed a photon gives the energy back as a new photon, usually of the same energy, nanoseconds later. The new photon has no memory of the old one: it leaves in a random direction with a random phase, which is what separates re-emission from reflection (Section 5.5). (This is the "semiclassical" picture. Full quantum electrodynamics describes the photon as a quantum of the electromagnetic field, but it keeps the key result: the photon's frequency is the sloshing frequency $\Delta E/h$.)

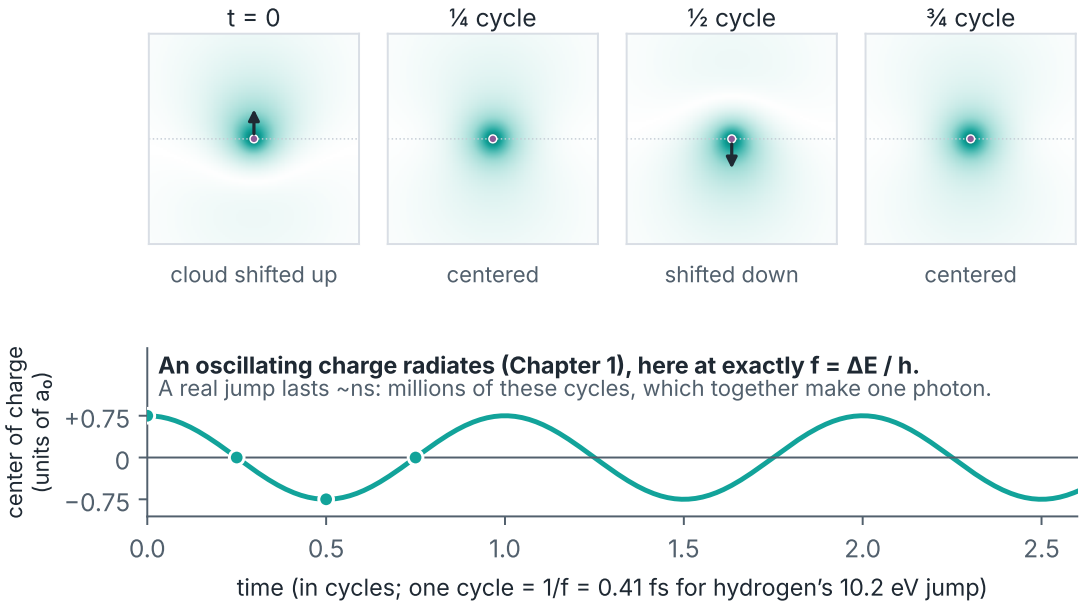


Figure 5.2 How an atom makes a photon, computed for hydrogen. During a 2p → 1s jump the electron is in a mixture of the two orbitals, and its cloud sloshes up and down at $f = \Delta E/h = 2.47 \times 10^{15}$ Hz. That oscillating charge radiates a 122 nm ultraviolet photon. (The arrows exaggerate the shift for visibility.)

A single, pure orbital is a steady cloud with nothing oscillating, so it does not radiate; emission happens only *during* a jump.

RABBIT HOLE **(optional) — What makes the electron decide to drop?**

Safe to skip. Even empty space has an electromagnetic field with unavoidable quantum jitter, and that jitter nudges an excited atom into the sloshing mixture. Some pairs of orbitals have shapes that barely slosh at all: “forbidden” transitions, whose upper state lingers. Hydrogen’s 2s state lasts about 0.12 s, some 75 million times longer than 2p. Such long-lived **metastable** states make phosphorescence and lasers possible.

5.3 Spectral lines: every element’s fingerprint

Each element has its own ladder, so each emits and absorbs its own **spectral lines**. For hydrogen, Equation 5.1 with $E_n = -13.6\ \text{eV}/n^2$ gives every line:

$$\Delta E = 13.6\ \text{eV}\left(\frac{1}{n_{\text{lower}}^2} - \frac{1}{n_{\text{upper}}^2}\right) \tag{5.3}$$

Drops ending on $n = 1$ (the **Lyman series**) are ultraviolet, on $n = 2$ (**Balmer**) visible, and on $n = 3$ (**Paschen**) infrared (Figure 5.3).

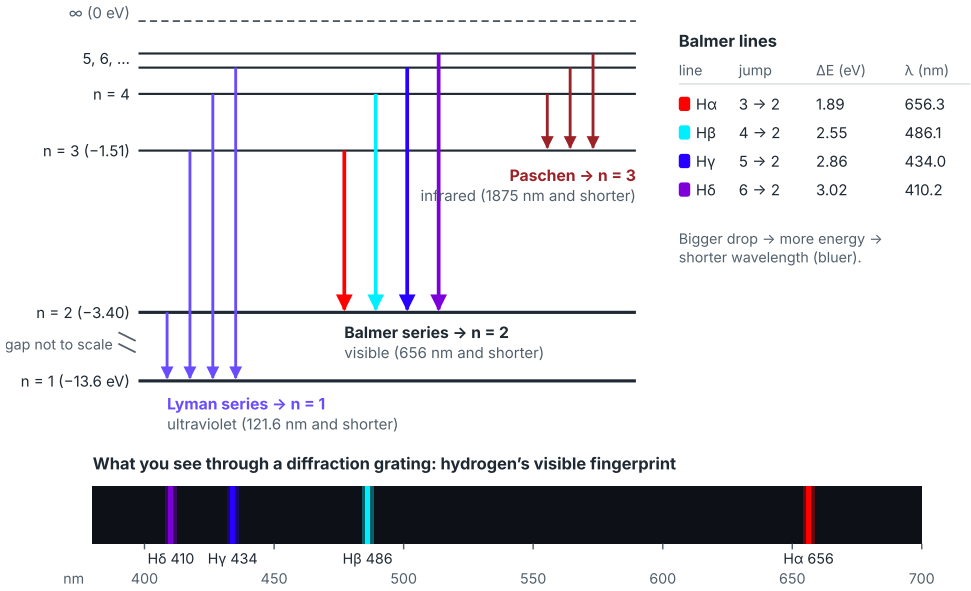


Figure 5.3 Hydrogen’s emission lines. Only the Balmer series falls in the visible; its four brightest lines (bottom) are hydrogen’s visible fingerprint. Bigger drops give bluer light.

WORKED EXAMPLE 5.2 **— Hydrogen’s red line**

Given: an electron in hydrogen drops from $n = 3$ to $n = 2$.

Find: the photon’s energy, wavelength and color.

Think: the photon carries exactly the gap (Equation 5.3); Equation 5.2 converts it to wavelength.

$$\Delta E = 13.6\ \text{eV}\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 13.6\ \text{eV}\left(\frac{5}{36}\right) = 1.89\ \text{eV}$$

$$\lambda = \frac{1240\ \text{eV}\cdot\text{nm}}{1.89\ \text{eV}} = 656\ \text{nm}$$

Answer: 1.89 eV, 656 nm, deep red: the H-alpha line (measured 656.3 nm) that colors glowing hydrogen clouds in astronomy photos.

The same fingerprints identify elements everywhere:

- **Neon signs:** lines such as 640.2 nm (red) and 585.2 nm (yellow) mix to neon's red-orange; other colors use other gases or phosphor coatings.
- **Sodium:** the yellow pair at 589.0 and 589.6 nm colors old streetlights and salt sprinkled in a flame.
- **Fireworks:** strontium salts burn red, barium green, copper blue.
- **Stars:** helium was discovered in the Sun's spectrum in 1868, 27 years before it was found on Earth.

5.4 Continuous, emission and absorption spectra

Spread light into its colors and you will see one of three patterns (Figure 5.4):

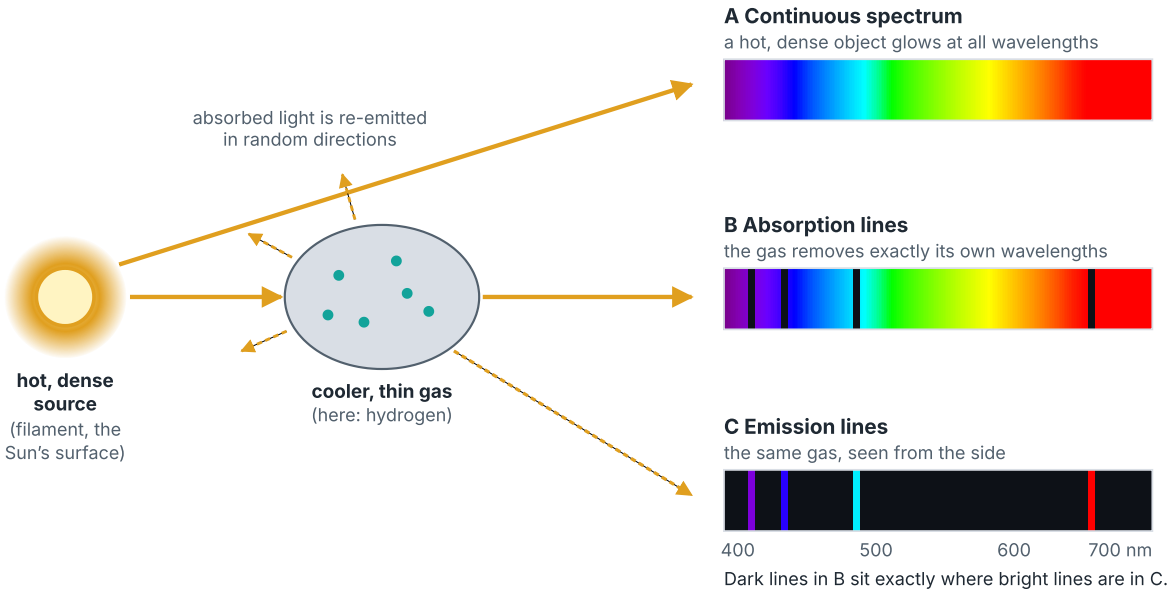


Figure 5.4 One source and one gas cloud make all three kinds of spectrum. The cloud absorbs its own wavelengths from the beam (B) and re-emits them in all directions, so from the side it glows at exactly those wavelengths (C).

- **Continuous:** a hot, *dense* object (a filament, a red-hot stovetop, the Sun's surface) glows at all wavelengths, because its crowded, jostling atoms offer energy gaps of every size. This is thermal radiation (Chapter 9).
- **Emission lines:** a hot, *thin* gas of separate atoms emits only its own jumps: bright lines on black.
- **Absorption lines:** a cooler gas in front of a continuous source removes exactly its own wavelengths. Thousands of these dark Fraunhofer lines cross the Sun's spectrum, revealing its composition.

If the gas re-emits what it absorbs, why are the lines dark? Re-emission goes in random directions, so almost none of it continues along the beam toward you; the beam stays depleted at those wavelengths, while the cloud glows at them from the side.

5.5 What happens to absorbed energy?

An absorbed photon no longer exists; its energy now belongs to the atom or molecule, and has three possible fates (Figure 5.5):

- **Re-emission:** a new photon leaves after nanoseconds, in a random direction. This dominates in thin gases, where nothing disturbs the atom first.
- **Heat:** in liquids and solids, neighbors jostle an excited atom far more often than once per nanosecond, so the energy usually drains into vibrations (temperature) before a photon can leave. That is why a black car gets hot in sunlight (Chapter 9).

- **Chemistry:** the energy breaks or rearranges a bond: photosynthesis, vision, fading dyes, sunburn (Chapter 10).

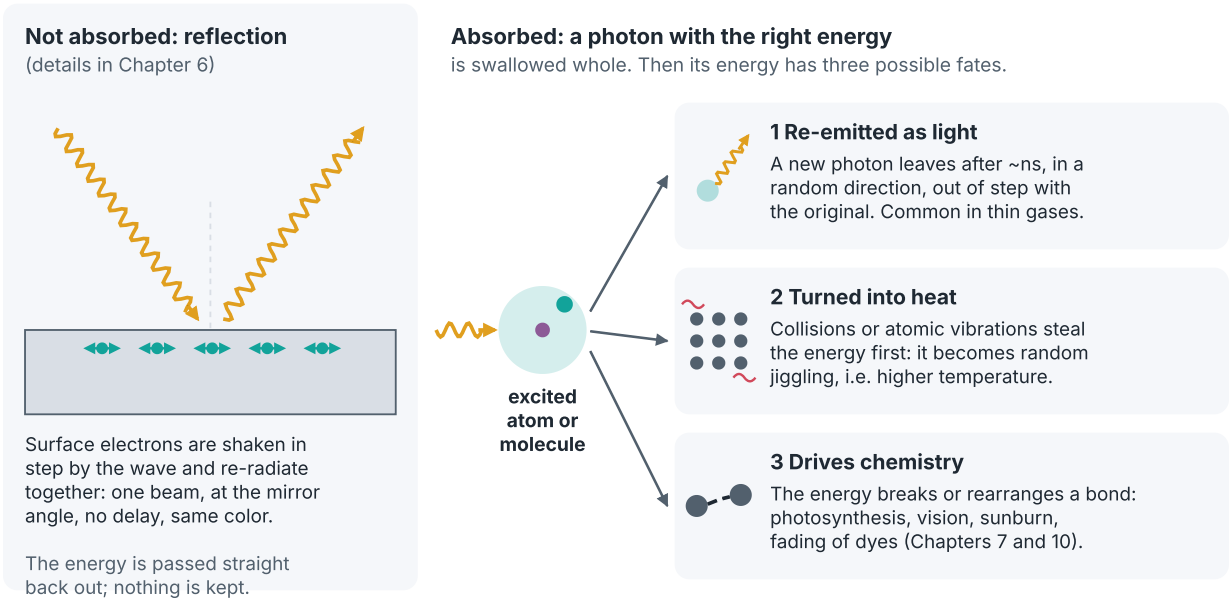


Figure 5.5 Reflection (left) passes energy straight back without absorbing it. Absorption (right) hands the energy to the atom, which re-emits it, turns it into heat, or uses it for chemistry.

KEY IDEA

Re-emission is not reflection. In reflection the wave shakes electrons that do *not* absorb it; they re-radiate in step with the wave and instantly, sending back one orderly beam at the mirror angle, same color. Re-emission follows a true absorption: delayed, random in direction and phase, sometimes a different color. Chapter 6 covers reflection.

MYTH VS. REALITY

Myth: A mirror (or a window) works by absorbing each photon and then re-emitting it.

Reality: Absorb-and-re-emit would scatter light in random directions, so a mirror would look like frosted glass instead of showing your face. Reflection and transmission are coherent; the light is never absorbed. Light “slows down” in glass by in-step re-radiation too, not absorb-and-wait (Chapter 6).

5.6 Fluorescence and phosphorescence

Molecules and crystals have **vibrational sub-levels** stacked on each electronic level. A molecule that absorbs an ultraviolet photon lands high on the excited state’s stack, slides to the bottom within picoseconds (shedding a little heat), then emits a visible photon on the way down. This is **fluorescence**: absorb short wavelengths, emit longer ones, within nanoseconds (Figure 5.6). The energy lost in between is the **Stokes shift**.

- **Highlighters** glow under a black light (365 nm UV): the dye (often pyranine) turns invisible UV into yellow-green.
- **Fluorescent lamps:** mercury vapor emits mostly 254 nm UV; a phosphor coating turns it into visible light.
- **White LEDs:** a blue chip excites a yellow phosphor; blue plus yellow looks white (Section 5.9).

Sometimes the electron side-steps into a “stuck” state (a molecular triplet state, or a defect trap in a crystal) from which the drop is forbidden (Section 5.2). It leaks out slowly, so the material glows long after the light goes off: **phosphorescence**, as in glow-in-the-dark stars and exit signs.

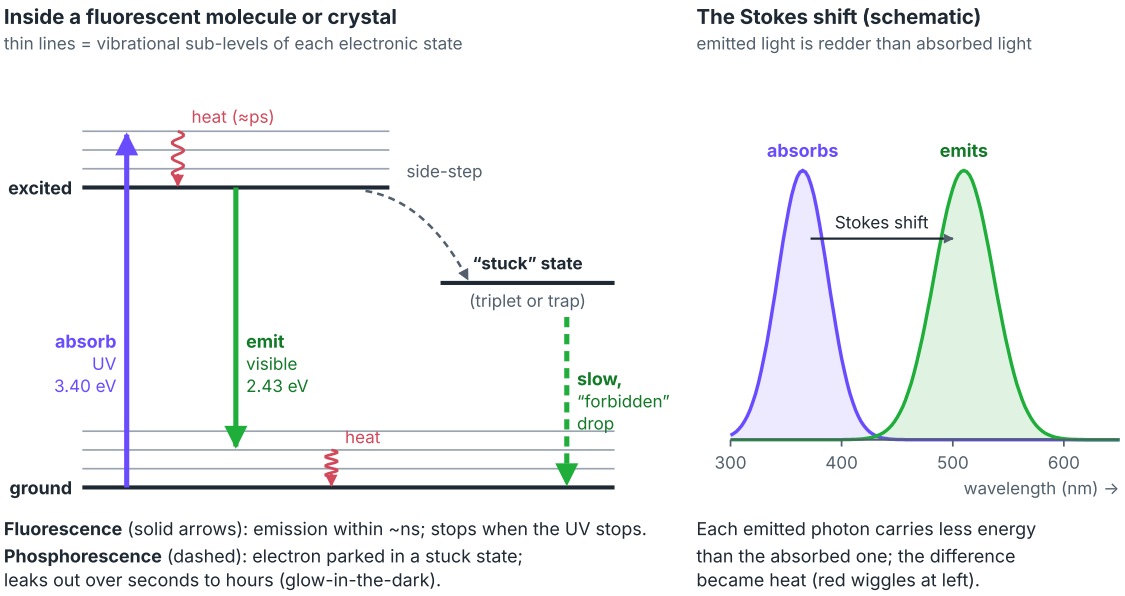


Figure 5.6 Fluorescence (solid arrows) and phosphorescence (dashed). Part of each absorbed photon's energy becomes heat, so the emitted light is always redder than the absorbed light (the Stokes shift).

WORKED EXAMPLE 5.3 — A highlighter's energy budget

Given: a highlighter dye absorbs 365 nm black-light UV and fluoresces near 510 nm.

Find: the energy of each photon, and the fraction of each absorbed photon's energy lost as heat.

Think: convert both wavelengths with Equation 5.2; the difference became vibration (heat).

$$E_{\text{in}} = \frac{1240}{365} = 3.40\ \text{eV}, \qquad E_{\text{out}} = \frac{1240}{510} = 2.43\ \text{eV}$$

$$\frac{E_{\text{in}} - E_{\text{out}}}{E_{\text{in}}} = \frac{0.97}{3.40} = 0.29$$

Answer: 3.40 eV in, 2.43 eV out; about 29% becomes heat. In daylight the dye adds UV-converted light to what it reflects, which is why highlighter ink looks unnaturally bright.

MYTH VS. REALITY

Myth: Glow-in-the-dark materials must be radioactive.

Reality: Modern glow-in-the-dark products store energy from ordinary light in trapped electrons and release it slowly; that is why they fade and need recharging under a lamp. The myth has a real origin: early-1900s watch dials used radium paint, which glowed from radioactivity and badly harmed the workers who applied it.

MATERIALS SCIENCE CONNECTION

Most phosphors are ceramics: a wide-gap oxide host, transparent by itself, doped with an ion that adds its own levels inside the gap. Glow-in-the-dark pigment is strontium aluminate ($SrAl_2O_4$) doped with Eu^{2+} (the emitter) and Dy^{3+} (which creates traps); it can glow for hours. White LEDs use yttrium aluminum garnet ($Y_3Al_5O_{12}$) doped with Ce^{3+}. Ruby, the first laser material, is sapphire (Al_2O_3) with a trace of Cr^{3+}. Same lesson as doping silicon: tiny impurity levels, big property changes.

5.7 Molecules: vibrations (infrared) and rotations (microwaves)

Molecules can store energy in two more quantized ways (Figure 5.7):

- **Vibration:** bonds stretch and bend like springs. Vibrational levels are spaced about 0.05–0.5 eV apart, matching **infrared** photons (about 2.5–25 μm).
- **Rotation:** the whole molecule spins. Rotational levels are only about 0.0001–0.01 eV apart, matching **microwave** and far-infrared photons.

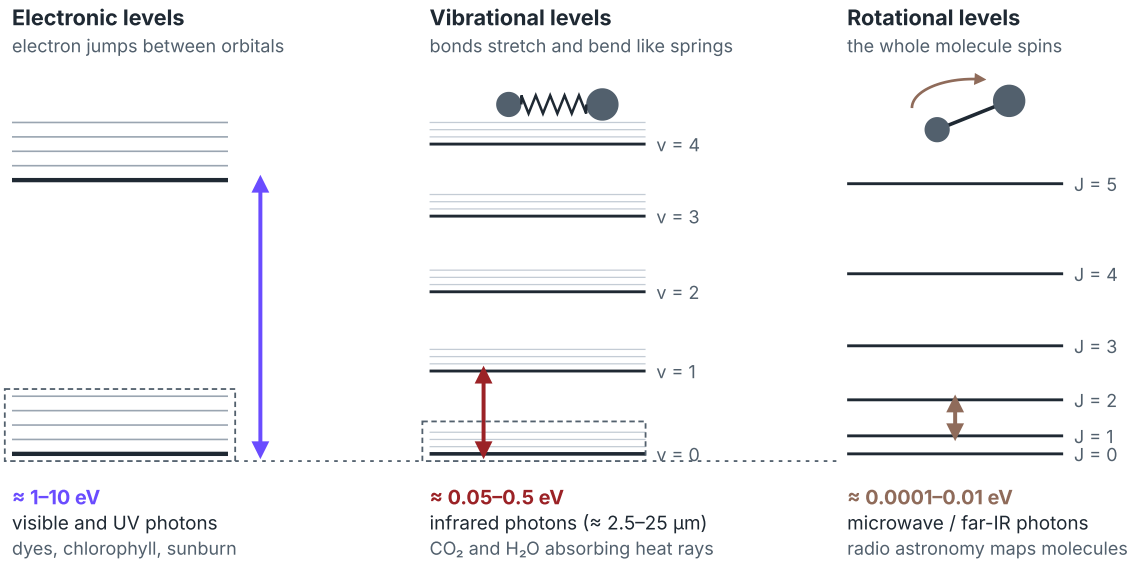


Figure 5.7 A molecule's three nested ladders. Each electronic level carries a stack of vibrational levels, and each of those a stack of rotational levels, each roughly 10–100 times finer than the last.

Absorption still needs an oscillating charge: a vibration absorbs infrared only if it makes the molecule's charge swing back and forth (a changing dipole). Carbon dioxide's bending vibration does, absorbing near 15 μm (0.083 eV), as does its asymmetric stretch near 4.3 μm. Nitrogen and oxygen, about 99% of air, are symmetric: stretching them shifts no charge, so they ignore infrared. That difference is the heart of the greenhouse effect (Chapter 9).

Rotations give sharp lines in gases; radio astronomers identify molecules in space by them. In liquid water, jammed molecules can't rotate freely, so microwave absorption is broad; a microwave oven is not tuned to a special water resonance (Chapter 9). And because electronic jumps in molecules drag vibrations along, dyes and chlorophyll absorb broad bands of color (Chapter 7).

5.8 Solids: energy bands and band gaps

Bring N atoms together into a crystal and their outer orbitals overlap. The Pauli exclusion principle forbids identical states, so each atomic level splits into N closely spaced levels, and with about 10^{23} atoms these blur into continuous **energy bands** separated by **band gaps** (Figure 5.8). The highest filled band is the **valence band**; the next one up, empty at low temperature, is the **conduction band**. The gap between them, E_g, controls almost everything.

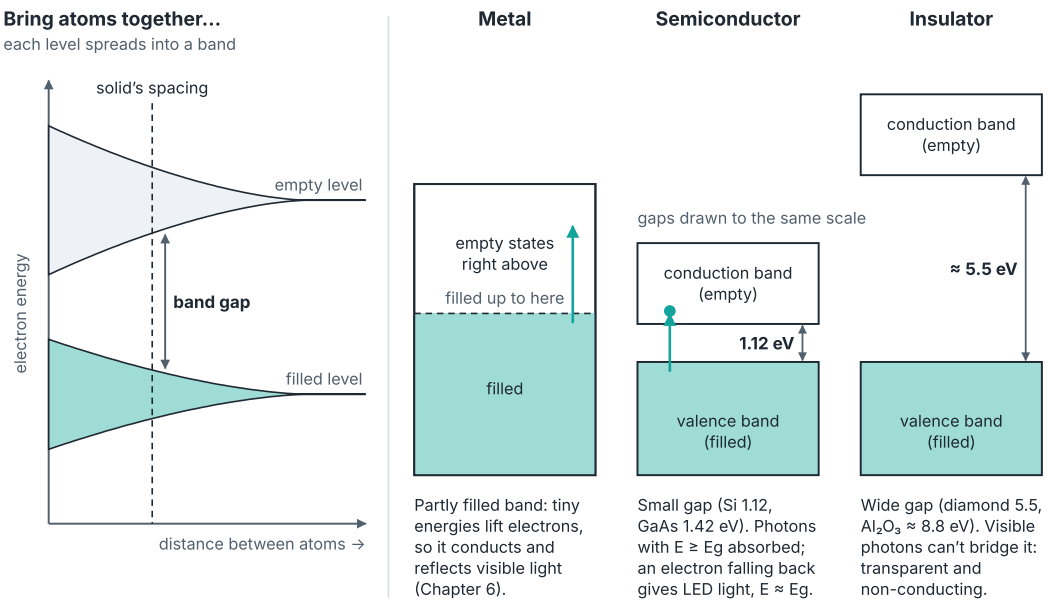


Figure 5.8 Energy bands. In a metal the top band is only partly filled, so electrons can absorb any small amount of energy. Semiconductors and insulators have a full valence band and an empty conduction band; the band gap sets which photons they absorb.

- **Metals:** the top band is only partly filled, so electrons can absorb any small energy. They move freely (conduction) and respond to all visible light, which makes metals shiny (Chapter 6).
- **Semiconductors:** a modest gap (silicon 1.12 eV, GaAs 1.42 eV). Room-temperature thermal energy ($k_B T \approx 0.026$ eV) lifts only a few electrons across, so they conduct a little.
- **Insulators:** a wide gap (diamond about 5.5 eV, sapphire about 8.8 eV). Essentially no electrons cross, and visible photons can't bridge it.

ANALOGY

A filled band is a completely full parking level: no car can move. Lift one car to the empty level above (the conduction band) and it can drive around, while the space it left (a "hole") lets cars below shuffle too.

For light the rule is simple: a photon with energy at or above E_g can be absorbed (any energy above works, because the bands are continuous); a photon below E_g has nothing to excite and passes through (Table 5.1).

Table 5.1 Band gaps near room temperature and what they mean. The wavelength $1240/E_g$ is the longest wavelength the material absorbs strongly.

Material	E_g (eV)	$1240/E_g$ (nm)	Consequence
Germanium (Ge)	0.66	1880 (IR)	absorbs all visible; infrared detectors
Silicon (Si)	1.12	1110 (IR)	absorbs all visible, so it looks gray and opaque; solar cells, camera sensors
Gallium arsenide (GaAs)	1.42	873 (IR)	infrared LEDs and lasers, high-efficiency solar cells
Gallium phosphide (GaP)	2.26	549 (green)	absorbs blue and violet, so it looks orange
Gallium nitride (GaN)	3.4	365 (UV)	transparent to visible; base of blue and UV LEDs
Diamond (C)	≈ 5.5	225 (UV)	transparent and colorless when pure; insulator
Sapphire (Al_2O_3)	≈ 8.8	141 (UV)	transparent; excellent insulator (spark-plug ceramics)
Fused silica (SiO_2)	≈ 9	138 (UV)	transparent glass; insulator (Chapter 6)

MATERIALS SCIENCE CONNECTION

Callister's electrical-properties chapter (Chapter 18) uses these same band pictures: metals have partly filled or overlapping bands; semiconductors and insulators have a full valence band and an empty conduction band, with a gap "relatively narrow (< 2 eV)" for semiconductors and "relatively wide (> 2 eV)" for insulators (a convention, not a law). The gap reflects bonding: in ionic ceramics such as Al_2O_3, MgO and NaCl, electrons sit in the filled shells of O^{2-} or Cl^- ions, held tightly, so gaps reach about 8–9 eV. Hence excellent insulators, transparent as pore-free crystals. Callister's optical-properties chapter (Chapter 21) uses the same gaps to predict color (Chapter 7).

5.9 LEDs and lasers

LEDs: light from a band gap

In a **light-emitting diode** (a junction between two differently doped regions of a semiconductor), an applied voltage drives conduction-band electrons to meet valence-band holes. Each drops across the gap and emits a photon:

$$E_{\text{photon}} \approx E_g \qquad \Rightarrow \qquad \lambda \approx \frac{1240\ \text{eV}\cdot\text{nm}}{E_g} \tag{5.4}$$

Engineers set the color by tuning the alloy: AlGaInP for red and amber, InGaN for green and blue. Silicon makes a poor LED: its "indirect" gap means a falling electron must also create a lattice vibration, so the energy usually becomes heat. White LEDs pair a blue InGaN chip (2014 Nobel Prize in Physics) with a yellow phosphor. And since each electron must be given at least E_g, a red LED needs about 2 volts and a blue one about 3.

WORKED EXAMPLE 5.4 — Designing an LED

Given: GaAs has $E_g = 1.42$ eV.

Find: (a) the wavelength a GaAs LED emits; (b) the band gap needed for a 450 nm blue LED.

Think: the photon energy is about the band gap (Equation 5.4).

(a) $\lambda = 1240/1.42 = 873$ nm: near-infrared, invisible to you.

(b) $E_g = 1240/450 = 2.76$ eV, well above silicon's or GaAs's gap: an InGaN alloy.

Answer: (a) 873 nm infrared, the kind of light GaAs-family LEDs send from a TV remote; (b) about 2.8 eV.

Lasers: copying photons

Einstein predicted in 1916–17 that a photon of exactly the right energy passing an *excited* atom can trigger it to emit: **stimulated emission** (Figure 5.1C). The new photon copies the trigger exactly (energy, direction, phase, polarization), so one photon becomes two, then four: a **laser** (light amplification by stimulated emission of radiation). It needs three ingredients:

- **Population inversion:** more atoms in the upper level than the lower, or absorption wins. This never happens on its own, so the atoms are *pumped* (flash lamp, current, another laser), usually into a long-lived metastable level.
- **A cavity:** mirrors bounce photons back and forth, copying them on every pass.
- **An exit:** one mirror lets a small fraction leak out as the beam.

The first laser (Theodore Maiman, 1960) was a flash-lamp-pumped ruby crystal emitting at 694.3 nm. Diode lasers in barcode scanners and fiber-optic links are essentially LEDs with mirrors, and high-power infrared lasers cut, weld and 3D-print metal parts.

> **MYTH VS. REALITY**
>
> **Myth:** Laser light is made of especially powerful photons.
>
> **Reality:** A laser's photon energy is set by its transition like any other light; a red laser pointer's photons each carry less energy than a blue LED's. What is special is that the photons are identical and in step, so the beam stays narrow and focuses to a tiny, intense spot.

Chapter summary

- One rule: $hf = E_{\text{upper}} - E_{\text{lower}}$. Absorption is resonance: the frequency must match, whatever the atom's size.
- A photon is created during a downward jump, as the mixed-state cloud sloshes at $f = \Delta E/h$. Re-emitted photons leave in random directions.
- Hydrogen: $\Delta E = 13.6\ \text{eV}(1/n_{\text{lower}}^2 - 1/n_{\text{upper}}^2)$; $3 \to 2$ gives the 656 nm red line.
- Hot dense objects: continuous spectra. Hot thin gases: emission lines. Cool gas before a hot source: dark lines at the same wavelengths.
- Absorbed energy is re-emitted, turned into heat, or used for chemistry. Reflection is coherent and involves no absorption.
- Fluorescence: absorb short, emit longer, within ns (Stokes shift). Phosphorescence: a stuck state releases light slowly.
- Vibrations absorb infrared (≈ 0.05–0.5 eV), rotations microwaves (≈ 0.001 eV), if they move charge.
- The band gap separates metals, semiconductors and insulators and sets an LED's photon energy. Lasers amplify by stimulated emission from a population inversion.

Check yourself

1. CONCEPT An 11.0 eV photon passes through ground-state hydrogen without being absorbed, even though it carries more energy than the 10.2 eV needed for $1 \to 2$. Why? Would a 14.0 eV photon be absorbed?
2. CALC Find the energy and wavelength of the H-beta line ($n = 4 \to 2$) and name its color.
3. CALC Find the wavelength of the Lyman-alpha line ($n = 2 \to 1$). Why can't you see it, and why doesn't it reach the ground from space?
4. CONCEPT A cloud of gas absorbs light from a star and then re-emits all of it. Explain why the star's spectrum still shows dark lines.
5. CONCEPT Give three differences between re-emission after absorption and reflection from a mirror.
6. CALC In a fluorescent lamp, mercury emits 254 nm UV, and a phosphor converts it to 545 nm green light. Find both photon energies and the fraction of each UV photon's energy lost as heat in the phosphor.
7. CALC CO_2's bending vibration absorbs at 15 µm. Find the photon energy in eV and compare it with $k_B T$ at room temperature (0.026 eV).

8. **CONCEPT** Why do CO_2 and H_2O absorb infrared strongly while N_2 and O_2 do not?
9. **APPLY** An LED is made from GaP ($E_g = 2.26$ eV). Predict its wavelength and color, and the minimum forward voltage you would expect.
10. **APPLY** Using band gaps, explain why a sapphire (Al_2O_3) watch crystal is transparent while a silicon wafer is opaque. Which one conducts electricity better at room temperature?
11. **CONCEPT** Why can’t you build a laser from atoms in ordinary thermal equilibrium? What is a population inversion and how is it created?

Full solutions are in the Answer Key at the back of the book.

CHAPTER 6

Reflection, Transmission, Absorption, and Refraction

A mirror, a window, a black T-shirt, and a diamond are all built from the same parts: nuclei and electrons. Which one reflects, which lets light through, which soaks it up, and which sparkles comes down to how those electrons move when a light wave shakes them.

THE BIG PICTURE

- Light hitting matter is reflected, absorbed, or transmitted (some of it scattered on the way): R + A + T = 1.
- One mechanism drives all of it: the wave's electric field shakes electrons, and shaken electrons radiate.
- Reflection and refraction are *coherent* re-radiation, not "absorb, wait, re-emit."
- In a material light keeps its frequency but its speed and wavelength drop by a factor n, which bends light at boundaries.

AFTER THIS CHAPTER YOU CAN

- Explain what physically reflects light, and why a mirror can form an image.
- Explain why metals are shiny and opaque while glass is transparent.
- Use $n = c/v$, Snell's law, and the normal-incidence reflectance formula.
- Find a critical angle and explain how optical fibers trap light.
- Explain prisms, rainbows, simple lenses, and why some ceramics are opaque.

6.1 The four fates of light

Light hitting an object can go back (**reflected**), through (**transmitted**), or into the material (**absorbed**). **Scattering** redirects some of it into many new directions; it counts as diffuse reflection or transmission (Figure 6.1).

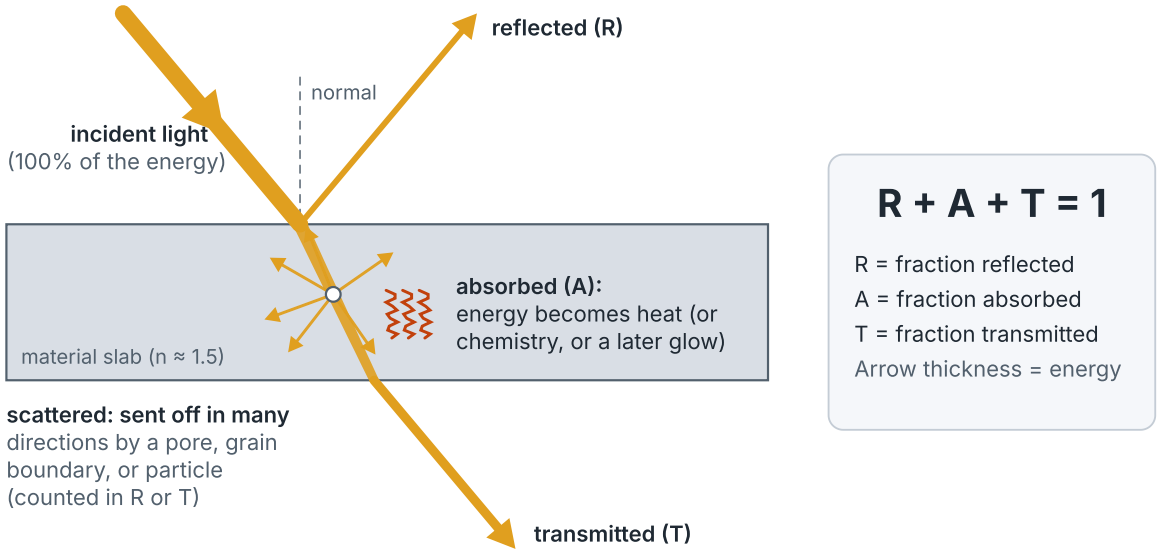


Figure 6.1 The four fates of light at a slab. Arrow thickness shows energy. Energy is conserved, so the reflected, absorbed, and transmitted fractions must add to 1.

$$R + A + T = 1 \tag{6.1}$$

where R, A, and T are the fractions of the incoming energy reflected, absorbed, and transmitted (scattered light is included in R or T).

Absorbed energy usually becomes heat (Chapter 9). Inside an absorber, intensity falls off exponentially with depth x:

$$I = I_0\, e^{-\alpha x} \tag{6.2}$$

where I_0 is the intensity just inside the surface and α is the absorption coefficient (m^{-1}). After a depth of $1/\alpha$, 37% remains.

KEY IDEA

R, A, and T depend on the material *and the wavelength*. Window glass transmits about 90% of visible light but almost none of the thermal infrared a warm object emits.

6.2 What really happens: light shakes electrons

Light is an oscillating electric field (Chapter 1), so it pushes every charge in its path back and forth at the light's frequency. Nuclei are thousands of times heavier, so the electrons do nearly all the moving.

A shaken electron is an accelerating charge, and accelerating charges radiate. Each electron becomes a tiny antenna broadcasting at *exactly* the driving frequency. Every effect in this chapter is the sum of these small waves plus the original one (Figure 6.2a).

The response depends on how close the light's frequency is to the electron's natural frequencies. Far from them, the electron jiggles gently in step with the wave, and the light passes with only a slowdown. At a natural frequency the response is large and energy is absorbed (the gap-matching rule of Chapter 5).

ANALOGY

Push a swing at a random rhythm and it just wiggles along with your hand; push at its natural rhythm and you pour energy in. Off-resonance means transparent; on-resonance means absorption.

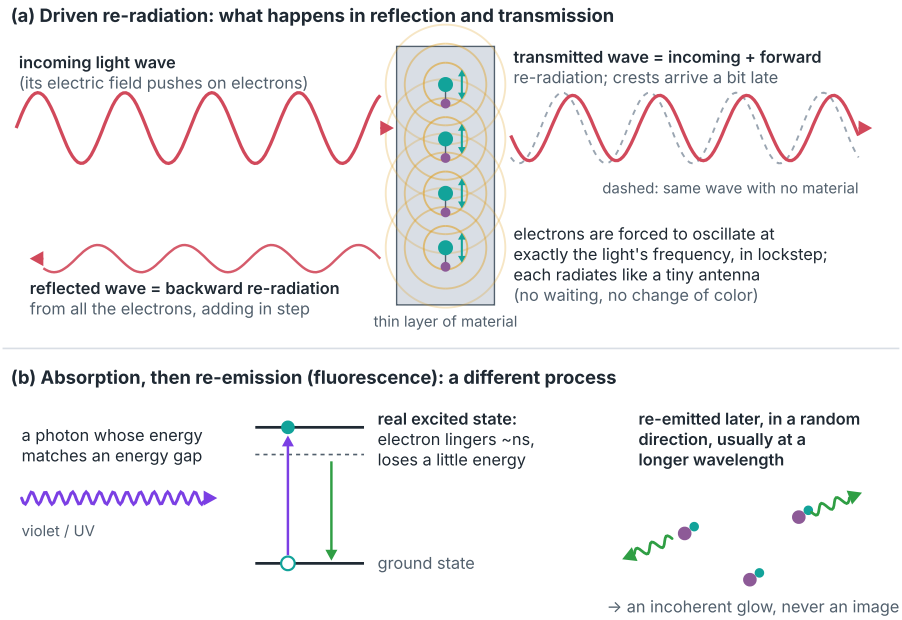


Figure 6.2 (a) Driven re-radiation: electrons forced to oscillate by the wave radiate in lockstep with it. Backward wavelets form the reflected wave; forward wavelets combine with the incoming wave to make a transmitted wave whose crests are slightly delayed. (b) Fluorescence, for contrast: a real excited state, a delay, a random direction, and a longer wavelength.

YOU ASKED

"How does light work? What actually reflects the wave when it hits something?"

The electrons in the material do: the incoming wave shakes them, and the waves they radiate add up to the reflected wave (and, going forward, to the transmitted wave).

Deep inside a uniform material, the sideways and backward wavelets from neighboring electrons cancel. At a boundary, where the material suddenly begins, the backward wavelets have nothing to cancel against, so they add up to a reflected wave. Stronger electron response means stronger reflection: about 4% for glass, over 90% for metals, whose free electrons respond to every visible frequency (Section 6.4).

YOU ASKED

"Is reflection the same thing as the material absorbing and re-emitting the EM wave?"

No. Reflection is coherent re-radiation by electrons that are *driven* by the wave; it is not a photon being absorbed into an excited state and re-emitted later. The two processes leave different fingerprints:

- **Driven re-radiation** (reflection, transmission, refraction): works at any frequency; comes out at the same frequency; the electrons oscillate locked in phase with the wave; no excited state, so no waiting; direction fixed by geometry.
- **Absorption, then re-emission** (fluorescence, Chapter 5): only when the photon energy matches a gap; the electron sits in a real excited state for nanoseconds; it emits at a random moment, in a random direction, usually at a longer wavelength.

The locked phase is what makes a mirror image possible. Billions of electrons re-radiate in step with the incoming wave, so their wavelets reinforce in one direction (the angle of reflection) and cancel in all others. If atoms re-emitted at random times in random directions, a mirror would merely glow, like a highlighter under a black light, with no image.

6.3 Reflection: which way, and how much

Measured from the **normal** (the line perpendicular to the surface), the angle of reflection equals the angle of incidence: the one direction in which all the driven electrons' wavelets arrive in step.

$$\theta_r = \theta_i \tag{6.3}$$

Bumps much smaller than the wavelength give **specular** (mirror-like) reflection. With bumps comparable to the wavelength or larger, every ray still obeys Equation 6.3 about its *local* normal, but the normals point every which way: **diffuse** reflection (Figure 6.3). White paper reflects most visible light, just diffusely.

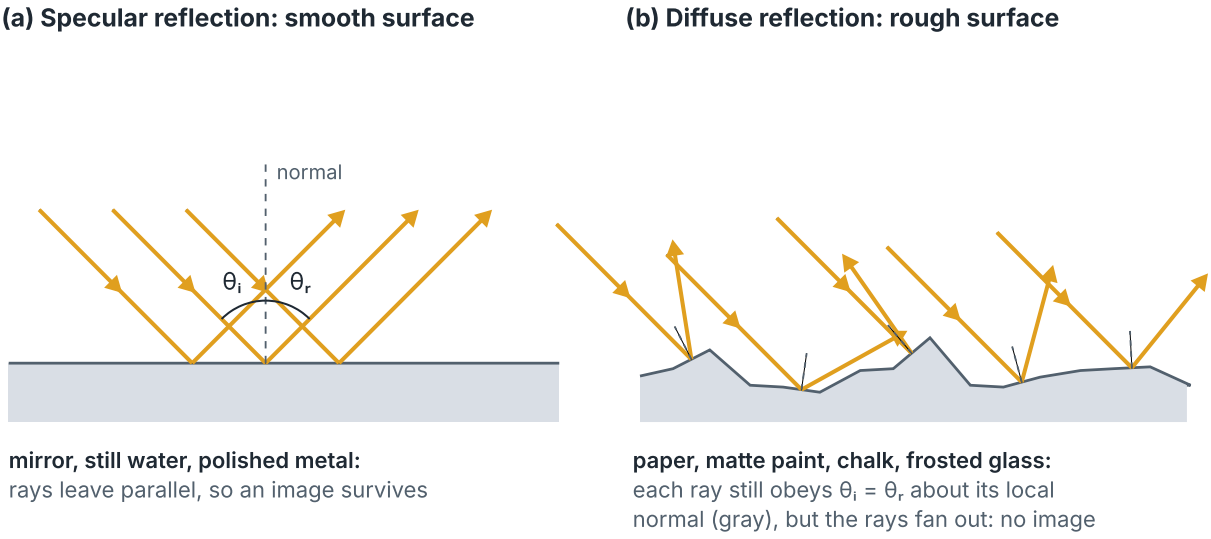


Figure 6.3 Specular vs. diffuse reflection. The law of reflection holds for every ray in both cases; only the surface geometry differs.

MYTH VS. REALITY

Myth: A mirror reflects light because it is very smooth.

Reality: Smoothness decides *where* reflected light goes; the electrons decide *how much*. Polished black glass is mirror-smooth but reflects about 4%. A mirror needs both a smooth surface and a metal coating.

At normal incidence, the reflected fraction depends only on the jump in refractive index n (Section 6.6) across the boundary:

$$R = \left(\frac{n_2 - n_1}{n_2 + n_1}\right)^2 \tag{6.4}$$

WORKED EXAMPLE 6.1 — Seeing yourself in a window at night

Given: crown glass, $n = 1.52$, in air; light at normal incidence.

Find: the reflectance of one surface, and the fraction transmitted by a single pane (2 surfaces) and a double-pane window (4 surfaces). Ignore absorption and repeated bounces.

Think: each air–glass surface is an index jump, so each one reflects according to Equation 6.4 and passes $1 - R$.

$$R = \left(\frac{1.52 - 1.00}{1.52 + 1.00}\right)^2 = (0.206)^2 = 0.043$$

Single pane: $T = (0.957)^2 = 0.92$. Double pane: $T = (0.957)^4 = 0.84$.

Answer: about 4% per surface; a single pane transmits 92% and a double pane 84%. At night the 8% reflected from your bright room outshines the dim scene outside, so you see yourself.

6.4 Why metals are shiny and opaque

In a metal, one to a few electrons per atom are free to roam (the conduction band, Chapter 5). A free electron has no spring holding it in place, so it has no natural frequency to be "far from." It responds strongly to every frequency below a cutoff called the **plasma frequency**, which for most metals lies in the ultraviolet.

The free electrons' re-radiated wave cancels the incoming wave within a thin surface layer, the **skin depth** (tens of nanometers for visible light), and sends nearly all the energy back out: metals reflect about 90–95% of visible light. The remainder is absorbed in that layer, so nothing gets through.

MATERIALS SCIENCE CONNECTION

The "sea of electrons" from metallic bonding explains electrical conductivity, thermal conductivity, and shine at once. Callister notes that visible light is absorbed within about 0.1 µm of a metal's surface, so only thinner films transmit. Gold leaf, about 0.1 µm thick, glows greenish when held up to a bright light.

RABBIT HOLE (optional) — The plasma frequency

Safe to skip. Free electrons keep up with the field only below $f_p = \frac{1}{2\pi}\sqrt{n_e e^2/(\varepsilon_0 m_e)}$, where n_e is the free-electron density. For aluminum ($n_e \approx 1.8 \times 10^{29}\ \mathrm{m}^{-3}$) the matching photon energy is about 15 eV (80 nm, far UV), so aluminum mirrors work into the ultraviolet. Above f_p a metal turns transparent; X-rays pass through thin foils (Chapter 11). The ionosphere is a dilute electron "metal" with a plasma frequency in the radio range: AM and shortwave radio bounce off it, while FM and GPS pass through.

6.5 Why glass is transparent (and where it isn't)

Glass is an insulator: pure silica has a band gap of roughly 9 eV, far above the 1.8–3.1 eV of visible photons. Visible light can only drive its tightly bound electrons off-resonance, so it is transmitted (and slowed), not absorbed. Glass is also uniform, with no pores or grain boundaries to scatter light. Outside the visible, things change (Table 6.1).

Table 6.1 What ordinary window glass does across the spectrum.

Band	What happens	Why
UVB (below about 315 nm)	Mostly absorbed	Photons energetic enough to excite electrons in the glass and its impurities (such as iron). Little sunburn through a window, though much UVA passes (Chapter 10).
Visible (400–700 nm)	About 90% transmitted	Below the gap and far from resonances; about 4% reflects at each surface.
Near infrared (0.7–3 µm)	Mostly transmitted	Still no matching electron or vibration energies.
Thermal infrared (about 5–15 µm)	Absorbed	Matches the vibration frequencies of Si–O bonds (strong band near 9 µm). This is why a closed car heats up in the sun (Chapter 9).

MYTH VS. REALITY

Myth: Transparent materials let light slip through the gaps between their atoms.

Reality: Atoms are mostly empty space (Chapter 4), but a 500 nm wave is thousands of times wider than an atom and drives every electron it passes. Transparency means those electrons cannot absorb at that frequency and nothing scatters. Diamond (gap 5.5 eV) and graphite (no gap) are both pure carbon; one is clear, the other black.

6.6 Refractive index: why light "slows down"

Inside a material, the forward wavelets from driven electrons add to the incoming wave. Each wavelet lags slightly in phase, so after every layer of atoms the combined crests sit a little farther back (Figure 6.2a) and advance more slowly than c. The **index of refraction** n measures how much:

$$n = \frac{c}{v} \tag{6.5}$$

where v is the speed of the wave crests (the phase velocity) in the material.

The electrons oscillate at the source's frequency, so frequency never changes. The wavelength shrinks instead:

$$\lambda = \frac{\lambda_0}{n} \tag{6.6}$$

where λ_0 is the vacuum wavelength. Photon energy hf is unchanged, so red light is still red underwater.

Table 6.2 Typical refractive indices for yellow light (about 589 nm).

Material	n	Material	n
Air	1.0003	Crown (window) glass	1.52
Ice	1.31	Quartz, crystalline SiO_2	1.54–1.55
Water	1.333	Polycarbonate	1.58
Fused silica (SiO_2 glass)	1.458	Sapphire, Al_2O_3	1.77
Acrylic (PMMA)	1.49	Diamond	2.42

KEY IDEA

Crossing into a material changes light's speed and wavelength by the factor n, never its frequency. Bending (refraction), dispersion, lenses, and fibers all follow from that one fact.

MYTH VS. REALITY

Myth: Light slows down in glass because atoms absorb it, hold it for a moment, and re-emit it.

Reality: Glass has no visible-frequency transition to absorb into, and absorb-and-re-emit would scramble direction and phase (Figure 6.2b), leaving glass cloudy. Between atoms, light travels at c; the slower speed belongs to the *combined* crest pattern of incoming and re-radiated waves. Clincher: for X-rays, n is slightly *less* than 1, so crests move slightly faster than c. No delay could do that; superposition does it naturally (energy and signals still never beat c).

MATERIALS SCIENCE CONNECTION

The index rises with electrons per volume and with how easily they shift (electronic polarizability). Same SiO_2, denser packing: crystalline quartz (2.65 g/cm³) has $n \approx 1.54$; fused silica glass (2.20 g/cm³) has 1.458. Large, easily polarized ions such as Pb^{2+} and Ba^{2+} raise n: leaded "flint" glasses reach about 1.7. Non-cubic crystals (quartz, calcite, sapphire) also have direction-dependent n, called **birefringence**; cubic crystals and glasses do not (see Section 6.11).

6.7 Snell's law: bending at a boundary

When a wave enters glass at an angle, one end of each wavefront reaches the slow medium first and lags. To keep crests continuous across the surface, the wavefront pivots and the ray bends (Figure 6.4):

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \tag{6.7}$$

with both angles measured from the normal. Entering a higher-n material, light bends toward the normal; entering a lower-n material, it bends away.

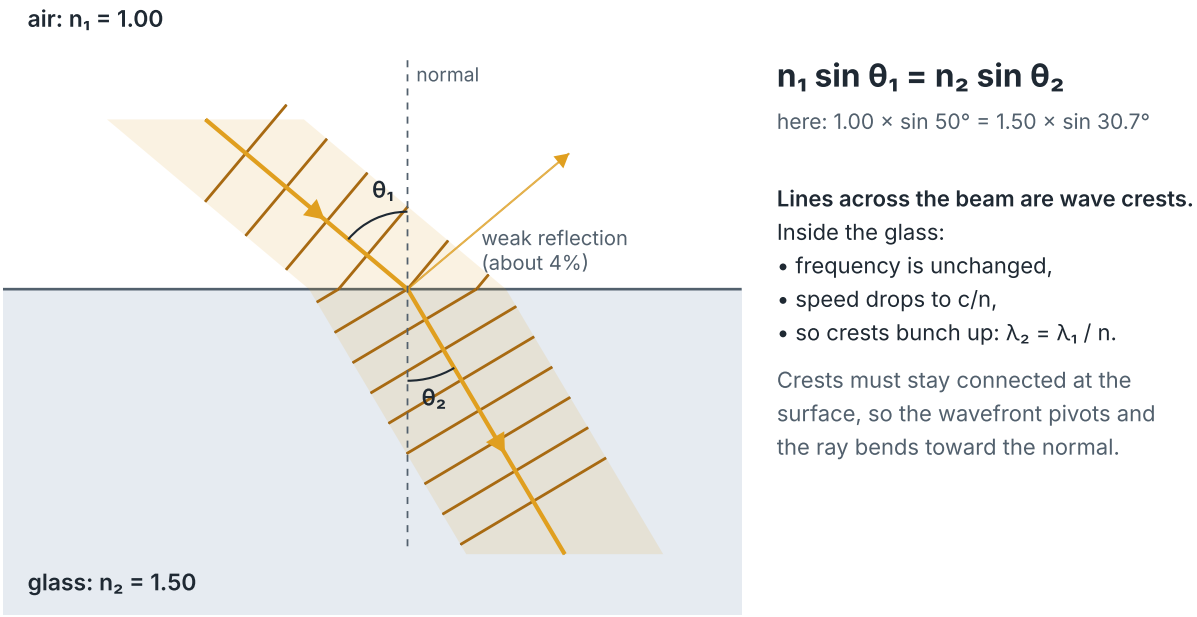


Figure 6.4 Refraction drawn with wave crests. Crests bunch up by the factor n in the glass and must stay connected at the surface, which forces the beam to bend.

ANALOGY

A marching band crosses from pavement onto mud at an angle. The marchers who reach the mud first slow down, and the whole line swings around toward the normal.

WORKED EXAMPLE 6.2 — A red laser enters a glass block

Given: $\lambda_0 = 633$ nm, $\theta_1 = 45.0^\circ$ in air, glass $n = 1.52$.

Find: the refraction angle, and the speed and wavelength inside the glass.

Think: Snell's law gives the angle; Equations 6.5 and 6.6 give speed and wavelength.

$$\sin\theta_2 = \frac{1.00 \times \sin 45.0^\circ}{1.52} = 0.465 \quad \Rightarrow \quad \theta_2 = 27.7^\circ$$

$$v = \frac{3.00 \times 10^8\,\text{m/s}}{1.52} = 1.97 \times 10^8\,\text{m/s}, \qquad \lambda = \frac{633\,\text{nm}}{1.52} = 416\,\text{nm}$$

Answer: 27.7° from the normal; 1.97×10^8 m/s; 416 nm. The beam is still red: its frequency, 4.74×10^{14} Hz, is unchanged.

6.8 Total internal reflection and optical fibers

Going from high n to low n, light bends away from the normal. At the **critical angle** θ_c the refracted ray grazes the surface; beyond it, no light escapes and 100% reflects. This is **total internal reflection** (Figure 6.5a):

$$\sin\theta_c = \frac{n_2}{n_1} \qquad (n_1 > n_2) \tag{6.8}$$

Nothing is absorbed, so total internal reflection beats any metal mirror. Diamond's small critical angle (24.4°) traps light for many bounces, which is why diamonds sparkle; binocular prisms use the same trick.

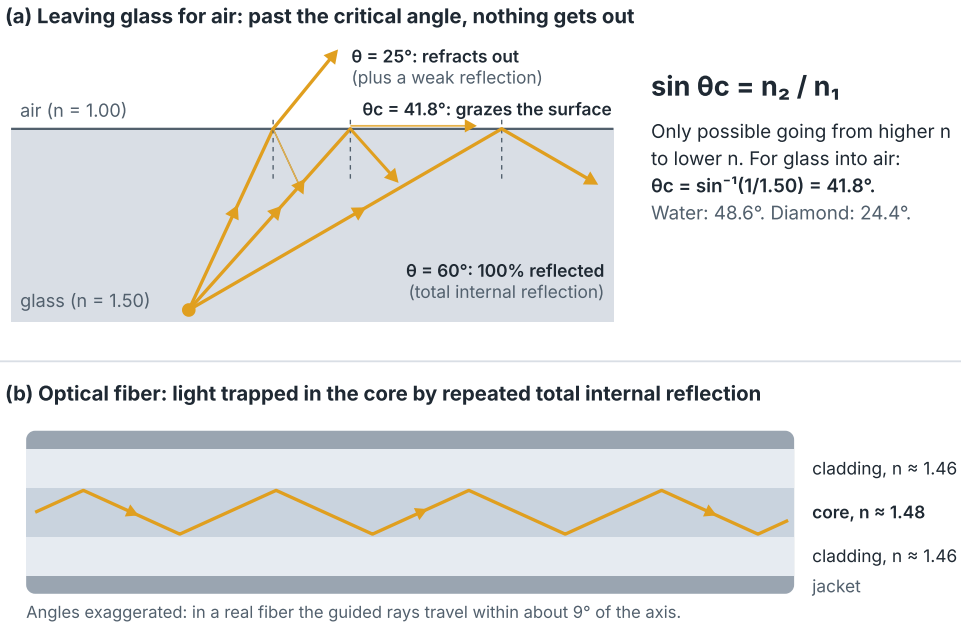


Figure 6.5 (a) Past the critical angle, light cannot leave the denser medium. (b) An optical fiber keeps light in its core by repeated total internal reflection at the core–cladding boundary.

An **optical fiber** is a glass core inside cladding of slightly lower index; light bounces along the core by total internal reflection. Good fibers lose only about 0.2 dB per kilometer at 1550 nm, so light travels about 15 km before losing half its power.

WORKED EXAMPLE 6.3 — How steeply can light travel in a fiber?

Given: core $n_1 = 1.48$, cladding $n_2 = 1.46$.

Find: the critical angle at the core wall, and the largest angle from the axis at which light can enter the fiber end from air.

Think: rays must hit the wall beyond θ_c, so inside the core they travel within $90^\circ - \theta_c$ of the axis. Snell's law at the flat end converts that to an angle in air.

$$\theta_c = \sin^{-1}\left(\frac{1.46}{1.48}\right) = 80.6^\circ \quad \Rightarrow \quad 90^\circ - 80.6^\circ = 9.4^\circ \text{ from the axis}$$

$$\sin\theta_{\text{air}} = 1.48 \sin 9.4^\circ = 0.242 \quad \Rightarrow \quad \theta_{\text{air}} = 14.0^\circ$$

Answer: $\theta_c = 80.6^\circ$; light entering within about 14° of the axis is trapped.

6.9 Dispersion: prisms and rainbows

The index depends slightly on wavelength: **dispersion**. In glass and water, n is larger for violet than red, because violet's frequency is closer to the electrons' ultraviolet resonances. Crown glass (BK7) has $n = 1.522$ at 486 nm (blue) and 1.514 at 656 nm (red); water goes from about 1.343 at 400 nm to 1.331 at 700 nm.

So a prism bends each color by a different angle (Figure 6.6a). In a rainbow, sunlight refracts into each raindrop, reflects once off the back, and refracts out, piling up near 42° from the **antisolar point** (the shadow of your head) for red and about 41° for violet. Red reaches you from slightly higher drops, so it forms the outer edge of the bow.

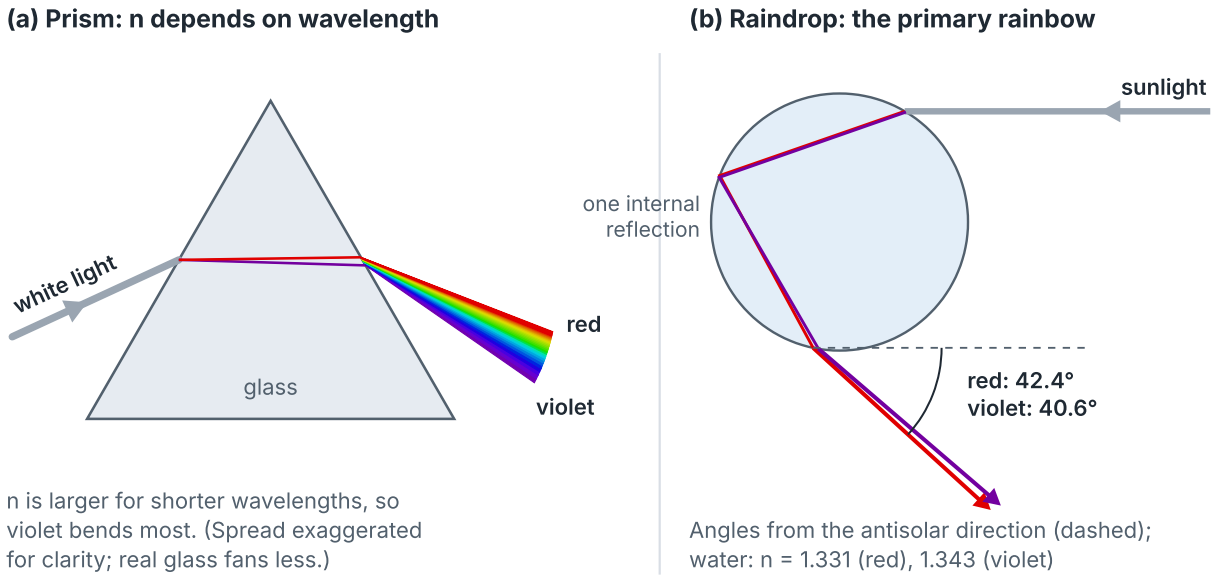


Figure 6.6 Dispersion at work. (a) A prism; the color spread is exaggerated for visibility. (b) A raindrop, ray-traced with water's real indices: one internal reflection sends red out at 42.4° and violet at 40.6°.

6.10 Lenses and focusing

A lens is shaped so that refraction steers rays to a common point. A **converging** lens (thicker in the middle) bends parallel rays through the **focal point** F, a distance f (the **focal length**) away. A **diverging** lens (thinner in the middle) spreads them as if from a point behind it (Figure 6.7). For a thin lens:

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \tag{6.9}$$

where d_o is the object distance and d_i the image distance ($f > 0$ for converging lenses). Eyeglass strength is quoted as power $P = 1/f$ in diopters (m^{-1}).

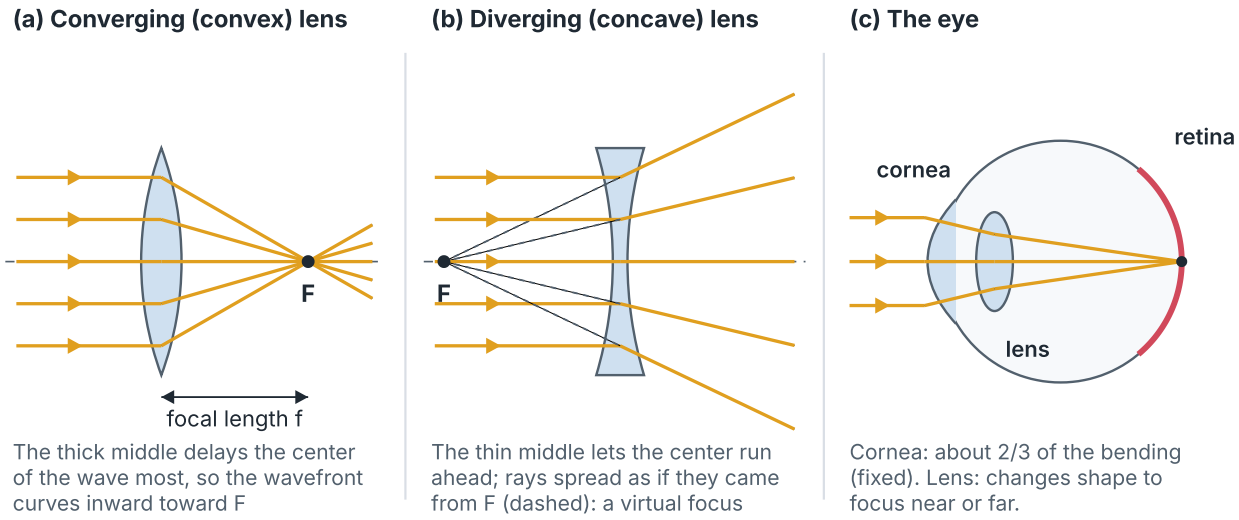


Figure 6.7 (a) A converging lens. (b) A diverging lens with its virtual focus. (c) The eye: the cornea does most of the bending; the lens fine-tunes the focus.

Your eye has about 60 diopters of power. About 43 (two-thirds) come from the cornea, because the biggest index jump is from air into the cornea; the lens adds an adjustable 15–20 diopters by changing shape.

6.11 Opacity and translucency in ceramics and polymers

A transparent material can still look white and opaque. Every internal boundary where n changes reflects a little light (Equation 6.4) and redirects it; with enough boundaries, most light comes back out diffusely. That is why snow and frosted glass look white though ice and glass are clear.

In ceramics the main scatterers are (Figure 6.8):

- **Pores**: a jump from $n \approx 1.77$ to 1.00 scatters strongly; a few percent porosity makes a ceramic opaque.
- **Grain boundaries in non-cubic crystals**: neighboring grains are oriented differently, so a birefringent material presents an index mismatch at every boundary.
- **Second-phase particles** with a different n, added on purpose to glazes as opacifiers.

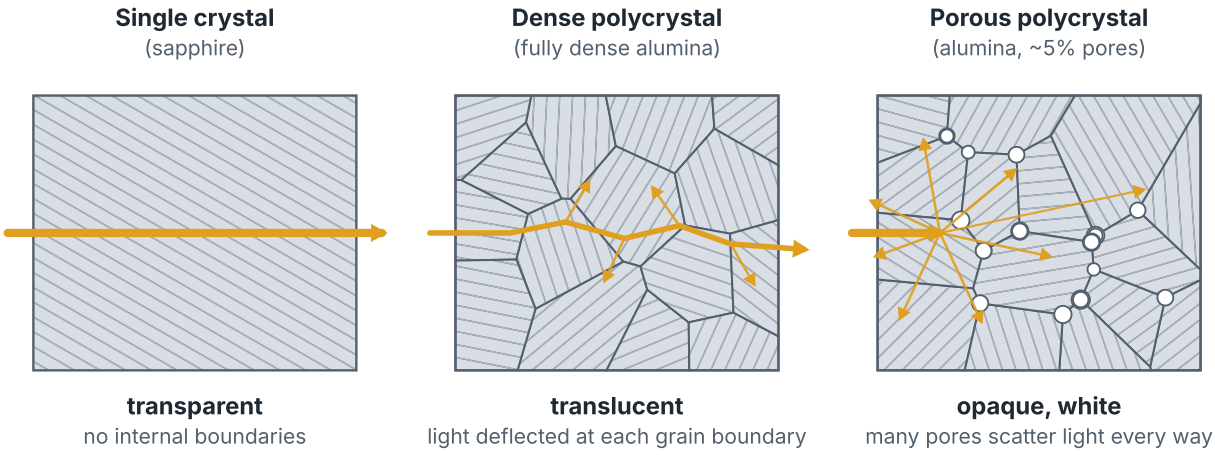


Figure 6.8 One compound, three appearances. Hatching shows crystal orientation. Light is redirected at grain boundaries (because Al_2O_3 is birefringent) and, much more strongly, at pores.

MATERIALS SCIENCE CONNECTION

Sapphire vs. alumina. Both are Al_2O_3 in the corundum structure (O^{2-} in a hexagonal close-packed array, Al^{3+} in two-thirds of the octahedral sites). A single crystal (sapphire) is a transparent window for watches and barcode scanners. Fully dense polycrystalline alumina is translucent (GE's Lucalox sodium-lamp tubes). With about 5% porosity it is opaque white, like a spark-plug insulator. Cubic spinel ($MgAl_2O_4$) has no birefringence, so fully dense polycrystalline spinel can be transparent.

Polymers. Amorphous polymers such as acrylic (PMMA) and polycarbonate are clear. Semicrystalline HDPE (milk jugs) is cloudy: its denser crystalline regions have a higher n than the amorphous ones. PET bottles stay clear because processing keeps their crystals smaller than the wavelength of light.

Chapter summary

- Light meeting matter is reflected, absorbed, or transmitted (some scattered): $R + A + T = 1$, all wavelength-dependent.
- The wave drives electrons, which re-radiate at the same frequency and in fixed phase. Reflection and refraction are this coherent re-radiation, not absorb-and-re-emit (fluorescence), which is delayed, random in direction, and usually red-shifted.
- Roughness compared with the wavelength sets specular vs. diffuse reflection; $R = [(n_2 - n_1)/(n_2 + n_1)]^2$ gives about 4% per glass surface.
- Metals' free electrons respond below the plasma frequency: they reflect 90–95% and absorb the rest within about 0.1 µm. Glass passes visible light (gap far above 3.1 eV) but absorbs UVB and thermal IR.
- $n = c/v$; frequency is unchanged and wavelength becomes λ_0/n. Snell's law gives the bending; beyond $\sin\theta_c = n_2/n_1$, total internal reflection traps light in fibers.

- Dispersion separates colors in prisms and rainbows; lenses focus by refraction. Pores and birefringent grain boundaries scatter light, making ceramics translucent or opaque.

Check yourself

1. CALC A sheet of tinted plastic reflects 6% and absorbs 22% of the light falling on it. What fraction is transmitted?
2. CONCEPT A clear glass marble is transparent, yet the same glass ground into powder looks white. Explain using what happens at each surface.
3. CONCEPT A highlighter pen glows under UV light but you cannot see your reflection in that glow. A mirror shows a sharp image. What is different about how the two re-emit light?
4. CALC Find the normal-incidence reflectance of a diamond surface ($n = 2.42$) in air and compare it with glass ($n = 1.52$).
5. CALC Sodium-yellow light (589 nm in vacuum) enters water ($n = 1.333$). Find its speed, wavelength, and frequency in the water. Does a diver see it as a different color?
6. CALC A light ray in air hits a calm lake at 60.0° from the normal. At what angle does it travel in the water?
7. CALC Find the critical angle for light going from water ($n = 1.333$) into air. What does a fish looking up see outside that cone?
8. CONCEPT Why are metals opaque to visible light even though a sheet of gold leaf about 0.1 µm thick lets some light through? Why do X-rays pass through thin metal foils?
9. APPLY Why is a closed car parked in the sun much hotter inside than the outside air, even though the windows are clear? Use Table 6.1.
10. APPLY The cornea has $n \approx 1.376$. Explain why your vision is blurry when you open your eyes underwater without goggles.
11. APPLY Both are Al_2O_3, yet a sapphire watch crystal is transparent while an alumina spark-plug insulator is opaque white. Explain, and explain why an HDPE milk jug is cloudy while a PMMA sheet is clear.
12. CALC A camera lens with $f = 50.0$ mm photographs a person 4.00 m away. How far behind the lens does the image form, and how tall is the image of a 1.80 m person?

Full solutions are in the Answer Key at the back of the book.

CHAPTER 7

Color

A ripe tomato, a gold ring, the blue sky, and a soap bubble are colored for four completely different physical reasons. Once you can name the mechanism, you can predict the color.

THE BIG PICTURE

- Color is your visual system's summary of which visible wavelengths (400–700 nm, photon energies 1.8–3.1 eV) reach your eye.
- Three types of cone cells turn any spectrum into just three signals, so very different spectra can look identical.
- Most objects are colored by subtraction: they absorb some wavelengths and send back the rest.
- Absorption happens where an electron energy gap matches a visible photon: band gaps, dye molecules, impurity ions, and metals' bands.
- Scattering, interference, and emission make color with no pigment at all.

AFTER THIS CHAPTER YOU CAN

- Explain color in terms of wavelengths, photon energies, and cone responses.
- Distinguish additive (light) from subtractive (pigment) color mixing.
- Predict a material's appearance from its band gap using the 1.8 eV / 3.1 eV rule.
- Explain the colors of ruby, sapphire, gold, copper, and silver.
- Compute relative Rayleigh scattering and explain blue skies, red sunsets, and white clouds.

7.1 Color is light plus a brain

Visible light spans wavelengths of about 400 nm (violet) to 700 nm (red). Using the shortcut from Chapter 2, those are photon energies from 3.1 eV down to 1.8 eV:

$$E = \frac{hc}{\lambda} = \frac{1240\ \text{eV} \cdot \text{nm}}{\lambda} \tag{7.1}$$

Light from almost any object is a mixture of many wavelengths, its **spectrum**. "Color" is the label your eye and brain attach to that mixture.

YOU ASKED

"How does color work in terms of EM radiation?"

Color is how your visual system labels the mix of visible wavelengths (photon energies of about 1.8 to 3.1 eV) entering your eye; an object's color is whatever part of the light falling on it the object sends toward you.

Three steps set what you see:

- **The source** provides a spectrum: sunlight contains every visible wavelength; a sodium streetlight is almost pure 589 nm yellow.
- **The object** reflects, absorbs, transmits, or scatters each wavelength differently (Chapter 6). What survives travels to you.
- **Your eye** measures the arriving light with three kinds of cones, and your brain compares the three signals (Section 7.2).

Change any step and the color changes. That is why a red car looks muddy brown under a sodium lamp: no red light is there for it to send back.

7.2 Three cones, three numbers

Your retina has three types of **cone cells**, each with a pigment tuned to a different range (Figure 7.1): **S** (short-wavelength) cones peak near 420 nm, **M** near 534 nm, and **L** near 564 nm. Rods, which peak near 498 nm, handle dim light and do not distinguish color.

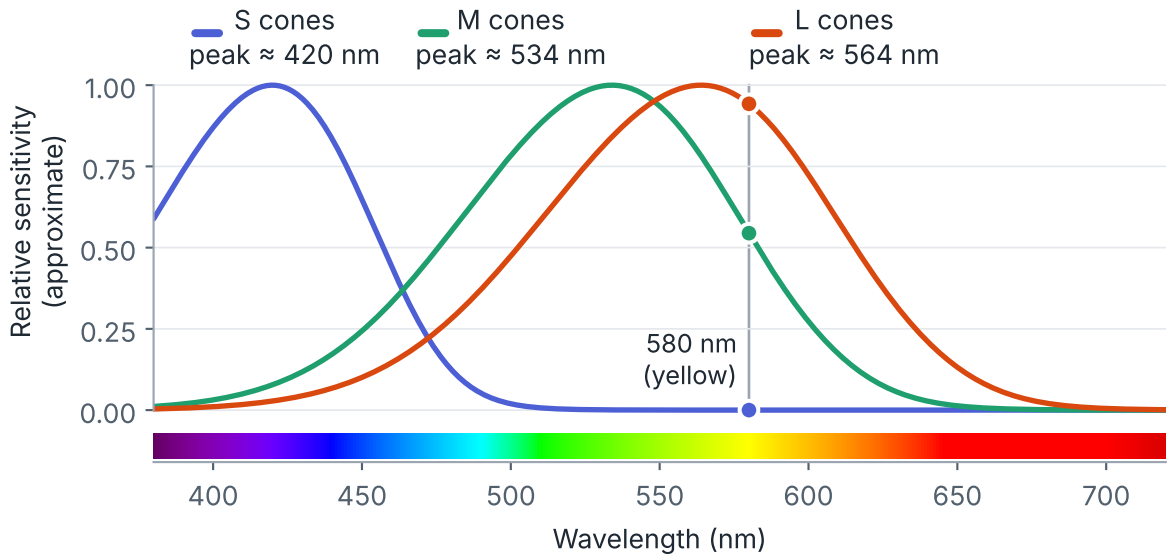


Figure 7.1 Approximate relative sensitivity of the three cone types, each normalized to 1 at its peak (curves from a standard pigment template). Yellow light at 580 nm excites L strongly and M about half as much. Red light plus green light in the right ratio produces the same two signals, and looks the same yellow.

Your brain never sees the full spectrum, only three numbers: how strongly S, M, and L respond. Any two spectra that produce the same three numbers look identical. Such spectra are called **metamers**, and they are why a screen can show yellow using only red and green light.

ANALOGY

Describing a spectrum with three cone signals is like describing a song only by the loudness of its bass, midrange, and treble. Very different songs can give the same three numbers.

MYTH VS. REALITY

Myth: Every color you can see corresponds to some wavelength of light.

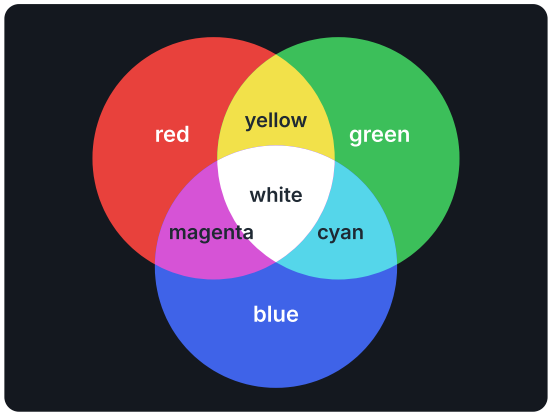
Reality: Magenta, pink, brown, and white appear nowhere in the rainbow. Magenta is what your brain reports when L and S cones fire but M cones do not, a combination no single wavelength can produce.

7.3 Mixing color: adding light vs. subtracting it

There are two opposite ways to mix color (Figure 7.2):

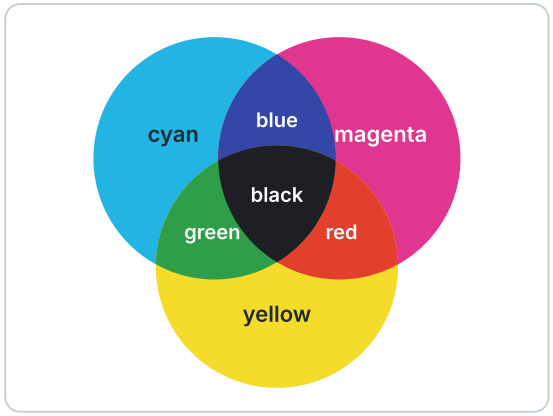
- **Additive mixing** combines lights. Start dark and add red, green, and blue (RGB) light in different amounts. Phone and TV screens work this way.
- **Subtractive mixing** combines absorbers: inks, paints, filters. Start with white light; each layer removes part of the spectrum. Cyan ink absorbs red, magenta absorbs green, and yellow absorbs blue (CMY).

(a) Additive: mixing light



Start dark and add light. Screens and stage lights work this way: red + green looks yellow.

(b) Subtractive: mixing inks and filters



Start white; each ink removes a band. Cyan absorbs red, magenta absorbs green, yellow absorbs blue.

(Real inks are imperfect, so printers add black ink: CMYK.)

Figure 7.2 (a) Additive mixing of light. (b) Subtractive mixing of inks or filters. Each mode's "primaries" are the other mode's two-color mixtures.

KEY IDEA

Pigments don't add color; they take it away. Yellow plus cyan paint looks green because yellow removes blue and cyan removes red, so only green survives both.

7.4 Objects are colored by what they remove

Under white light, an object absorbs some wavelengths and reflects or transmits the rest. The leftovers set its color (Figure 7.3). A red apple absorbs violet through yellow. A leaf's chlorophyll absorbs strongly in the blue (near 430 nm) and red (near 662 nm), leaving mostly green to come back.

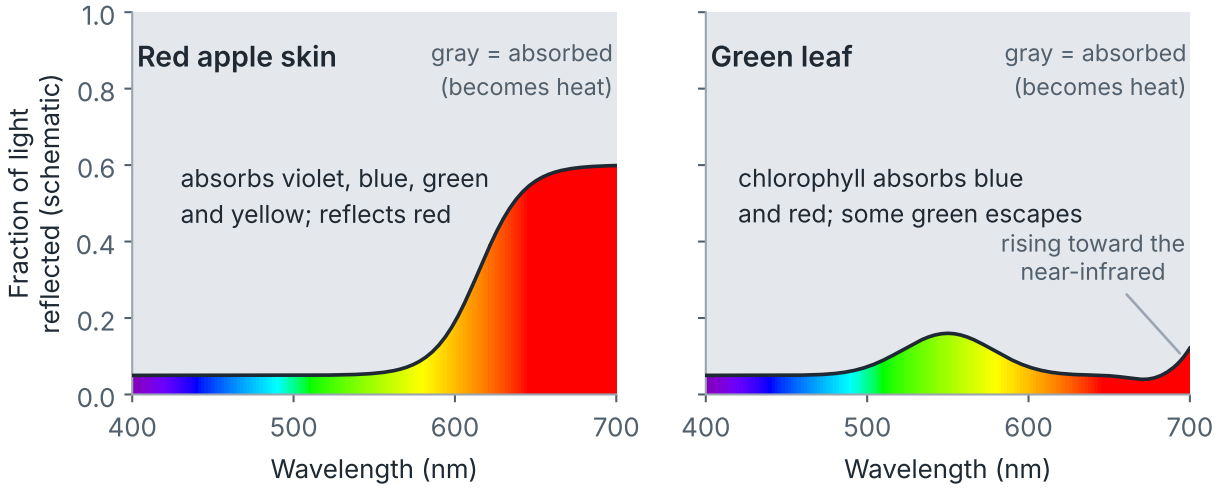


Figure 7.3 White light in, leftovers out (schematic reflectance curves). The colored area is the light that reaches your eye; the gray area is absorbed and becomes heat.

Because color is leftover light, illumination matters. Under pure blue light, a red apple has nothing to reflect and looks nearly black.

> **KEY IDEA**
>
> An object doesn't "have" a color the way it has a mass. It has an absorption spectrum. The color you see is the illumination minus what the object absorbs, as read by your three cones.

Dyes and pigments absorb because their molecules have electron energy gaps in the 1.8–3.1 eV range. The usual culprit is a **conjugated** chain of alternating single and double bonds, along which some electrons are spread out. Like a longer guitar string, a longer chain gives lower energy levels and smaller gaps, so it absorbs longer wavelengths. Beta-carotene (carrots) has 11 conjugated double bonds and absorbs blue light near 450 nm, so carrots look orange. Molecules with short conjugated chains absorb only UV, which is why most plastics and sugars are colorless.

Absorbed energy mostly turns into heat within picoseconds as the molecule shakes its neighbors. Occasionally it is re-emitted as fluorescence (Chapter 5).

7.5 Band gaps and the 1.8–3.1 eV rule

> **YOU ASKED**
>
> *"What are the electron energy gaps that give things their color?"*
>
> They are the energy differences between states an electron can occupy, and a material is colored when one of its gaps falls between about 1.8 and 3.1 eV, the energies of visible photons. Photons that can bridge the gap are absorbed; the rest come back to you.
>
> Four kinds of gaps do most of the coloring in the world:
>
> - **Band gaps in solids** (semiconductors, minerals, pigments): the gap between the filled valence band and the empty conduction band (Chapter 5). Rule below.
> - **Gaps between molecular orbitals** in dyes, chlorophyll, and carotene; longer conjugated chains give smaller gaps (Section 7.4).
> - **Gaps between d-orbital levels of transition-metal ions** in a crystal, such as Cr^{3+} in Al_2O_3 (ruby), split by the surrounding ions (Section 7.6).
> - **Metals** have no gap for their free electrons, so they reflect strongly; their color comes from where deeper electrons begin absorbing, near 2.4 eV in gold (Section 7.7).

For a pure solid with band gap E_g, photons with $E > E_g$ are absorbed and lower-energy photons are not. The **absorption edge** is at

$$\lambda_{\text{edge}} = \frac{1240\ \text{eV} \cdot \text{nm}}{E_g} \tag{7.2}$$

Wavelengths shorter than λ_{edge} are absorbed; longer ones pass or reflect.

Comparing E_g with the visible window gives Callister's rule (Figure 7.4):

- $E_g > 3.1$ **eV**: no visible light is absorbed. Pure crystals are colorless and transparent (diamond, 5.5 eV; sapphire, about 9 eV; ZnS, 3.6 eV). Their powders look white.
- $E_g < 1.8$ **eV**: every visible photon is absorbed, so the material is opaque. Silicon (1.1 eV) and GaAs (1.4 eV) look gray and metallic, because with $n \approx 4$ their surfaces still reflect about 35% of the light (Equation 6.4).
- $1.8 < E_g < 3.1$ **eV**: only the high-energy (blue) end is absorbed. The material looks yellow, orange, or red, shifting toward red as E_g shrinks.

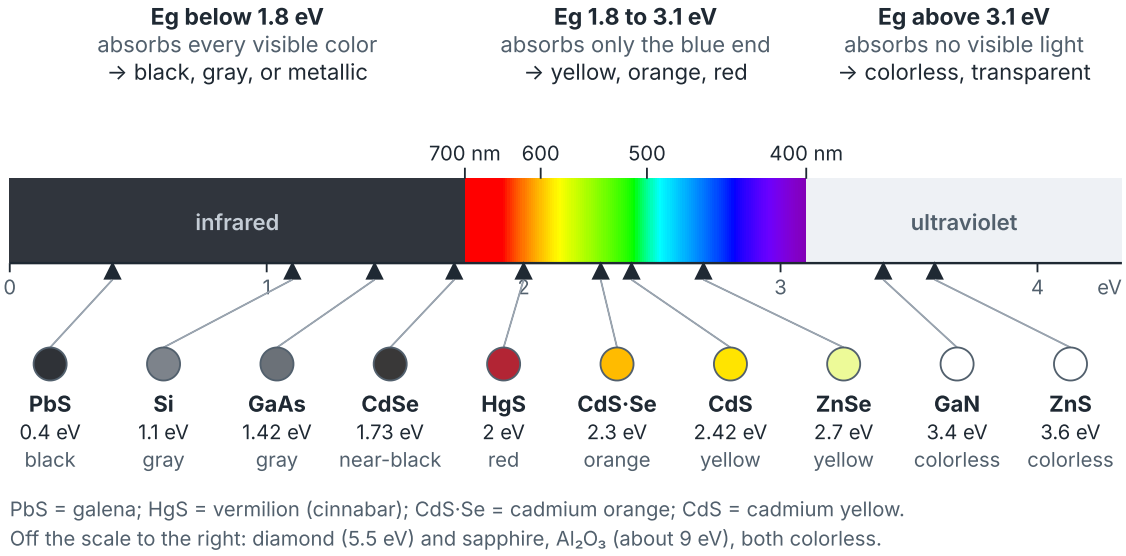


Figure 7.4 The band-gap color rule. Ticks mark each material's gap. Swatches for the colored materials are computed from the absorption edge $1240/E_g$ under daylight. Band gaps from HyperPhysics and WebExhibits.

WORKED EXAMPLE 7.1 — The color of cadmium sulfide

Given: CdS has a band gap $E_g = 2.42$ eV.

Find: its absorption edge and its color in white light.

Think: photons above E_g (wavelengths below the edge) are absorbed; the leftovers set the color.

$$\lambda_{\text{edge}} = \frac{1240\ \text{eV} \cdot \text{nm}}{2.42\ \text{eV}} = 512\ \text{nm}$$

CdS absorbs violet, blue, and blue-green (below 512 nm) and returns green, yellow, orange, and red. With blue missing, the mixture looks yellow to yellow-orange.

Answer: edge at 512 nm; CdS looks yellow-orange. It is the classic artist's pigment "cadmium yellow."

MATERIALS SCIENCE CONNECTION

Tuning a band gap. Within a family of compounds, heavier and larger atoms give weaker bonds and smaller gaps: ZnS (3.6 eV, white) → ZnSe (2.7 eV, yellow); CdS (2.4 eV, yellow) → CdSe (1.7 eV, near black). Alloying CdS with CdSe sets the gap anywhere in between, which is how cadmium orange and cadmium red pigments are made. LEDs use the same trick: the emitted photon energy is roughly the band gap (Chapter 5). GaN's 3.4 eV gap would emit 365 nm UV; alloying in indium lowers the gap to about 2.7 eV for the blue (about 450 nm) LEDs in white lights.

7.6 Impurity ions: ruby, sapphire, and colored glass

Pure Al_2O_3 has a gap near 9 eV, so it is colorless. Replace up to about 1% of the Al^{3+} ions with Cr^{3+} and you have ruby. Chromium's partly filled 3d orbitals are split into levels about 2–3 eV apart by the surrounding O^{2-} ions (the **crystal field**). Ruby absorbs violet (near 410 nm) and yellow-green (near 550 nm), transmitting red plus a little blue. Chromium in ruby also fluoresces deep red at 694 nm, the line used in the first laser.

Blue sapphire works differently. Neighboring Fe^{2+} and Ti^{4+} ions can swap an electron when a photon supplies about 2 eV (**charge transfer**). That absorbs a broad band of yellow and orange light, leaving blue. Glassmakers use the same physics: cobalt (Co^{2+}) gives deep blue glass, and the iron (Fe^{2+}) in ordinary window glass gives its edges a green tint.

MATERIALS SCIENCE CONNECTION

In corundum (Al_2O_3), O^{2-} ions form a hexagonal close-packed array and Al^{3+} fills two-thirds of the octahedral interstitial sites. Cr^{3+} has the same charge and a similar size to Al^{3+}, so it slips onto Al sites as a substitutional impurity. Callister describes ruby as single-crystal sapphire with about 0.5–2% Cr_2O_3 added. A fraction of a percent of the right ion turns a colorless ceramic into a gemstone, the optical version of how small dopant levels transform a semiconductor's conductivity. The same Cr^{3+} in beryl, where the surrounding field is slightly weaker, shifts the absorption bands and gives emerald green.

7.7 Metals: why gold is yellow and silver isn't

All metals reflect strongly because their free electrons respond to every visible frequency (Chapter 6). A metal's color comes from where that reflection *drops*, which happens where more tightly bound electrons in the filled d band can start absorbing (Figure 7.5):

- **Silver**: d-band absorption starts near 4 eV (about 320 nm, in the UV), so all visible colors reflect equally. Silver looks colorless and bright.
- **Gold**: absorption starts near 2.4 eV (about 520 nm), removing much of the blue and violet. The remaining red, orange, yellow, and green look yellow.
- **Copper**: absorption starts lower still (near 2 eV, about 590 nm), taking out more green and leaving copper's reddish orange.

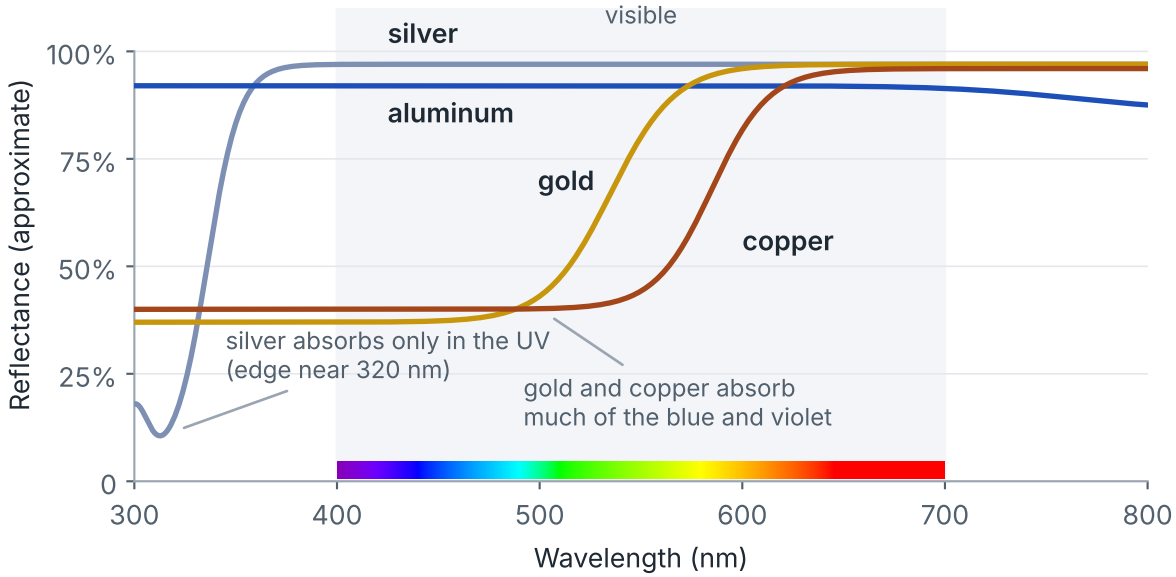


Figure 7.5 Approximate reflectance of polished metals (smooth curves that follow the familiar shapes; not measured data). Where a curve is low, that metal absorbs those wavelengths.

MYTH VS. REALITY

Myth: Gold is yellow because it reflects only yellow light.

Reality: Gold reflects red, orange, yellow, and most green almost perfectly. It looks yellow because blue and violet are *missing*, and white light minus blue looks yellow.

RABBIT HOLE (optional) — Gold is yellow because of relativity

Safe to skip. Gold sits directly below silver in the periodic table, so why is its d-band edge at 2.4 eV instead of about 4 eV? Gold's nucleus (Z = 79) is so highly charged that its inner electrons move at a sizable fraction of c. Einstein's relativity then lowers the energy of gold's 6s orbitals and raises its 5d orbitals, shrinking the gap between them from the UV into the blue. Without relativity, gold would look like silver.

7.8 Scattering: blue sky, red sunsets, white clouds

Air molecules (N_2 and O_2) are about a thousand times smaller than visible wavelengths. Light drives their electrons, which re-radiate (Chapter 6). Because the molecules are randomly placed in a gas, the sideways wavelets do not fully cancel, and some light is scattered in all directions. For scatterers much smaller than the wavelength, this **Rayleigh scattering** depends steeply on wavelength (Chapter 8 explains why):

$$I_{\text{scattered}} \propto \frac{1}{\lambda^4} \tag{7.3}$$

WORKED EXAMPLE 7.2 — How much more is blue scattered than red?

Given: blue light at 450 nm and red light at 700 nm, equally bright, passing through the same air.

Find: the ratio of the scattered intensities.

Think: air molecules are far smaller than both wavelengths, so Equation 7.3 applies and the constant cancels in a ratio.

$$\frac{I_{450}}{I_{700}} = \left(\frac{700\ \text{nm}}{450\ \text{nm}}\right)^4 = (1.556)^4 = 5.86$$

Answer: blue is scattered about 6 times more strongly than red (and 400 nm violet about 9.4 times).

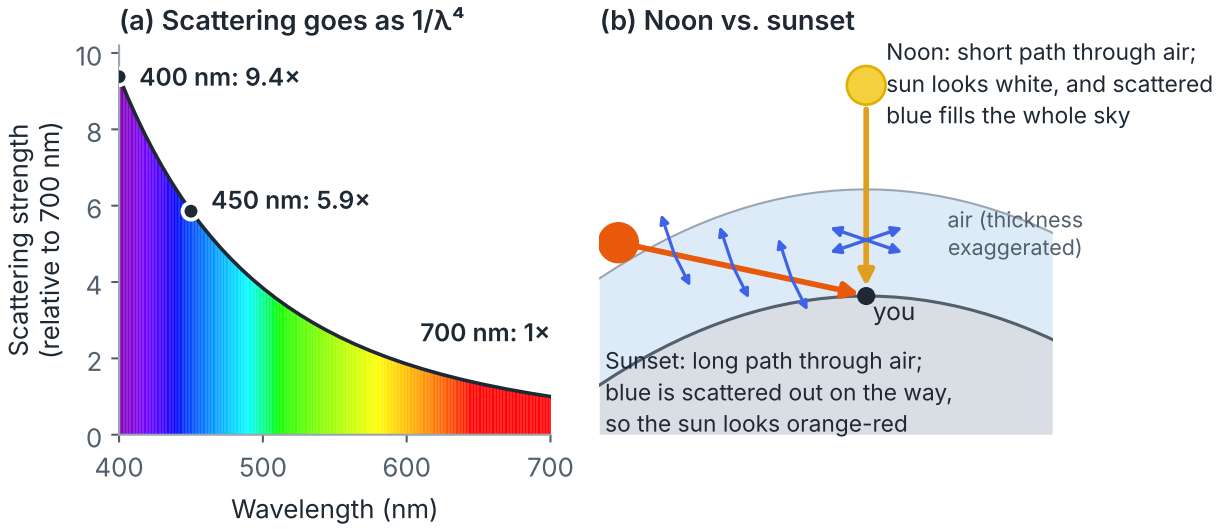


Figure 7.6 (a) Rayleigh scattering strength, relative to 700 nm. (b) At noon the sunlight's path through air is short, and the scattered blue fills the sky. At sunset the path is many times longer, and so much blue is scattered out that the sun looks orange-red.

When you look at any patch of sky away from the sun, you see sunlight scattered toward you, which is rich in blue. At sunset, sunlight crosses far more air to reach you; most of the blue is scattered out of the beam on the way, and what arrives directly is orange and red (Figure 7.6b). Dust and smoke add extra scattering and can make sunsets more vivid.

Clouds are white because their water droplets (around 10–20 μm across) are much *larger* than visible wavelengths. Large droplets scatter all colors about equally, and light bounces among many droplets, so clouds return white light. Very thick clouds look gray because little light makes it through to their bottoms.

MYTH VS. REALITY

Myth: The sky is blue because it reflects the ocean.

Reality: The sky is just as blue over deserts. Its blue is sunlight scattered by air molecules, following Equation 7.3.

RABBIT HOLE (optional) — Why isn't the sky violet?

Safe to skip. Violet scatters even more than blue, so why isn't the sky violet? Three reasons add up: sunlight contains less violet than blue, some violet is absorbed high in the atmosphere, and your eyes are much less sensitive to violet (Figure 7.1). Your cones' response to the scattered skylight adds up to sky blue.

7.9 Structural color: soap bubbles and butterflies

Some colors come from no pigment at all, only from structure. A soap film reflects light from both its front and back surfaces (Figure 7.7a). The back-surface wave travels an extra distance of about $2nt$, where t is the film thickness. Wavelengths whose crests line up are reinforced and others cancel, a wave effect called **interference** (Chapter 8). As t varies across the film, so does the color (Figure 7.7b). Where the film thins to a few tens of nanometers, the two reflections cancel for every color and the film looks black, a sign it is about to pop.

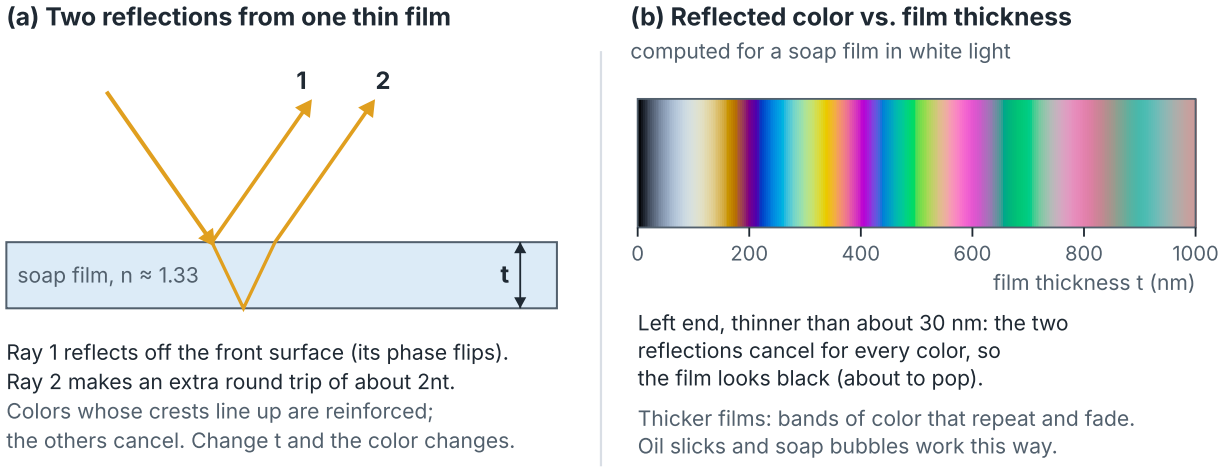


Figure 7.7 Thin-film interference. (a) Two reflections from one film. (b) Reflected color versus thickness, computed for a soap film in white light (brightness boosted; real films reflect only a few percent).

Oil slicks on wet pavement work the same way. The brilliant blue of *Morpho* butterflies comes from ridged nanostructures in their wing scales, with spacing comparable to a quarter of a blue wavelength, rather than from blue pigment. Structural colors usually shift as you tilt your head, an effect called **iridescence**, because the extra path length changes with angle.

7.10 Color from emission: neon, fireworks, LEDs, screens

Things that make their own light have the color of their emission spectrum (Chapter 5):

- **Neon signs**: pure neon glows red-orange. Other "neon" colors use argon with a little mercury, whose UV light makes phosphor coatings on the tube glow.
- **Fireworks**: metal atoms emit their characteristic lines: strontium red, barium green, copper blue, sodium yellow (589 nm).
- **LEDs** emit photons with energy close to the band gap. Most white LEDs are blue LEDs coated with a yellow-emitting phosphor, which absorbs some blue and re-emits yellow (the incoherent absorb-and-re-emit process of Chapter 6). Blue plus yellow looks white.
- **Screens** build every pixel from red, green, and blue subpixels and rely on additive mixing and your three cones (Figure 7.2a).

7.11 Why black things get hot in the sun

A black surface absorbs most of the sunlight hitting it; a white surface reflects most of it. Absorbed light becomes heat (Section 7.4), so a black car seat can burn your legs while a white one stays tolerable. More than half of sunlight's energy is actually near-infrared, so a surface's heating also depends on wavelengths you can't see. "Cool roof" coatings in dark colors use pigments that reflect near-infrared.

WORKED EXAMPLE 7.3 — Black roof vs. white roof on a car

Given: sunlight of about 1000 W/m² hits a 3.0 m² car roof. A black roof absorbs about 95% of it; a white roof reflects about 70%, so it absorbs 30%.

Find: the absorbed power for each roof.

Think: absorbed power = absorbed fraction × intensity × area.

$$P_{\text{black}} = 0.95 \times 1000\ \text{W/m}^2 \times 3.0\ \text{m}^2 = 2850\ \text{W}$$

$$P_{\text{white}} = 0.30 \times 1000\ \text{W/m}^2 \times 3.0\ \text{m}^2 = 900\ \text{W}$$

> **Answer: about 2.9 kW vs. 0.9 kW. The 2 kW difference is more than a typical space heater. How hot each roof gets depends on how fast it loses heat, which is Chapter 9's topic.**

Chapter summary

- Visible light is 400–700 nm, photon energies 3.1–1.8 eV. Color is the brain's reading of three cone signals (S, M, L peaks near 420, 534, 564 nm), so different spectra (metamers) can look identical.
- Light mixes additively (RGB screens); pigments mix subtractively (CMY inks), because each pigment removes part of the spectrum.
- Most objects are colored by what they absorb. The absorbed light becomes heat; the leftovers reach your eye.
- Band-gap rule: $E_g > 3.1$ eV is colorless, $E_g < 1.8$ eV is opaque and dark or metallic, and gaps in between are yellow to red. The edge is at $\lambda = 1240/E_g$ nm.
- Impurity ions (Cr^{3+} in ruby, Fe^{2+}–Ti^{4+} in sapphire) and conjugated molecules (dyes, carotene, chlorophyll) create visible-energy gaps. Gold and copper are colored by d-band absorption starting near 2.4 and 2 eV; silver's starts in the UV.
- Rayleigh scattering ($\propto 1/\lambda^4$) makes the sky blue and sunsets red; large cloud droplets scatter all colors and look white.
- Thin films and nanostructures make structural color by interference; emitters (neon, fireworks, LEDs, screens) show their emission spectra.

Check yourself

1. CONCEPT A red apple is lit only by blue light in a dark room. What color does it look, and why?
2. CONCEPT Your phone shows a patch of pure yellow. Which wavelengths are actually leaving the screen, and why do you see yellow?
3. CALC ZnSe has a band gap of 2.7 eV. Find its absorption edge wavelength and predict its color.
4. CALC Use the band-gap rule to predict whether diamond ($E_g = 5.5$ eV) and silicon ($E_g = 1.1$ eV) are transparent, colored, or opaque. Why does silicon look shiny gray rather than black?
5. CALC How many times more strongly is 400 nm violet light Rayleigh-scattered than 650 nm red light?
6. CONCEPT Why are clouds white, while the clear sky around them is blue? Why are storm clouds gray?
7. APPLY Ruby and sapphire are both Al_2O_3. Why is pure sapphire colorless, and what makes ruby red and blue sapphire blue?
8. CONCEPT Explain, in terms of which colors each ink absorbs, why overlapping cyan and yellow ink looks green.
9. CONCEPT Gold looks yellow and silver looks white, though both reflect over 90% of red light. Which colors does gold absorb, and why doesn't silver absorb them?
10. CALC A red LED emits at 630 nm. What band gap does its semiconductor need? What color would a GaN LED ($E_g = 3.4$ eV) emit?
11. APPLY In 1000 W/m² sunlight, about 0.30 m² of your shirt faces the sun. Compare the absorbed power for a black shirt (absorbs 90%) and a white shirt (absorbs 30%).
12. CONCEPT Why does a soap bubble show swirling colors, and why does a spot on it turn black just before it pops?

Full solutions are in the Answer Key at the back of the book.

CHAPTER 8

Interference and Diffraction (and Why Size vs. Wavelength Matters)

Why does a CD throw rainbows, why can't a light microscope show you an atom, and how can an atom 5000 times smaller than a light wave still absorb it? It comes down to waves adding up, and to comparing size with wavelength.

THE BIG PICTURE

- Overlapping waves add: in step they reinforce, out of step they cancel. That is interference.
- Waves spread out after passing through an opening, by an angle of about λ/a. That is diffraction.
- Electrons and other matter diffract too: quantum objects travel as waves.
- Diffraction blurs every image: no lens resolves details much below half a wavelength.
- Size vs. wavelength controls *scattering* and *imaging*; *absorption and emission* are controlled by resonance instead.

AFTER THIS CHAPTER YOU CAN

- Predict bright and dark fringes from path differences, using $d \sin\theta = m\lambda$.
- Explain diffraction with Huygens' principle and estimate how much a beam spreads.
- Compute a de Broglie wavelength and explain what electron diffraction shows.
- Estimate the resolution limit of an eye, telescope, or microscope.
- Sort an object into the Rayleigh, Mie, or geometric regime, and explain why tiny atoms still absorb and emit long waves.

8.1 Superposition: waves add

When two waves overlap, the total field at each point is the sum of the two fields. This is **superposition**, and for light it is very nearly exact: two beams cross without disturbing each other.

What matters is the **phase**, meaning where each wave is in its cycle. If crest meets crest, the waves are in phase and the amplitude doubles: **constructive interference**. If crest meets trough, they are half a wavelength out of phase and cancel: **destructive interference** (Figure 8.1). Intensity goes as amplitude squared (Chapter 1), so the bright spots get four times brighter; the energy missing from dark spots shows up in bright ones.

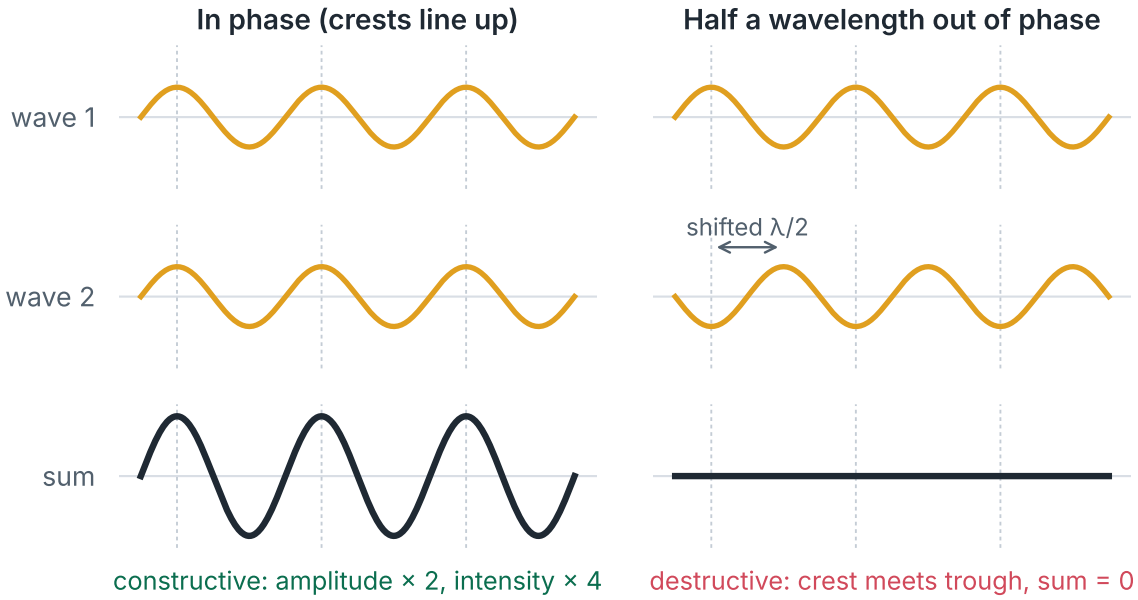


Figure 8.1 Superposition. Left: crests line up, so the sum has twice the amplitude. Right: the second wave is shifted by half a wavelength, so every crest meets a trough and the sum is zero.

The usual source of a phase shift is a **path difference**: one wave travels farther than the other to reach the same point. (Noise-canceling headphones add a half-cycle shift on purpose, playing an inverted copy of the noise.)

> **KEY IDEA**
>
> Path difference decides everything. A whole number of wavelengths extra, $(0,\ \lambda,\ 2\lambda,\ \ldots)$, gives a bright spot; an extra half wavelength, $(\lambda/2,\ 3\lambda/2,\ \ldots)$, gives a dark one. The pattern only holds still if the waves keep a fixed phase relationship (they are **coherent**), which is why lasers make such clean patterns.

8.2 Diffraction: waves spread through openings

Diffraction is what a wave does when part of it is blocked: it spreads into the region behind the obstacle instead of casting a perfectly sharp shadow. **Huygens' principle** (1678) explains why: every point on a wavefront acts as a source of small wavelets, and the next wavefront is their sum.

At an opening, only the wavelets from inside the gap survive. A narrow gap holds just a few, so the wave leaving it spreads like ripples from a single point. A wide gap holds many, and they rebuild a straight beam that spreads only at its edges (Figure 8.2).

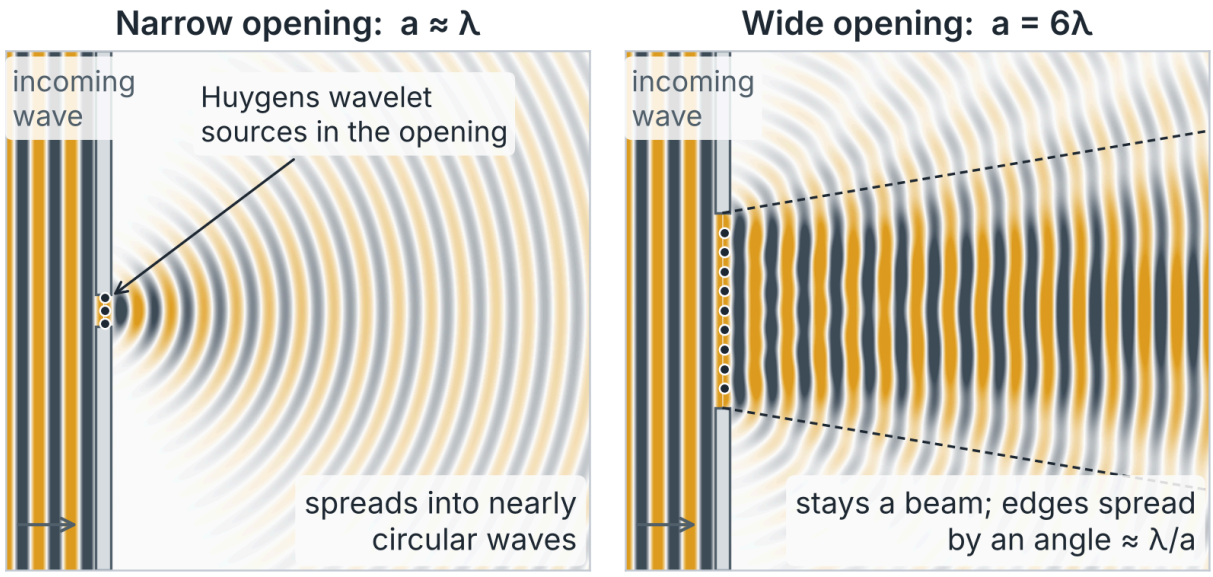


Figure 8.2 Huygens' principle at work, computed by adding up wavelets from every point in the opening (dots). A gap about one wavelength wide (left) sends out nearly circular waves. A gap six wavelengths wide (right) passes a beam whose edges spread by roughly λ/a (dashed lines).

For a single slit of width a, the wavelets from the two halves of the slit cancel at the angle where

$$\sin\theta_{\text{first dark}} = \frac{\lambda}{a} \tag{8.1}$$

where a is the slit width and λ the wavelength. Most of the light lands inside this angle.

KEY IDEA

A wave squeezed through an opening of width a spreads by an angle of about λ/a. Diffraction is dramatic when the opening is about the size of the wavelength and negligible when the opening is thousands of wavelengths across.

That is why you can hear someone around a corner but not see them. A voice has wavelengths around 1 m, about the width of a doorway, so it spreads widely. For visible light ($\lambda \approx 0.5\ \mu\text{m}$) the same doorway gives $\lambda/a \approx 6 \times 10^{-7}$: essentially no spreading.

ANALOGY

Ocean swells at a harbor entrance do the same thing. A narrow gap sends half-circle ripples across the whole harbor; a wide gap lets a straight band through and leaves calm water behind the breakwater.

MYTH VS. REALITY

Myth: Light travels in perfectly straight lines; only sound and water waves bend around corners.

Reality: Light diffracts around every edge; its wavelength is just tiny compared with everyday objects. Look at a distant streetlight through a fine mesh curtain and you will see a diffraction pattern.

8.3 Double slits, gratings, and thin films

Young's double slit

In the early 1800s Thomas Young sent light through two closely spaced slits and saw bright and dark bands, strong evidence that light is a wave. The two waves reach a point P on the screen by paths that differ by $d\sin\theta$ (Figure 8.3), so bright fringes appear where

$$d \sin\theta = m\lambda \qquad (m = 0, \pm 1, \pm 2, \dots) \tag{8.2}$$

where d is the slit spacing, θ the angle from the center line, and m the **order** of the fringe.

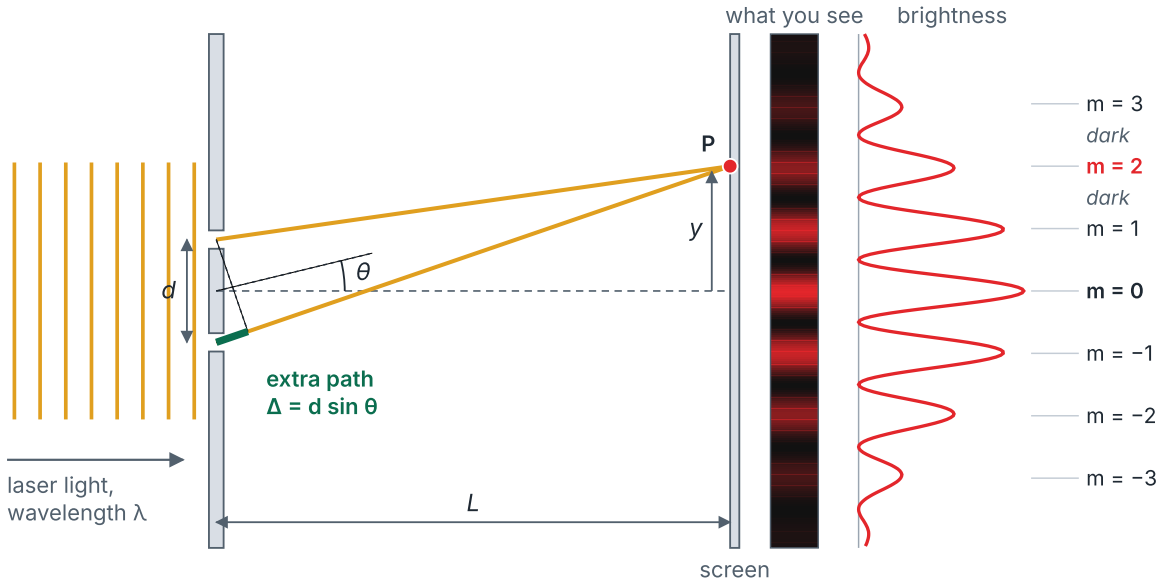


Figure 8.3 Double-slit interference. The ray from the lower slit travels an extra distance $\Delta = d\sin\theta$ (green). Where $\Delta = m\lambda$, the waves arrive in step and the screen is bright. Here P sits on the $m = 2$ fringe. The slow fade of the fringes comes from diffraction by each individual slit.

For small angles, $\sin\theta \approx \tan\theta = y/L$, so neighboring bright fringes on a screen a distance L away are spaced by

$$\Delta y = \frac{\lambda L}{d} \tag{8.3}$$

WORKED EXAMPLE 8.1 — A laser pointer through two slits

Given: red laser, $\lambda = 650$ nm; slit spacing $d = 0.100$ mm; screen $L = 2.00$ m away.

Find: the angle of the first bright fringe and the fringe spacing on the screen.

Think: first bright fringe means $m = 1$ in Eq. 8.2. The angle will be tiny, so Eq. 8.3 applies.

$$\sin\theta_1 = \frac{\lambda}{d} = \frac{650 \times 10^{-9}\,\text{m}}{1.00 \times 10^{-4}\,\text{m}} = 6.50 \times 10^{-3} \;\Rightarrow\; \theta_1 = 0.372^\circ$$

$$\Delta y = \frac{\lambda L}{d} = \frac{(650 \times 10^{-9}\,\text{m})(2.00\,\text{m})}{1.00 \times 10^{-4}\,\text{m}} = 1.30 \times 10^{-2}\,\text{m}$$

Answer: $\theta_1 \approx 0.37^\circ$; fringes about 1.3 cm apart. Slits a tenth of a millimeter apart bend light by only a third of a degree, which is how well light hides its wave nature.

Diffraction gratings and the CD rainbow

A **diffraction grating** has thousands of evenly spaced slits or grooves. Eq. 8.2 still gives the angles, but with so many sources the bright lines become very sharp. Because θ depends on λ, a grating fans white light into a spectrum, which makes it the heart of most **spectrometers** (the instruments that read Chapter 5's spectral fingerprints). A CD is a reflection grating whose track repeats every 1.6 µm (Figure 8.4a).

WORKED EXAMPLE 8.2 — Where the CD rainbow appears

Given: track spacing $d = 1.6\ \mu\text{m} = 1600$ nm; white light (400–700 nm) arriving straight on.

Find: the angles spanned by the first-order ($m = 1$) and second-order ($m = 2$) rainbows.

Think: apply $\sin\theta = m\lambda/d$ to each end of the visible range.

$$m = 1: \quad \sin\theta = \frac{400}{1600} = 0.250 \Rightarrow 14.5^\circ; \qquad \sin\theta = \frac{700}{1600} = 0.438 \Rightarrow 25.9^\circ$$

$$m = 2: \quad \sin\theta = \frac{800}{1600} = 0.500 \Rightarrow 30.0^\circ; \qquad \sin\theta = \frac{1400}{1600} = 0.875 \Rightarrow 61.0^\circ$$

Answer: first-order rainbow from 14.5° (violet) to 25.9° (red); second-order from 30° to 61°. Tilt a CD under a lamp to find both.

Thin films: soap bubbles and oil slicks

A soap film a few hundred nanometers thick reflects light from both surfaces (Figure 8.4b). The back-surface ray travels about $2t$ farther (effectively $2nt$, since light is slower in the film; Chapter 6). Some colors return in step and look bright while others cancel, so white light comes back tinted, and the tint shifts as the film thins. The front reflection is also flipped over, so a film much thinner than a wavelength cancels every color and looks black just before it pops.

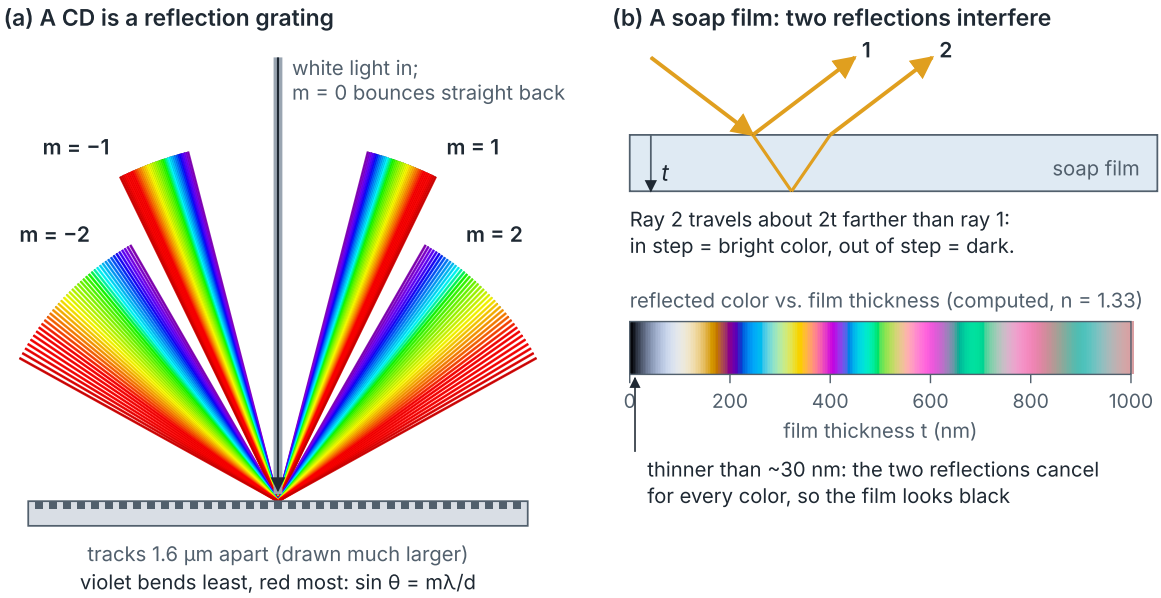


Figure 8.4 Two everyday rainbows from interference. (a) Ray angles computed from $d\sin\theta = m\lambda$ for a 1.6 µm track spacing. (b) Thin-film interference, with the reflected color computed for a soap film versus its thickness. Below about 30 nm the film looks black.

The same physics colors oil on a wet road, anti-reflection coatings on lenses, butterfly wings (structural color, Chapter 7), and heat-tinted steel (Chapter 9).

8.4 Diffraction in quantum physics

In 1924 Louis de Broglie proposed that matter has a wavelength too, linked to its momentum the same way a photon's is (Chapter 2):

$$\lambda = \frac{h}{p} \tag{8.4}$$

where $h = 6.626 \times 10^{-34}$ J·s and p is momentum ($p = \sqrt{2mK}$ for a slow particle of mass m and kinetic energy K).

Everyday objects have absurdly tiny wavelengths and never visibly diffract. But electrons from a modest voltage have wavelengths about the spacing between atoms, so a crystal acts as a natural grating, as Davisson and Germer showed in 1927.

WORKED EXAMPLE 8.3 — An electron behaves like a wave

Given: electrons with $K = 54$ eV (as in Davisson and Germer's experiment); $m_e = 9.109 \times 10^{-31}$ kg; rows of atoms on the nickel surface 0.215 nm apart.

Find: the electron wavelength and the angle of the first diffracted beam.

Think: 54 eV is far below the electron's rest energy, so $p = \sqrt{2mK}$. Then treat the rows of atoms as a grating (Eq. 8.2 with $m = 1$).

$$p = \sqrt{2(9.109 \times 10^{-31}\,\text{kg})(54 \times 1.602 \times 10^{-19}\,\text{J})} = 3.97 \times 10^{-24}\,\text{kg}\cdot\text{m/s}$$

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{3.97 \times 10^{-24}}\,\text{m} = 1.67 \times 10^{-10}\,\text{m} = 0.167\,\text{nm}, \qquad \sin\theta = \frac{0.167}{0.215} = 0.777 \Rightarrow \theta = 51^\circ$$

Answer: $\lambda \approx 0.167$ nm and a diffracted beam near 51°. Davisson and Germer found a strong beam at 50°, matching de Broglie's prediction.

The strangest version is the one-at-a-time double slit. Send electrons (or photons) so weakly that only one is in the apparatus at a time. Each lands at a single spot, like a particle, yet thousands of them build the interference pattern dot by dot (Figure 2.5 in Chapter 2 shows the buildup with photons; electrons look the same). This was done with electrons in Bologna in the 1970s and at Hitachi in 1989; even whole C_{60} molecules have been diffracted.

Quantum mechanics describes this with a **probability amplitude**: a wave that passes through both slits and interferes with itself, whose intensity at each spot gives the chance of detecting the particle there (wave–particle duality, Chapter 2).

MYTH VS. REALITY

Myth: Each photon or electron secretly goes through one slit or the other; we just don't know which.

Reality: If each particle used only one slit, there would be nothing to interfere with and no fringes. Add a detector that records which slit each particle takes, and the fringes do vanish, leaving two plain blobs.

YOU ASKED

"What is diffraction — and what does it mean in quantum physics?"

Diffraction is the spreading of a wave as it passes through an opening or around an edge, by an angle of roughly λ/a; it is strongest when the opening is about a wavelength wide (Section 8.2). Every point of the wave launches wavelets, and an opening keeps only some of them.

In quantum physics, diffraction is the proof that particles travel as waves. Electrons, neutrons, and even molecules diffract with $\lambda = h/p$, and particles sent one at a time still build an interference pattern. Each is *detected* at one spot but *travels* as a wave whose intensity sets the odds of where that spot will be.

8.5 The diffraction limit

A lens is also an opening, so it diffracts. Even a perfect lens images a point as a small blurred disk with faint rings, the **Airy disk**, and points closer than about one disk radius merge (Figure 8.5). For an aperture of diameter D, the **Rayleigh criterion** says two points are just resolved at an angular separation of

$$\theta_{\text{min}} \approx 1.22\,\frac{\lambda}{D} \quad \text{(radians)} \tag{8.5}$$

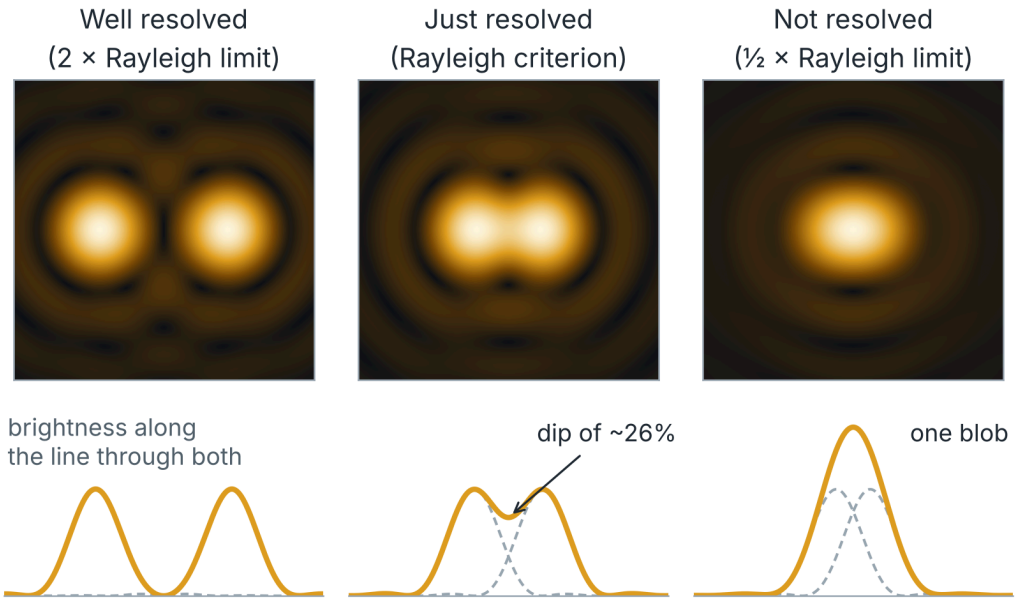


Figure 8.5 Computed Airy patterns of two point sources. At the Rayleigh separation (center), the peak of one sits on the first dark ring of the other and the combined brightness dips by about 26% between them, just enough to tell them apart. Any closer and they blur into one.

WORKED EXAMPLE 8.4 — Your eye vs. the Hubble Space Telescope

Given: $\lambda = 550$ nm. Daytime pupil $D = 3.0$ mm; car headlights 1.5 m apart. Hubble's mirror $D = 2.4$ m; the Moon is 3.84×10^{8} m away.

Find: (a) how far away the headlights must be before a diffraction-limited eye sees them as one light; (b) the smallest detail Hubble can resolve on the Moon.

Think: find θ_{min} from Eq. 8.5, then use (separation) = (distance) × θ.

$$\text{(a) } \theta_{\text{min}} = \frac{1.22(550 \times 10^{-9})}{3.0 \times 10^{-3}} = 2.24 \times 10^{-4}\ \text{rad}, \qquad \text{distance} = \frac{1.5\ \text{m}}{2.24 \times 10^{-4}} = 6.7\ \text{km}$$

$$\text{(b) } \theta_{\text{min}} = \frac{1.22(550 \times 10^{-9})}{2.4} = 2.8 \times 10^{-7}\ \text{rad}, \qquad s = (3.84 \times 10^{8})(2.8 \times 10^{-7}) \approx 110\ \text{m}$$

Answer: (a) about 7 km; (b) about 110 m, so Hubble cannot see the few-meter Apollo landers. At a given wavelength, only a bigger aperture does better.

A microscope faces the same limit. The smallest separation it can resolve is about

$$d_{\text{min}} \approx \frac{\lambda}{2\,\text{NA}} \tag{8.6}$$

where NA, the **numerical aperture**, measures how wide a cone of light the objective collects. The best oil-immersion objectives reach NA ≈ 1.4–1.5.

With green light, $d_{\text{min}} \approx 550/(2 \times 1.4) \approx 200$ nm. Bacteria (about 1000 nm) are easy to see; most viruses (20–300 nm) are fuzzy dots at best; atoms (0.1–0.3 nm, Chapter 4) are a thousand times too small. The fix is a shorter wavelength. Electrons accelerated through 100 kV have $\lambda \approx 0.0037$ nm, so electron microscopes can resolve rows of atoms. Their advantage is wavelength, not that electrons are "smaller"; in practice imperfect magnetic lenses limit them to about 0.1 nm. X-rays with $\lambda \sim 0.1$ nm reveal crystal structure through diffraction (Chapter 11).

KEY IDEA

You can't resolve details much smaller than about half the wavelength you look with. To see smaller things, use a shorter wavelength or a bigger aperture.

YOU ASKED

"What is the diffraction limit?"

The diffraction limit is the smallest detail any lens-based instrument can resolve, set by the wavelength: about $\lambda/(2\,\mathrm{NA})$, or roughly half a wavelength for the best microscopes. For a telescope or eye, the limit is an angle, $1.22\lambda/D$.

The cause is diffraction at the lens itself: every point is imaged as a small Airy disk, and details closer than the disk size blur together. That is why visible-light microscopes bottom out near 200 nm (no atoms; most viruses only as blurs), why telescopes keep getting bigger, and why we use electrons or X-rays, with wavelengths thousands of times shorter, to see atomic structure.

RABBIT HOLE **(optional) — Beating the limit with blinking molecules**

Safe to skip. "Super-resolution" fluorescence microscopy (Nobel Prize in Chemistry, 2014) never lets neighboring molecules glow at the same time. Each still makes a blurry Airy spot, but an isolated spot's center can be located to tens of nanometers. Light up a sparse handful, locate them, repeat thousands of times, and assemble the map. Diffraction is unchanged; the trick is what you light up.

8.6 Size versus wavelength: the master rule

Whenever a wave meets an object or a hole, compare the size d with λ. The ratio sorts almost every situation into three regimes (Table 8.1, Figure 8.6).

Table 8.1 What happens when a wave meets an object of size d.

Regime	What the wave does	Examples
$d \ll \lambda$ (Rayleigh)	Scatters only weakly, much more for short wavelengths; can't image the object; holes this small block the wave	Air molecules in sunlight (blue sky, Chapter 7); microwave-oven door holes; AM radio meeting a building
$d \sim \lambda$ (Mie)	Strong, complicated scattering and diffraction; colors from interference	Haze and smoke; milk's fat droplets; CD tracks; Astrophage in its own infrared light
$d \gg \lambda$ (geometric)	Travels in straight rays: sharp shadows, reflection, refraction (Chapter 6)	Sunlight on buildings; cloud droplets (white clouds); FM radio and hills

A particle much smaller than λ has its charges shaken together as one tiny dipole, which re-radiates feebly. This is **Rayleigh scattering**:

$$\sigma_{\text{scatter}} \propto \frac{d^6}{\lambda^4} \qquad (d \ll \lambda) \tag{8.7}$$

where σ is the scattering cross-section, the effective area that removes light from the beam. Per unit of the particle's own area ($\propto d^2$) this goes as $(d/\lambda)^4$: halve d/λ and the scattering per area drops 16-fold.

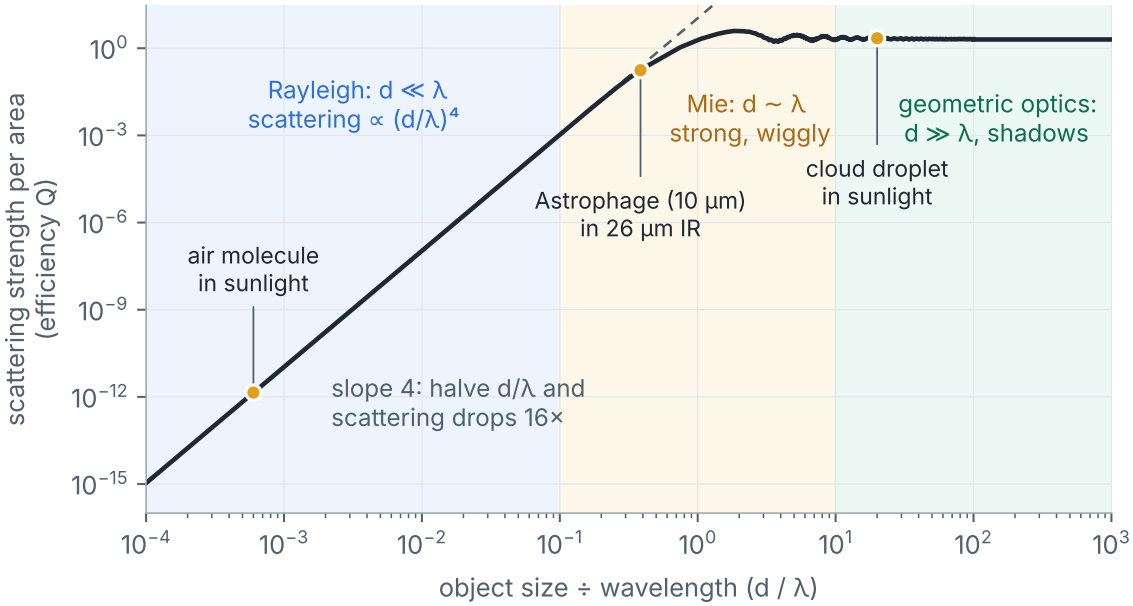


Figure 8.6 How strongly a water droplet scatters light, per unit of its cross-sectional area, versus size ÷ wavelength (computed with Mie theory). Tiny objects fall on the steep $(d/\lambda)^4$ line: an air molecule in sunlight scatters roughly a trillion times less effectively, per area, than a cloud droplet. Near $d \sim \lambda$ scattering is strong; for large objects it levels off.

The microwave-oven door

A microwave oven runs at 2.45 GHz, so $\lambda = c/f = 12.2$ cm. The door's metal mesh has holes about 1–2 mm across, nearly a hundred times smaller. To a 12 cm wave the mesh acts like solid metal: the field drives currents that flow around the tiny holes, and the wave is reflected back inside. Visible light sees each hole as thousands of wavelengths wide and passes straight through, so you can watch your food.

Radio waves around buildings

AM radio wavelengths (about 180–560 m) are longer than buildings and comparable to hills, so AM diffracts around them and fills in the "shadows." FM (about 3 m) and cell signals (tens of centimeters) are far smaller than a hill, so they act more like light: they cast shadows and need line of sight or reflections. That is part of why FM fades behind a ridge while AM carries on.

YOU ASKED

"Why can big wavelengths interact with small objects?"

They can because a wave is an oscillating electric field, and anything containing charges feels it, whatever its size. Size controls *which kind* of interaction is strong:

- **Scattering, reflecting, shadowing, imaging:** these really do fade for objects much smaller than λ, as $(d/\lambda)^4$ per unit area. Here "too small to notice" is basically right (the microwave door).
- **Absorbing and emitting:** these depend on *resonance*, a photon energy matching an energy gap, not on size. A 0.1 nm atom absorbs 500 nm light easily.
- **Antennas:** a small antenna still works, but radiates weakly unless it is resonant.

Section 8.7 unpacks the last two.

8.7 How tiny atoms handle huge waves

An atom is about 0.1 nm across; green light has $\lambda = 500$ nm, 5000 times bigger. Across the atom the light's field is almost perfectly uniform (Figure 8.7a): it points up everywhere, then down everywhere, 6×10^{14} times per second. That push rocks the electron cloud against the nucleus, making an oscillating **electric dipole** (the picture introduced in Chapter 5). This is the

standard starting point for calculating atomic absorption and emission, and it works *because* the atom is tiny.

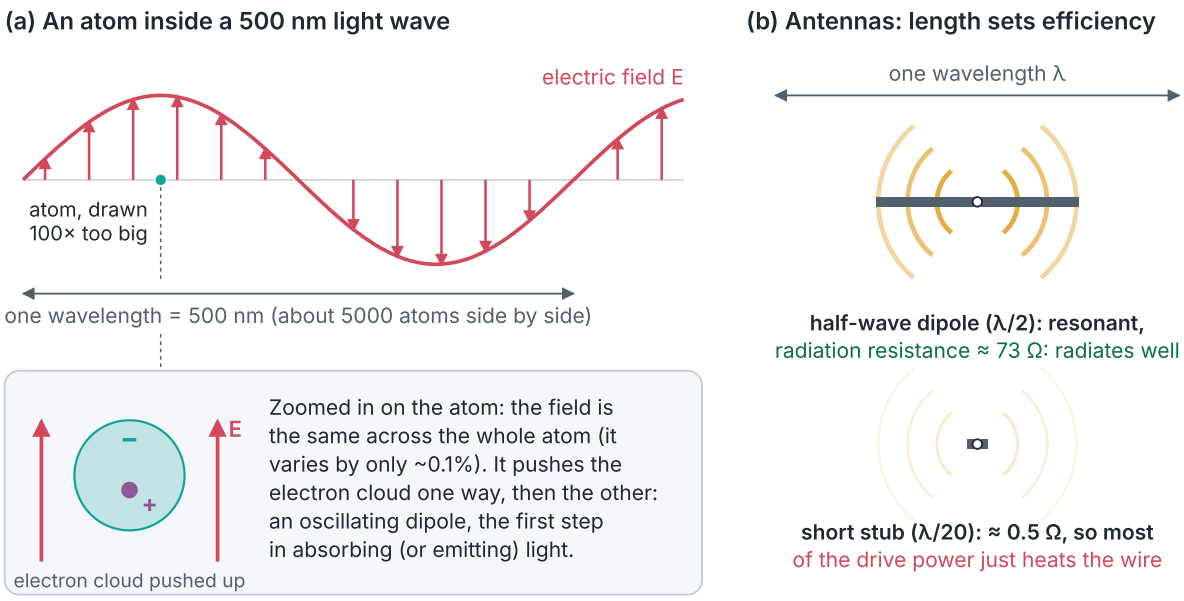


Figure 8.7 (a) Even drawn 100 times too large, an atom is a dot inside one wavelength of visible light, so the field is uniform across it and simply rocks the electron cloud. (b) For antennas, length relative to λ sets efficiency: a resonant half-wave dipole radiates well, while a $\lambda/20$ stub (radiation resistance about 0.5 Ω) wastes most of its drive power as heat.

Whether the rocking turns into absorption depends on **resonance**. If the light's frequency matches a natural frequency of the atom, so that hf equals the gap between two energy levels (Chapter 5), the atom absorbs a photon. Off resonance, the cloud just jiggles slightly and re-radiates the energy, which gives refraction (Chapter 6) and the weak Rayleigh scattering above.

ANALOGY

Pushing a swing works if your pushes match its rhythm, no matter how small your hands are. An atom's energy levels set its rhythm; its size doesn't enter.

YOU ASKED

"Something so small (an atom) shouldn't be able to interact with such a long wavelength — the wave should be unnoticeable to the tiny atom. What's the idea here?"

Your intuition is right for *scattering and imaging*, but absorption and emission don't depend on size at all; they depend on resonance. Separate three cases:

(a) Scattering, reflecting, resolving: size matters. An object much smaller than λ scatters weakly (efficiency $\propto (d/\lambda)^4$), can't reflect the wave or cast a shadow, and can't be imaged with it (Section 8.5). Here the wave really does "not notice" the atom, which is why the sky is only faintly blue and why light can't show you atoms.

(b) Absorption and emission: resonance matters, not size. A photon is absorbed when its energy matches an energy gap (Chapter 5). Atoms about 0.1 nm across absorb and emit 500 nm light all the time: neon signs and sodium streetlights are atoms emitting light thousands of times their size. Being tiny keeps it simple, because the field is uniform across the atom and just rocks it as a dipole (Figure 8.7a).

(c) Antennas: why size *feels* like it should matter. An antenna much shorter than λ is a weak, inefficient radiator: a half-wave dipole has a radiation resistance near 73 Ω, a $\lambda/20$ stub only about 0.5 Ω, so most of its power becomes heat (Figure 8.7b). A length near $\lambda/2$ makes it resonant. An atom is a tiny antenna already tuned by its energy levels, so it doesn't need to be wavelength-sized.

RABBIT HOLE **(optional) — On resonance, an atom "looks" as big as the wavelength**

Safe to skip. An isolated atom driven exactly at resonance absorbs as if it were a disk of area about $3\lambda^2/2\pi$, set by the wavelength, not the atom. For sodium's 589 nm line that is a disk about 0.2 µm in radius, millions of times the atom's geometric cross-section. (Atomic motion and collisions in real gases broaden the line and lower this peak.)

PROJECT HAIL MARY CONNECTION

"In Project Hail Mary*, what does Grace say about Astrophage interacting with a wavelength that's too big for it?"*

Grace finds that an Astrophage cell is about 10 µm across, yet it emits and responds to infrared light at the Petrova wavelength, 25.984 µm, about 2.6 times larger than the cell. His narration uses the rule of thumb from Section 8.6, that light basically ignores things much smaller than its wavelength (with the microwave-oven door as the example), so by that rule the cell shouldn't interact with 26 µm light at all. His tests also show the cells are black to every wavelength he tries.

What real physics says:

- **Emission and absorption aren't gated by size; resonance is.** A CO_2 molecule about 0.2 nm long absorbs 15 µm infrared, tens of thousands of times its length (that is the greenhouse effect, Chapter 9).
- **2.6× isn't "much smaller."** At $d/\lambda \approx 0.4$, Astrophage sits at the edge of the Mie regime (Figure 8.6), where interaction is strong.
- **Where Grace's instinct holds:** ordinary matter much smaller than a wavelength scatters and blocks it only weakly. A cell that stayed perfectly black at wavelengths far larger than itself would be genuinely strange physics, which is part of what makes Astrophage such a puzzle.

Chapter summary

- Superposition: overlapping waves add. A path difference of $m\lambda$ gives a bright spot; $(m + \frac{1}{2})\lambda$ gives a dark one.
- Huygens' principle explains diffraction. An opening of width a spreads a wave by about λ/a, which is dramatic when $a \sim \lambda$.
- Double slits and gratings: bright fringes where $d \sin\theta = m\lambda$. Gratings (and CDs) split white light into spectra; thin films color soap bubbles.
- Matter diffracts with $\lambda = h/p$. Particles sent one at a time still build an interference pattern, so they travel as waves and are detected as particles.
- The diffraction limit: angular resolution $1.22\lambda/D$; microscope resolution $\approx \lambda/(2\,\mathrm{NA}) \approx 200$ nm for visible light. Electrons and X-rays reach atomic scales.
- Size vs. wavelength: Rayleigh ($d \ll \lambda$, scattering $\propto d^6/\lambda^4$), Mie ($d \sim \lambda$), geometric ($d \gg \lambda$). This explains the microwave door and AM vs. FM.
- Atoms absorb and emit waves thousands of times larger than themselves because absorption is set by resonance, not size; small antennas are inefficient unless resonant.

Check yourself

1. CONCEPT Two speakers play the same steady tone. At one spot in the room the sound almost vanishes. What can you say about the difference in distance from that spot to the two speakers?
2. CALC A green laser (532 nm) shines on two slits 0.25 mm apart. How far apart are the bright fringes on a wall 3.0 m away?
3. CALC A DVD's tracks are 0.74 µm apart. At what angles does first-order light of 400 nm and 650 nm leave the disc (light arriving straight on)? Why is there no second-order rainbow?
4. CONCEPT Why can you hear a friend talking in the next room through an open door, but not see them unless you are in line with the door?

5. CALC Find the de Broglie wavelength of an electron with kinetic energy 100 eV. Would a crystal with atoms 0.25 nm apart diffract it noticeably?
6. CALC A backyard telescope has a 0.20 m mirror. Using 550 nm light, what is its diffraction-limited angular resolution, and what is the smallest crater it could resolve on the Moon (3.84×10^8 m away)?
7. APPLY A biologist wants to see the shape of a 100 nm virus with an oil-immersion objective (NA = 1.4) and 450 nm blue light. Can she? If not, what instrument should she use, and why does it work?
8. CALC By what factor are air molecules better at scattering 450 nm blue light than 700 nm red light?
9. CONCEPT The holes in a microwave-oven door are about 1–2 mm wide. What would happen if a manufacturer used 3 cm holes to save metal? Explain using size vs. wavelength.
10. APPLY In materials science, X-rays with $\lambda = 0.154$ nm are used to measure crystal structures (Chapter 11), but visible light is useless for this. Use $d \sin\theta = m\lambda$ to explain why, for atomic planes about 0.2 nm apart.
11. APPLY Astrophage is about 10 µm across and emits at 25.984 µm. Compute d/λ, place it on Figure 8.6, and explain in two sentences why its emitting this wavelength does not by itself break physics.

Full solutions are in the Answer Key at the back of the book.

CHAPTER 9

Infrared, Heat, and Thermal Radiation

You are glowing right now, about 700 watts' worth, just not in a color your eyes can see. Thermal radiation explains why a stove glows red, why a thermal camera can't see through a window, and how a spacecraft gets rid of heat with nothing around it.

THE BIG PICTURE

- Everything above absolute zero emits light because its jiggling charges radiate. Hotter objects emit far more ($\propto T^4$) and at shorter wavelengths ($\propto 1/T$).
- At everyday temperatures that glow is infrared, peaking near 10 µm. Objects only glow visibly above about 800 K.
- Infrared is light, not heat. It earns the nickname "heat radiation" because warm things emit it and molecules absorb it eagerly.
- Emissivity sets how well a surface emits and absorbs. That one property explains thermal cameras, thermoses, space blankets, and spacecraft radiators.

AFTER THIS CHAPTER YOU CAN

- Distinguish heat, temperature, and thermal radiation.
- Use Wien's law and the Stefan–Boltzmann law to find peak wavelengths and radiated power.
- Explain the red-hot to white-hot color sequence and the color temperature of light bulbs.
- Explain why infrared is called heat radiation, and why that is only half true.
- Explain thermal cameras, emissivity, the greenhouse effect, microwave ovens, and radiative cooling in space.

9.1 Heat, temperature, and radiation

These three words get blurred together in everyday speech, but they mean different things:

- **Thermal energy** is the energy of the random jiggling of atoms and molecules.
- **Temperature** measures how vigorous that jiggling is on average. At 300 K, a typical molecule's share of thermal energy is about $k_B T \approx 0.026$ eV.
- **Heat** is energy *in transit* because of a temperature difference. It moves by conduction (contact), convection (moving fluid), or radiation (light).

Thermal radiation is the only one of the three that works through empty space. It is how the Sun warms Earth, and the only way a spacecraft can shed heat.

9.2 Everything glows

Every atom contains charges, and in a warm object those charges are constantly being jostled and accelerated. Accelerating charges radiate (Chapter 1). Because thermal jiggling is random, it contains a whole spread of frequencies, so the emitted light is a smooth, continuous spectrum rather than the sharp lines of Chapter 5.

ANALOGY

Thermal radiation is like the roar of a stadium crowd: no single note, just random voices at every pitch. A more excited crowd is louder and its roar shifts higher, and a hotter object likewise glows brighter and bluer.

An ideal **blackbody** absorbs every wavelength that hits it. It turns out that a perfect absorber is also the best possible emitter, and its spectrum depends on nothing but its temperature. Max Planck's formula for that spectrum (1900), shown in Figure 9.1, was the first step toward quantum physics.

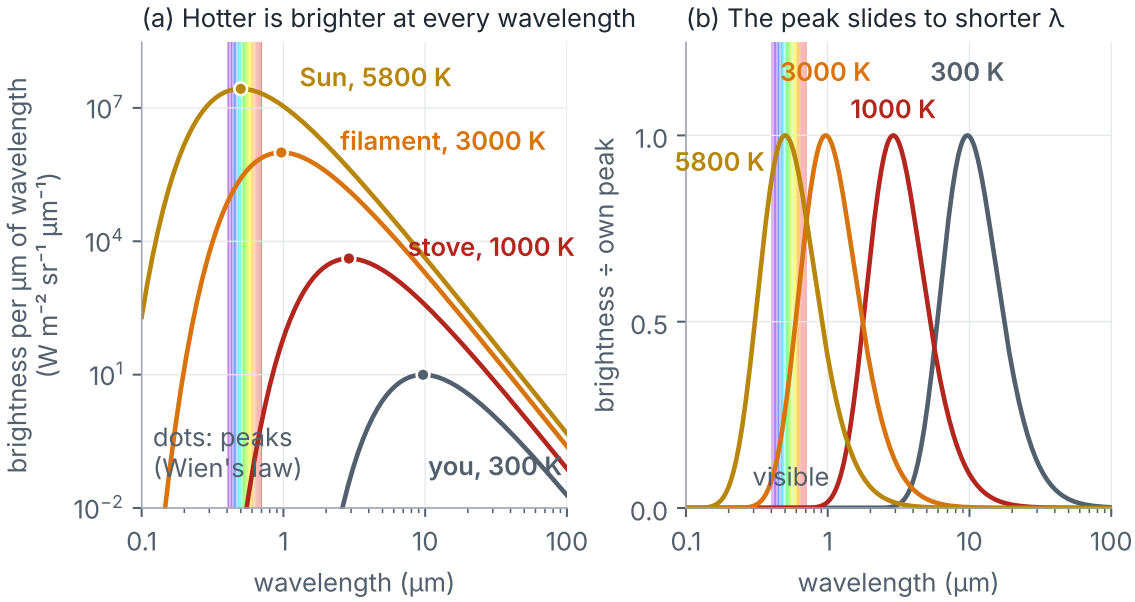


Figure 9.1 Blackbody spectra, computed from Planck's law, with the visible band shaded. (a) A hotter object is brighter at *every* wavelength (note the log scale: the Sun's surface outshines you per square meter by about a million times at the peak). (b) Each curve divided by its own peak. The peak slides from about 10 µm for you to 0.5 µm for the Sun, and only the Sun's curve really overlaps the visible band.

Two rules capture Figure 9.1. **Wien's displacement law** gives where the peak sits:

$$\lambda_{\text{peak}} = \frac{2898\,\mu\text{m}\cdot\text{K}}{T} \tag{9.1}$$

The **Stefan–Boltzmann law** gives the total power radiated:

$$P = \varepsilon\,\sigma A T^4 \tag{9.2}$$

where T is in kelvins, A is the surface area, $\sigma = 5.67 \times 10^{-8}$ W/(m²K⁴), and ε is the **emissivity** (1 for a blackbody, less for real surfaces; Section 9.5).

Since your surroundings radiate back at you, the *net* loss for an object at T in surroundings at T_{s} is

$$P_{\text{net}} = \varepsilon\,\sigma A\left(T^4 - T_{\text{s}}^4\right) \tag{9.3}$$

WORKED EXAMPLE 9.1 — Where you, a stove, and the Sun peak

Given: human body $T = 310$ K; red-hot stove element $T \approx 1000$ K; Sun's surface $T \approx 5800$ K.

Find: the peak wavelength of each.

Think: Wien's law (Eq. 9.1), with temperatures in kelvins.

$$\lambda_{\text{you}} = \frac{2898}{310}\,\mu\text{m} = 9.35\,\mu\text{m}, \qquad \lambda_{\text{stove}} = \frac{2898}{1000}\,\mu\text{m} = 2.9\,\mu\text{m}, \qquad \lambda_{\text{Sun}} = \frac{2898}{5800}\,\mu\text{m} = 0.500\,\mu\text{m}$$

Answer: about 9.3 μm (deep infrared), 2.9 μm (infrared), and 500 nm (blue-green, the middle of the visible band). Even a glowing stove puts out most of its energy as invisible infrared; the red you see is just the tail of its spectrum.

WORKED EXAMPLE 9.2 — How much you radiate

Given: skin at 33.0 °C (306 K), area $A = 1.50$ m², emissivity of skin $\varepsilon = 0.97$; room walls at 22.0 °C (295 K). (These are OpenStax's example values.)

Find: the power you emit, the power you absorb from the room, and the net loss.

Think: Eq. 9.2 for each direction; Eq. 9.3 for the difference.

$$P_{\text{out}} = (0.97)(5.67 \times 10^{-8})(1.50)(306)^4 \approx 723 \text{ W}, \qquad P_{\text{in}} = (0.97)(5.67 \times 10^{-8})(1.50)(295)^4 \approx 625 \text{ W}$$

$$P_{\text{net}} = 723 \text{ W} - 625 \text{ W} \approx 99 \text{ W}$$

Answer: you emit about 720 W of infrared and absorb about 620 W back from the room, a net loss of about 100 W. That is comparable to your resting metabolic power (roughly 100 W), which is why radiation is one of the main ways your body sheds heat.

RABBIT HOLE (optional) — The blackbody puzzle that started quantum physics

Safe to skip. Classical physics predicted that a hot object should emit ever more power at ever shorter wavelengths, without limit, which would make every warm object blaze with ultraviolet. (This was later nicknamed the "ultraviolet catastrophe.") Planck fixed it in 1900 by assuming energy is emitted in chunks of hf. A high-frequency chunk costs far more than the typical thermal energy $k_B T$, so those modes are rarely excited, and the spectrum turns over and dies away. That is why your 310 K body emits infrared and essentially no visible light or X-rays. The chunks became photons (Chapter 2).

KEY IDEA

Every object emits thermal radiation all the time. Double the absolute temperature and it emits 16 times the power, with its peak at half the wavelength. Whether something warms or cools depends on the *net* exchange with its surroundings.

MYTH VS. REALITY

Myth: Cold objects don't give off radiation, so a cold window must be "radiating cold" at you.

Reality: Ice at 0 °C still emits about 315 W per square meter. There is no such thing as "cold rays." You feel chilly near a cold window because you radiate more energy to it than it radiates back to you.

9.3 Red-hot, white-hot, and the color of light bulbs

As something heats up, the short-wavelength tail of its spectrum climbs into the visible range red end first (Figure 9.1b). Around 800 K (about 525 °C, the **Draper point**) the glow first becomes visible in a dark room as a dull red. Hotter still, more of the visible range fills in: the glow turns orange, then yellow, then white once the whole range is well covered (Figure 9.2). The brightness also rockets upward, by a factor of more than a million between 800 K and 1500 K in the visible range.

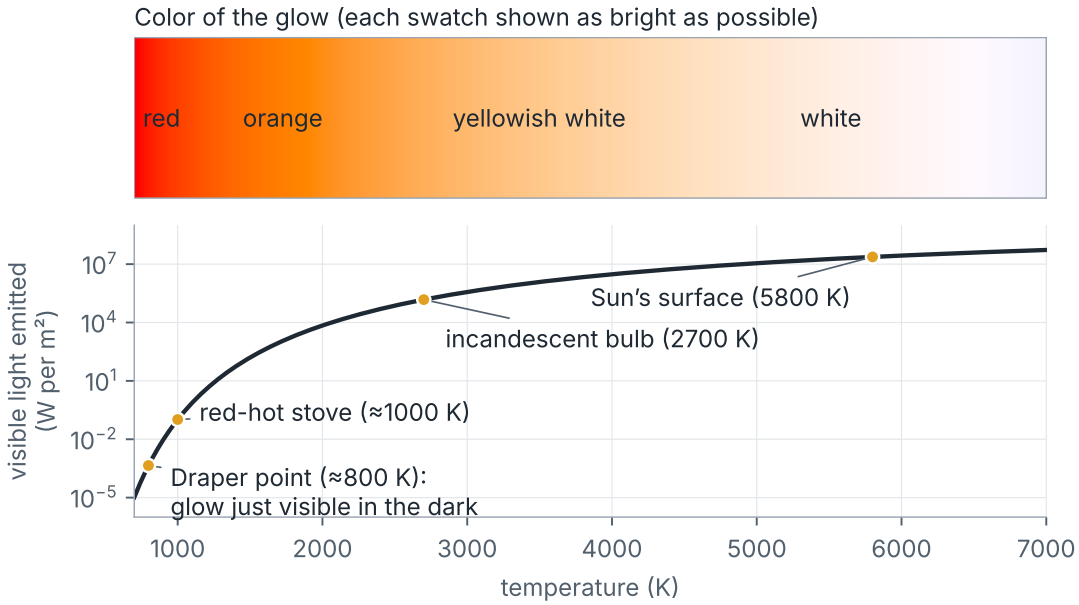


Figure 9.2 Top: the color of a blackbody's glow, computed from Planck's law and the eye's color response. Each swatch is shown at full brightness, though the real brightness varies enormously. Bottom: visible light emitted per square meter. A 1000 K stove gives off about a million times less visible light than a 2700 K filament, which is why "red-hot" looks dim.

Table 9.1 Approximate heat colors of steel (seen in dim light; charts vary).

Color	Temperature (°C)	Temperature (K)
Faint dull red, dark room only	≈ 500–550	≈ 800
Cherry red	≈ 800–900	≈ 1100–1150
Orange	≈ 1000–1100	≈ 1250–1350
Yellow	≈ 1100–1250	≈ 1350–1500
White	≈ 1300 and up	≈ 1550 and up

Light bulbs are labeled by **color temperature**: the temperature of a blackbody with the same color. An incandescent filament really is that hot, about 2700–3300 K. Even so, its Wien peak is near 1 µm, in the infrared, and only about 5–8% of its output falls in the visible band. That is why incandescent bulbs waste most of their energy as heat. An LED marked "2700 K warm white" only mimics that color; the LED itself stays near room temperature. Daylight is around 5000–6500 K. Note the naming is backwards from the physics: "warm" light matches a *cooler* glow than "cool" light.

MATERIALS SCIENCE CONNECTION

Glow colors vs. temper colors. A blacksmith reads two color scales made by different physics. *Glow* colors (Table 9.1) are thermal emission and only appear above about 500 °C. *Temper* colors appear on cleaned steel heated to just 200–350 °C and are seen by reflected light after it cools. A thin oxide film (tens to a couple of hundred nanometers thick) colors the surface by thin-film interference (Chapter 8), and the color roughly gauges the tempering temperature that sets hardness and toughness: light straw near 200 °C, brown near 260 °C, purple near 280 °C, blue near 300–340 °C (time at temperature and alloy matter too).

Why ceramics glow brightly. Kirchhoff's law: a surface's emissivity equals its absorptivity at each wavelength, so good absorbers are good emitters. Dark, rough oxides and ceramics glow brightly when hot, while polished metal at the same temperature glows dimmer. Refractories such as alumina (melting point above 2000 °C) can sit at white heat. Inside a hot kiln everything is near one temperature and glows alike, so details nearly vanish: the kiln acts as a blackbody cavity. Optical pyrometers in steel mills read temperature from exactly this glow.

9.4 Why infrared gets called “heat radiation”

In 1800 William Herschel spread sunlight into a spectrum and moved thermometers through the colors. The biggest temperature rise came just *beyond* the red, where he could see nothing. These “heat rays” were infrared light, and the name stuck for two real reasons.

1. **Warm things emit infrared.** Anything between about 250 K and 500 K, including you, your house, the ground, and a hot pan, emits almost all its thermal radiation in the infrared (Wien peaks of about 6–12 µm). The radiation you feel from a warm object is infrared.
2. **Molecules absorb infrared and turn it straight into heat.** Infrared photons (about 0.01–1 eV) match the energy steps of molecular vibrations (Chapter 5). Water, skin, glass, and CO_2 absorb them strongly, often within micrometers to millimeters, and the vibration is quickly shared by collisions as random motion (Figure 9.3).

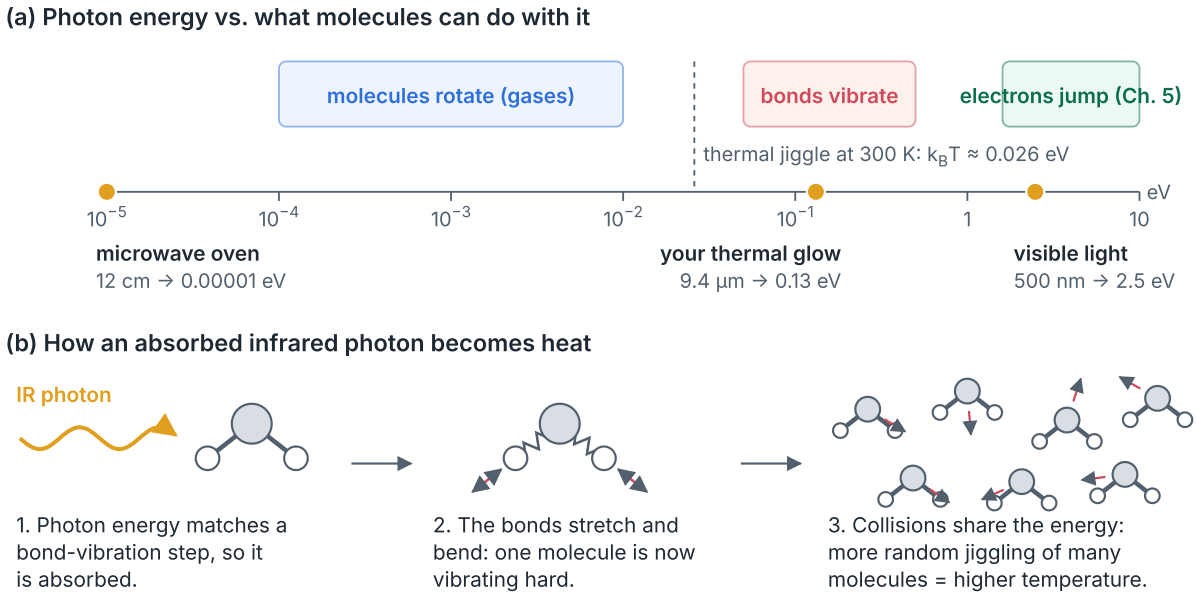


Figure 9.3 (a) Thermal-infrared photons (about 0.1 eV) land right in the energy range of bond vibrations; visible photons match electron jumps instead. (b) Once a bond is vibrating, collisions spread that energy into the random motion of many molecules, which is exactly what a temperature rise is.

But “heat radiation” is only half true. *Any* light that is absorbed ends up as thermal energy. A black car in the sun gets hot mostly from visible and near-infrared sunlight (Chapter 7), and microwaves heat your food (Section 9.7). Sunlight reaching the ground is roughly half infrared and half visible, with a few percent UV.

YOU ASKED

“Is infrared the same thing as heat?”

No. Infrared is light, an electromagnetic wave just like visible light but with longer wavelengths (about 0.7 µm to 1 mm). Heat is energy being transferred because of a temperature difference, and thermal energy is the random jiggling of atoms.

They get linked for two reasons: objects at everyday temperatures emit mostly infrared (Wien’s law), and infrared photons match molecular vibration energies, so materials like water, skin, glass, and CO_2 absorb them and turn them directly into random thermal motion. The link isn’t exclusive. Visible light heats a black car, and microwaves heat food. Infrared can also carry information with no noticeable heating, like the near-infrared beam from your TV remote.

YOU ASKED

“Why are infrared waves often called ‘heat waves’?”

Because infrared is the radiation warm things give off and the radiation your skin feels as warmth. The name dates to Herschel’s 1800 thermometer experiment. Physically, it is light that happens to match the energies of molecular vibrations, so it is absorbed strongly and becomes heat almost immediately (see the previous box and Figure 9.3). “Heat radiation” describes what infrared usually *does*, not what it *is*.

9.5 Thermal cameras and emissivity

YOU ASKED

“Why does infrared show heat?”

Because at everyday temperatures the light an object emits by itself is almost entirely infrared, and the amount rises steeply with temperature ($\propto T^4$). Measure an object’s infrared glow and you have measured its temperature, from a distance. A normal camera records *reflected* visible light, which tells you about color, not temperature. A thermal camera records each object’s *own* emission, so its image is essentially a temperature map.

Most thermal cameras detect the 8–14 µm band. That band sits right on the emission peak of room-temperature objects, and it coincides with a window where air is nearly transparent. Each pixel is a tiny **microbolometer**, a speck of material that warms slightly when infrared lands on it and changes its electrical resistance. The lenses are made of germanium, not glass, for a reason that shows up in Figure 9.4b.

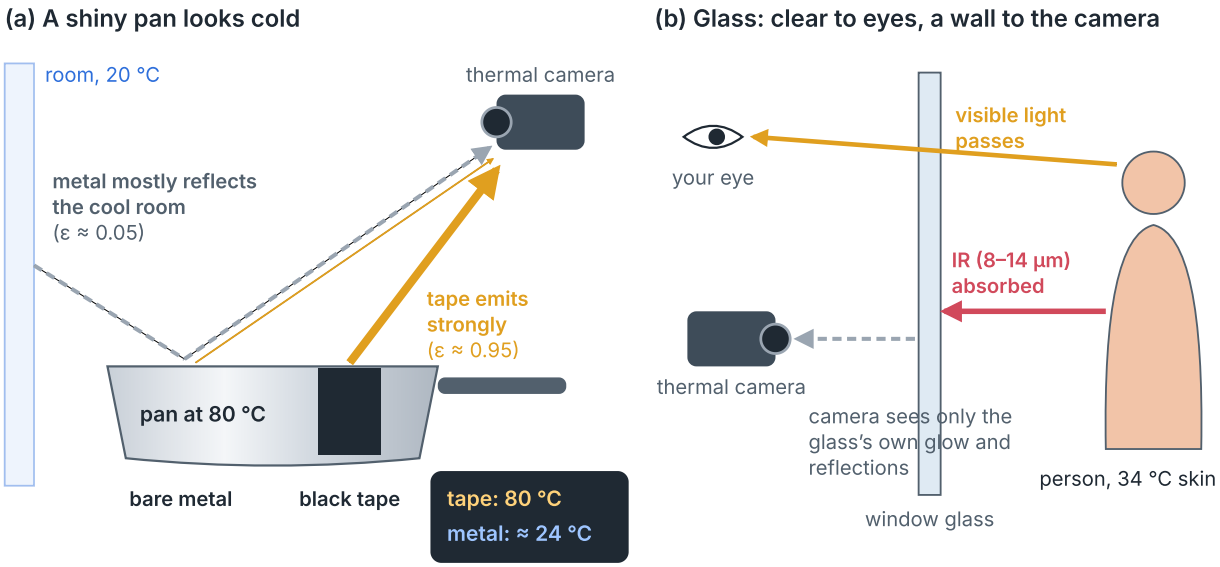


Figure 9.4 What a thermal camera really measures. (a) Polished metal ($\varepsilon \approx 0.05$) emits little and mostly reflects the cool room, so a camera set for $\varepsilon = 0.95$ reads it as roughly 24 °C even though the pan is at 80 °C. Matte black tape reads true. (b) Window glass is transparent to visible light but absorbs 8–14 µm infrared, so the camera sees the glass’s own glow and your reflection, not the person behind it.

Why glass blocks thermal cameras. Glass is transparent to visible light because visible photons can’t excite its electrons (Chapter 6). But its silicon–oxygen bonds vibrate at frequencies matching light near 9 µm (Chapter 6), so it absorbs strongly in exactly the thermal camera’s band. By Kirchhoff’s law (Section 9.3), a strong absorber is also a strong emitter. The camera sees the glass glowing at its own temperature, with reflections on top.

Why you can't see through walls. Walls are opaque to infrared too, so a thermal camera only ever sees a wall's surface temperature. It can reveal a hot pipe, a stud, or missing insulation, but only because those change how heat conducts to the surface. It cannot see people in the next room.

Emissivity (ε, from 0 to 1) measures how well a surface emits compared with a blackbody. For an opaque surface, whatever it doesn't absorb it reflects, so a low-emissivity surface is a good mirror for infrared (Table 9.2). That is why a shiny pan "looks cold" to a thermal camera (Figure 9.4a). Technicians fix this with a strip of matte tape or paint.

> **KEY IDEA**
>
> Good absorbers are good emitters, and good reflectors are poor emitters (Kirchhoff's law). A thermal camera reads a surface's own glow only when its emissivity is high; on shiny metal it mostly sees reflections.

Table 9.2 Typical thermal-infrared emissivities (approximate; they vary with surface finish and temperature).

Surface	Emissivity ε	What a thermal camera sees
Polished silver, aluminum, aluminum foil	≈ 0.02–0.05	Mostly reflections of the surroundings
Oxidized (rusty) steel	≈ 0.8	Mostly its own temperature
Window glass, brick, concrete	≈ 0.9	Its own temperature
Matte black paint	≈ 0.95–0.98	Its true temperature
Human skin, water	≈ 0.96–0.98	Its true temperature (nearly a blackbody in the IR)

9.6 The greenhouse effect

Earth absorbs sunlight, mostly visible and near-infrared, and must radiate the same power back to space as thermal infrared near 10 µm. The two bands barely overlap (Figure 9.5), so the atmosphere can treat them very differently. Nitrogen and oxygen, 99% of dry air, are symmetric two-atom molecules: stretching them shifts no charge, so they are essentially transparent to infrared (Chapter 5). CO_2 and H_2O have vibrations that absorb thermal infrared strongly, CO_2 especially around 15 µm, close to the peak of Earth's glow.

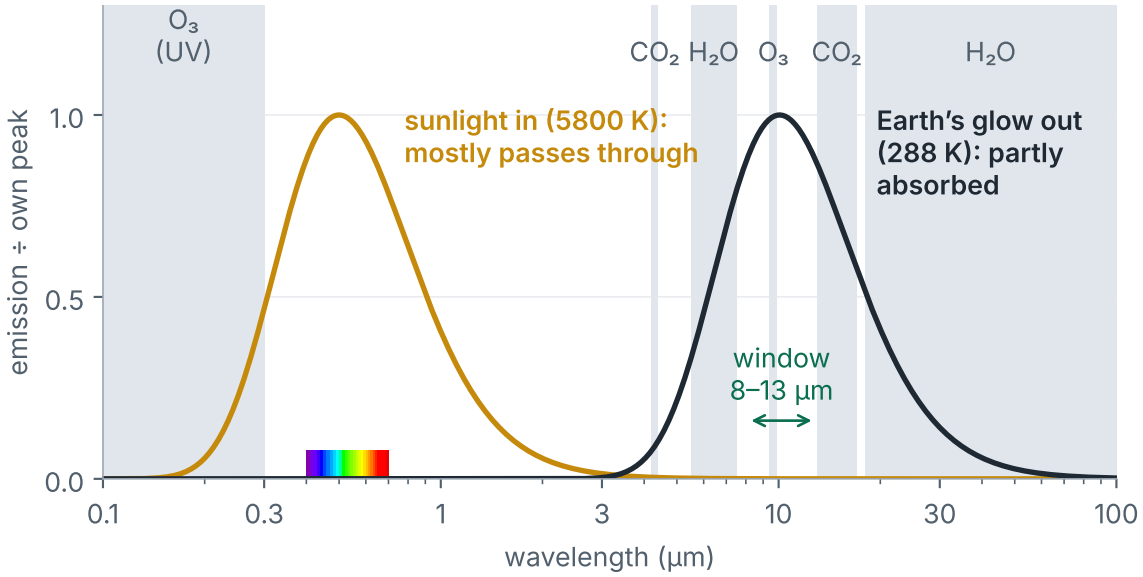


Figure 9.5 Sunlight comes in at short wavelengths, and air lets most of it through. Earth's glow goes out at long wavelengths, where CO_2 and water vapor absorb across wide bands (shaded; edges simplified). Only the 8–13 µm window lets surface radiation escape straight to space.

The absorbing molecules re-emit in all directions, so part of Earth's outgoing infrared is sent back down. The surface has to warm up until enough radiation leaks out to balance the incoming sunlight. In the simplest energy-balance model, Earth without greenhouse gases would average about 255 K (−18 °C); the actual average is about 288 K (15 °C). Adding CO_2 widens and deepens the absorbed bands, so the balance point moves warmer.

MYTH VS. REALITY

Myth: The atmosphere warms Earth the same way a glass greenhouse warms plants.

Reality: A real greenhouse stays warm mainly because its walls stop warm air from rising and blowing away; it blocks convection. The atmospheric greenhouse effect is radiative: gases absorb and re-emit outgoing infrared. Same name, different mechanism.

9.7 Microwave ovens

A microwave oven fills its cavity with 2.45 GHz waves ($\lambda = 12.2$ cm). Each photon carries only about 10^{-5} eV, far too little to break a chemical bond or ionize anything (Chapter 10). The heating works through the oscillating electric field instead. Water molecules are electric dipoles, and the field tries to twist them back and forth 2.45 billion times per second.

In liquid water, each molecule is caught in a jostling network of hydrogen bonds, so it can't rotate freely and lags behind the field. Because of that lag, the field keeps doing work on the molecules, and the energy is lost as random motion, which is heat. This is **dielectric heating**. Its strength varies smoothly with frequency, with a broad maximum near 20 GHz at room temperature (Figure 9.6). Ice heats poorly because its molecules are locked in a crystal and can barely turn, which is why defrost settings cycle on and off to let heat conduct inward.

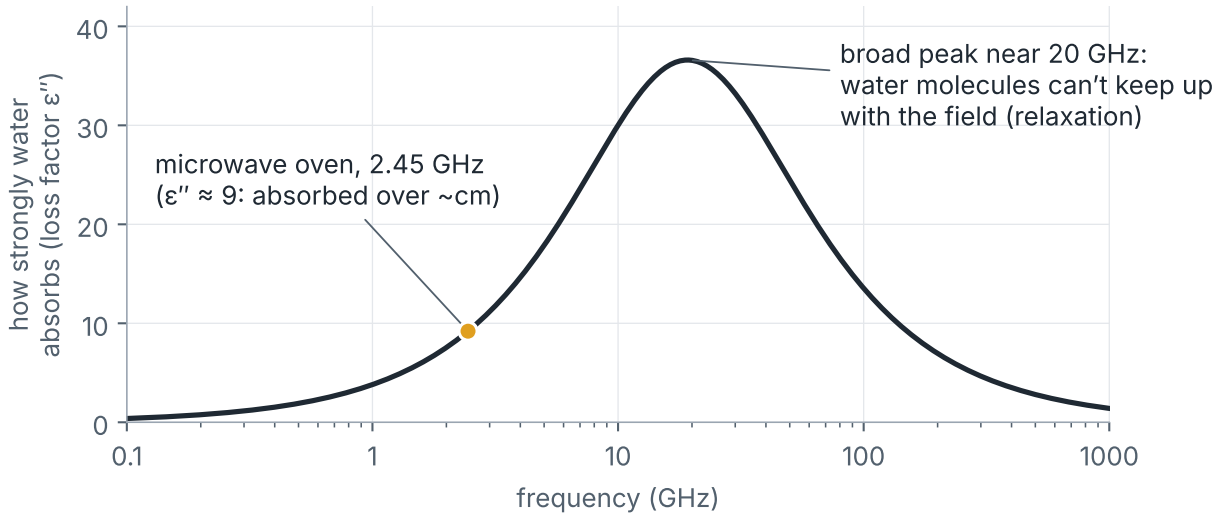


Figure 9.6 How strongly liquid water absorbs microwaves versus frequency (single-relaxation Debye model at about 25 °C). The curve is a broad hump spanning a factor of 100 in frequency, not a sharp resonance line. The oven's 2.45 GHz sits on the gentle rising side.

MYTH VS. REALITY

Myth: Microwave ovens are tuned to the resonant frequency of water molecules.

Reality: Liquid water has no sharp resonance at 2.45 GHz. Its absorption is a broad relaxation hump that peaks near 20 GHz. Ovens use 2.45 GHz because that band is allocated for industrial and home heating, and because water absorbs it only moderately, so the waves penetrate a centimeter or more into food. At 20 GHz the outer skin would absorb nearly everything and the inside would stay cold.

9.8 Controlling thermal radiation: thermoses, space blankets, and radiators

Keeping heat in. A thermos (vacuum flask) blocks all three heat paths. The vacuum gap stops conduction and convection, and the silvered, low-emissivity walls barely emit infrared and reflect back what reaches them. A space blanket is plastic film coated with a thin layer of aluminum, with ε of only a few percent. It reflects your infrared back toward you and emits very little outward. Spacecraft wrap themselves in **multilayer insulation**, many such aluminized layers separated by vacuum.

Getting heat out. In space there is no air, so a spacecraft can only lose heat by radiating it. The deep-space background is only about 3 K. Spacecraft pipe waste heat (on the International Space Station, through liquid ammonia loops) to **radiator** panels coated with high-emissivity white paint. White paint seems like a strange choice for a radiator, but it reflects most *visible* sunlight while emitting strongly in the *infrared*. That is possible because emissivity can be very different at 0.5 μm and at 10 μm.

WORKED EXAMPLE 9.3 — Sizing a spacecraft radiator

Given: electronics produce 1.0 kW of waste heat; a radiator at 300 K with $\varepsilon = 0.90$ faces deep space (background ≈ 3 K) on one side; ignore sunlight.

Find: the radiator area needed.

Think: in steady state, radiated power equals waste heat. The 3 K background radiates back almost nothing (3^4 is negligible next to 300^4), so Eq. 9.2 alone is enough.

$$\frac{P}{A} = \varepsilon\sigma T^4 = (0.90)(5.67\times10^{-8})(300)^4 = 413\ \text{W/m}^2 \qquad \Rightarrow \qquad A = \frac{1000\ \text{W}}{413\ \text{W/m}^2} = 2.4\ \text{m}^2$$

Answer: about 2.4 m², the size of a large tabletop for just 1 kW. That is why spacecraft with big power systems carry huge radiator wings.

MYTH VS. REALITY

Myth: Space is so cold that an unprotected person would freeze solid instantly.

Reality: Vacuum is an excellent insulator. With no air for conduction or convection, a body loses heat only by radiation, a few hundred watts, so cooling takes a long time. In direct sunlight, overheating is the bigger problem. Spacecraft thermal design is mostly about dumping heat, not keeping warm.

Chapter summary

- Thermal energy is random atomic motion, temperature measures its average, and heat is energy transferred by a temperature difference. Only radiation works through vacuum.
- Jiggling charges make every object emit a continuous thermal spectrum. Wien's law gives $\lambda_{\text{peak}} = 2898\ \mu\text{m}\cdot\text{K}/T$; Stefan–Boltzmann gives $P = \varepsilon\sigma A T^4$. You peak near 9–10 μm and radiate about 100 W net.
- Visible glow starts near 800 K (dull red), then goes orange, yellow, and white. Bulbs are labeled by color temperature, and "warm" light means a lower temperature.
- Infrared is light, not heat. It is called heat radiation because everyday objects emit it and molecular vibrations absorb it, but any absorbed light, visible or microwave, becomes heat.
- Thermal cameras read 8–14 μm emission. Glass and walls are opaque there, and low-emissivity shiny surfaces mostly show reflections. Kirchhoff's law: good absorbers are good emitters.
- Greenhouse gases (CO_2, H_2O) pass sunlight but absorb Earth's outgoing infrared: about 255 K without them, about 288 K with them.
- Microwave ovens heat by dielectric relaxation of water (a broad hump near 20 GHz), not a sharp resonance. Low-ε surfaces keep heat in; high-ε radiators dump heat in space.

Check yourself

1. CONCEPT A cup of coffee cools on your desk. Use the words *thermal energy*, *temperature*, and *heat* correctly to describe what happens, and name the three ways the heat leaves.
2. CALC A piece of steel in a forge is at 1500 K. Where is its spectrum's peak? Is most of its radiated energy visible?
3. CALC If a stove element's absolute temperature doubles from 1000 K to 2000 K, by what factor does its radiated power change, and where does its peak move?
4. CALC A person (area 1.8 m², $\varepsilon = 0.98$, skin at 33 °C) stands in a room whose walls are at 15 °C. Estimate the net radiative loss. Compare with Worked Example 9.2.
5. CONCEPT Pointing a thermal camera at a window, you see your own reflection and not the street outside. Explain both observations.
6. APPLY An engine bracket at 80 °C has a polished aluminum section and a black-anodized section. Which reads hotter on a thermal camera, and why? How would you measure the polished section correctly?
7. CONCEPT Give two counterexamples to the claim "infrared is heat."
8. APPLY Why are N_2 and O_2, 99% of dry air, not greenhouse gases, while CO_2 and H_2O are? Why does it matter that CO_2 absorbs near 15 μm in particular?
9. CALC A satellite must reject 3.0 kW of heat using a radiator at 280 K with $\varepsilon = 0.85$, one side facing deep space. What area does it need (ignore sunlight)?
10. APPLY In a blacksmith's shop you see a bar glowing bright orange in the forge and a finished chisel with a blue sheen near its edge. Which color is emitted light and which is interference? Roughly what temperatures does each indicate?
11. CONCEPT Why does a frozen block of soup heat so unevenly in a microwave, and why do defrost settings pulse on and off?

Full solutions are in the Answer Key at the back of the book.

CHAPTER 10

Ionizing Radiation, Shielding, and Space

Your Wi-Fi router and a dental X-ray machine both emit "radiation." Only one of them can break a DNA strand, and the reason comes down to one number per photon.

THE BIG PICTURE

- "Ionizing" means one photon or particle carries enough energy (about 10 eV) to knock an electron out of an atom.
- Damage means broken bonds in DNA, made directly or through free radicals.
- Dose is measured in gray (energy per kilogram) and sievert (gray weighted for biological effect).
- Distance and mass are the shields. Magnetic fields steer charged particles but do nothing to photons or neutrons.

AFTER THIS CHAPTER YOU CAN

- Decide whether a photon is ionizing from its wavelength or frequency.
- Compare how far alpha, beta, gamma, and neutron radiation go, and what stops each.
- Read doses in mSv and put everyday exposures on one scale.
- Use the inverse-square law, exponential attenuation, and half-value layers.
- Compute the spiral radius of a charged particle in a magnetic field, and explain what a magnetosphere does and does not block.

10.1 Two meanings of "radiation"

Radiation means energy traveling outward from a source. It comes in two kinds:

- **Electromagnetic radiation**: photons, from radio waves to gamma rays (Chapter 3).
- **Particle radiation**: fast particles with mass, such as alpha particles, electrons, neutrons, and the protons and nuclei of cosmic rays.

Either kind is **ionizing radiation** only if a *single* photon or particle carries enough energy to rip electrons off atoms.

> **KEY IDEA**
> What matters is the energy per packet, not the total energy. A beam of a trillion weak photons can warm you, but it cannot ionize. One energetic photon can.

10.2 The ionizing threshold

To **ionize** an atom, you have to supply at least its ionization energy (Chapter 4). For atoms this ranges from 3.9 eV (cesium) to 24.6 eV (helium); water, most of your body, needs 12.6 eV. So the conventional dividing line is about **10 eV**, which corresponds to a wavelength of

$$\lambda = \frac{hc}{E} = \frac{1240\ \text{eV}\cdot\text{nm}}{10\ \text{eV}} = 124\ \text{nm}, \tag{10.1}$$

deep in the ultraviolet. Different agencies put the line anywhere from 10 to 33 eV, so treat it as a fuzzy boundary (Figure 10.1).

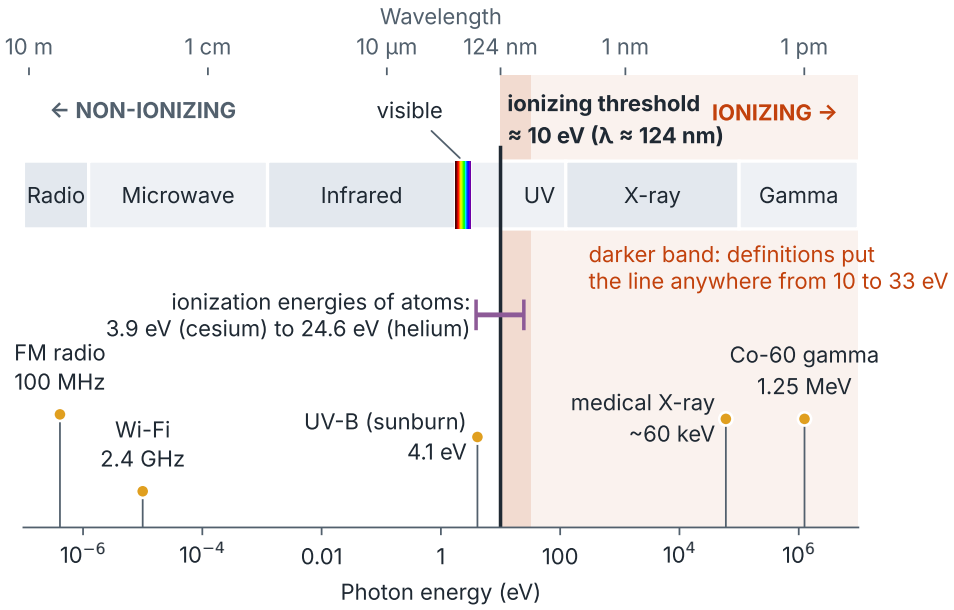


Figure 10.1 The ionizing threshold sits in the far ultraviolet. Everything to its left (radio, microwaves, infrared, visible, near-UV) cannot ionize, however intense. Medical X-rays and gamma rays are thousands to millions of times above the line.

Why can't a brighter lamp make up the difference? An electron absorbs one photon at a time (Chapter 2), and it sheds a 2 eV kick as heat long before a second photon arrives. Brightness changes how *many* electrons get kicked, not how hard each kick is.

MYTH VS. REALITY

Myth: A powerful enough Wi-Fi router or cell tower could ionize your cells.

Reality: A 2.4 GHz photon carries about 10^{-5} eV, a million times too little. Strong microwaves can *heat* tissue (a microwave oven), which is a different mechanism with a different safety question.

WORKED EXAMPLE 10.1 — Which photons can ionize?

Given: Wi-Fi at $f = 2.4$ GHz, UV-B at $\lambda = 300$ nm, a 60 keV dental X-ray photon.

Find: Each photon's energy, and whether it can ionize.

Think: Compare $E = hf = hc/\lambda$ with the ~10 eV line.

$$E_{\text{Wi-Fi}} = \frac{(6.626 \times 10^{-34})(2.4 \times 10^{9})}{1.602 \times 10^{-19}}\ \text{eV} = 9.9 \times 10^{-6}\ \text{eV}$$

$$E_{\text{UV-B}} = \frac{1240\ \text{eV} \cdot \text{nm}}{300\ \text{nm}} = 4.1\ \text{eV}$$

The X-ray photon is 60,000 eV. As it slows, it frees an electron that knocks loose many more; one ion pair costs about 34 eV on average, so one photon creates roughly $60{,}000/34 \approx 1800$ ions.

Answer: Wi-Fi $\approx 10^{-5}$ eV (no); UV-B 4.1 eV (no, but see Section 10.3); X-ray 60 keV (yes, about 1800 ionizations).

10.3 UV-B and sunburn: damage below the line

UV-B (280–315 nm, 3.9–4.4 eV) is *not* ionizing, yet it gives you sunburn and skin cancer. The reason is **photochemistry**: a photon does not need to remove an electron to cause trouble; it only needs to be absorbed by a molecule and rearrange its bonds.

DNA bases absorb UV strongly. A UV-B photon absorbed where two thymine bases sit side by side can fuse them into a **thymine dimer** (Figure 10.3c), a kink that stalls DNA copying. Cells sense it and trigger inflammation (the redness); badly damaged cells self-destruct (the peeling). Dimers repaired wrongly become mutations.

> **KEY IDEA**
> "Non-ionizing" does not mean "harmless." Visible and IR photons are too weak to break most bonds, but UV photons (3–10 eV) are in the bond-breaking range even though they cannot ionize.

10.4 The particle zoo: alpha, beta, gamma, neutrons, cosmic rays

Unstable nuclei, reactors, and space produce five kinds of ionizing radiation:

- **Alpha (α)**: a helium nucleus (charge +2), typically about 5 MeV. It ionizes densely and stops in a few cm of air or in the dead outer layer of skin, so it is dangerous mainly when inhaled or swallowed (radon gas).
- **Beta (β)**: an electron (or positron) shot out of a nucleus. It travels a few millimeters in tissue and is stopped by a few millimeters of aluminum or plastic.
- **Gamma (γ) and X-rays**: high-energy photons; the name records the source (nucleus vs. electrons). Very penetrating; lead or thick concrete cuts them down.
- **Neutrons (n)**: uncharged, so they ignore electrons and hit only nuclei. They pass through lead, are slowed by hydrogen-rich materials, and can make materials radioactive.
- **Cosmic rays**: nuclei from deep space near light speed, about 85% protons, 14% helium, and 1% heavier "HZE" nuclei such as iron. The Sun adds bursts of protons during storms.

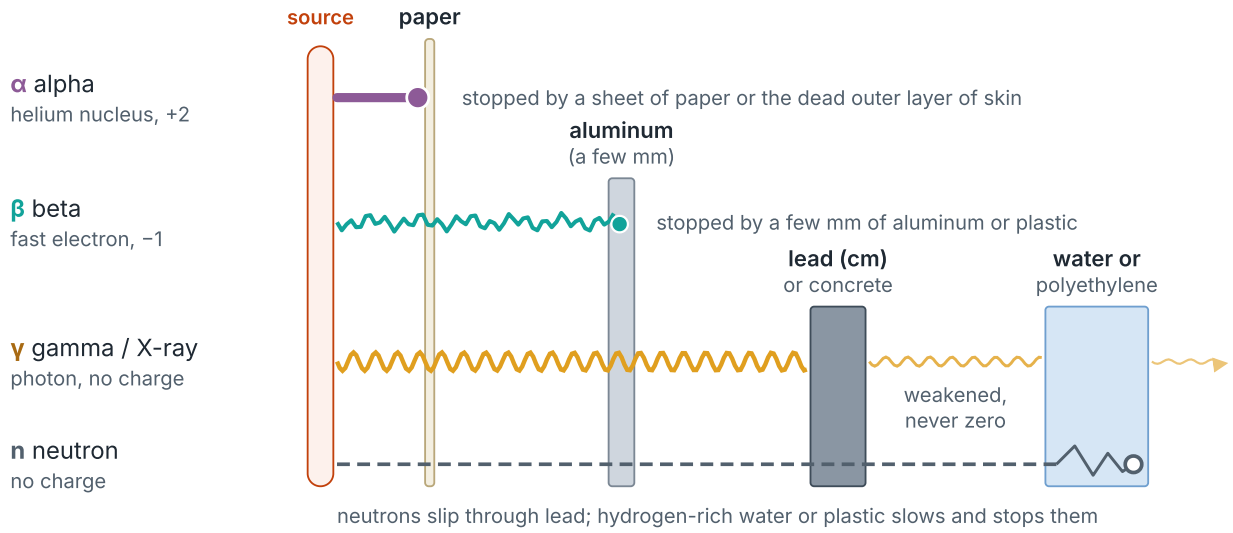


Figure 10.2 What stops what. Charged particles (α, β) have a definite range. Photons are only ever *reduced*, never stopped completely. Neutrons need hydrogen.

> **KEY IDEA**
> A charged particle tugs on every electron it passes, so it loses energy steadily and stops at a definite **range**. A photon or neutron travels untouched until one chance collision, so photons fall off exponentially (Section 10.8) instead of stopping at a wall.

ANALOGY

An alpha particle is a bowling ball rolled down a crowded hallway: it hits everyone and stops fast. A gamma ray is a single pellet fired down the same hallway: it usually flies through, but now and then it hits someone hard.

10.5 How radiation damages cells

Radiation harms a cell by breaking chemical bonds, and the bonds that matter most are in DNA. Damage happens in two ways (Figure 10.3):

- **Direct**: the track ionizes the DNA molecule itself and breaks a strand.
- **Indirect**: the track ionizes water, most of a cell, making **free radicals** such as hydroxyl (•OH) that drift a few nanometers and attack DNA. For X-rays and gamma rays, most damage is indirect.

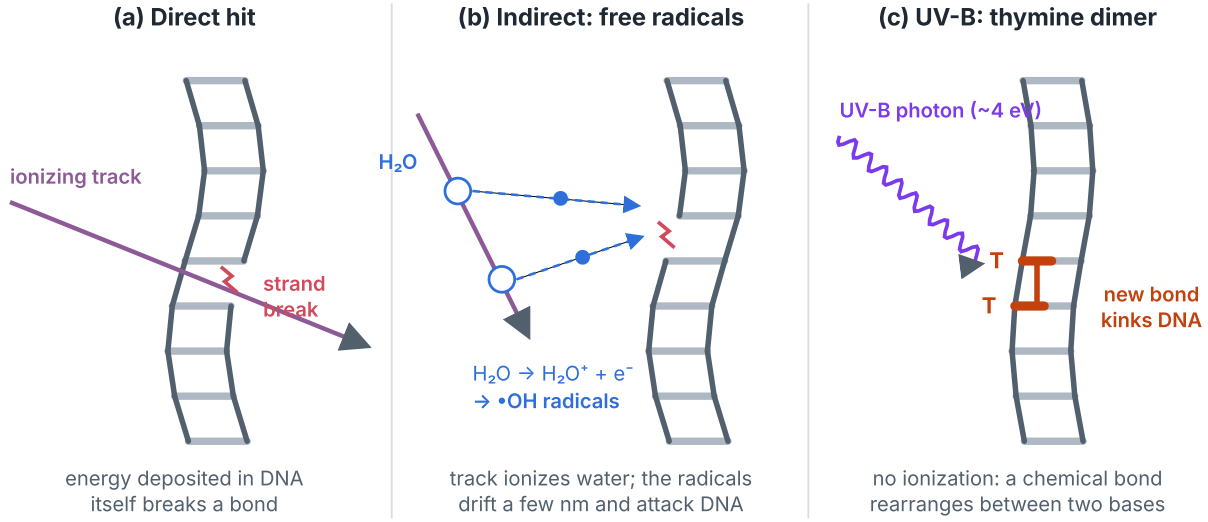


Figure 10.3 Three routes to DNA damage. (a) and (b) need ionizing radiation. (c), the thymine dimer behind sunburn, needs only a UV-B photon's chemistry.

Cells fix single-strand breaks easily, copying from the other strand. **Double-strand breaks** are harder, and tight clusters of breaks hardest; dense alpha and heavy-nucleus tracks make exactly those clusters, so they do more harm per joule. The outcome is repair, cell death, or a mutation that may raise cancer risk years later. Rapidly dividing cells (bone marrow, gut lining, tumors) are the most sensitive.

10.6 Measuring dose: gray and sievert

Absorbed dose D is energy deposited per kilogram of tissue, measured in **gray**: 1 Gy = 1 J/kg. Because a joule from alpha particles does more harm than a joule from X-rays, we also use the **equivalent dose** H, in **sieverts** (Sv):

$$H = w_R \, D \tag{10.2}$$

where the radiation weighting factor w_R is 1 for photons and electrons, 2 for protons, 20 for alpha particles and heavy nuclei, and 2.5–20 for neutrons depending on energy. Whole-body risk uses the **effective dose**, also in Sv, which further weights each organ's sensitivity. Older US units: 1 Gy = 100 rad, 1 Sv = 100 rem.

KEY IDEA

A gray is a tiny amount of heat: 1 J/kg warms water by 0.00024 °C. A sudden whole-body 4–5 Sv kills half of those exposed, yet warms the body by about a thousandth of a degree. What matters is *where* the energy goes (DNA bonds), not how much.

Table 10.1 Typical effective doses (mSv). Medical values vary with the patient and machine.

Exposure	Dose (mSv)	Source of value
Dental X-ray (intraoral)	0.005	ACR/RSNA RadiologyInfo
Cross-country flight, one way	0.02–0.05	EPA
Chest X-ray	0.1	RadiologyInfo
US natural background, per year (about three-quarters from radon)	≈3.1	NCRP 160 via NRC
US average from all sources, per year (natural + medical + other)	≈6.2	NCRP 160 via NRC
CT scan, abdomen and pelvis	7.7	RadiologyInfo
One day on the ISS	≈0.2–0.9	NASA
Six months on the ISS	≈40–160	NASA
Mars round trip, cruise only (measured by Curiosity's RAD)	≈660	NASA/JPL
Full Mars mission (180 d out, 500 d on surface, 180 d back)	≈1000	NASA/JPL
Limits: public 1/yr (above background, NRC); radiation workers 50/yr (NRC); NASA astronaut career 600	1 / 50 / 600	NRC; NASA-STD-3001
Acute radiation sickness possible (whole body, at once)	above ≈700 (0.7 Gy)	CDC

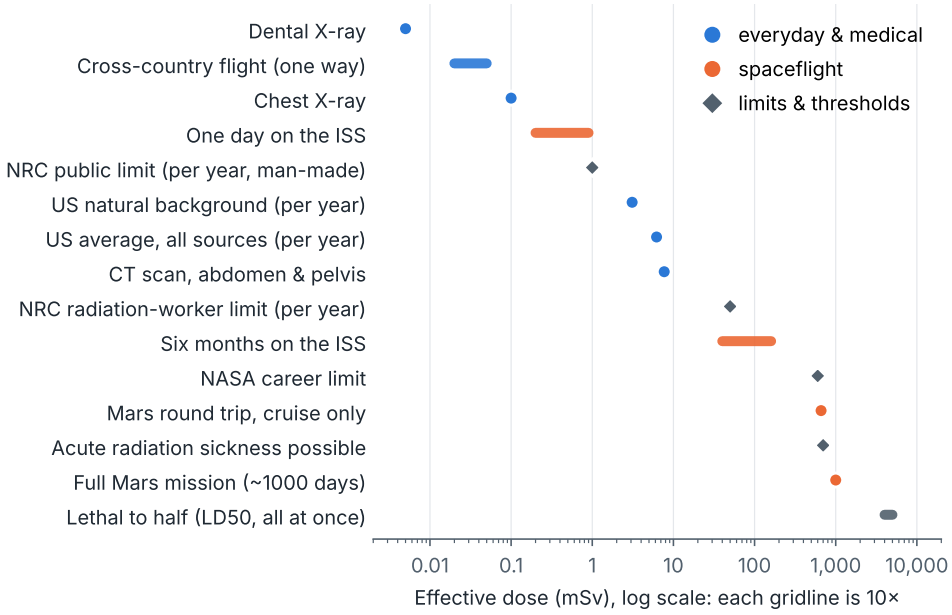


Figure 10.4 The dose ladder on a log scale (each gridline is ×10). A chest X-ray is about ten days of natural background. A six-month ISS stay equals one to three years at a radiation worker's annual limit, and a full Mars mission would exceed NASA's whole-career limit.

10.7 Distance: the inverse-square law

A small source sends radiation out evenly in all directions. At distance r the same energy is spread over a sphere of area $4\pi r^2$, so the intensity is

$$I = \frac{P}{4\pi r^2} \quad \Longrightarrow \quad \frac{I_2}{I_1} = \left(\frac{r_1}{r_2}\right)^2 \tag{10.3}$$

Valid for a point-like source with nothing absorbing in between. Close to a large source (a big tank, the ground), dose falls off more slowly.

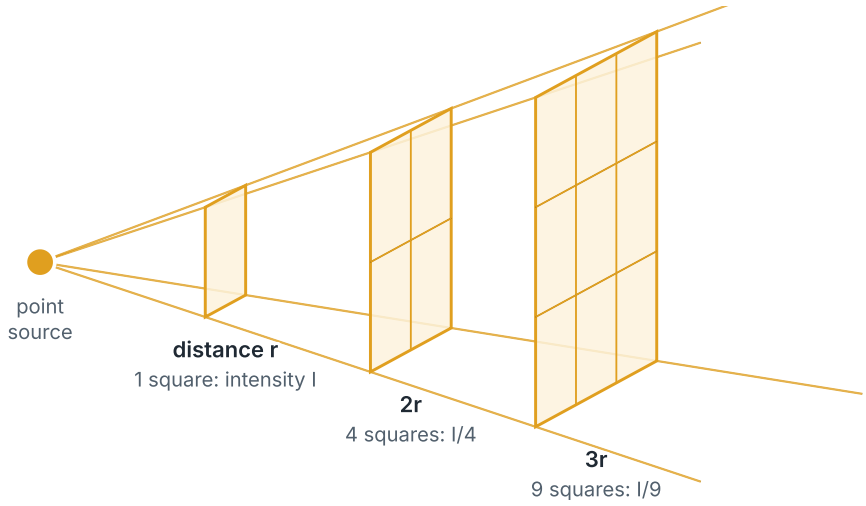


Figure 10.5 The same photons spread over 4× the area at twice the distance and 9× at three times the distance.

WORKED EXAMPLE 10.2 — Stepping back from a source

Given: A small lab source gives a dose rate of 20 µSv/h at 0.50 m.

Find: (a) The rate at 2.0 m. (b) The distance where it drops to 0.50 µSv/h.

Think: Point source, air barely absorbs gamma rays over meters, so use Eq. 10.3.

(a) $\dot{H}_2 = 20\ \mu\text{Sv/h} \times (0.50/2.0)^2 = 1.25\ \mu\text{Sv/h}$

(b) $r_2 = r_1\sqrt{\dot{H}_1/\dot{H}_2} = 0.50\ \text{m} \times \sqrt{20/0.50} = 0.50 \times 6.32 = 3.2\ \text{m}$

Answer: (a) 1.25 µSv/h, a factor of 16 lower for 4× the distance. (b) about 3.2 m.

10.8 Shielding: why mass, and why hydrogen

Photons: exponential attenuation

Each photon has a fixed chance per centimeter of interacting, so every layer removes the same *fraction* of whatever arrives:

$$I = I_0\, e^{-\mu x}, \qquad \mu = \left(\frac{\mu}{\rho}\right)\rho \tag{10.4}$$

where μ is the linear attenuation coefficient (cm⁻¹), μ/ρ the mass attenuation coefficient (cm²/g, tabulated by NIST), ρ the density, and x the thickness. This "narrow-beam" law slightly underestimates real transmission, since scattered photons can still reach you.

The thickness that halves the beam is the **half-value layer**:

$$\text{HVL} = \frac{\ln 2}{\mu} \approx \frac{0.693}{\mu} \tag{10.5}$$

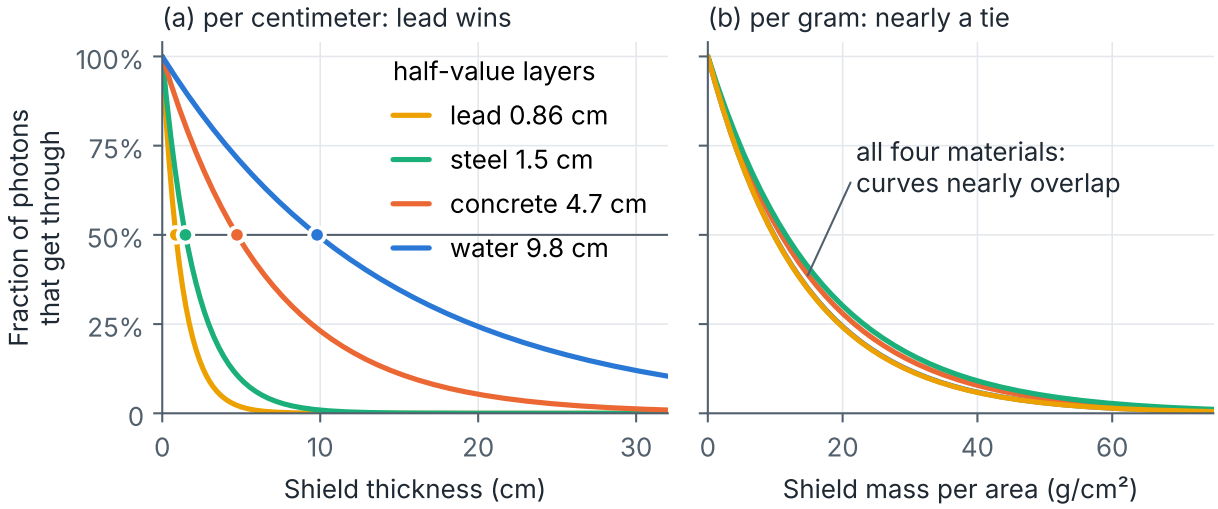


Figure 10.6 1 MeV gamma rays in four shields (NIST data). (a) Per centimeter, lead is about 11× better than water. (b) Per gram, they are almost identical. At these energies, photons mostly scatter off electrons (Compton scattering), and every material has about the same number of electrons per gram.

WORKED EXAMPLE 10.3 — Lead vs. water against 1 MeV gamma rays

Given: NIST mass attenuation coefficients at 1.0 MeV: lead 0.0710 cm²/g ($\rho = 11.35$ g/cm³), water 0.0707 cm²/g ($\rho = 1.00$ g/cm³).

Find: The HVL of each, and the thickness that cuts the beam to 1%.

Think: Eq. 10.4 with $I/I_0 = 0.01$ gives $x = \ln(100)/\mu$.

Lead: $\mu = 0.0710 \times 11.35 = 0.806\ \text{cm}^{-1}$, so HVL $= 0.693/0.806 = 0.86$ cm, and $x_{1\%} = 4.61/0.806 = 5.7$ cm.

Water: $\mu = 0.0707\ \text{cm}^{-1}$, so HVL $= 9.8$ cm, and $x_{1\%} = 65$ cm.

Mass per area: lead $5.7 \times 11.35 \approx 65\ \text{g/cm}^2$; water $65 \times 1.00 = 65\ \text{g/cm}^2$.

Answer: HVL 0.86 cm (lead) vs. 9.8 cm (water); 1% needs 5.7 cm of lead or 65 cm of water, which is the same mass per area. Lead wins on space, not on weight.

At lower energies (medical X-rays, tens of keV) the story changes: photoelectric absorption grows steeply with atomic number, so lead becomes far better per gram. That is why the dentist's apron is lead (see Chapter 11).

Neutrons and cosmic rays: hydrogen wins

A neutron loses energy by bouncing off nuclei like a billiard ball. A head-on hit on a hydrogen nucleus (one proton, the same mass) can stop it dead; off lead, 207 times heavier, it keeps at least 98% of its energy, like a ping-pong ball off a bowling ball. So neutron shields are water, polyethylene (CH_2), or concrete, often with boron to absorb the slowed neutrons.

Cosmic rays add a second reason. A fast heavy nucleus hitting a heavy shield nucleus can shatter both into a spray of secondary neutrons, protons, and fragments, so a thin metal shield can make things worse. Hydrogen cannot shatter, light nuclei make fewer secondaries, and hydrogen has the most electrons per gram, so it slows charged particles best per kilogram launched. That is why NASA favors water and polyethylene.

MYTH VS. REALITY

Myth: Lead is the best radiation shield, period.

Reality: Lead is excellent against X-rays and compact against gamma rays, but poor against neutrons and it worsens cosmic-ray showers. The right shield depends on the radiation.

MATERIALS SCIENCE CONNECTION

Shield design is a materials-selection problem, like those in ME2340. The deciding properties are density (thickness), atomic number Z (low-energy photons), electrons per gram via Z/A (1 for hydrogen, 0.4 for lead), hydrogen content (neutrons), and fragmentation. Mass-limited designs (spacecraft) push you toward polymers and water; space-limited ones (an X-ray room) toward lead or dense concrete.

10.9 Magnetic fields: steering charged particles

A magnetic field exerts a force only on a *moving charge*:

$$\vec{F} = q\,\vec{v} \times \vec{B} \tag{10.6}$$

The force is perpendicular to both $\vec{v}$ and $\vec{B}$, so it changes the particle's direction but never its speed.

A force that always pushes sideways bends the path into a circle (Figure 10.7a). Setting $qvB = mv^2/r$ gives the **gyroradius**:

$$r = \frac{mv}{qB} \tag{10.7}$$

Any motion along the field stretches the circle into a **helix** around the field line (Figure 10.7b), so charged particles are guided along field lines, not across them. Photons and neutrons have $q = 0$ and go straight.

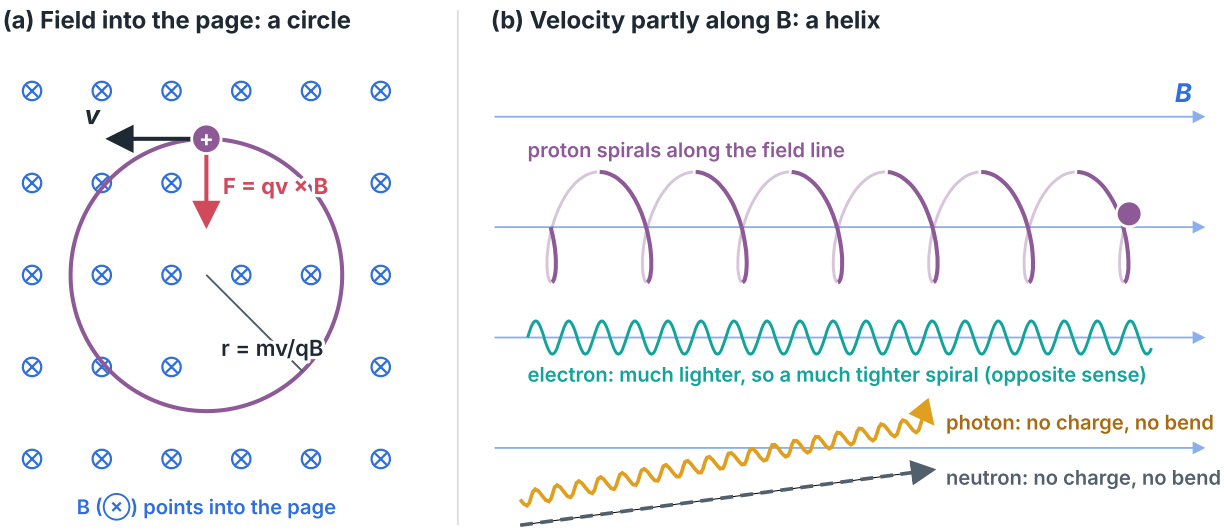


Figure 10.7 The Lorentz force. (a) Perpendicular motion gives a circle of radius mv/qB. (b) Motion with a component along B gives a helix. Uncharged radiation is not bent at all.

WORKED EXAMPLE 10.4 — A 1 MeV proton in Earth's field

Given: A 1.0 MeV proton ($m = 1.673 \times 10^{-27}$ kg) moving perpendicular to $B = 30\ \mu\text{T}$, about Earth's field at the equator.

Find: Its speed and gyroradius.

Think: 1 MeV is far below the proton's 938 MeV rest energy, so $KE = \frac{1}{2}mv^2$ is fine. Then use Eq. 10.7.

$$v = \sqrt{\frac{2(1.0 \times 10^{6})(1.602 \times 10^{-19}\ \text{J})}{1.673 \times 10^{-27}\ \text{kg}}} = 1.38 \times 10^{7}\ \text{m/s}$$

$$r = \frac{(1.673 \times 10^{-27})(1.38 \times 10^{7})}{(1.602 \times 10^{-19})(3.0 \times 10^{-5})} = 4.8 \times 10^{3}\ \text{m}$$

Answer: $v \approx 1.4 \times 10^7$ m/s (about 5% of c), $r \approx 4.8$ km, tiny compared with Earth's 6371 km radius. The proton is captured into spiraling along a field line instead of flying straight in.

YOU ASKED

"How does a magnetic field help protect against radiation?"

A magnetic field deflects *charged* particles (protons, electrons, alpha particles, cosmic-ray nuclei) so they spiral along field lines instead of reaching the surface, but it does nothing at all to photons (UV, X-rays, gamma rays) or neutrons.

The Lorentz force $\vec{F} = q\vec{v} \times \vec{B}$ is always sideways, so a charged particle curls into a helix around a field line (Figure 10.7). Earth's field turns most of the solar wind aside at the magnetopause and traps other particles in the Van Allen belts (Figure 10.8). With $q = 0$, photons and neutrons feel no force. Earth stops those, and the most energetic cosmic rays, with its **atmosphere**: about 1 kg of air over every square centimeter, the mass of a 10 m column of water. Gases high up absorb the UV-C and X-rays.

Why not give spacecraft a magnetic shield? A 1 GeV cosmic-ray proton curls with a radius of about 6 m even in a 1 tesla field (30,000 times Earth's). Deflecting it means a strong field over meters around the crew: big superconducting magnets, cryogenics, heavy structure, and huge stored energy, and the highest-energy particles still get through. For now, shielding is mass.

Earth's magnetosphere, Van Allen belts, and aurora

Earth's field, about 25–65 μT at the surface, is roughly a dipole. The solar wind compresses it on the day side and stretches it into a long tail on the night side (Figure 10.8). Its outer boundary, the **magnetopause**, deflects most of the solar wind around Earth.

Field lines crowd together near the poles, and a spiraling particle heading into the stronger field is pushed back (a **magnetic mirror**). Trapped particles bounce pole to pole, forming two doughnut-shaped zones of intense radiation, the **Van Allen belts**: an inner belt (roughly 1,000–6,000 km up, mostly protons) and an outer belt (roughly 13,000–60,000 km, mostly electrons). The ISS, about 400 km up, stays below both except over the South Atlantic, where the inner belt dips low.

Near the magnetic poles, field lines dip into the atmosphere. Particles (mainly electrons) sliding down them excite oxygen and nitrogen 100–300 km up, which then emit light (Chapter 5): green 557.7 nm and red 630.0 nm from oxygen, blue-violet from nitrogen. That is the **aurora**.

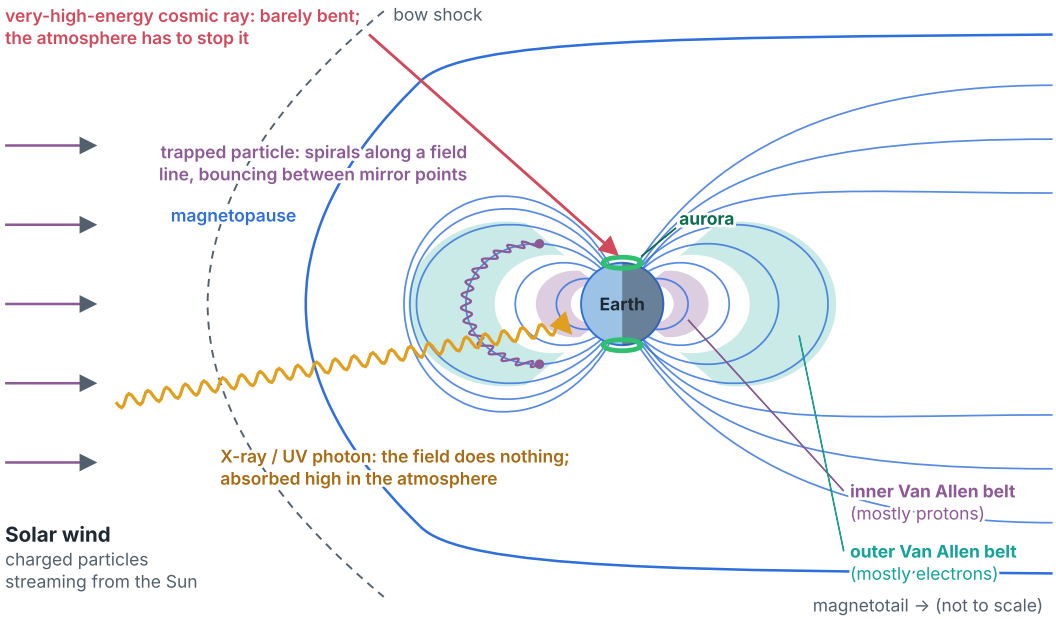


Figure 10.8 Earth's magnetosphere (not to scale). Charged particles are deflected or trapped; photons pass straight through and are absorbed by the atmosphere; the most energetic cosmic rays get through near the poles and are stopped by the air.

10.10 Space radiation and spacecraft shielding

Astronauts face **galactic cosmic rays** (steady, extremely energetic, hard to stop), **solar particle events** (sudden storms of lower-energy protons, easier to shield), and the trapped belts. The ISS crew gets about 0.2–0.9 mSv per day; Curiosity measured about 1.8 mSv per day on its way to Mars. Defenses:

- hydrogen-rich mass around crew quarters (water, polyethylene, even food and supplies);
- a heavily shielded "storm shelter" for solar events;
- shorter trips, and timing missions around the solar cycle (an active Sun pushes some galactic cosmic rays away).

PROJECT HAIL MARY CONNECTION

You asked: "What are the radiation concerns about the tunnel connecting the two ships?"

In the book. When Rocky describes his crewmates dying of unexplained cell damage, Grace realizes it was radiation, something Eridians had never heard of. By Grace's estimate Erid's magnetic field is at least 25 times Earth's and its atmosphere is far thicker than ours, so essentially no space radiation reaches the surface and Eridians never had reason to discover it. Rocky survived because, as engineer, he worked near the Astrophage fuel, and Astrophage blocks radiation. The Hail Mary has a layer of Astrophage built into its hull as a shield. The tunnel, and Rocky's side of it, lie outside that protection, so the fix in the book is the obvious one: put Astrophage between Rocky and space.

The real physics. Grace's "near-light-speed hydrogen ions" are galactic cosmic-ray protons. A thin tunnel wall, however strong, is almost no *mass*, and mass per area is what stops them (Section 10.8). Erid's strong field would deflect many charged particles (Section 10.9), but its thick atmosphere would do the heavy lifting against the energetic ones and all photons, as on Earth. No real material absorbs perfectly like fictional Astrophage; the best real choice is thick, hydrogen-rich mass such as water or polyethylene.

RABBIT HOLE **(optional) — Why a 1 GeV proton is so hard to steer**

Safe to skip. Near light speed, use relativistic momentum: for 1 GeV of kinetic energy, $pc = \sqrt{KE^2 + 2\,KE\,mc^2} = \sqrt{1 + 2(0.938)}$ GeV $= 1.70$ GeV, so $p = 9.1 \times 10^{-19}$ kg·m/s. Then $r = p/(qB)$ is 5.7 m in 1 T and about 190 km in Earth's 30 μT. Earth's field works because it extends over thousands of kilometers; a spacecraft magnet must do the job within a few meters.

Chapter summary

- Ionizing means about 10 eV or more per photon or particle ($\lambda \lesssim 124$ nm). Brightness cannot substitute for photon energy.
- UV-B is below the threshold but damages DNA by photochemistry (thymine dimers): sunburn.
- Alpha: stopped by paper or skin. Beta: by millimeters of aluminum or plastic. Gamma/X-rays: reduced exponentially, best by dense mass. Neutrons: slowed by hydrogen.
- DNA is damaged directly or by •OH radicals; double-strand and clustered breaks are the worst.
- Gray = J/kg. Sievert = gray × biological weighting. The US average is about 6.2 mSv per year, half of it natural.
- Inverse-square law: $I \propto 1/r^2$. Attenuation: $I = I_0 e^{-\mu x}$, HVL $= 0.693/\mu$. For MeV gamma rays, mass per area is what counts.
- Magnetic fields bend charged particles into helices ($r = mv/qB$) and do nothing to photons or neutrons. Earth relies on both its magnetosphere and its atmosphere (about 10 m of water equivalent).

Check yourself

1. CONCEPT A 1000 W infrared heat lamp (photons near 1 μm) delivers far more power than a dental X-ray tube. Which one can ionize atoms, and why does the power rating not decide it?
2. CALC What is the photon energy of 254 nm UV-C light? Is it above the 10 eV threshold? Can it still damage DNA? Explain.
3. CONCEPT Why is an alpha emitter nearly harmless outside the body but very dangerous if inhaled?
4. CONCEPT Explain why a neutron loses much more energy in a collision with a hydrogen nucleus than with a lead nucleus.
5. CALC A worker receives an absorbed dose of 2.0 mGy from gamma rays and, separately, 0.10 mGy from alpha particles in the lungs. Find the equivalent dose from each in mSv.
6. CALC A survey meter reads 12 μSv/h at 1.5 m from a small source. What will it read at 0.5 m? At 6.0 m?
7. CALC For 1 MeV gamma rays, concrete ($\rho = 2.3$ g/cm³) has $\mu/\rho \approx 0.064$ cm²/g. Find its HVL and the thickness needed to reduce the beam to 1/1000.
8. CALC An electron with speed 2.0×10^7 m/s moves perpendicular to a 30 μT field. Find its gyroradius and compare it with the 1 MeV proton in Worked Example 10.4.
9. CONCEPT Earth's magnetic field weakens and the magnetic poles wander. Would this change the amount of solar UV reaching the ground? Would it change the cosmic-ray particle flux at high altitude? Explain.
10. APPLY You are designing a crew cabin for a Mars transit with a fixed mass budget. Rank aluminum, lead, and polyethylene for shielding against galactic cosmic rays and justify the ranking.
11. APPLY Using Table 10.1, how many chest X-rays equal (a) one year of US natural background and (b) a six-month ISS stay at 160 mSv?

Full solutions are in the Answer Key at the back of the book.

CHAPTER 11

X-Rays and Materials Science

The same invisible light that shows your dentist a cavity also told us how the atoms in a copper wire are stacked. Both tricks work because an X-ray's wavelength is about the size of an atom.

THE BIG PICTURE

- X-ray tubes make a continuous "braking" spectrum plus sharp characteristic lines that fingerprint the target metal.
- An X-ray image is a shadow: absorption rises steeply with atomic number, so bone and metal block far more than flesh.
- Crystal planes scatter X-rays in step only at angles satisfying Bragg's law, $n\lambda = 2d\sin\theta$.
- Peak positions give d-spacings, which give the lattice parameter and the crystal structure (FCC vs. BCC) through selection rules.

AFTER THIS CHAPTER YOU CAN

- Explain how an X-ray tube makes bremsstrahlung and Kα/Kβ lines, and find $\lambda_{\min}$ from the voltage.
- Explain why bones and metal show up on X-rays, using the Z-dependence of absorption.
- Derive and use Bragg's law and $d_{hkl} = a/\sqrt{h^2 + k^2 + l^2}$.
- Relate (110) and (220), and explain why (100) is missing for BCC and FCC.
- Connect interplanar spacing with planar density, and index a powder pattern to find the structure and lattice parameter.

11.1 Making X-rays: the X-ray tube

An **X-ray tube** is a vacuum tube with two electrodes (Figure 11.1). A hot filament, the **cathode**, boils off electrons. A high voltage V (about 20–150 kV) accelerates them toward a metal **anode**, or target. Each electron arrives with kinetic energy eV, so 30 kV gives 30 keV electrons. When they slam into the target, about 99% of their energy becomes heat (which is why anodes rotate or are water-cooled) and about 1% comes out as X-rays.

11.2 The X-ray spectrum: braking radiation and fingerprint lines

Two processes make the photons (Figure 11.2):

- **Bremsstrahlung** ("braking radiation"). An electron swinging past a nucleus is sharply deflected. Accelerating charges radiate (Chapter 1), so it emits a photon. It can lose any fraction of its energy, so the spectrum is continuous, up to a hard limit where one electron gives *all* of its energy eV to one photon:

$$\lambda_{\min} = \frac{hc}{eV} = \frac{1240\ \text{eV}\cdot\text{nm}}{eV} \qquad \text{e.g. } 35\ \text{kV} \Rightarrow 0.0354\ \text{nm} \tag{11.1}$$

- **Characteristic lines**. An incoming electron knocks a K-shell (n = 1) electron out of a target atom. An electron from the L shell drops into the hole and emits a **Kα** photon; one from the M shell emits **Kβ**. These energies are set by the target atom's energy levels (Chapter 5), so they are a fingerprint of the element. They appear only when eV exceeds the K-shell binding energy (8.98 keV for copper).

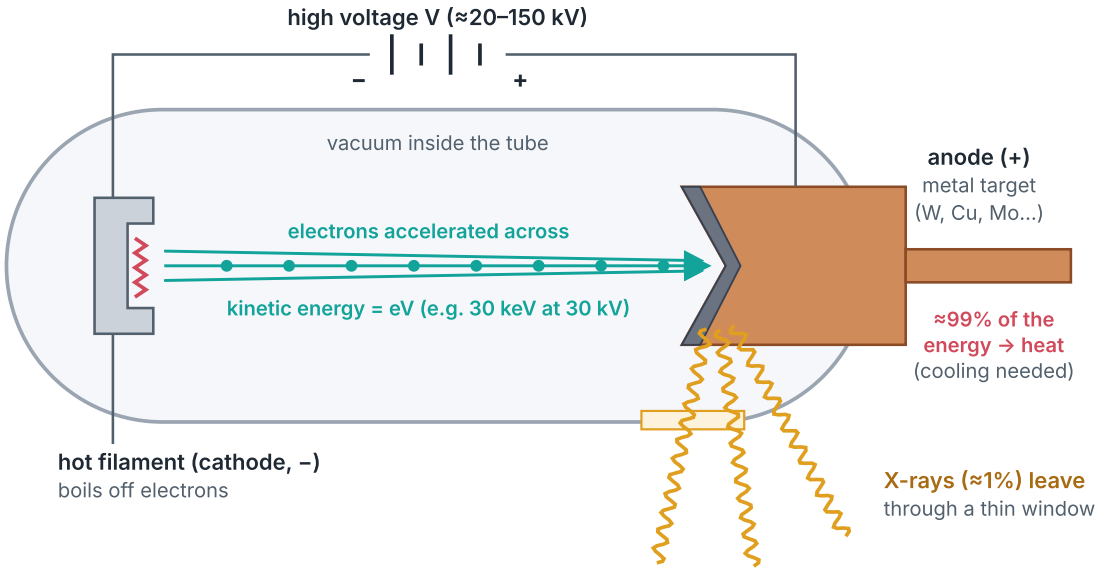


Figure 11.1 An X-ray tube. Electrons gain energy eV crossing the vacuum and stop in the target. Most of the energy becomes heat; a small fraction becomes X-ray photons.

Useful target lines: Cu Kα 0.1542 nm (8.04 keV), Co Kα 0.1790 nm, Mo Kα 0.0709 nm, W Kα 0.0209 nm (59 keV). Diffraction labs want one wavelength, so they filter out Kβ. For copper a thin nickel foil does it: nickel's K-edge (8.33 keV) lies between Cu Kα (8.05 keV) and Kβ (8.90 keV), so Kβ is absorbed and Kα passes.

RABBIT HOLE (optional) — Moseley's law

Safe to skip. Treat the K-shell like hydrogen (Chapter 4) with nuclear charge $Z - 1$, since the one remaining K electron screens one proton. An L → K drop then gives $E_{K\alpha} \approx 13.6\ \text{eV}\,(Z-1)^2(1-\frac{1}{4}) = 10.2\ \text{eV}\,(Z-1)^2$. For copper (Z = 29) that is 8.0 keV, against a measured 8.05 keV. In 1913 Henry Moseley used this pattern to show that the periodic table is ordered by atomic number, not atomic mass.

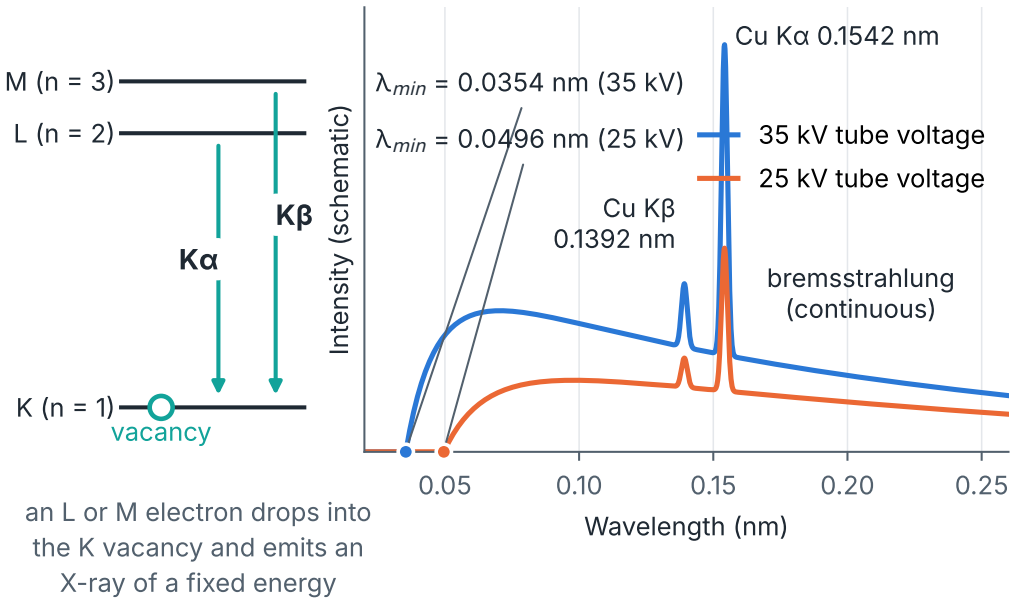


Figure 11.2 Schematic X-ray spectrum of a copper-target tube (peak heights illustrative). Raising the voltage moves $\lambda_{\min}$ shorter and raises everything, but the Kα and Kβ lines stay at the same wavelengths: they belong to copper, not to the voltage.

11.3 X-ray imaging: shadows, not reflections

YOU ASKED

"What does 'X-ray vision' mean — how do X-rays see through things?"

X-rays don't "see" anything: an X-ray image is a **shadow**. Photons pass straight through low-density, low-atomic-number material such as skin and muscle, bone and metal absorb them, and a detector records where they got through.

At medical energies (about 20–150 keV), the dominant absorption is photoelectric. Its probability rises very steeply with atomic number (roughly Z^3 to Z^4) and falls steeply with energy (roughly $1/E^3$). Soft tissue is mostly H, C, N, and O (Z ≤ 8). Bone contains calcium (Z = 20) and phosphorus (Z = 15) and is nearly twice as dense. So at 30 keV, a centimeter of bone stops about 8 times more photons than a centimeter of muscle (Worked Example 11.1), and metal stops almost all of them. On a radiograph, bright means "few photons arrived," so bones look white. Comic-book X-ray vision, seeing X-rays bounced back off hidden objects, is not how it works (see the Myth box below).

MYTH VS. REALITY

Myth: "X-ray vision" means seeing X-rays that bounce off hidden objects.

Reality: Ordinary objects reflect almost no X-rays (for X-rays, a material's refractive index differs from 1 by only about 10^{-5} or less), and nothing around you is lit by X-rays anyway. Real X-ray imaging always needs a source on one side and a detector on the other.

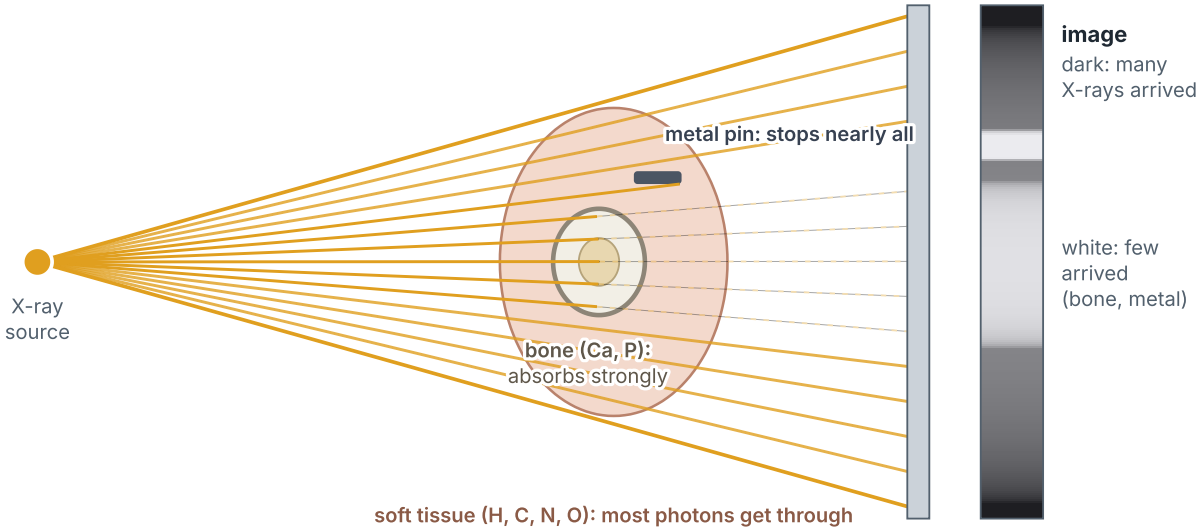


Figure 11.3 An X-ray image is a shadow. The image strip is computed from the drawing: transmission through each path sets the gray level (radiograph convention: white = few X-rays arrived).

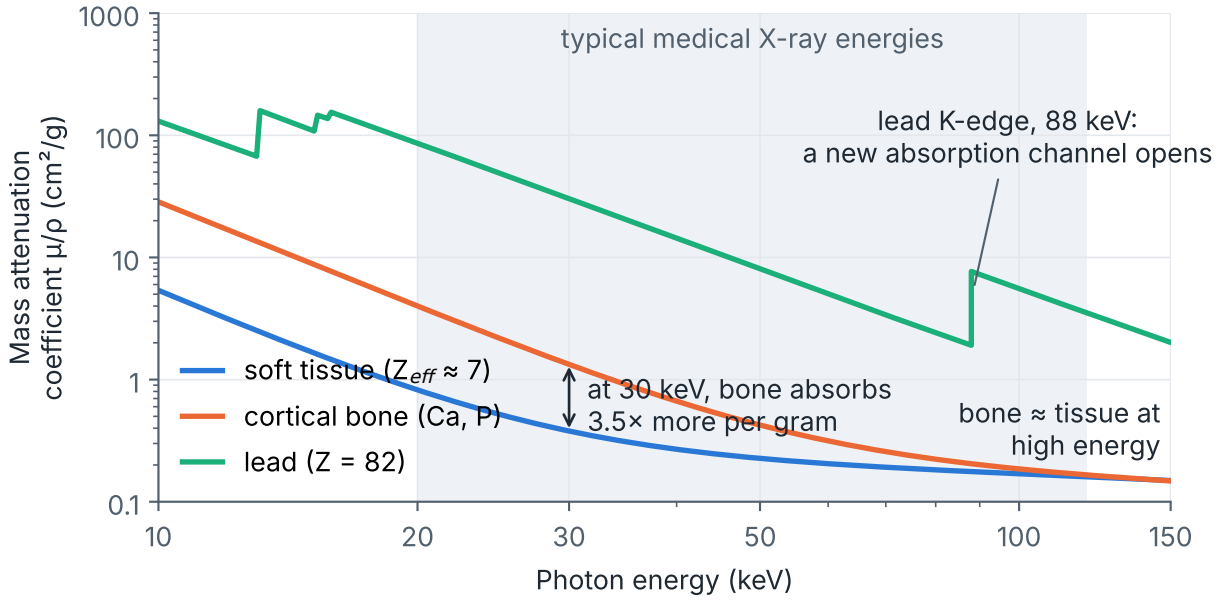


Figure 11.4 Absorption per gram versus photon energy (NIST-based data). At low energies the higher-Z elements in bone make a big difference. At higher energies Compton scattering, which barely depends on Z, takes over and the contrast fades. Lead's absorption jumps at its K-edge, where photons become able to eject its K electrons.

WORKED EXAMPLE 11.1 — Bone vs. muscle contrast

Given: At 30 keV, μ/ρ = 0.379 cm²/g for soft tissue ($\rho = 1.06$ g/cm³) and 1.331 cm²/g for cortical bone ($\rho = 1.92$ g/cm³). At 100 keV: 0.169 and 0.186 cm²/g.

Find: The fraction transmitted through 1.0 cm of each, at both energies.

Think: $T = e^{-\mu x}$ with $\mu = (\mu/\rho)\rho$ (Eq. 10.4).

30 keV: tissue $\mu = 0.402\ \text{cm}^{-1}$, $T = e^{-0.402} = 0.67$. Bone $\mu = 2.56\ \text{cm}^{-1}$, $T = e^{-2.56} = 0.078$.

100 keV: tissue $\mu = 0.179\ \text{cm}^{-1}$, $T = 0.84$. Bone $\mu = 0.356\ \text{cm}^{-1}$, $T = 0.70$.

Answer: At 30 keV, 67% vs. 8% get through, so the contrast is 8.6:1. At 100 keV it is 84% vs. 70%, only 1.2:1. Lower energy gives more contrast but more absorbed dose, so radiologists pick tube voltage to balance the two.

MYTH VS. REALITY

Myth: An X-ray machine makes you or the room radioactive.

Reality: X-ray photons are absorbed or pass through, and they are gone the moment the tube switches off. Medical X-ray energies are far too low to change nuclei, so nothing is left behind.

Contrast agents and CT

Soft organs look alike on an X-ray, so doctors add high-Z **contrast agents**, such as iodine (Z = 53) in blood vessels or barium (Z = 56) in the gut. Their K-edges (33 and 37 keV) sit right in the diagnostic range.

A single radiograph squashes 3D anatomy into one shadow. **Computed tomography (CT)** rotates the tube and detectors around the body, records hundreds of projections, and lets a computer reconstruct a map of μ slice by slice. The values are reported in Hounsfield units (air −1000, water 0, dense bone around +1000 or higher). The price is dose: about 7.7 mSv for an abdominal CT versus 0.1 mSv for a chest X-ray (Table 10.1). Engineers use the same method, industrial CT, to find porosity in castings, voids in welds, and defects in 3D-printed parts without cutting them open.

11.4 Why X-rays can "see" atoms

Atoms in a solid sit about 0.1–0.4 nm apart. Visible light (400–700 nm) is thousands of times too long to resolve that (Chapter 8). X-ray wavelengths of 0.05–0.25 nm match it. When an X-ray passes through a crystal, its electric field shakes every atom's electrons, and each shaken electron re-radiates a weak spherical wavelet (Chapter 6). A crystal is a 3D array of these sources, a natural **diffraction grating**. The wavelets cancel almost everywhere and add up only in a few special directions.

> **KEY IDEA**
>
> Diffraction needs a wavelength comparable to the spacing being probed (Chapter 8). Atoms are about 0.1–0.4 nm apart, so the probe must be X-rays (or electrons or neutrons) with wavelengths of a fraction of a nanometer.

11.5 Bragg's law

W. L. Bragg found the simplest way to predict those directions (1913). Treat the crystal as stacks of parallel atomic planes a distance d apart. A ray scattered by the second plane travels farther than one scattered by the first, by $d\sin\theta$ on the way in plus $d\sin\theta$ on the way out (Figure 11.5). The scattered waves add in step when that extra path is a whole number of wavelengths:

$$n\lambda = 2d\sin\theta \tag{11.2}$$

where θ is measured from the planes (not from the normal), $n = 1, 2, 3, \ldots$ is the order, and λ is the X-ray wavelength.

> **KEY IDEA**
>
> A Bragg "reflection" is not a mirror bouncing light. It is interference: the planes scatter X-rays at every angle, but the scattered waves add up only at angles where $2d\sin\theta = n\lambda$. A diffractometer swings its detector through the angle 2θ between the incoming and outgoing beams, so patterns are plotted against 2θ.

> **ANALOGY**
>
> Clap in front of a tall stone staircase and the echo comes back as a chirp: each step returns a little echo, spaced in time by the step depth. Crystal planes do the same with X-rays, except the "pitch" shows up as an angle.

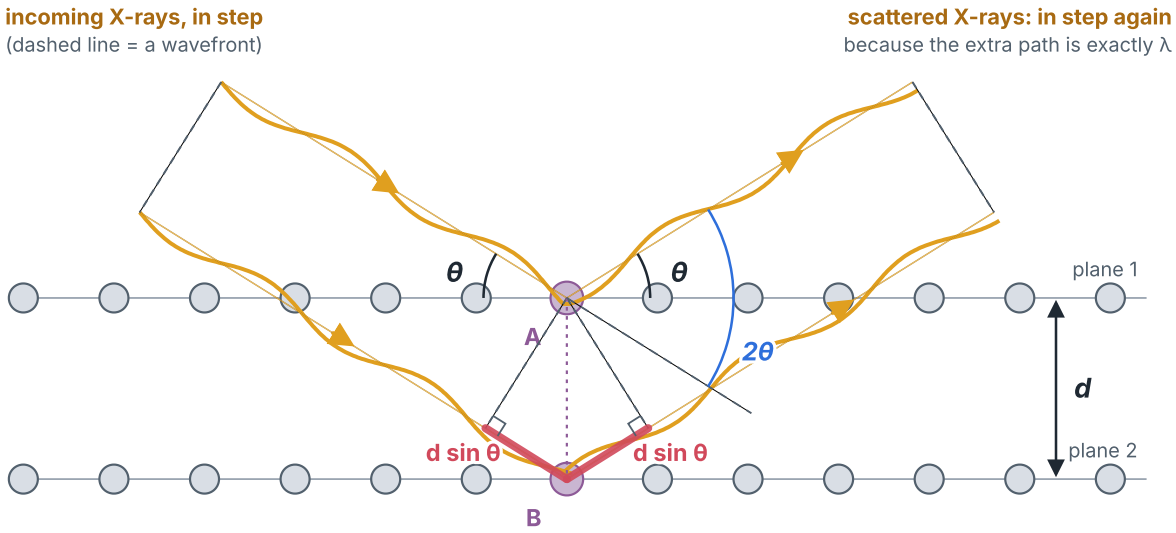


Figure 11.5 Bragg's law, drawn to scale for $n = 1$. Before scattering, the dashed wavefront through A meets ray 2 at the foot of a perpendicular; after scattering, a second perpendicular from A marks where ray 2 has caught up. The two red segments, each $d \sin\theta$, are ray 2's extra path. Here $2d \sin\theta$ equals exactly one wavelength, so the outgoing waves are crest-to-crest.

11.6 Planes, Miller indices, and d-spacings

For a cubic crystal, the spacing between adjacent (hkl) planes is

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \tag{11.3}$$

So $d_{100} = a$, $d_{110} = a/\sqrt{2}$, $d_{111} = a/\sqrt{3}$, $d_{200} = a/2$, $d_{220} = a/\sqrt{8}$. Larger indices mean more closely spaced planes.

MATERIALS SCIENCE CONNECTION

Miller indices, the ME2340 version. For a plane: take its intercepts with x, y, z in units of the lattice parameter a, take reciprocals, clear fractions, and write (hkl), with a bar for negatives. A direction $[uvw]$ is the vector components, reduced to smallest integers. Braces $\{hkl\}$ and angle brackets $\langle uvw \rangle$ denote whole families. In cubic crystals, $[hkl]$ is perpendicular to (hkl).

Lattice parameter from atomic radius. Atoms touch along the face diagonal in FCC ($4R = a\sqrt{2}$, so $a = 2R\sqrt{2}$) and along the body diagonal in BCC ($4R = a\sqrt{3}$, so $a = 4R/\sqrt{3}$). Copper (FCC, R = 0.1278 nm) has $a = 0.3615$ nm; α-iron (BCC, R = 0.1241 nm) has $a = 0.2866$ nm. An X-ray pattern measures a directly, which is how those radii were found in the first place.

YOU ASKED

"How can parallel planes have different Miller indices — what's the deal with (110) vs (220), or (100) vs (200)?"

(220) planes have the same orientation as (110) planes but are packed twice as closely: $d_{220} = d_{110}/2$. The (220) set includes every (110) plane plus an extra plane halfway between each pair (Figure 11.6a).

Geometrically, a (220) plane cuts the axes at $a/2$, $a/2$, ∞ instead of a, a, ∞. In plain geometry you would reduce (220) to (110). X-ray crystallographers keep the extra factor deliberately, because it matches Bragg's law: $2\lambda = 2d_{110}\sin\theta$ is the same equation as $\lambda = 2(d_{110}/2)\sin\theta = 2d_{220}\sin\theta$. A "(220) reflection" *is* the second-order ($n = 2$) reflection from the (110) planes. Labeling it (220) lets us always write Bragg's law with $n = 1$.

The (100)/(200) case adds a twist. In BCC, the body-center atoms lie on planes exactly halfway between the (100) planes of corner atoms (Figure 11.6b). At the angle where corner planes scatter in step (path difference λ), the halfway planes are off by λ/2. They hold the same number of atoms, so they cancel the corner planes exactly, and no (100) peak appears. At the (200) angle, all planes, spaced $a/2$, are in step. FCC behaves the same way because its face-center atoms also sit on the halfway planes. That is why BCC and FCC patterns list (200), never (100).

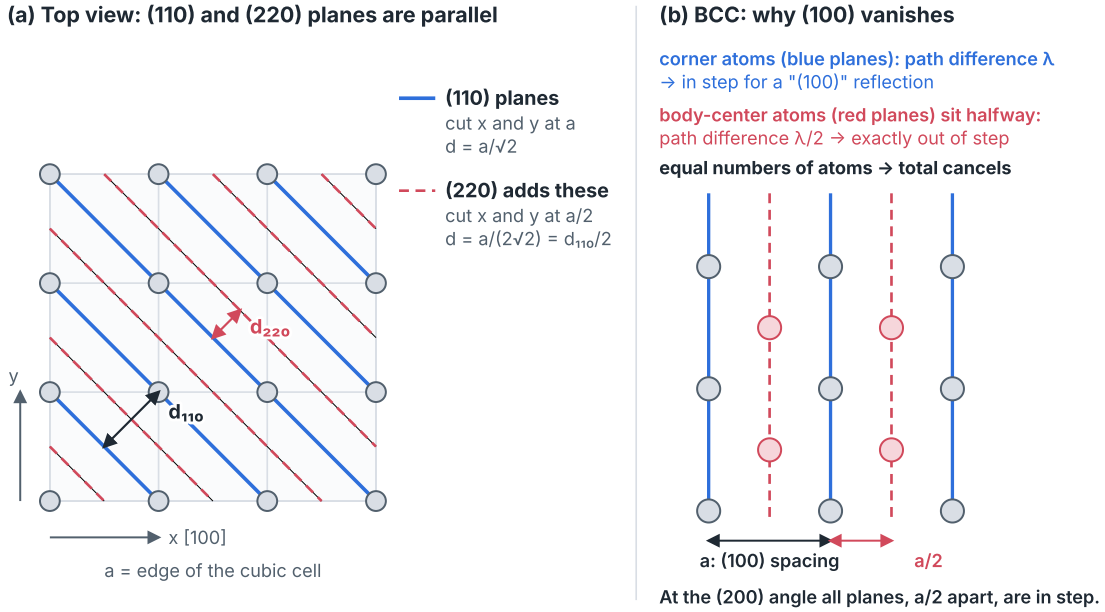


Figure 11.6 (a) The (220) set is the (110) set plus a plane halfway between each pair, so $d_{220} = d_{110}/2$. (b) In BCC, body-center atoms fill the halfway planes. They scatter exactly out of step at the (100) angle and cancel it.

Table 11.1 Which (hkl) reflections appear (cubic crystals, one atom per lattice point). $N = h^2 + k^2 + l^2$.

Lattice	Rule for a reflection to appear	First allowed reflections	N values
Simple cubic	all h, k, l	100, 110, 111, 200, 210, 211, 220	1, 2, 3, 4, 5, 6, 8 (never 7)
BCC (Fe, Cr, W)	$h + k + l$ even	110, 200, 211, 220, 310, 222, 321	2, 4, 6, 8, 10, 12, 14
FCC (Cu, Al, Ni)	h, k, l all odd or all even	111, 200, 220, 311, 222, 400, 331	3, 4, 8, 11, 12, 16, 19

RABBIT HOLE **(optional) — The structure factor**

Safe to skip. Add the wave from every atom j in the unit cell, at fractional position (x_j, y_j, z_j), with phase $2\pi(hx_j + ky_j + lz_j)$: $F_{hkl} = \sum_j f_j\, e^{2\pi i(hx_j+ky_j+lz_j)}$. For BCC (atoms at 0,0,0 and ½,½,½), $F = f[1 + e^{i\pi(h+k+l)}]$, which is $2f$ when $h + k + l$ is even and 0 when it is odd. FCC's four atoms give $4f$ or 0 by the same logic. Intensity goes as $|F|^2$, and Table 11.1 falls right out.

11.7 Interplanar spacing and planar density

YOU ASKED

"If the interplanar spacing is higher, does that mean the density is higher?"

Within one crystal, yes for the *planar* density: the families of planes with the largest spacing are the most densely packed planes. It says nothing about the material's mass density (g/cm³), which is the same however you slice it.

The reason is bookkeeping. Every atom lies on some plane of a given family, and the number of atoms per unit volume n_v is fixed. Stack planes with PD atoms per unit area at spacing d, and a slab of area A and thickness L holds $PD \cdot A \cdot (L/d)$ atoms. Setting that equal to $n_v AL$:

$$PD = n_v\, d \tag{11.4}$$

So widely spaced planes must be crowded and closely spaced planes sparse. Picture slicing a loaf into thick slices or thin ones: same bread, different amount per slice. For FCC, (111) has the largest spacing ($a/\sqrt{3}$) and is the close-packed plane; for BCC, (110) is both. Figure 11.7 and Worked Example 11.2 check this for copper.

Two caveats. First, use the distance between *adjacent atom-bearing* planes. For FCC (100) that is $a/2$ (d_{200}), because face-center atoms sit halfway. Second, the relation is exact for crystals with one atom per lattice point (FCC and BCC metals). In crystals with several atoms per lattice point (NaCl, diamond, HCP), the planes in a family may not be identical or evenly spaced, so it becomes an approximation.

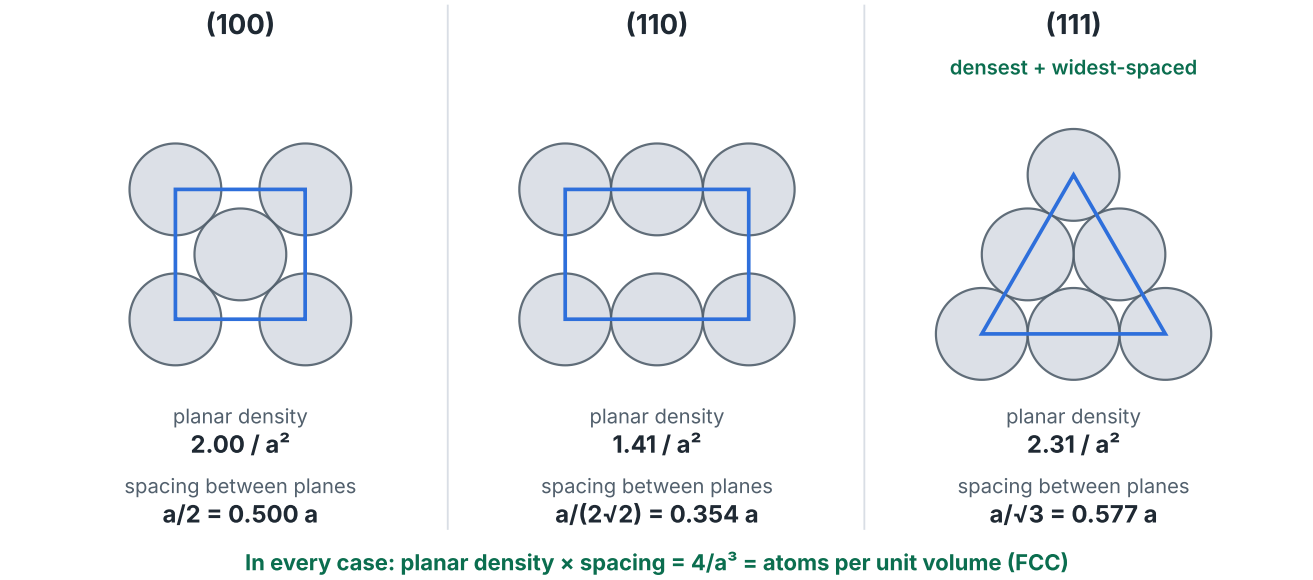


Figure 11.7 Three planes in an FCC crystal (hard spheres). The (111) plane is the densest and the most widely spaced; (110) is the sparsest and most closely spaced. Planar density × spacing is the same for all three.

WORKED EXAMPLE 11.2 — Planar density vs. spacing in copper

Given: Copper is FCC with $R = 0.1278$ nm.

Find: Planar density PD and adjacent-plane spacing d for (100), (110), (111), and check that $PD/d = n_v$.

Think: $a = 2R\sqrt{2}$. Count atoms centered on each plane section (Callister style), divide by its area, then compare with $n_v = 4/a^3$.

$a = 2(0.1278)\sqrt{2} = 0.3615$ nm, so $n_v = 4/a^3 = 84.7$ atoms/nm^3.

(100): 2 atoms on a^2: $PD = 2/a^2 = 15.3$ nm^{-2}; $d = a/2 = 0.181$ nm; $PD/d = 84.7$.

(110): 2 atoms on $a \times a\sqrt{2}$: $PD = 10.8$ nm^{-2}; $d = a/(2\sqrt{2}) = 0.128$ nm; $PD/d = 84.7$.

(111): 2 atoms on a triangle of area $(\sqrt{3}/2)a^2$: $PD = 17.7$ nm^{-2}; $d = a/\sqrt{3} = 0.209$ nm; $PD/d = 84.7$.

Answer: (111) has the largest d (0.209 nm) and the highest PD (17.7 atoms/nm²); (110) has the smallest of both. All three give $PD/d = n_v = 84.7$ atoms/nm³.

MATERIALS SCIENCE CONNECTION

This is why metals slip on their densest planes. Close-packed planes are the farthest apart and the smoothest, so they slide past each other most easily. FCC metals slip on {111} planes and BCC metals mostly on {110}, which you'll see as "slip systems" in the dislocations chapter of Callister. In a diffraction pattern, those same planes usually give the first strong peak, (111) for FCC and (110) for BCC, because the largest d gives the smallest angle.

11.8 Reading a diffraction pattern

In **powder diffraction**, the sample contains millions of randomly oriented grains, so every plane family finds some grains at its Bragg angle. The detector sweeps 2θ and records a peak for each allowed family (Figure 11.8).

Peak positions are only the start. A pattern also reveals:

- **Phase identification**: each crystalline compound has a fingerprint pattern. One example is telling FCC austenite from BCC ferrite in a steel.
- **Composition and strain**: alloying or residual stress shifts a slightly, so peaks shift.
- **Grain size**: very small crystallites (below about 100 nm) broaden the peaks.
- **Amorphous material**: glass and many polymers give broad humps instead of sharp peaks, because they have no long-range order.

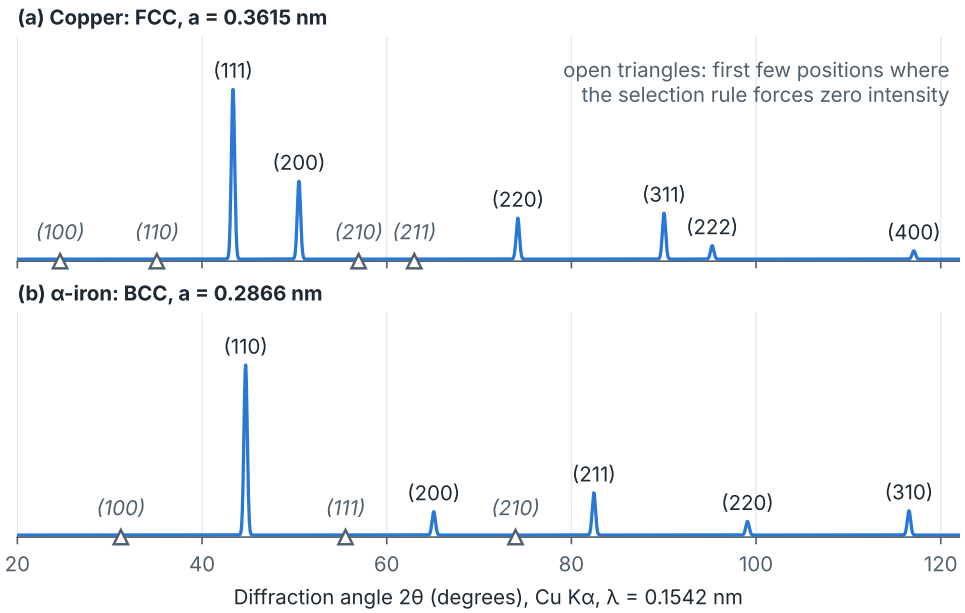


Figure 11.8 Simulated powder patterns with Cu Kα. Peak positions come from Bragg's law and Table 11.1; heights are approximate (computed from multiplicity, scattering factors, and geometry factors). The FCC pairs (111)/(200) and (311)/(222) look nothing like BCC's evenly stepped sequence, so the structure can be read at a glance.

WORKED EXAMPLE 11.3 — BCC iron, (220) planes (Callister-style)

Given: α-iron is BCC with $a = 0.2866$ nm. Monochromatic X-rays with $\lambda = 0.1790$ nm (Co Kα); first-order reflection ($n = 1$) from (220) planes.

Find: d_{220} and the diffraction angle 2θ.

Think: Eq. 11.3 for d, then Bragg's law for θ. (220) is allowed for BCC since 2 + 2 + 0 is even.

$$d_{220} = \frac{0.2866\ \text{nm}}{\sqrt{2^2 + 2^2 + 0^2}} = \frac{0.2866}{2.828} = 0.1013\ \text{nm}$$

$$\sin\theta = \frac{n\lambda}{2d} = \frac{0.1790}{2(0.1013)} = 0.8833 \ \Rightarrow\ \theta = 62.04^\circ, \quad 2\theta = 124.08^\circ$$

Answer: $d_{220} = 0.1013$ nm and $2\theta \approx 124.1^\circ$. (Callister's printed 124.26° comes from rounding $\sin\theta$ to 0.884 before taking the arcsine. Carry four digits and you get 124.08°.)

WORKED EXAMPLE 11.4 — Identify an unknown cubic metal

Given: Cu Kα ($\lambda = 0.1542$ nm) peaks at 2θ = 38.5°, 44.7°, 65.1°, 78.2°, 82.4°.

Find: The structure and lattice parameter.

Think: Bragg plus Eq. 11.3 give $\sin^2\theta = (\lambda^2/4a^2)\,N$, so the $\sin^2\theta$ values must be in the ratio of the allowed N values in Table 11.1.

- $\sin^2\theta$: 0.1087, 0.1446, 0.2895, 0.3978, 0.4339
- Divided by the first: 1, 1.330, 2.663, 3.659, 3.992. These are not integers, so try ×2 (BCC) or ×3 (FCC).
- ×3: 3.00, 3.99, 7.99, 10.98, 11.97 → N = 3, 4, 8, 11, 12. That is FCC: (111), (200), (220), (311), (222).
- $a = \lambda\sqrt{N}/(2\sin\theta)$: from (111), $0.1542\sqrt{3}/(2 \times 0.3297) = 0.405$ nm. The other peaks agree within 0.1%.

Answer: FCC with $a \approx 0.405$ nm, which matches aluminum ($a = 0.4049$ nm).

11.9 Electron microscopes and electron diffraction

Electrons are waves too, with $\lambda = h/p$ (Chapter 8). Accelerate one through 100 kV and, including a small relativistic correction,

$$\lambda_e = \frac{h}{\sqrt{2m_e eV\left(1 + \dfrac{eV}{2m_e c^2}\right)}} = 0.0037\ \text{nm} \quad (100\ \text{kV}) \tag{11.5}$$

That is about 40 times shorter than Cu Kα, so electron Bragg angles are tiny. For copper's (111) planes, $\sin\theta = 0.0037/(2 \times 0.2087)$, so $\theta \approx 0.5^\circ$. Electrons interact with matter far more strongly than X-rays, so samples must be very thin (under about 100 nm) and held in vacuum.

- **TEM** (transmission electron microscope): shines electrons through a thin foil. It can image dislocations and even columns of atoms, and it records electron diffraction patterns from regions a few nanometers across.
- **SEM** (scanning electron microscope): scans a focused beam over a surface and collects knocked-out electrons to image fracture surfaces and microstructure.
- **EDS**: the beam knocks out inner-shell electrons, just as in an X-ray tube, so the sample emits its own characteristic X-rays. Their energies (Section 11.2) identify the elements present.

Chapter summary

- X-ray tubes: bremsstrahlung gives a continuum down to $\lambda_{\min} = hc/eV$; K-shell vacancies give element-specific Kα/Kβ lines (Cu Kα = 0.1542 nm).
- X-ray images are shadows: photoelectric absorption scales roughly as Z^3–Z^4/E^3, so bone and metal absorb far more than soft tissue. CT maps μ in 3D.
- Bragg's law $n\lambda = 2d\sin\theta$, with θ from the planes; patterns are plotted against 2θ.

- Cubic spacing: $d_{hkl} = a/\sqrt{h^2 + k^2 + l^2}$. (220) planes are parallel to (110) at half the spacing, and a (220) reflection is the second order of (110).
- BCC needs $h + k + l$ even; FCC needs all odd or all even. (100) vanishes in both because halfway-plane atoms scatter exactly out of step.
- Planar density = $n_v \times d$: the widest-spaced planes are the densest (FCC {111}, BCC {110}), and they are also the slip planes.
- 100 kV electrons have λ ≈ 0.0037 nm, enabling TEM, electron diffraction, and EDS.

Check yourself

1. CALC A dental X-ray tube runs at 70 kV. What is the shortest wavelength it produces, and what is the highest photon energy?
2. CONCEPT Why does raising a copper tube from 25 kV to 35 kV change the continuum but not the Kα position?
3. CALC At 30 keV, lead has $\mu/\rho = 30.3$ cm²/g ($\rho = 11.35$ g/cm³). What fraction of 30 keV photons gets through a 0.25 mm lead apron?
4. CONCEPT Why do mammograms use low tube voltages (about 25–30 kV)?
5. CONCEPT A friend says diffraction peaks are mirror "reflections" from atomic planes. What is right and wrong about that?
6. CALC Find d_{111} and d_{200} for aluminum (FCC, $a = 0.4049$ nm), and the 2θ angle of each with Cu Kα ($\lambda = 0.1542$ nm).
7. CALC Using $\lambda = 0.1542$ nm and $d_{110} = 0.2027$ nm for iron, find the angle of the first-order (110) reflection and of the second-order (110) reflection. Show that the second matches the (220) peak.
8. CONCEPT For BCC iron, which of (100), (110), (111), (200), (210), (211) appear? Explain the rule physically.
9. APPLY For BCC iron ($a = 0.2866$ nm), compute the planar density of (110) and (100) from atom counting, and verify $PD = n_v d$ for each. Which plane would you expect to be the slip plane?
10. APPLY An unknown metal gives Cu Kα peaks at 2θ = 40.3°, 58.3°, 73.3°, 87.1°, 100.8°, 115.1°, 131.4°. Show that it is BCC, not simple cubic, and find a. Which metal could it be?
11. CALC Find the wavelength of 200 kV TEM electrons with Eq. 11.5, and compare with the nonrelativistic $h/\sqrt{2m_e eV}$.

Full solutions are in the Answer Key at the back of the book.

Formula Sheet & Constants

Constant / handy number	Value
Speed of light in vacuum	$c = 2.998 \times 10^{8}\ \mathrm{m/s} \approx 3.00 \times 10^{8}\ \mathrm{m/s}$
Planck's constant	$h = 6.626 \times 10^{-34}\ \mathrm{J \cdot s} = 4.136 \times 10^{-15}\ \mathrm{eV \cdot s}$
Photon-energy shortcut	$hc = 1240\ \mathrm{eV \cdot nm}$
Electron-volt	$1\ \mathrm{eV} = 1.602 \times 10^{-19}\ \mathrm{J}$
Elementary charge	$e = 1.602 \times 10^{-19}\ \mathrm{C}$
Electron mass	$m_e = 9.109 \times 10^{-31}\ \mathrm{kg}$
Proton mass	$m_p = 1.673 \times 10^{-27}\ \mathrm{kg}$
Boltzmann constant	$k_B = 1.381 \times 10^{-23}\ \mathrm{J/K} = 8.617 \times 10^{-5}\ \mathrm{eV/K}$
Stefan–Boltzmann constant	$\sigma = 5.670 \times 10^{-8}\ \mathrm{W/(m^2 K^4)}$
Wien's displacement constant	$b = 2898\ \mu\mathrm{m \cdot K}$
Avogadro's number	$N_A = 6.022 \times 10^{23}\ \mathrm{mol^{-1}}$
Hydrogen ground-state energy	$E_1 = -13.6\ \mathrm{eV}$
Bohr radius	$a_0 = 0.0529\ \mathrm{nm}$
Cu Kα X-ray wavelength	$\lambda_{\mathrm{Cu\ K}\alpha} = 0.1542\ \mathrm{nm}$

Electric field definition Ch. 1

$$\vec{E} = \frac{\vec{F}}{q}$$

Electric field is the force per unit charge on a small test charge q. Units N/C = V/m.

Speed of light from electromagnetism Ch. 1

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3.00 \times 10^{8}\ \mathrm{m/s}$$

The speed of EM waves in vacuum follows from the electric constant ε0 = 8.85×10⁻¹² C²/(N·m²) and the magnetic constant μ0 = 1.257×10⁻⁶ T·m/A.

Wave speed relation Ch. 1

$$c = \lambda f$$

Speed of light (m/s) equals wavelength (m) times frequency (Hz). Long wavelength means low frequency.

Period and frequency Ch. 1

$$T = \frac{1}{f}$$

The period (s) is the time for one cycle; frequency (Hz) is cycles per second.

Field ratio in an EM wave Ch. 1

$$B_0 = \frac{E_0}{c}$$

In an EM wave in vacuum, the magnetic field amplitude (T) equals the electric field amplitude (V/m) divided by c.

Intensity of an EM wave Ch. 1

$$I = \tfrac{1}{2}\, c\, \varepsilon_0\, E_0^2$$

Average power per area (W/m²) carried by a wave of electric field amplitude E0 (V/m). Intensity grows as amplitude squared.

Unpolarized light through a polarizer Ch. 1

$$I = \tfrac{1}{2} I_0$$

An ideal polarizer transmits half the intensity of unpolarized light.

Malus's law Ch. 1

$$I = I_0 \cos^2 \theta$$

Polarized light of intensity I0 hitting a polarizer at angle θ between its polarization and the axis emerges with intensity I.

Photon energy Ch. 2

$$E = hf = \frac{hc}{\lambda}$$

Energy of one photon (J or eV) from its frequency f (Hz) or wavelength λ (m); h = 6.626×10⁻³⁴ J·s.

Photon energy shortcut Ch. 2

$$E\ (\mathrm{eV}) = \frac{1240}{\lambda\ (\mathrm{nm})}$$

Quick photon energy in electron volts from wavelength in nanometers, since hc = 1240 eV·nm.

Einstein's photoelectric equation Ch. 2

$$KE_{\max} = hf - \phi$$

Maximum kinetic energy (eV) of electrons ejected by photons of energy hf from a surface with work function φ. No emission if hf is less than φ.

Threshold frequency and wavelength Ch. 2

$$f_0 = \frac{\phi}{h}, \qquad \lambda_0 = \frac{hc}{\phi}$$

Lowest frequency (longest wavelength) of light that can eject electrons from a material with work function φ.

Stopping voltage Ch. 2

$$KE_{\max} = eV_s$$

The reverse voltage V_s (V) that stops the fastest photoelectrons measures their kinetic energy; in eV the number equals V_s in volts.

Photon rate from power Ch. 2

$$\dot{N} = \frac{P}{hf} = \frac{P\lambda}{hc}$$

Number of photons per second in a beam of power P (W) at frequency f or wavelength λ.

Photon momentum Ch. 2

$$p = \frac{h}{\lambda} = \frac{E}{c}$$

Momentum (kg·m/s) carried by one photon of wavelength λ or energy E.

Radiation force on an absorber Ch. 2

$$F = \frac{P}{c}$$

Force (N) exerted by a light beam of power P (W) that is completely absorbed.

Photon energy from wavelength Ch. 3

$$E\,(\text{eV}) = \frac{1240}{\lambda\,(\text{nm})}$$

Converts any band's wavelength to its photon energy; the key to what that band can do to matter.

Nuclear radius Ch. 4

$$R \approx (1.2\,\text{fm})\,A^{1/3}$$

Radius of a nucleus with A protons plus neutrons, in femtometers (10^{-15} m).

de Broglie wavelength Ch. 4

$$\lambda = \frac{h}{mv}$$

Wavelength (m) of a particle of mass m (kg) moving at speed v (m/s); h = 6.626×10⁻³⁴ J·s.

Standing-wave condition (de Broglie model) Ch. 4

$$n\lambda = 2\pi r_n$$

A whole number n of electron wavelengths must fit around an allowed orbit of radius r_n; a helpful model, not the full quantum picture.

Hydrogen energy levels Ch. 4

$$E_n = -\frac{13.6\ \text{eV}}{n^2}$$

Allowed energies of hydrogen's electron (eV); n = 1 is the ground state, 0 eV means free.

Bohr-model orbit radius Ch. 4

$$r_n = n^2 a_0$$

Size of hydrogen's nth state grows as n²; a_0 = 0.0529 nm.

Volume per atom in a solid Ch. 4

$$V_{\text{atom}} = \frac{M}{\rho N_A}$$

Volume each atom occupies, from molar mass M (g/mol), density ρ (g/cm³) and Avogadro's number; its cube root estimates atomic spacing.

FCC lattice parameter (hard spheres) Ch. 4

$$a = 2R\sqrt{2}$$

Cube edge of a face-centered cubic cell in terms of atomic radius R (Callister); R is half the equilibrium spacing.

Photon–gap matching rule Ch. 5

$$E_{\text{photon}} = hf = \frac{hc}{\lambda} = E_{\text{upper}} - E_{\text{lower}}$$

A photon is absorbed or emitted only with energy equal to the gap between two levels (J or eV).

Wavelength from an energy gap Ch. 5

$$\lambda\,(\text{nm}) = \frac{1240\ \text{eV}\cdot\text{nm}}{\Delta E\,(\text{eV})}$$

Converts a transition energy in eV to the photon wavelength in nm.

Hydrogen transition energy Ch. 5

$$\Delta E = 13.6\ \text{eV}\left(\frac{1}{n_{\text{lower}}^2} - \frac{1}{n_{\text{upper}}^2}\right)$$

Energy of the photon emitted or absorbed when hydrogen's electron moves between levels; 3 → 2 gives 1.89 eV (656 nm).

Natural (resonant) frequency of a transition Ch. 5

$$f = \frac{\Delta E}{h}$$

Frequency (Hz) at which a mixed-state electron cloud oscillates, and the frequency of light the transition absorbs or emits.

LED photon energy Ch. 5

$$E_{\text{photon}} \approx E_g, \quad \lambda \approx \frac{1240\ \text{eV}\cdot\text{nm}}{E_g}$$

An LED emits photons with energy about equal to its semiconductor's band gap (eV).

Fluorescence energy loss Ch. 5

$$\frac{E_{\text{in}} - E_{\text{out}}}{E_{\text{in}}}$$

Fraction of each absorbed photon's energy turned into heat by the Stokes shift (dimensionless).

Thermal energy scale Ch. 5

$$k_B T \approx 0.026\ \text{eV at } 300\ \text{K}$$

Typical thermal energy per particle at room temperature; compare with band gaps and vibrational spacings.

Conservation of light energy at a material Ch. 6

$$R + A + T = 1$$

The fractions of incoming light energy reflected, absorbed, and transmitted add to 1 (scattered light is counted in R or T). Dimensionless.

Exponential absorption Ch. 6

$$I = I_0\, e^{-\alpha x}$$

Intensity remaining after depth x (m) in an absorber with absorption coefficient α (m⁻¹); I_0 is the intensity just inside the surface.

Law of reflection Ch. 6

$$\theta_r = \theta_i$$

The angle of reflection equals the angle of incidence, both measured from the surface normal.

Normal-incidence reflectance Ch. 6

$$R = \left(\tfrac{n_2 - n_1}{n_2 + n_1}\right)^2$$

Fraction of light reflected at a boundary between indices n_1 and n_2 when light hits straight on; about 0.04 for glass in air.

Index of refraction Ch. 6

$$n = \tfrac{c}{v}$$

Speed of light in vacuum c divided by the speed v of wave crests in the material. Dimensionless, always about 1 or more for visible light.

Wavelength in a material Ch. 6

$$\lambda = \tfrac{\lambda_0}{n}$$

The wavelength shrinks by the factor n inside a material; the frequency (and photon energy) is unchanged.

Snell's law Ch. 6

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Relates the angles (from the normal) on each side of a boundary between indices n_1 and n_2.

Critical angle Ch. 6

$\sin\theta_c = \frac{n_2}{n_1} \quad (n_1 > n_2)$

Beyond this angle of incidence, light in the higher-index material is totally internally reflected.

Thin-lens equation Ch. 6

$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

Relates focal length f, object distance d_o, and image distance d_i (all in m); f is positive for a converging lens.

Lens power Ch. 6

$P = \frac{1}{f}$

Optical power in diopters (m⁻¹) for focal length f in meters.

Photon energy from wavelength Ch. 7

$E = \frac{hc}{\lambda} = \frac{1240\ \text{eV·nm}}{\lambda}$

Energy of one photon in eV for wavelength λ in nm; visible light (400–700 nm) spans 3.1 to 1.8 eV.

Absorption edge Ch. 7

$\lambda_{\text{edge}} = \frac{1240\ \text{eV·nm}}{E_g}$

For band gap Eg in eV, wavelengths shorter than λ_edge (nm) are absorbed and longer ones pass; sets a semiconductor's or pigment's color.

Band-gap color rule Ch. 7

$E_g > 3.1\ \text{eV}$: colorless; $\quad 1.8 < E_g < 3.1\ \text{eV}$: colored; $\quad E_g < 1.8\ \text{eV}$: opaque

Compares a pure material's band gap with visible photon energies to predict whether it looks transparent, yellow-to-red, or dark/metallic.

Rayleigh scattering Ch. 7

$I_{\text{scattered}} \propto \frac{1}{\lambda^4}$

For scatterers much smaller than the wavelength, scattered intensity rises steeply for shorter wavelengths; 450 nm light scatters about 5.9 times more than 700 nm light.

Thin-film extra path Ch. 7

$\Delta \approx 2nt$

Extra optical path traveled by light reflected from the back of a film of thickness t and index n (normal incidence); with phase flips it sets which colors are reinforced.

Absorbed solar power Ch. 7

$P = a\,I\,A$

Power absorbed (W) by a surface with absorbed fraction a, in sunlight of intensity I (W/m², about 1000 on a clear day), over area A (m²).

Single-slit first dark fringe Ch. 8

$\sin\theta = \frac{\lambda}{a}$

A slit of width a (m) spreads a wave of wavelength lambda (m) so that the first dark fringe is at angle theta; spreading is about lambda/a radians.

Double-slit / grating bright fringes Ch. 8

$d\sin\theta = m\lambda$

Bright fringes appear where the path difference d sin(theta) between neighboring slits (spacing d, in m) equals a whole number m of wavelengths.

Double-slit fringe spacing (small angles) Ch. 8

$\Delta y = \frac{\lambda L}{d}$

Distance between neighboring bright fringes on a screen a distance L away (all lengths in m).

de Broglie wavelength Ch. 8

$\lambda = \frac{h}{p}$

Wavelength of a particle with momentum p (kg·m/s); h = 6.626e-34 J·s.

Momentum of a slow particle Ch. 8

$p = \sqrt{2mK}$

Momentum (kg·m/s) of a particle of mass m (kg) and kinetic energy K (J), valid when K is much less than its rest energy.

Rayleigh criterion (circular aperture) Ch. 8

$\theta_{\min} \approx 1.22\,\frac{\lambda}{D}$

Smallest resolvable angle (radians) for an eye, camera, or telescope of aperture diameter D (m).

Microscope (Abbe) diffraction limit Ch. 8

$d_{\min} \approx \frac{\lambda}{2\,\text{NA}}$

Smallest resolvable separation (m) for a microscope objective of numerical aperture NA; about 200 nm for visible light.

Rayleigh scattering scaling Ch. 8

$\sigma_{\text{scatter}} \propto \frac{d^6}{\lambda^4}$

For particles much smaller than the wavelength, scattering grows steeply with size d and falls steeply with wavelength; per unit area it goes as (d/lambda)^4.

Wien's displacement law Ch. 9

$\lambda_{\text{peak}} = \frac{2898\ \mu\text{m}\cdot\text{K}}{T}$

Wavelength (micrometers) at which a blackbody at absolute temperature T (K) is brightest; 310 K gives about 9.3 micrometers, 5800 K about 500 nm.

Stefan-Boltzmann law Ch. 9

$P = \varepsilon\,\sigma\,A T^4$

Total thermal power (W) radiated by a surface of area A (m^2), emissivity epsilon (0-1), and temperature T (K); sigma = 5.67e-8 W/(m^2 K^4).

Net radiative exchange Ch. 9

$P_{\text{net}} = \varepsilon\,\sigma\,A\left(T^4 - T_{\text{s}}^4\right)$

Net power (W) lost by an object at T to surroundings at T_s (both in K); positive means the object is cooling by radiation.

Thermal energy scale Ch. 9

$E_{\text{thermal}} \sim k_B T$

Typical energy (J, or eV) of random motion per particle at temperature T; k_B = 1.381e-23 J/K, so about 0.026 eV at 300 K.

Photon energy from wavelength Ch. 9

$E = \frac{hc}{\lambda} = \frac{1240\ \text{eV}\cdot\text{nm}}{\lambda}$

Energy (eV) of a photon of wavelength lambda (nm); a 10 micrometer thermal-IR photon carries about 0.12 eV, matching molecular vibrations.

Ionizing threshold wavelength Ch. 10

$\lambda = \frac{hc}{E} = \frac{1240\ \text{eV·nm}}{10\ \text{eV}} \approx 124\ \text{nm}$

A photon must have at least about 10 eV (wavelength about 124 nm or shorter) to ionize; λ in nm, E in eV.

Equivalent dose Ch. 10

$H = w_R\,D$

Biologically weighted dose H (Sv) equals the radiation weighting factor w_R (dimensionless) times the absorbed dose D (Gy = J/kg).

Inverse-square law Ch. 10

$$I = \frac{P}{4\pi r^2}, \qquad \frac{I_2}{I_1} = \left(\frac{r_1}{r_2}\right)^2$$

Intensity (W/m² or dose rate) from a point source falls as 1/distance²; P is source power (W), r distance (m).

Exponential attenuation Ch. 10

$$I = I_0\, e^{-\mu x}, \qquad \mu = \left(\frac{\mu}{\rho}\right)\rho$$

Photon intensity after thickness x (cm) of shield; μ is the linear attenuation coefficient (cm⁻¹), μ/ρ the mass attenuation coefficient (cm²/g), ρ the density (g/cm³).

Half-value layer Ch. 10

$$\mathrm{HVL} = \frac{\ln 2}{\mu} \approx \frac{0.693}{\mu}$$

Thickness (cm) that halves a photon beam; n half-value layers transmit (1/2)^n.

Lorentz (magnetic) force Ch. 10

$$\vec{F} = q\,\vec{v} \times \vec{B}$$

Force (N) on charge q (C) moving with velocity v (m/s) in magnetic field B (T). It is perpendicular to v, so it changes direction but not speed; zero for uncharged photons and neutrons.

Gyroradius Ch. 10

$$r = \frac{mv}{qB}$$

Radius (m) of a charged particle's circular or helical path; m mass (kg), v speed perpendicular to B (m/s). Near light speed, replace mv with the relativistic momentum p.

Shortest tube wavelength (Duane–Hunt) Ch. 11

$$\lambda_{\min} = \frac{hc}{eV} = \frac{1240\ \text{eV·nm}}{eV}$$

Shortest X-ray wavelength (nm) from a tube at voltage V; eV is the electron kinetic energy in eV (numerically equal to V in volts).

Moseley's law (approximate) Ch. 11

$$E_{K\alpha} \approx 10.2\ \text{eV}\,(Z-1)^2$$

Approximate Kα photon energy for an element of atomic number Z; about 8.0 keV for copper (Z = 29).

Bragg's law Ch. 11

$$n\lambda = 2d\sin\theta$$

Condition for constructive X-ray diffraction from planes spaced d (nm) apart; λ wavelength (nm), θ angle from the planes, n = 1, 2, 3, … the order. Diffractometers report 2θ.

Cubic interplanar spacing Ch. 11

$$d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$$

Spacing (nm) between adjacent (hkl) planes in a cubic crystal with lattice parameter a (nm).

Lattice parameter vs atomic radius Ch. 11

$$a_{\mathrm{FCC}} = 2R\sqrt{2}, \qquad a_{\mathrm{BCC}} = \frac{4R}{\sqrt{3}}$$

Cube edge a from atomic radius R for hard-sphere atoms touching along the face diagonal (FCC) or body diagonal (BCC).

Indexing a cubic pattern Ch. 11

$$\sin^2\theta = \frac{\lambda^2}{4a^2}\,(h^2+k^2+l^2)$$

sin²θ values are proportional to N = h²+k²+l²; their ratios reveal SC (1,2,3,4,5,6,8), BCC (2,4,6,8,…), or FCC (3,4,8,11,12,…).

Planar density and spacing Ch. 11

$$PD = n_v\, d$$

Atoms per unit area on a plane (nm⁻²) equals atoms per unit volume (nm⁻³) times the spacing between adjacent atom-bearing planes (nm); exact for one atom per lattice point.

Relativistic electron wavelength Ch. 11

$$\lambda_e = \frac{h}{\sqrt{2m_e eV\left(1+\dfrac{eV}{2m_e c^2}\right)}}$$

de Broglie wavelength (m) of an electron accelerated through V volts; 0.0037 nm at 100 kV, 0.0025 nm at 200 kV.

Myth-Busting Roundup

Popular explanations of light and radiation are full of half-truths. Here are the ones this book corrects, with the page where each is explained.

MYTH The wavy line in drawings of light is the path the light travels, snaking up and down through space. p. 16

REALITY The wavy line is a graph. It plots the field's strength and direction at points along a perfectly straight beam.

MYTH Space must be filled with some invisible substance that carries light, the way air carries sound. p. 17

REALITY Nineteenth-century physicists believed exactly this and called the substance the "luminiferous aether." In 1887 Michelson and Morley tried to detect Earth moving through it and found nothing.

MYTH Brighter light travels faster, or carries "stronger" color. p. 19

REALITY All EM waves in vacuum travel at c, dim or bright.

MYTH Brighter light knocks electrons out harder, so they come out faster. p. 26

REALITY Brighter light of the same color knocks out more electrons, each with the same maximum energy.

MYTH A photon is a tiny ball of light that flies along a definite path, like a microscopic marble. p. 29

REALITY A photon has no size or surface, and between emission and detection it does not follow a single trajectory.

MYTH Light switches between being a wave and being a particle, depending on whether someone is watching. p. 30

REALITY Light is always the same quantum object.

MYTH X-rays and gamma rays are different kinds of radiation, and gamma rays always have more energy. p. 34

REALITY Both are high-energy photons, and their energy ranges overlap.

MYTH Wi-Fi and microwaves are "radiation," so they are dangerous in the same way as X-rays. p. 37

REALITY All light is electromagnetic radiation, including the light you read by.

MYTH Visible light is a special kind of radiation, different from the invisible kinds. p. 39

REALITY Visible light differs from infrared or ultraviolet only in wavelength.

MYTH Electrons orbit the nucleus like planets around the Sun, and the cloud is a blur of that fast-moving dot. p. 45

REALITY No electron follows a track.

MYTH Since atoms never really touch, you have never actually touched anything. p. 49

REALITY At the atomic scale, electron clouds pushing on each other at about 0.1–0.3 nm *is* what touching means.

MYTH A mirror (or a window) works by absorbing each photon and then re-emitting it. p. 56

REALITY Absorb-and-re-emit would scatter light in random directions, so a mirror would look like frosted glass instead of showing your face.

MYTH Glow-in-the-dark materials must be radioactive. p. 57

REALITY Modern glow-in-the-dark products store energy from ordinary light in trapped electrons and release it slowly; that is why they fade and need recharging under a lamp.

MYTH Laser light is made of especially powerful photons. p. **61**

REALITY A laser's photon energy is set by its transition like any other light; a red laser pointer's photons each carry less energy than a blue LED's.

MYTH A mirror reflects light because it is very smooth. p. **66**

REALITY Smoothness decides *where* reflected light goes; the electrons decide *how much* .

MYTH Transparent materials let light slip through the gaps between their atoms. p. **67**

REALITY Atoms are mostly empty space (Chapter 4), but a 500 nm wave is thousands of times wider than an atom and drives every electron it passes.

MYTH Light slows down in glass because atoms absorb it, hold it for a moment, and re-emit it. p. **68**

REALITY Glass has no visible-frequency transition to absorb into, and absorb-and-re-emit would scramble direction and phase (Figure 6.2b), leaving glass cloudy.

MYTH Every color you can see corresponds to some wavelength of light. p. **76**

REALITY Magenta, pink, brown, and white appear nowhere in the rainbow.

MYTH Gold is yellow because it reflects only yellow light. p. **80**

REALITY Gold reflects red, orange, yellow, and most green almost perfectly.

MYTH The sky is blue because it reflects the ocean. p. **81**

REALITY The sky is just as blue over deserts.

MYTH Light travels in perfectly straight lines; only sound and water waves bend around corners. p. **86**

REALITY Light diffracts around every edge; its wavelength is just tiny compared with everyday objects.

MYTH Each photon or electron secretly goes through one slit or the other; we just don't know which. p. **89**

REALITY If each particle used only one slit, there would be nothing to interfere with and no fringes.

MYTH Cold objects don't give off radiation, so a cold window must be "radiating cold" at you. p. **98**

REALITY Ice at 0 °C still emits about 315 W per square meter.

MYTH The atmosphere warms Earth the same way a glass greenhouse warms plants. p. **103**

REALITY A real greenhouse stays warm mainly because its walls stop warm air from rising and blowing away; it blocks convection.

MYTH Microwave ovens are tuned to the resonant frequency of water molecules. p. **103**

REALITY Liquid water has no sharp resonance at 2.45 GHz.

MYTH Space is so cold that an unprotected person would freeze solid instantly. p. **104**

REALITY Vacuum is an excellent insulator.

MYTH A powerful enough Wi-Fi router or cell tower could ionize your cells. p. **107**

REALITY A 2.4 GHz photon carries about 10^{-5} eV, a million times too little.

MYTH Lead is the best radiation shield, period. p. **112**

REALITY Lead is excellent against X-rays and compact against gamma rays, but poor against neutrons and it worsens cosmic-ray showers.

MYTH "X-ray vision" means seeing X-rays that bounce off hidden objects. p. **119**

REALITY Ordinary objects reflect almost no X-rays (for X-rays, a material's refractive index differs from 1 by only about 10^{-5} or less), and nothing around you is lit by X-rays anyway.

MYTH An X-ray machine makes you or the room radioactive. p. **120**

REALITY X-ray photons are absorbed or pass through, and they are gone the moment the tube switches off.

Glossary

Absorption — A photon disappears and its energy lifts an electron to a higher level; it happens only if the photon energy matches an energy gap (or exceeds the ionization energy). (Ch. 5, 6)

Absorption coefficient — The constant α in $I = I_0 e^{-\alpha x}$, in m^{-1}; after a depth of $1/\alpha$, about 37% of the light remains. (Ch. 6)

Absorption edge — The wavelength 1240 eV·nm / Eg below which a semiconductor or insulator absorbs light strongly. (Ch. 7)

Absorption spectrum — Dark lines in a continuous spectrum where a cooler gas in front of the source removed its own wavelengths. (Ch. 5)

Accelerating charge — A charge that is speeding up, slowing down or changing direction. Only accelerating charges radiate EM waves. (Ch. 1)

Additive mixing — Combining colored lights (usually red, green, and blue); more light added means brighter, and all three together look white. Screens work this way. (Ch. 7)

Airy disk — The small blurred disk with faint rings that a perfect circular lens makes from a single point of light, because of diffraction at the lens. (Ch. 8)

Alpha particle — A helium nucleus (two protons, two neutrons, charge +2) emitted by some radioactive nuclei. It ionizes densely and is stopped by paper or the outer layer of skin. (Ch. 10)

Amplitude (E0) — The peak strength of a wave's electric field, in V/m. It sets the wave's brightness, not its color. (Ch. 1)

Antenna — A conductor in which electrons are driven back and forth to launch EM waves, or are driven by arriving waves to detect them. (Ch. 1)

Atmospheric window — A band of wavelengths that the atmosphere lets through to the ground, chiefly visible/near-infrared light and radio waves. (Ch. 3, 9)

Atom — The smallest unit of an element: a tiny, dense nucleus surrounded by an electron cloud, about 0.1–0.5 nm across. (Ch. 4)

Balmer series — Hydrogen's emission lines from drops to n = 2; four are visible: 656.3, 486.1, 434.0 and 410.2 nm. (Ch. 5)

Band gap (E_g) — The forbidden energy range between valence and conduction bands; photons below E_g pass through, photons above are absorbed (Si 1.12 eV, diamond ≈ 5.5 eV). (Ch. 5)

Band gap (Eg) — The energy gap between a solid's filled valence band and empty conduction band; photons with energy above Eg can be absorbed. (Ch. 7)

Band-gap color rule — Eg above 3.1 eV: colorless and transparent; Eg below 1.8 eV: all visible light absorbed (opaque, dark or metallic); Eg between: only the blue end absorbed, so the material looks yellow, orange, or red. (Ch. 7)

Beta particle — A fast electron (or positron) emitted by a nucleus. It travels a few millimeters in tissue and is stopped by a few millimeters of aluminum or plastic. (Ch. 10)

Birefringence — A direction-dependent refractive index found in non-cubic crystals such as quartz, calcite, and sapphire; it makes grain boundaries in polycrystalline ceramics scatter light. (Ch. 6)

Blackbody — An ideal object that absorbs all radiation hitting it; it is also the best possible emitter, and its spectrum depends only on its temperature. (Ch. 9)

Bohr radius (a_0) — 0.0529 nm: the most probable distance of hydrogen's ground-state electron from the nucleus, and the radius of the n = 1 orbit in the Bohr model. (Ch. 4)

Bragg's law — $n\lambda = 2d \sin\theta$: X-rays scattered by parallel crystal planes spaced d apart add in step only when the extra path $2d \sin\theta$ is a whole number of wavelengths; θ is measured from the planes. (Ch. 11)

Bremsstrahlung — "Braking radiation": the continuous X-ray spectrum emitted when fast electrons are deflected and slowed by nuclei in the target. (Ch. 11)

Brewster's angle — The angle of incidence at which reflected light is completely polarized parallel to the surface; about 53° from the vertical for water. (Ch. 1)

Characteristic X-rays (Kα, Kβ) — Sharp X-ray lines emitted when an electron from the L shell (Kα) or M shell (Kβ) fills a K-shell vacancy; their energies identify the element (Cu Kα = 0.1542 nm). (Ch. 11)

Charge transfer — Absorption of a photon by moving an electron from one ion to a neighboring ion, such as Fe^{2+} to Ti^{4+} in blue sapphire; it produces strong colors from trace impurities. (Ch. 7)

Coherent — Describes waves that keep a fixed phase relationship with each other, which is needed for a stable interference pattern (lasers are highly coherent). (Ch. 8)

Coherent re-radiation — Radiation from electrons forced to oscillate by a light wave, at the same frequency and with a fixed phase relationship to it; the basis of reflection and refraction. (Ch. 6)

Color — The label your visual system gives to the mix of visible wavelengths (about 400–700 nm, photon energies 1.8–3.1 eV) reaching your eye; an object's color is the part of the incoming light it sends toward you. (Ch. 7)

Color temperature — The temperature of a blackbody whose glow has the same color as a given light source; warm-white bulbs are about 2700 K, daylight about 5000-6500 K. (Ch. 9)

Computed tomography (CT) — Imaging that records X-ray projections from many angles and computes a cross-sectional map of attenuation coefficient, reported in Hounsfield units (air −1000, water 0). (Ch. 11)

Cone cells — Color-sensing cells in the retina of three types, S, M, and L, with sensitivity peaks near 420, 534, and 564 nm. (Ch. 7)

Conjugated molecule — A molecule with a chain of alternating single and double bonds; longer chains have smaller electron energy gaps and absorb longer wavelengths, which is how many dyes and pigments get their color. (Ch. 7)

Constructive interference — Overlapping waves arriving in step (crest on crest), so their amplitudes add and the spot is bright. (Ch. 8)

Critical angle — The angle of incidence, given by $\sin\theta_c = n_2/n_1$, beyond which light traveling from a higher-index to a lower-index material cannot escape. (Ch. 6)

Crystal field — The electric field of the ions surrounding a transition-metal ion in a crystal, which splits its d-orbital energies into gaps that can absorb visible light (as with Cr^{3+} in ruby). (Ch. 7)

De Broglie wavelength — The wavelength of a moving particle, λ = h/(mv); for an electron in hydrogen's ground state it is 0.332 nm. (Ch. 4, 8)

Destructive interference — Overlapping waves arriving half a cycle out of step (crest on trough), so they cancel and the spot is dark. (Ch. 8)

Dielectric heating — Heating by an oscillating electric field that twists polar molecules such as water; because the molecules lag behind the field, energy is lost into random motion. It is how microwave ovens work. (Ch. 9)

Diffraction — The spreading of a wave as it passes through an opening or around an edge, by an angle of roughly wavelength divided by opening width. (Ch. 8)

Diffraction angle (2θ) — The angle between the incident and diffracted X-ray beams, which is what a diffractometer measures and plots. (Ch. 11)

Diffraction grating — A surface with many evenly spaced slits or grooves that sends each wavelength off at its own angle, d sin(theta) = m lambda, splitting light into a spectrum. (Ch. 8)

Diffraction limit — The smallest detail a lens-based instrument can resolve, set by wavelength: about lambda/(2 NA) for a microscope, or roughly half a wavelength at best. (Ch. 8)

Diffuse reflection — Reflection from a rough surface (bumps comparable to the wavelength or larger); each ray obeys the law of reflection locally, but rays leave in all directions. (Ch. 6)

Diopter — The unit of lens power, P = 1/f with f in meters; the human eye has roughly 60 diopters, about two-thirds of it from the cornea. (Ch. 6)

Dispersion — The dependence of refractive index on wavelength; in glass and water n is larger for violet than for red, which separates colors in prisms and rainbows. (Ch. 6)

Draper point — About 798 K (525 degrees C), the temperature at which a hot object's dull red glow first becomes visible in the dark. (Ch. 9)

Duane–Hunt limit — The shortest wavelength in a tube's spectrum, λ_min = hc/(eV), reached when one electron gives all its kinetic energy to one photon. (Ch. 11)

Electric charge — A property of particles such as electrons (negative) and protons (positive) that makes them push or pull on each other through electric and magnetic fields. (Ch. 1)

Electric field (E) — The force a small test charge would feel at a point, divided by its charge (E = F/q). Measured in N/C, equivalent to V/m. (Ch. 1)

Electromagnetic (EM) wave — A self-sustaining oscillation of electric and magnetic fields that travels through space at the speed of light. It is launched by accelerating charges. (Ch. 1)

Electromagnetic spectrum — The full range of electromagnetic waves, ordered by wavelength (or frequency or photon energy), from radio waves to gamma rays. All of it travels at c and is made of photons. (Ch. 3)

Electron cloud — A picture of an orbital's probability density as shading; it fades smoothly with distance and has no hard edge. (Ch. 4)

Electron volt (eV) — The energy an electron gains crossing a potential difference of 1 volt; 1 eV = 1.602×10^{-19} J. Convenient for photon and atomic energies. (Ch. 2)

Electron wavelength — The de Broglie wavelength λ = h/p of an electron; about 0.0037 nm at 100 kV, short enough for transmission electron microscopy and electron diffraction. (Ch. 11)

Emission spectrum — Bright lines on a dark background, emitted by a hot thin gas of separate atoms. (Ch. 5)

Emissivity — A number from 0 to 1 giving how well a surface emits thermal radiation compared with a blackbody; shiny metals are about 0.02-0.05, skin and matte black paint about 0.95-0.98. (Ch. 9)

Energy band — A nearly continuous range of allowed electron energies in a solid, formed when many atoms' levels overlap and split. (Ch. 5)

Energy level — One of the allowed energies of an electron bound in an atom or molecule; for hydrogen, E_n = −13.6 eV/n². (Ch. 4)

Energy-dispersive X-ray spectroscopy (EDS) — Chemical analysis in an electron microscope that measures the characteristic X-rays emitted when the beam creates inner-shell vacancies in the sample. (Ch. 11)

Equilibrium spacing (r_0) — The distance between neighboring nuclei at which attraction and repulsion balance, typically 0.2–0.3 nm in solids; it sets the atomic radius used in crystal models. (Ch. 4)

Excited state — Any state above the ground state; an atom gets there by absorbing energy and usually leaves within nanoseconds. (Ch. 4)

Femtometer (fm) — 10^{-15} m, the natural length unit for nuclei; a proton's charge radius is about 0.84 fm. (Ch. 4)

Field — A physical quantity with a value (and usually a direction) at every point in space, such as the electric or magnetic field. (Ch. 1)

Fluorescence — Absorption of shorter-wavelength light followed within nanoseconds by emission at a longer wavelength, after some energy is lost to vibrations. (Ch. 5)

Focal length — The distance from a lens to the point where it brings parallel rays to a focus (converging lens) or from which they appear to diverge (diverging lens). (Ch. 6)

Free radical — A highly reactive molecular fragment with an unpaired electron, such as the hydroxyl radical (•OH) formed when radiation ionizes water. It causes most X-ray and gamma-ray DNA damage. (Ch. 10)

Frequency (f) — The number of wave cycles passing a point each second, in hertz (Hz = 1/s). Related to wavelength by c = λf. (Ch. 1)

Galactic cosmic rays — Atomic nuclei from outside the solar system moving near light speed: about 85% protons, 14% helium nuclei, and 1% heavier (HZE) nuclei. (Ch. 10)

Gamma ray — A high-energy photon emitted by an atomic nucleus. It is physically the same as an X-ray of the same energy; only the origin differs. (Ch. 10)

Gamma rays — The highest-energy photons, usually defined by their origin in atomic nuclei (radioactive decay, nuclear reactions) rather than by a strict wavelength cutoff. (Ch. 3)

Gray (Gy) — SI unit of absorbed dose: one joule of radiation energy deposited per kilogram of material (1 Gy = 100 rad). (Ch. 10)

Greenhouse effect — Warming of Earth's surface because gases such as CO2 and water vapor pass incoming sunlight but absorb and re-emit outgoing thermal infrared, some of it back downward. (Ch. 9)

Ground state — The lowest-energy state of an atom (n = 1 for hydrogen), where an undisturbed atom normally sits. (Ch. 4)

Gyroradius — The radius r = mv/(qB) of the circle or helix a charged particle traces in a magnetic field, using the velocity component perpendicular to the field. (Ch. 10)

Half-value layer (HVL) — The thickness of a material that cuts a photon beam's intensity in half, equal to ln 2 / μ. (Ch. 10)

Half-wave dipole — An antenna about half a wavelength long that is resonant and radiates efficiently (radiation resistance near 73 ohms). (Ch. 8)

Heat — Energy in transit from one object to another because of a temperature difference, moved by conduction, convection, or radiation. (Ch. 9)

Higher-order reflection — A Bragg reflection with n > 1. The nth-order reflection from (hkl) planes is equivalent to the first-order reflection from (nh nk nl) planes, so (220) = second order of (110). (Ch. 11)

Huygens' principle — Every point on a wavefront acts as a source of small wavelets; the next wavefront is the sum of those wavelets. It explains diffraction. (Ch. 8)

Index of refraction (n) — The ratio n = c/v of the speed of light in vacuum to the speed of wave crests in a material; it also gives the factor by which the wavelength shrinks. (Ch. 6)

Infrared (IR) — EM waves from about 700 nm to 1 mm. Emitted strongly by objects at everyday temperatures and absorbed by molecular vibrations. (Ch. 3, 9)

Intensity (I) — The power a wave delivers per unit area, in W/m². For an EM wave, I = ½cε0E0², so intensity grows as amplitude squared. (Ch. 1)

Interference pattern — The stripes of bright and dark produced when waves from two or more paths overlap; built up dot by dot even when photons arrive one at a time. (Ch. 2)

Interplanar spacing (d-spacing) — The perpendicular distance between adjacent parallel lattice planes; for cubic crystals d_hkl = a/√(h²+k²+l²). (Ch. 11)

Inverse-square law — For a point source with no absorption, intensity falls as 1/r² because the same energy spreads over a sphere of area 4πr². (Ch. 10)

Ion — An atom (or molecule) that has lost or gained electrons and so carries a net electric charge. (Ch. 4)

Ionization energy — The minimum energy needed to remove the most loosely held electron from an isolated ground-state atom; 13.6 eV for hydrogen, 3.89–24.6 eV across the elements. (Ch. 4)

Ionizing radiation — Radiation whose individual photons or particles carry enough energy (conventionally about 10 eV or more) to knock electrons out of atoms or molecules. (Ch. 10)

Ionizing threshold — The conventional boundary between non-ionizing and ionizing photons, about 10 eV (124 nm, far ultraviolet). Different standards place it between 10 and 33 eV. (Ch. 10)

Iridescence — Color that changes with viewing angle, typical of structural color in soap bubbles, oil slicks, and butterfly wings. (Ch. 7)

K-edge — The photon energy at which absorption jumps because photons become able to eject an atom's K-shell electrons (88 keV for lead, 33 keV for iodine). (Ch. 11)

Kirchhoff's law of thermal radiation — At each wavelength a surface's emissivity equals its absorptivity: good absorbers are good emitters, and good reflectors are poor emitters. (Ch. 9)

Laser — Light amplification by stimulated emission of radiation: a pumped material with a population inversion inside a mirror cavity, giving identical, in-step photons. (Ch. 5)

Light-emitting diode (LED) — A semiconductor device in which electrons falling across the band gap emit photons of energy approximately E_g. (Ch. 5)

Linear attenuation coefficient — The fraction of photons removed per unit thickness of a material (μ, in cm^{-1}), so that intensity falls as $I = I_0 e^{(-\mu x)}$. It equals the mass attenuation coefficient times density. (Ch. 10)

Lorentz force — The magnetic force F = qv × B on a moving charge. It is perpendicular to the velocity, so it bends the path without changing the speed. (Ch. 10)

Magnetic field (B) — The field created by moving charges (currents) that pushes on other moving charges. Measured in teslas (T). (Ch. 1)

Magnetosphere — The region around a planet controlled by its magnetic field, bounded by the magnetopause, which deflects most solar-wind particles around the planet. (Ch. 10)

Malus's law — For polarized light hitting a polarizer whose axis is at angle θ to the light's polarization, the transmitted intensity is I = I0 cos²θ. (Ch. 1)

Metamers — Different spectra that produce the same three cone signals and therefore look the same color, such as 580 nm yellow and a red-plus-green mixture. (Ch. 7)

Microbolometer — A thermal-camera pixel that warms slightly when infrared lands on it, changing its electrical resistance; most such cameras work in the 8-14 micrometer band. (Ch. 9)

Microwaves — EM waves from about 1 mm to 30 cm. Used for Wi-Fi, radar and microwave ovens; they can set polar molecules such as water rotating. (Ch. 3)

Mie scattering — Strong, complicated scattering by objects about the same size as the wavelength, such as haze, smoke, and milk droplets in visible light. (Ch. 8)

Miller indices — Integers (hkl) naming a crystal plane: the reciprocals of its axis intercepts in units of the lattice parameter, cleared of fractions. [uvw] names a direction. (Ch. 11)

Multilayer insulation (MLI) — Spacecraft blankets made of many thin aluminized plastic layers separated by vacuum, which block heat flow by radiation. (Ch. 9)

Neutron radiation — Free neutrons, which carry no charge, pass easily through heavy materials, are slowed most effectively by collisions with hydrogen nuclei, and can make materials radioactive. (Ch. 10)

Non-ionizing radiation — EM radiation whose photons carry too little energy (below roughly 10 eV) to knock electrons out of atoms: radio, microwave, infrared, visible and most UV. (Ch. 3)

Nucleus — The central core of an atom, made of protons and neutrons; only femtometers across, it holds more than 99.9% of the atom's mass. (Ch. 4)

Numerical aperture (NA) — A measure of how wide a cone of light a microscope objective collects; the best oil-immersion objectives reach about 1.4 to 1.5. (Ch. 8)

Optical fiber — A thin glass core surrounded by lower-index cladding that guides light over long distances by repeated total internal reflection. (Ch. 6)

Orbital — A three-dimensional standing-wave pattern of an electron in an atom; its square gives the probability of finding the electron at each point. (Ch. 4)

Order (of a fringe) — The whole number m in d sin(theta) = m lambda that counts how many wavelengths of path difference produce a given bright fringe. (Ch. 8)

Path difference — The extra distance one wave travels compared with another to reach the same point; whole wavelengths give bright spots, odd half-wavelengths give dark ones. (Ch. 8)

Pauli exclusion principle — No two electrons can occupy the same quantum state; it forces overlapping electron clouds into higher energies and helps make matter solid. (Ch. 4)

Period (T) — The time for one full oscillation at a fixed point; T = 1/f. (Ch. 1)

Phase — Where a wave is in its cycle (crest, trough, or in between) at a given place and time; a half-wavelength shift puts crest against trough. (Ch. 8)

Phosphor — A material that absorbs light (often blue or UV) and re-emits it at a longer wavelength; white LEDs use a yellow phosphor on a blue LED. (Ch. 7)

Phosphorescence — Slow light emission (seconds to hours) from an electron parked in a long-lived 'stuck' state, as in glow-in-the-dark materials. (Ch. 5)

Photochemistry — Chemical change caused by absorbing a photon that rearranges bonds without removing an electron. UV-B sunburn is photochemistry, not ionization. (Ch. 10)

Photoelectric absorption — Absorption of an X-ray photon by an inner-shell electron, which is ejected. Its probability rises steeply with atomic number (about Z^3–Z^4) and falls steeply with energy (about $1/E^3$), making bone and metal visible on X-rays. (Ch. 11)

Photoelectric effect — The ejection of electrons from a material when light shines on it. Electrons are emitted only if each photon's energy exceeds the work function, regardless of brightness. (Ch. 2)

Photoelectron — An electron ejected from a material by absorbing a photon. (Ch. 2)

Photomultiplier tube — A light detector that uses the photoelectric effect at a low-work-function surface, then multiplies each freed electron into a measurable pulse, sensitive enough to count single photons. (Ch. 2)

Photon — The smallest possible unit (quantum) of energy in the electromagnetic field at a given frequency; an elementary particle with no mass, no charge and no internal parts, carrying energy hf and momentum h/λ. (Ch. 2)

Photon flux / photon rate — The number of photons arriving (or emitted) per second; for a beam of power P, $\dot{N} = P/(hf)$. (Ch. 2)

Photon momentum — The momentum carried by a photon, p = h/λ = E/c, which is why light exerts a (tiny) pressure on surfaces it hits. (Ch. 2)

Planar density — Number of atoms centered on a crystal plane per unit area. For one-atom-per-lattice-point crystals it equals atoms per volume × interplanar spacing, so the widest-spaced planes are the densest. (Ch. 11)

Planck's constant (h) — The constant linking a photon's energy to its frequency, E = hf; h = 6.626 × 10^{-34} J·s = 4.136 × 10^{-15} eV·s. (Ch. 2)

Planck's law — The formula (1900) giving a blackbody's brightness at each wavelength for a given temperature; deriving it required assuming energy comes in chunks of hf. (Ch. 9)

Plane wave — An idealized wave whose crests are flat planes, a good model for light far from its source. (Ch. 1)

Plasma frequency — The frequency below which a metal's free electrons can follow an EM wave and reflect it; for most metals it lies in the ultraviolet, and above it the metal becomes transparent. (Ch. 6)

Polarization — The direction in which a wave's electric field oscillates. (Ch. 1)

Polarizer — A material that transmits only the component of the electric field along its transmission axis and absorbs the perpendicular component. (Ch. 1)

Population inversion — More atoms in an upper level than in the lower level of a transition, required for a laser; created by pumping energy in. (Ch. 5)

Powder diffraction — X-ray diffraction from a sample of many randomly oriented grains, which gives a peak at the Bragg angle of every allowed plane family; used to identify phases and measure lattice parameters. (Ch. 11)

Principal quantum number (n) — The whole number 1, 2, 3, … that labels hydrogen's energy levels and sets their energy and size. (Ch. 4)

Probability amplitude — The quantum wave that travels through an apparatus (for example, both slits); its intensity at a spot gives the chance of detecting the particle there. (Ch. 8)

Quantization — The restriction of a quantity, such as a bound electron's energy, to certain discrete values; it arises because confined waves allow only certain patterns. (Ch. 4)

Quantum — The smallest indivisible amount of some quantity; for light of frequency f, one quantum of energy is hf. (Ch. 2)

Radiation weighting factor — A multiplier w_R that converts gray to sievert: 1 for photons and electrons, 2 for protons, 20 for alpha particles and heavy ions, and 2.5–20 for neutrons depending on energy. (Ch. 10)

Radio waves — The longest-wavelength EM waves (roughly 30 cm and longer). Made and detected by oscillating currents in antennas; photon energies far too small to affect atoms. (Ch. 3)

Radiograph — An X-ray shadow image; bright regions are where few X-rays reached the detector (bone, metal), dark regions where many did. (Ch. 11)

Rayleigh criterion — Two points are just resolved when their angular separation is about 1.22 lambda/D, so one's Airy peak falls on the other's first dark ring. (Ch. 8)

Rayleigh scattering — Scattering of light by particles much smaller than the wavelength, with intensity proportional to $1/\lambda^4$; it makes the sky blue and sunsets red. (Ch. 7, 8)

Re-emission — Release of absorbed energy as a new photon, delayed by nanoseconds and sent in a random direction with random phase; unlike reflection, it follows a true absorption. (Ch. 5)

Reflection — Light sent back from a surface. Physically, it is the backward wave radiated in step by electrons that the incoming light drives at a boundary. (Ch. 6)

Resonance — Strong response of a system driven at one of its natural frequencies; atoms absorb light when its frequency equals ΔE/h. (Ch. 5)

Rotational level — A quantized energy of a whole molecule's spin, spaced about 0.0001–0.01 eV apart (microwave and far-infrared photons). (Ch. 5)

Scattering — Redirection of light into many directions by small particles, pores, boundaries, or density fluctuations; it is counted as diffuse reflection or diffuse transmission. (Ch. 6)

Selection rules (systematic absences) — Rules for which reflections have nonzero intensity: all hkl for simple cubic; h+k+l even for BCC; h, k, l all odd or all even for FCC. (Ch. 11)

Sievert (Sv) — SI unit of equivalent or effective dose: absorbed dose multiplied by weighting factors for radiation type (and, for effective dose, organ sensitivity). 1 Sv = 100 rem. (Ch. 10)

Skin depth — The thin surface layer (tens of nanometers for visible light in metals) within which a metal cancels and absorbs an incoming EM wave. (Ch. 6)

Snell's law — The rule $n_1 \sin \theta_1 = n_2 \sin \theta_2$ for how much light bends when it crosses a boundary between materials, with angles measured from the normal. (Ch. 6)

Spacecraft radiator — A panel with a high-infrared-emissivity (often white) coating that rejects waste heat to space by thermal radiation, the only cooling route available in vacuum. (Ch. 9)

Spectral line — A sharp, specific wavelength emitted or absorbed by an atom, set by one energy gap; each element has its own set. (Ch. 5)

Spectrum — The amount of light at each wavelength in a beam; most real light is a mixture of many wavelengths. (Ch. 7)

Specular reflection — Mirror-like reflection from a surface whose bumps are much smaller than the wavelength, so parallel rays stay parallel and an image survives. (Ch. 6)

Speed of light (c) — The speed of all EM waves in vacuum: exactly 299,792,458 m/s, rounded to 3.00 × 10^8 m/s in this book. (Ch. 1)

Spontaneous emission — An excited atom drops to a lower level on its own, typically within nanoseconds, creating a photon of energy equal to the gap. (Ch. 5)

Standing wave — A wave pattern confined to a region, such as a plucked string, that vibrates in place; only certain wavelengths fit the boundaries. (Ch. 4)

Stefan-Boltzmann law — The total power radiated by a surface is P = emissivity x sigma x area x T^4, so doubling the absolute temperature multiplies the power by 16. (Ch. 9)

Stimulated emission — A passing photon of exactly the right energy triggers an excited atom to emit an identical photon (same energy, direction and phase); the basis of lasers. (Ch. 5)

Stokes shift — The difference between absorbed and emitted wavelengths (or energies) in fluorescence; the missing energy becomes heat. (Ch. 5)

Stopping voltage (V_s) — The reverse voltage that just stops the fastest photoelectrons from reaching the collector; it measures their kinetic energy, KE_max = eV_s. (Ch. 2)

Structural color — Color produced by interference from structures about the size of light's wavelength (thin films, layered nanostructures) rather than by pigment absorption. (Ch. 7)

Structure factor — The sum of waves scattered by all atoms in the unit cell, F_hkl = Σ f_j exp[2πi(hx_j + ky_j + lz_j)]; reflection intensity is proportional to $|F|^2$, and F = 0 gives an absent reflection. (Ch. 11)

Subtractive mixing — Combining absorbers such as inks, paints, or filters (usually cyan, magenta, and yellow); each removes part of the spectrum, and all three together look black. (Ch. 7)

Superposition — The rule that when waves overlap, the total field at each point is the sum of the individual fields. (Ch. 8)

Temper colors — Straw, brown, purple, and blue colors on steel heated to about 200-350 degrees C, caused by thin-film interference in a growing oxide layer, not by glowing. (Ch. 9)

Temperature — A measure of how vigorous the random atomic motion is on average; at 300 K the typical thermal energy scale is k_B T, about 0.026 eV. (Ch. 9)

Thermal energy — The energy of the random jiggling motion of atoms and molecules in a material. (Ch. 9)

Thermal radiation — The continuous spectrum of electromagnetic waves emitted by every object above absolute zero because its randomly moving charges accelerate. (Ch. 9)

Thin-film interference — Reinforcement or cancellation of light reflected from the front and back surfaces of a thin layer; the result depends on thickness, index, and viewing angle. (Ch. 7, 8)

Threshold frequency (f_0) — The lowest light frequency that can eject electrons from a given material, $f_0 = \varphi/h$. The corresponding threshold wavelength is $\lambda_0 = hc/\varphi$. (Ch. 2)

Thymine dimer — A defect in which two adjacent thymine bases on one DNA strand become chemically bonded after absorbing UV light, kinking the DNA and blocking copying. (Ch. 10)

Total internal reflection — Complete (100%) reflection of light inside a higher-index material when it strikes the boundary beyond the critical angle; it guides light in optical fibers. (Ch. 6)

Translucent — Letting light through only diffusely, so objects behind cannot be seen clearly; caused by internal scattering, as in dense polycrystalline alumina. (Ch. 6)

Transmission — Light passing through a material. The transmitted wave is the incoming wave plus the forward waves re-radiated by the material's driven electrons. (Ch. 6)

Transverse wave — A wave whose oscillation is perpendicular to its direction of travel. In an EM wave, both E and B are perpendicular to the direction of travel. (Ch. 1)

Ultraviolet (UV) — EM waves from about 10 nm to 400 nm. Energetic enough to break chemical bonds (sunburn); the shortest UV can ionize atoms. (Ch. 3)

Unpolarized light — Light whose electric field direction is a random, rapidly changing mix of all directions perpendicular to the beam, such as sunlight or bulb light. (Ch. 1)

Valence band / conduction band — The highest filled band and the next, normally empty, band above it in a semiconductor or insulator. (Ch. 5)

Van Allen belts — Two doughnut-shaped zones of charged particles trapped by Earth's magnetic field: an inner belt dominated by protons and an outer belt dominated by electrons. (Ch. 10)

Vibrational level — A quantized energy of a molecule's stretching or bending bonds, spaced about 0.05–0.5 eV apart (infrared photons). (Ch. 5)

Visible light — The narrow band from about 400 nm (violet) to 700 nm (red), photon energies about 1.8 to 3.1 eV, that human eyes detect. (Ch. 3)

Wavefunction — The mathematical wave that describes a quantum particle; its square is the probability density. (Ch. 4)

Wavelength (λ) — The distance between neighboring crests of a wave, in meters. (Ch. 1)

Wave–particle duality — The fact that light (and matter) shows wave behavior in how it travels and particle behavior in how it is absorbed or emitted. (Ch. 2)

Wien's displacement law — The peak wavelength of a blackbody's spectrum is 2898 micrometer-kelvins divided by its temperature, so hotter objects peak at shorter wavelengths. (Ch. 9)

Work function (φ) — The minimum energy needed to remove an electron from the surface of a material; about 2–6 eV for metals. It depends on the material, its surface condition and the exposed crystal face. (Ch. 2)

X-ray tube — A vacuum tube in which electrons from a heated cathode are accelerated by a high voltage into a metal anode (target), producing X-rays (about 1% of the energy) and heat (about 99%). (Ch. 11)

X-rays — EM waves from about 0.01 nm to 10 nm, photon energies of roughly 100 eV to 100 keV. Pass through soft tissue and diffract from crystal planes. (Ch. 3)

Answer Key

Full solutions to every Check Yourself question. Try each question before you look.

Chapter 1 — Electromagnetic Waves: Ripples in Fields

1. The electric and magnetic fields oscillate. At every point along the beam, the strength and direction of $\vec{E}$ (and $\vec{B}$ with it) swing back and forth at the wave's frequency. Nothing material travels along the beam: no air, no particles of "stuff." What travels is the pattern of field oscillation, together with the energy and momentum it carries. If you placed a charge in the beam, you would see it shoved back and forth; that shove is the wave.
2. A change in a charge's field spreads outward at speed c. A charge at rest has a static field, and a charge coasting at constant velocity carries a smoothly moving field with it, so no kink ever forms and nothing is radiated. When a charge changes its velocity (speeds up, slows down or reverses), the field near it updates to the new motion while the field farther out still reflects the old motion. The kink connecting the two regions travels outward as radiation. A charge oscillating back and forth is always accelerating, so it launches a continuous train of kinks: an EM wave.
3. Sound is a pressure wave carried by collisions between molecules, so it needs a material medium. The space between the Sun and Earth is nearly empty, so there is nothing to carry sound. Light is an oscillation of the electric and magnetic fields, which exist in empty space, and each changing field regenerates the other. So light crosses the vacuum and sound does not.
4. $\lambda = c/f = (3.00 \times 10^8\ \text{m/s})/(5.0 \times 10^9\ \text{Hz}) = 0.060\ \text{m} = 6.0\ \text{cm}$. That is about half the 12.5 cm wavelength of 2.4 GHz Wi-Fi, since the frequency is about twice as high ($\lambda \propto 1/f$).
5. $f = c/\lambda = (3.00 \times 10^8)/(650 \times 10^{-9}) = 4.62 \times 10^{14}\ \text{Hz}$. The period is $T = 1/f = 2.17 \times 10^{-15}$ s, about 2.2 femtoseconds. Red light has a lower frequency (longer period) than the green light of Worked Example 1.1 because its wavelength is longer.
6. $t = d/c = (1.50 \times 10^{11}\ \text{m})/(3.00 \times 10^8\ \text{m/s}) = 500\ \text{s} \approx 8.3\ \text{minutes}$. You always see the Sun as it was about 8 minutes ago.
7. (a) Intensity is proportional to amplitude squared, $I \propto E_0^2$. Tripling E_0 multiplies I by $3^2 = 9$. (b) If I falls to $\frac{1}{4}$, then E_0^2 falls to $\frac{1}{4}$, so E_0 falls to $\sqrt{1/4} = \frac{1}{2}$: the amplitude halves.
8. Solve $I = \frac{1}{2}c\varepsilon_0 E_0^2$ for E_0:

$$E_0 = \sqrt{\frac{2I}{c\varepsilon_0}} = \sqrt{\frac{2(1361)}{(3.00 \times 10^8)(8.85 \times 10^{-12})}}\ \text{V/m} = 1.01 \times 10^3\ \text{V/m}$$

   Then $B_0 = E_0/c = (1.01 \times 10^3)/(3.00 \times 10^8) = 3.4 \times 10^{-6}\ \text{T} = 3.4\ \mu\text{T}$. These are a bit larger than the ground-level values in Worked Example 1.2 because the atmosphere absorbs and scatters part of the sunlight.
9. The wavelength is $\lambda = c/f = (3.00 \times 10^8)/(1.00 \times 10^8) = 3.00\ \text{m}$. A quarter wavelength is $3.00/4 = 0.75\ \text{m}$, so the antenna is about 75 cm long. Electrons sloshing along a rod of this length resonate with the wave, which is why it picks up the signal efficiently.
10. The first polarizer passes half of unpolarized light: $I_1 = 800/2 = 400\ \text{W/m}^2$, now polarized along the first axis. The second polarizer is at 45° to that polarization, so by Malus's law $I_2 = I_1 \cos^2 45^\circ = 400 \times 0.50 = 200\ \text{W/m}^2$.
11. Sunlight reflecting off the lake is mostly polarized horizontally (parallel to the water), because electrons in the water that are shaken parallel to the surface radiate strongly into the reflected direction. Polarized sunglasses have a vertical transmission axis, so they absorb most of that horizontally polarized glare while still passing roughly half of the unpolarized light from everything else, which lets you see into the water. If you tilt your head 90°, the lens axis becomes horizontal. Now it passes the glare and blocks the vertical component instead, so the glare returns.

Chapter 2 — Photons: Light as Packets of Energy

1. An electron absorbs light one photon at a time, and the photon's energy hf depends only on the light's frequency. A red photon (about 1.9 eV) has less energy than sodium's work function (2.28 eV), so it cannot free an electron, and brighter red light just supplies more photons that each fall short. An electron also can't pool energy from several red photons, because it loses any extra energy to the metal almost immediately. A violet photon (3.1 eV) exceeds the work function, so even a few violet photons each free an electron at once.
2. Red: $E = 1240/650 = 1.91\ \text{eV} = 1.91 \times 1.602 \times 10^{-19} = 3.06 \times 10^{-19}\ \text{J}$. Blue: $E = 1240/450 = 2.76\ \text{eV} = 4.42 \times 10^{-19}\ \text{J}$. Shorter wavelength means more energy per photon.
3. (a) $\lambda_0 = 1240/2.1 = 590\ \text{nm}$ (yellow-orange). (b) A 650 nm photon carries $1240/650 = 1.91\ \text{eV} < 2.1\ \text{eV}$, so no electrons are ejected. Equivalently, 650 nm is longer than the 590 nm threshold. (c) A 532 nm photon carries 2.33 eV, which exceeds 2.1 eV, so electrons are ejected with $KE_{\text{max}} = 2.33 - 2.1 = 0.23\ \text{eV}$.

4. Photon energy $E = 1240/254 = 4.88$ eV. Then $KE_{\max} = 4.88 - 4.7 = 0.18$ eV, and the stopping voltage is $V_s = KE_{\max}/e = 0.18$ V. Copper responds only to ultraviolet, and even this energetic UV gives slow electrons.
5. (a) Doubling the intensity at fixed frequency doubles the number of photons per second, so the number of electrons ejected per second (the current) doubles. (b) Each electron still absorbs one photon of the same energy hf, so $KE_{\max} = hf - \phi$ is unchanged.
6. Photon energy: $E = 1240/550 = 2.25$ eV $= 3.61 \times 10^{-19}$ J. Photon rate: $\dot{N} = P/E = 2.5/(3.61 \times 10^{-19}) = 6.9 \times 10^{18}$ photons per second, roughly 7×10^{18}. (The 2.5 W figure is an assumed round number; real bulbs vary, but the answer is in the 10^{18}–10^{19} range.)
7. One green photon: 2.33 eV. One Wi-Fi photon: $hf = (4.136 \times 10^{-15})(2.4 \times 10^{9}) = 9.9 \times 10^{-6}$ eV. Ratio: $2.33/(9.9 \times 10^{-6}) \approx 2.4 \times 10^{5}$, so about 240,000 Wi-Fi photons. They can't do what one green photon does because molecules and electrons absorb photons one at a time. Each Wi-Fi photon has far too little energy to lift an electron to a higher level, so the energy can't be collected into a single electron. At most, Wi-Fi photons make molecules jiggle slightly, which is heating.
8. A photon has zero (rest) mass and no charge. It can never be at rest: in vacuum it always travels at *c*, and it still carries energy $E = hf$ and momentum $p = h/\lambda$. It is not made of anything smaller; as far as all experiments show, it is an elementary particle, the smallest unit (quantum) of energy in the electromagnetic field at its frequency.
9. (a) $p = h/\lambda = (6.626 \times 10^{-34})/(532 \times 10^{-9}) = 1.25 \times 10^{-27}$ kg · m/s. (b) $F = P/c = (5.0 \times 10^{-3})/(3.00 \times 10^{8}) = 1.7 \times 10^{-11}$ N. Check: 1.3×10^{16} photons/s times 1.25×10^{-27} kg·m/s each gives the same force. It is far too small to feel, which is why light pressure matters only for things like solar sails.
10. $\lambda_{\max} = 1240/1.1 \approx 1130$ nm, in the near infrared. Sunlight with wavelengths longer than about 1130 nm, which is a substantial part of the Sun's infrared, carries too little energy per photon to free electrons and is not converted. (Photons well above 1.1 eV, like blue light, are also partly wasted: their extra energy becomes heat.)
11. The photon's wave, which passes through both slits and interferes with itself, determines the probability of detection: where the wave's intensity (amplitude squared) is large, the photon is likely to land; where it is zero, it never lands. With only 20 photons you see 20 random samples from that probability pattern, too few to reveal its shape, just as 20 coin flips can't reliably show a 50/50 split. With thousands of photons the stripes become obvious.

Chapter 3 — The Electromagnetic Spectrum

1. In common: both are electromagnetic waves, oscillating E and B fields perpendicular to each other and to the direction of travel, moving at c in vacuum and delivered as photons of energy hf. Different: the wavelength (radio waves are longer than about 30 cm; gamma rays are shorter than about 0.01 nm), and therefore the frequency and photon energy (micro-electron-volts versus MeV). That in turn changes how they are made (electrons sloshing in antennas versus nuclei rearranging) and what they do to matter (driving currents and gentle heating versus ionizing atoms).
2. $\lambda = c/f = (3.00 \times 10^{8})/(1.00 \times 10^{6}) = 300$ m, about three football fields. $E = hf = (4.136 \times 10^{-15}$ eV · s$)(1.00 \times 10^{6}$ Hz$) = 4.1 \times 10^{-9}$ eV.
3. $f = c/\lambda = (3.00 \times 10^{8})/(940 \times 10^{-9}) = 3.19 \times 10^{14}$ Hz, and $E = 1240/940 = 1.32$ eV. It is near infrared, just beyond the red end of the visible (700 nm, 1.8 eV).
4. $E = 1240/13.5 = 91.9$ eV, about 92 eV. That is far above the ~10 eV ionization threshold, so yes, this extreme-UV light is ionizing. It sits near the boundary between UV and soft X-rays, and it is absorbed by air, which is why EUV lithography machines operate in vacuum.
5. $\lambda = 1240$ eV · nm$/(1.17 \times 10^{6}$ eV$) = 1.06 \times 10^{-3}$ nm $= 1.06$ pm. An atom (~0.1 nm = 100 pm) is about 100 times larger. The wavelength is closer to the size of the nucleus that emitted it.
6. Absorption happens one photon at a time. Infrared photons carry less than 1.8 eV (typically much less), not enough to break the chemical bonds in DNA and proteins; absorbed IR just makes molecules vibrate faster, which is heat. A very intense heat lamp can still burn you by heating, but that takes a lot of total energy. UV photons carry 3–4 eV or more, enough to break or rearrange bonds directly in skin-cell DNA. That photochemical damage, not heat, is sunburn, and it takes far less total energy.
7. X-rays are absorbed high in the atmosphere, so essentially none reach the ground; X-ray telescopes must be on satellites. Ground-based telescopes can use the two main atmospheric windows: visible light (with near-infrared windows) and radio waves from about 1 cm to about 30 m.
8. $E = 1240/0.1542 = 8.04 \times 10^{3}$ eV $= 8.04$ keV. To produce diffraction from planes of atoms, the wavelength must be comparable to the plane spacing; Bragg's law $n\lambda = 2d \sin\theta$ (Chapter 11) has no solution unless $\lambda \leq 2d$. For BCC iron, spacings are a few tenths of a nanometer, similar to 0.154 nm. Green light at 530 nm is about 1800 times longer than the 0.2866 nm lattice parameter, so it cannot diffract from atomic planes and cannot reveal the crystal structure.

9. Any three of: (1) Sunlight is plentiful in 400–700 nm and the atmosphere is transparent there. (2) Water, which fills the eye and in which eyes first evolved, is transparent in this band but absorbs UV and IR. (3) Photons of 1.8–3.1 eV can reliably flip the shape of the retinal molecule, the first step of vision, without breaking other bonds. (4) UV photons damage molecules, so the cornea and lens filter UV out. (5) IR photons are too weak for a reliable molecular switch; a pigment that sensitive would be triggered by thermal jostling at body temperature, and the body itself glows in the infrared.
10. Frequency ratio = wavelength ratio $= 30\ \text{m}/(0.03 \times 10^{-9}\ \text{m}) = 10^{12}$. Octaves $= \log_2(10^{12}) = 12 \times 3.32 \approx 40$. Visible light spans a frequency ratio of $700/400 = 1.75$, which is $\log_2 1.75 = 0.81$ octave, less than one octave.
11. The remote's photons are 940 nm, about 1.32 eV. That is below the energy range your retina's pigments respond to (roughly 1.8–3.1 eV, 400–700 nm), so you see nothing. It is above silicon's 1.1 eV band gap, so each photon can free an electron in the camera sensor, and the phone displays that signal as light (often purple-white). Many cameras include an infrared-blocking filter, so the spot may look faint, but it is usually still visible.

Chapter 4 — Inside the Atom: Scale, Orbitals, and Quantized Energy

1. An atom has no surface. Its electron cloud thins out smoothly with distance and never reaches exactly zero, so any "diameter" depends on where you choose to draw the line (for example, the sphere enclosing 90% of the probability). In practice, radii are measured as half the equilibrium distance between neighboring nuclei, and that distance depends on the bonding environment. The same element therefore has a metallic radius, a covalent radius and (as an ion) an ionic radius, all slightly different.
2. Diameter $d = 2(0.143\ \text{nm}) = 0.286\ \text{nm}$. Number of atoms in 1 mm: $N = (1 \times 10^{-3}\ \text{m})/(0.286 \times 10^{-9}\ \text{m}) = 3.5 \times 10^{6}$. About 3.5 million aluminum atoms span a millimeter.
3. $R = 1.2\ \text{fm} \times 56^{1/3} = 1.2 \times 3.83 = 4.59\ \text{fm}$, so the nuclear diameter is about 9.2 fm. Ratio of diameters: $0.248 \times 10^{-9}/9.18 \times 10^{-15} \approx 27{,}000$. Scaled up: $1.5\ \text{cm} \times 27{,}000 \approx 405\ \text{m}$. The iron atom would be about 400 m across, nearly four football fields.
4. Any two of: (1) *Stability*: a charge moving in a circle is accelerating and must radiate (Chapter 1), so a planetary electron would lose energy and spiral into the nucleus almost instantly, yet atoms are stable. (2) *Discrete energies*: a planet can orbit at any distance with any energy, but atoms absorb and emit only specific photon energies, revealing a ladder of allowed levels. (3) *Shape*: hydrogen's ground state is spherically symmetric with zero orbital angular momentum, not a flat circular track; quantum mechanics gives a probability cloud, not a trajectory.
5. A string fixed at both ends can only vibrate in patterns where a whole number of half-wavelengths fits between the ends, so only certain frequencies (notes) are possible. A bound electron is a wave confined by the electric attraction of the nucleus, so it too can exist only in standing-wave patterns that "fit," and each pattern has a definite energy. A free electron is not confined; any wavelength is allowed, so its energy can take any value (the continuum above 0 eV in Figure 4.5).
6. $\lambda = h/(mv) = (6.626 \times 10^{-34})/[(9.11 \times 10^{-31})(1.09 \times 10^{6})] = 6.67 \times 10^{-10}\ \text{m} = 0.667\ \text{nm}$. Circumference: $2\pi(0.212\ \text{nm}) = 1.33\ \text{nm}$. Ratio: $1.33/0.667 = 2.0$. Exactly two wavelengths fit, as Equation 4.2 requires for $n = 2$.
7. $E_3 = -13.6/9 = -1.51\ \text{eV}$, so 1.51 eV ionizes it. A 2.0 eV photon has more than enough: it frees the electron, and the extra $2.0 - 1.51 = 0.49\ \text{eV}$ becomes the electron's kinetic energy. A ground-state atom needs 13.6 eV, so the same photon cannot ionize it. This is why visible light is "non-ionizing" for ordinary (ground-state) atoms.
8. Neon (21.6 eV) > magnesium (7.65 eV) > sodium (5.14 eV). Neon's outer shell ($n = 2$) is completely full and held close to the nucleus, so removing an electron is very hard. Sodium has a single 3s electron alone in a new, larger shell, screened from its nucleus by the ten inner electrons, so it leaves easily. Magnesium also has its outer electrons in the 3s shell, but it has one more proton pulling on them than sodium does, so they are held somewhat more tightly, though still far less tightly than neon's.
9. When your hand's atoms approach the table's atoms, their electron clouds start to overlap. (1) Electric repulsion: the negative clouds repel each other, and each nucleus becomes less shielded from the other nucleus. (2) The Pauli exclusion principle: two electrons cannot share the same quantum state, so squeezing filled clouds together forces electrons into higher-energy states. Together these make the energy climb steeply at separations below r_0 (Figure 4.7). The resulting repulsive force stops your hand when your atoms are still about 0.1–0.3 nm from the table's.
10. $a = 2R\sqrt{2} = 2(0.128\ \text{nm})(1.414) = 0.362\ \text{nm}$, within about 0.1% of the measured 0.3615 nm. The spheres "touch" only in the sense that nearest-neighbor nuclei sit at the equilibrium spacing $2R = 0.256\ \text{nm}$, where attraction and repulsion balance (the minimum of Figure 4.7). The electron clouds actually overlap somewhat at that distance; there are no surfaces to meet. The hard-sphere model works because the repulsive wall is so steep that atoms behave almost like rigid balls of radius R.

Chapter 5 — Absorption and Emission: Electrons Jumping Up and Down

1. Absorption is all or nothing: the photon must vanish completely and leave the electron on an allowed level. From $n = 1$ the nearest gaps are 10.20 eV (to $n = 2$) and 12.09 eV (to $n = 3$). An 11.0 eV photon matches neither, and an isolated atom cannot take 10.2 eV and hand back 0.8 eV, so the photon passes through. A 14.0 eV photon *is* absorbed: it exceeds the 13.6 eV ionization energy, and above that

point the freed electron can have any energy. The electron leaves with $14.0 - 13.6 = 0.4$ eV of kinetic energy.

2. $\Delta E = 13.6\text{ eV}\,(1/2^2 - 1/4^2) = 13.6 \times 3/16 = 2.55$ eV. $\lambda = 1240/2.55 = 486$ nm: blue-green (cyan), the H-beta line.
3. $\Delta E = 13.6\text{ eV}\,(1 - 1/4) = 10.2$ eV, so $\lambda = 1240/10.2 = 122$ nm (measured 121.6 nm). That is far ultraviolet, well outside the 400–700 nm your eyes detect. It also never reaches the ground: air (especially oxygen) absorbs such short-wavelength UV high in the atmosphere, so astronomers observe it from space.
4. Re-emission sends photons in random directions. Of the light the gas removed from the beam, only a tiny fraction happens to be re-emitted along the original line toward you, so the beam you see is depleted at exactly the gas's wavelengths: dark lines. The missing light has not vanished; it went sideways (and some was lost to collisions as heat). Viewed from the side, the same gas glows at those wavelengths (Figure 5.4C).
5. Any three of: (1) *Direction*: reflection follows the law of reflection (one beam at the mirror angle); re-emission goes in random directions. (2) *Timing*: reflection is effectively instantaneous; re-emission follows absorption by nanoseconds. (3) *Phase*: reflected light stays in step with the incoming wave, which is why a mirror forms an image; re-emitted photons have random phases. (4) *Absorption*: reflection involves no absorption at all, while re-emission happens only after a real, resonant absorption. (5) *Color*: reflected light keeps its wavelength; re-emitted light can be at a longer wavelength (fluorescence).
6. $E_{\text{UV}} = 1240/254 = 4.88$ eV; $E_{\text{green}} = 1240/545 = 2.28$ eV. Lost per photon: $4.88 - 2.28 = 2.60$ eV, or $2.60/4.88 = 53\%$. Over half of the UV energy becomes heat in the phosphor, a built-in efficiency limit of fluorescent lamps.
7. $E = 1240\text{ eV} \cdot \text{nm}/15{,}000\text{ nm} = 0.083$ eV, only about $0.083/0.026 \approx 3$ times $k_B T$ at room temperature. Photons this weak are exactly what room-temperature objects (including Earth's surface) emit in large numbers (Chapter 9), which is why CO_2 intercepts outgoing heat radiation.
8. A molecule absorbs infrared only if the vibration makes its charge distribution oscillate (a changing dipole), because that oscillating charge is what couples to the wave's electric field. CO_2's bending and asymmetric-stretch vibrations shift charge from side to side, and bent, polar H_2O changes its dipole in every vibration. N_2 and O_2 are made of two identical atoms; stretching them keeps the charge perfectly symmetric, so there is no oscillating dipole and no infrared absorption.
9. $\lambda \approx 1240/2.26 = 549$ nm: green to yellow-green. Each electron must be given at least E_g, so the forward voltage must be at least about 2.3 V. (In practice GaP's gap is indirect, so pure GaP LEDs are dim; early green indicator LEDs used doped GaP.)
10. Visible photons carry 1.8–3.1 eV. Sapphire's gap (about 8.8 eV) is far larger, so no visible photon can lift an electron across it: visible light passes straight through, and the crystal is transparent. Silicon's gap (1.12 eV) is smaller than *every* visible photon energy, so all visible light can be absorbed: a wafer is opaque (its gray, mirror-like look is the light it reflects). Silicon also conducts far better: at room temperature thermal energy lifts some electrons across a 1.12 eV gap, but essentially none across 8.8 eV, which is why alumina is used as an electrical insulator.
11. In thermal equilibrium the lower level always holds more atoms than the upper one (for a 2 eV gap at room temperature, the upper level is essentially empty). A passing photon is then far more likely to be absorbed by a lower-level atom than to stimulate emission from an upper-level atom, so light is absorbed rather than amplified. A population inversion means more atoms in the upper level than the lower. It is created by pumping energy in (a flash lamp, an electric current, or another laser), usually into a level that drops quickly into a long-lived metastable upper level, where atoms pile up faster than they leave.

Chapter 6 — Reflection, Transmission, Absorption, and Refraction

1. Energy is conserved, so $R + A + T = 1$ (Equation 6.1). With $R = 0.06$ and $A = 0.22$: $T = 1 - 0.06 - 0.22 = 0.72$. The sheet transmits **72%** of the light.
2. A marble has only two surfaces in the beam's path, each reflecting about 4%, so most light goes straight through and you see clearly through it. Powder has millions of tiny surfaces at random angles. At each one, about 4% reflects and the rest refracts into a new direction. After many such encounters the light's direction is randomized and much of it comes back out diffusely. Glass absorbs no visible color, so all colors are returned equally: the powder looks **white**. Snow vs. ice and frosted vs. clear glass work the same way.
3. The highlighter's dye **absorbs** UV or blue photons into real excited states, waits a few nanoseconds, and re-emits at a longer wavelength in a random direction with a random phase (fluorescence). Waves from different molecules have no fixed phase relationship, so they cannot add up in any particular direction: you get a glow, not an image. The mirror's electrons are **driven** by the wave and re-radiate immediately at the same frequency, phase-locked to it. Their wavelets reinforce only in the direction $\theta_r = \theta_i$, so the arrangement of the incoming rays (the image) is preserved.
4. Equation 6.4 with $n_1 = 1.00$: diamond $R = \left(\frac{2.42-1}{2.42+1}\right)^2 = (0.415)^2 = 0.172$, about **17%**. Glass: $R = (0.52/2.52)^2 = 0.043$, about 4%. Diamond's surface reflects roughly four times as much light as glass, which (together with its small critical angle) gives it its brilliance.

5. Speed: $v = c/n = (3.00 \times 10^8)/1.333 = \mathbf{2.25 \times 10^8}$ **m/s**. Wavelength: $\lambda = \lambda_0/n = 589/1.333 = \mathbf{442}$ **nm**. Frequency is unchanged: $f = c/\lambda_0 = (3.00 \times 10^8)/(589 \times 10^{-9}) = \mathbf{5.09 \times 10^{14}}$ **Hz**. The diver sees the same yellow. Color depends on photon energy hf, and the frequency never changes when light crosses a boundary (once inside the diver's eye, the wavelength is set by the eye's fluids anyway).
6. Snell's law: $\sin\theta_2 = \frac{1.00 \times \sin 60.0^\circ}{1.333} = \frac{0.866}{1.333} = 0.650$, so $\theta_2 = \mathbf{40.5}^\circ$ from the normal. The ray bends toward the normal because it enters a higher-index medium.
7. $\sin\theta_c = n_2/n_1 = 1.000/1.333 = 0.750$, so $\theta_c = \mathbf{48.6}^\circ$. Light from the entire sky (every direction above the surface) is squeezed into a cone of half-angle 48.6° (about 97° across) overhead, called Snell's window. Looking up at angles outside that cone, the fish sees the underside of the surface acting as a perfect mirror (total internal reflection), reflecting the bottom and underwater scene.
8. A metal's free electrons respond to every frequency below the plasma frequency (in the UV for most metals). Their re-radiation cancels the wave within a skin depth of tens of nanometers, reflecting 90–95% and absorbing the rest, so a thick piece transmits nothing. Gold leaf (about 0.1 μm) is about as thin as that absorption depth, so a little light, greenish for gold, leaks through. X-ray frequencies are far above the plasma frequency: the electrons cannot keep up, the metal stops acting like a mirror, and thin foils transmit X-rays.
9. Sunlight's visible and near-infrared light passes through the glass (Table 6.1) and is absorbed by seats and the dashboard, becoming heat. The warm interior radiates mainly thermal infrared near 10 μm (Chapter 9), which glass absorbs (Si–O vibrations), so that energy cannot simply radiate back out through the windows. The closed car also traps the warm air. Energy enters faster than it leaves until the interior is much hotter than outside.
10. The cornea's focusing comes from refraction at its front surface, and how strongly a surface bends light depends on the index jump. In air the jump is $1.376 - 1.000 = 0.376$; in water it is only $1.376 - 1.333 = 0.043$, about 11% as much. The cornea loses most of its roughly 43 diopters, far more than the lens can make up, so images focus well behind the retina and look blurry. Goggles restore an air layer in front of the cornea.
11. Sapphire is a single crystal: one uniform index throughout and no internal boundaries, so light passes straight through. A spark-plug insulator is polycrystalline alumina with pores. Al_2O_3 is non-cubic and birefringent, so randomly oriented grains present index mismatches at every grain boundary, and pores (n from 1.77 to 1.00) reflect and redirect light strongly. Multiple scattering sends light back out diffusely: opaque white. Similarly, HDPE is semicrystalline: its denser crystalline regions have a different index from the amorphous regions, and the many interfaces scatter light (cloudy). PMMA is amorphous with a uniform index, so it is clear.
12. Thin-lens equation: $\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{0.0500\,\text{m}} - \frac{1}{4.00\,\text{m}} = 20.00 - 0.25 = 19.75\,\text{m}^{-1}$, so $d_i = 0.0506\,\text{m} = \mathbf{50.6}$ **mm**, just beyond the focal length. Magnification: $m = -d_i/d_o = -0.0506/4.00 = -0.0127$. Image height: $1.80\,\text{m} \times 0.0127 = 0.0228\,\text{m} \approx \mathbf{23}$ **mm**, inverted, which fits on a 24 mm tall sensor. Focusing on this person means moving the lens about 0.6 mm away from its infinity position.

Chapter 7 — Color

1. It looks nearly **black**. An apple's color is leftover light: its skin absorbs violet through yellow and reflects mainly red (Figure 7.3). Blue light contains no red for it to reflect, and the small fraction of blue it does reflect (a few percent) makes it only very dark, not blue.
2. Only the **red and green** subpixels are lit, so the screen emits red light (around 610–630 nm) and green light (around 530 nm), with essentially no 580 nm light. That mixture excites your L cones strongly and your M cones moderately, the same ratio that 580 nm yellow light produces (Figure 7.1). Your brain receives the same three cone signals, so it reports yellow. The two spectra are metamers.
3. $\lambda_{\text{edge}} = 1240/2.7 = \mathbf{459}$ **nm**. ZnSe absorbs violet and the bluest part of the blue (below 459 nm) and returns everything from blue-green to red. White light minus violet and deep blue looks **yellow** (pale yellow), matching Figure 7.4.
4. Diamond: $\lambda_{\text{edge}} = 1240/5.5 = 225$ nm, in the UV. $E_g > 3.1$ eV, so no visible photon is absorbed: **transparent and colorless**. Silicon: $\lambda_{\text{edge}} = 1240/1.1 = 1130$ nm, in the infrared. $E_g < 1.8$ eV, so every visible photon is absorbed: **opaque**. Silicon still looks shiny gray because its high index ($n \approx 4$) makes its surface reflect $R = (3/5)^2 \approx 0.36$ of the light, roughly equally at all visible wavelengths. Equal reflection of all colors at roughly 35% looks like a neutral, metallic gray. (Silicon is transparent to infrared beyond about 1.1 μm.)
5. $\dfrac{I_{400}}{I_{650}} = \left(\dfrac{650}{400}\right)^4 = (1.625)^4 = 2.64^2 = \mathbf{6.97}$, so violet is scattered about **7 times** more strongly than red.
6. Air molecules are far smaller than visible wavelengths, so they follow Rayleigh's $1/\lambda^4$ law and scatter mostly blue: the clear sky is blue. Cloud droplets (10–20 μm) are much larger than visible wavelengths and scatter all colors about equally; light also bounces among many droplets, so clouds return **white** light. Storm clouds are very thick: most light is scattered back out of the top or lost over many bounces, so little reaches the bottom, which looks **gray** from below.
7. Pure Al_2O_3 has a band gap of about 9 eV, far above 3.1 eV, so no visible light is absorbed: sapphire is colorless. In ruby, Cr^{3+} substitutes for up to about 1% of the Al^{3+} ions. The crystal field of the surrounding O^{2-} ions splits chromium's 3d levels into gaps of about 2–3 eV, so ruby absorbs violet (near 410 nm) and yellow-green (near 550 nm) and transmits **red**. In blue sapphire, neighboring Fe^{2+} and Ti^{4+} ions transfer an electron when a photon supplies about 2 eV, absorbing yellow and orange light and leaving **blue**. A fraction of a percent of the right impurity sets the color.

8. Think of white light as red + green + blue. **Cyan** ink absorbs red and passes green and blue. **Yellow** ink absorbs blue and passes red and green. Light that passes through both layers (reflecting off the paper beneath) must survive both: red is removed by cyan and blue by yellow, so only **green** remains.
9. Both metals' free electrons reflect visible light strongly. The difference is where deeper d-band electrons begin to absorb. In gold, transitions from the filled 5d band into the conduction band start near 2.4 eV (about 520 nm), so gold absorbs **violet and much of the blue** while reflecting red, orange, yellow, and green: it looks yellow. In silver the corresponding transitions need about 4 eV (about 320 nm, ultraviolet), so silver reflects all visible colors equally and looks white or colorless. (Relativistic effects in heavy gold are why its d band sits so much closer; see the Rabbit Hole in Section 7.7.)
10. Photon energy $\approx E_g$: $E_g = 1240/630 = \mathbf{1.97\ eV}$, about 2.0 eV. For GaN, $\lambda = 1240/3.4 = \mathbf{365\ nm}$, in the near ultraviolet, so a GaN LED emits invisible UV light. Blue LEDs alloy indium into GaN to lower the gap to about 2.7 eV.
11. Absorbed power = absorbed fraction × intensity × area. Black: $0.90 \times 1000 \times 0.30 = \mathbf{270\ W}$. White: $0.30 \times 1000 \times 0.30 = \mathbf{90\ W}$. The black shirt absorbs an extra 180 W, nearly twice your body's resting heat output (about 100 W), which is why it feels so much hotter.
12. Light reflects from both the front and back surfaces of the film, and the back-surface wave travels an extra distance of about $2nt$. Wavelengths whose crests line up are reinforced and others cancel. The film's thickness varies from place to place and changes as it drains and swirls, so the reinforced color varies: swirling bands. Just before popping, the film thins to a few tens of nanometers, much less than any visible wavelength, so the extra path is negligible. The front reflection (air into soapy water, higher n) flips the wave's phase, but the back reflection does not, so the two reflected waves are half a cycle out of step for **every** color and cancel. The film reflects almost nothing and looks black.

Chapter 8 — Interference and Diffraction (and Why Size vs. Wavelength Matters)

1. Near-silence means destructive interference: the two sound waves arrive half a cycle out of step. So the distances from that spot to the two speakers differ by an odd number of half wavelengths, $\Delta = (m + \tfrac{1}{2})\lambda$: $\lambda/2$, $3\lambda/2$, and so on. (It is rarely perfectly silent because reflections from the walls add other paths.)
2. Use Eq. 8.3: $\Delta y = \lambda L/d = (532 \times 10^{-9}\,\text{m})(3.0\,\text{m})/(0.25 \times 10^{-3}\,\text{m}) = 6.4 \times 10^{-3}$ m. The bright fringes are about **6.4 mm apart**. Check: the angle is tiny ($\lambda/d \approx 0.002$), so the small-angle form is fine.
3. First order: $\sin\theta = \lambda/d$. For 400 nm, $\sin\theta = 400/740 = 0.541$, so $\theta = 32.7°$. For 650 nm, $\sin\theta = 650/740 = 0.878$, so $\theta = 61.4°$. Second order would need $\sin\theta = 2\lambda/d$; even for violet, $2(400)/740 = 1.08 > 1$, which no angle satisfies. With the DVD's finer track spacing, the second-order beams simply don't exist for visible light, so you only see one steep rainbow.
4. Diffraction depends on wavelength compared with the opening. Voices have wavelengths from about 0.3 m to 3 m (1000 Hz to 100 Hz), comparable to a 0.9 m doorway, so λ/a is near 1 and sound spreads widely into the next room (wall reflections help too). Light has $\lambda \approx 5 \times 10^{-7}$ m, so $\lambda/a \approx 6 \times 10^{-7}$. It goes essentially straight through the door, and you only see what is in line with it.
5. $K = 100$ eV $= 1.602 \times 10^{-17}$ J. Then $p = \sqrt{2mK} = \sqrt{2(9.109 \times 10^{-31})(1.602 \times 10^{-17})} = 5.40 \times 10^{-24}$ kg·m/s, and $\lambda = h/p = 6.626 \times 10^{-34}/5.40 \times 10^{-24} = 1.23 \times 10^{-10}$ m $= 0.123$ nm. That is about half the 0.25 nm atomic spacing, so $\sin\theta = \lambda/d \approx 0.49$ and the first diffracted beam comes out near 29°. **Yes**, the crystal diffracts it strongly, which is the basis of low-energy electron diffraction used to study surfaces.
6. $\theta_{\text{min}} = 1.22\lambda/D = 1.22(550 \times 10^{-9})/0.20 = 3.4 \times 10^{-6}$ rad. On the Moon, $s = (3.84 \times 10^{8}\,\text{m})(3.4 \times 10^{-6}) \approx 1.3 \times 10^{3}$ m. The telescope can resolve craters about **1.3 km** across. In practice, air turbulence often blurs backyard views to about this level or worse.
7. The Abbe limit is $d_{\text{min}} = \lambda/(2\,\text{NA}) = 450/(2 \times 1.4) \approx 160$ nm, larger than the 100 nm virus, so **she cannot resolve its shape**. At best a fluorescently labeled virus shows up as a blurry diffraction-limited spot: detected, not imaged. She should use an electron microscope. Electrons accelerated through 100 kV have $\lambda \approx 0.004$ nm, so diffraction blur is no longer the limit, and resolution reaches the nanometer scale or better.
8. Rayleigh scattering goes as $1/\lambda^4$, so the ratio is $(700/450)^4 = (1.556)^4 \approx 5.9$. Blue light is scattered about **6 times** more strongly than red, which is why the sky is blue (Chapter 7).
9. The oven's wavelength is 12.2 cm, so a 3 cm hole is about $\lambda/4$, no longer much smaller than the wavelength. The mesh would stop acting like a solid metal sheet, and a significant amount of microwave power could leak out through the holes, which would be unsafe and waste energy. The 1–2 mm holes are roughly $\lambda/100$, so almost nothing gets through. Visible light passes either way, because both hole sizes are thousands of wavelengths across.
10. For diffraction you need $\sin\theta = m\lambda/d \leq 1$, with angles big enough to measure. Visible light ($\lambda \approx 500$ nm) on planes 0.2 nm apart would need $\sin\theta = 2500$, which is impossible, so there are no diffracted beams at all and the crystal acts like a uniform medium. X-rays with $\lambda = 0.154$ nm give $\lambda/d \approx 0.77$, a large, easily measured angle (about 50° in this simple model; Bragg's law in Chapter 11 handles the reflection geometry). The wavelength must be comparable to the spacing you want to measure.

11. $d/\lambda = 10/25.984 \approx 0.38$. On Figure 8.6 that sits at the lower edge of the Mie regime (the curve gives $Q \approx 0.2$ for a water sphere), not deep in the weak Rayleigh region, so the cell interacts substantially with this light. More importantly, emission and absorption are set by resonance, not size: molecules a fraction of a nanometer across emit and absorb infrared wavelengths of many micrometers. What would be strange is a cell staying perfectly black at wavelengths far larger than itself.

Chapter 9 — Infrared, Heat, and Thermal Radiation

1. The hot coffee's molecules jiggle more vigorously than the room's, so the coffee has a higher **temperature**. Because of that temperature difference, energy flows out as **heat**, and the coffee's **thermal energy** (the total random motion of its molecules) decreases until its temperature matches the room. The heat leaves by *conduction* (through the mug into the desk and the air touching it), *convection* (warm air and vapor rising away; evaporation carries off a lot of energy too), and *radiation* (the mug and coffee surface emit infrared, more than they absorb from the cooler room, per Eq. 9.3).
2. Wien's law: $\lambda_{\text{peak}} = 2898/1500\ \mu\text{m} = 1.93\ \mu\text{m}$, in the infrared. **No**, most of its energy is not visible. Integrating Planck's law shows less than 0.1% of a 1500 K blackbody's output falls between 400 and 700 nm. The steel looks bright yellow-orange (Table 9.1) only because the eye is so sensitive; over 99.9% of what it radiates is infrared, which is why you feel the forge's heat from across the room.
3. Power goes as T^4: $(2000/1000)^4 = 16$, so it radiates **16 times** as much. The peak moves to half the wavelength: from $2898/1000 = 2.9\ \mu\text{m}$ to $2898/2000 = 1.45\ \mu\text{m}$, still infrared, but now much more of the tail spills into the visible, so it glows yellow-white instead of dull red.
4. Skin $T = 306$ K; walls $T_s = 288$ K. $P_{\text{net}} = \varepsilon\sigma A(T^4 - T_s^4) = (0.98)(5.67 \times 10^{-8})(1.8)(306^4 - 288^4) = (1.00 \times 10^{-7})(8.77 \times 10^9 - 6.88 \times 10^9) \approx 190\ \text{W}$. That is about **twice** the 99 W of Worked Example 9.2: the area is 20% larger and, more importantly, the colder walls send back much less radiation, so $T^4 - T_s^4$ is about 60% bigger. This is why a room with cold walls feels chilly even when the air is warm.
5. Window glass strongly absorbs 8–14 µm infrared, because its Si–O bonds vibrate at those frequencies. So the infrared from the street never gets through, and the camera sees only the glass, glowing at its own fairly uniform temperature. Glass is also smooth, so it reflects some of the long-wave infrared hitting it (its emissivity is about 0.9, not 1). You are warmer than the glass, so your reflected infrared stands out as a visible "thermal reflection" of yourself.
6. The **black-anodized** section reads hotter, close to the true 80 °C, because its emissivity is high, so the camera is really seeing its own emission. Polished aluminum has $\varepsilon \approx 0.05$: it emits very little and mostly reflects the cooler surroundings, so it reads near room temperature (or shows reflections of warm objects nearby). To measure it, stick on matte black tape or paint a small spot with a known high emissivity, wait for it to reach the part's temperature, and read that spot. A contact thermocouple also works.
7. Examples: (1) A black car gets hot in sunlight, mostly from absorbed visible and near-infrared light, so visible light also becomes heat when absorbed. (2) A microwave oven heats food with 12 cm microwaves, not infrared. (3) A TV remote's near-infrared beam heats nothing noticeably. Infrared is light; it only becomes heat when something absorbs it. (Also, heat moves by conduction and convection with no infrared involved at all.)
8. A molecule absorbs infrared only if its vibration makes its charge distribution oscillate (a changing dipole), so the wave's electric field has something to push on. N_2 and O_2 are symmetric two-atom molecules: stretching them shifts no charge, so they are transparent to thermal infrared. CO_2's bending and asymmetric-stretch vibrations and H_2O's vibrations and rotations do shift charge, so they absorb. The 15 µm band matters because it sits near the peak of Earth's ~10 µm glow and just outside the 8–13 µm window, so it intercepts a large share of the outgoing radiation (Figure 9.5).
9. Power per area: $\varepsilon\sigma T^4 = (0.85)(5.67 \times 10^{-8})(280)^4 = (0.85)(5.67 \times 10^{-8})(6.15 \times 10^9) \approx 296\ \text{W/m}^2$ (the 3 K sky sends back essentially nothing). Area: $A = 3000/296 \approx$ **10 m²**. A cooler radiator needs a lot more area, because power drops as T^4.
10. The **orange glow** is emitted thermal radiation. The bar is hot enough that the short-wavelength tail of its Planck spectrum is bright in the visible, which means roughly 1000–1100 °C (Table 9.1). The **blue sheen** is thin-film interference in an oxide layer, seen by reflected room light after the chisel cooled (Chapter 8). It indicates the edge was tempered at roughly 300 °C, far below the ~525 °C Draper point, so steel at that temperature gives off no visible glow at all.
11. Microwave heating needs water molecules that can twist back and forth with the field (dielectric relaxation). In ice they are locked into a crystal and can barely rotate, so ice absorbs microwaves far more weakly than liquid water. Once one spot thaws, that liquid absorbs strongly and heats even faster, so you get boiling edges around a frozen core. Defrost settings pulse the power on and off so that, during the off periods, heat can conduct from the thawed regions into the frozen ones and even things out.

Chapter 10 — Ionizing Radiation, Shielding, and Space

1. **The X-ray tube.** Ionization is decided photon by photon. An infrared photon near 1 µm has $E = 1240/1000 \approx 1.2$ eV, far below the ~10 eV needed, and an electron cannot save up energy from several photons. The heat lamp's 1000 W just means many more 1.2 eV photons, which warm you. A dental X-ray photon carries tens of keV, thousands of times the threshold, so even a weak X-ray beam ionizes.

2. $E = 1240\ \text{eV}\cdot\text{nm}/254\ \text{nm} = 4.9$ eV. That is below 10 eV, so UV-C at 254 nm is *not* ionizing. It still damages DNA, because DNA bases absorb very strongly near 260 nm and the absorbed energy rearranges bonds (thymine dimers, Section 10.3). That photochemistry is exactly why 254 nm lamps kill germs, and why you should never look at one or expose skin to it.
3. Outside the body, a typical 5 MeV alpha particle stops in a few centimeters of air or in the dead outer layer of skin, so it never reaches living cells. Inhaled (radon, for example), the emitter sits right next to living lung cells and deposits all of its energy in a very short, dense track. It makes clustered DNA damage, which is why alpha radiation has a weighting factor of 20.
4. In an elastic collision, the most energy a moving particle can hand to a target at rest is a fraction $4A/(1+A)^2$ of its kinetic energy, where A is the target-to-projectile mass ratio. For hydrogen ($A \approx 1$) this is $4/4 = 100\%$: a head-on hit can stop the neutron dead. For lead ($A \approx 207$) it is $4(207)/208^2 = 1.9\%$, so the neutron keeps at least 98% of its energy each bounce. Hundreds of bounces off lead do what a few bounces off hydrogen do.
5. Gamma: $H = w_R D = 1 \times 2.0\ \text{mGy} = 2.0$ mSv. Alpha: $H = 20 \times 0.10\ \text{mGy} = 2.0$ mSv. One-twentieth the absorbed energy from alpha particles gives the same biological dose, because their dense tracks make harder-to-repair damage.
6. Inverse-square: $\dot{H}_2 = \dot{H}_1 (r_1/r_2)^2$. At 0.5 m: $12 \times (1.5/0.5)^2 = 12 \times 9 = 108$ μSv/h. At 6.0 m: $12 \times (1.5/6.0)^2 = 12/16 = 0.75$ μSv/h. Tripling your closeness multiplies the rate by 9; quadrupling the distance divides it by 16.
7. $\mu = (\mu/\rho)\rho = 0.064 \times 2.3 = 0.147\ \text{cm}^{-1}$. HVL $= 0.693/0.147 = 4.7$ cm. For $I/I_0 = 10^{-3}$: $x = \ln(1000)/\mu = 6.91/0.147 = 47$ cm. (Check: 1/1000 is about 10 halvings, since $2^{10} = 1024$, and $10 \times 4.7 = 47$ cm.) This narrow-beam estimate ignores scattered photons, so a real shield would be made somewhat thicker.
8. $r = mv/(qB) = (9.109 \times 10^{-31})(2.0 \times 10^{7})/[(1.602 \times 10^{-19})(3.0 \times 10^{-5})] = 1.82 \times 10^{-23}/4.81 \times 10^{-24} = 3.8$ m. (Its speed is only 7% of c, so the nonrelativistic formula is fine; its kinetic energy is about 1.1 keV.) The 1 MeV proton's radius was 4.8 km, about 1300 times larger, mostly because the proton is 1836 times heavier. Electrons in Earth's field spiral in tight meter-scale helices.
9. Solar UV: **no change**. Photons carry no charge, so the magnetic field never affected them; UV reaching the ground is controlled by ozone and the rest of the atmosphere. Cosmic-ray particles: **yes**. A weaker or shifted field deflects fewer charged particles, so more reach the upper atmosphere, especially at latitudes that used to be well protected. The effect on dose is largest at high altitude (aircraft, the ISS) and smaller at the ground, because the atmosphere's ~10 m-water-equivalent of mass still absorbs most of the energy.
10. Best to worst per kilogram: **polyethylene, then aluminum, then lead**. Polyethylene (CH_2) is rich in hydrogen. Hydrogen nuclei cannot fragment, carbon makes relatively few secondaries, and hydrogen gives the most electrons per gram (highest Z/A), so it slows charged particles best per unit mass and also moderates secondary neutrons. Aluminum is light but fragments heavy nuclei and produces secondary neutrons. Lead has the lowest Z/A (about 0.4), so it has the fewest electrons per gram, and its large nuclei shatter incoming heavy ions into showers of secondaries. Its density is no help when the constraint is mass, not thickness.
11. (a) $3.1\ \text{mSv}/0.1\ \text{mSv} = 31$ chest X-rays per year of natural background. (b) $160/0.1 = 1600$ chest X-rays. These comparisons are rough: space radiation includes heavy ions and neutrons whose biology differs from X-rays, which is part of why NASA studies it separately.

Chapter 11 — X-Rays and Materials Science

1. The most energetic photon takes all of one electron's kinetic energy: $E_{\text{max}} = eV = 70$ keV. Then $\lambda_{\text{min}} = 1240\ \text{eV}\cdot\text{nm}/70{,}000\ \text{eV} = 0.0177$ nm. No photon in the beam can have a shorter wavelength, whatever the current.
2. The continuum comes from electrons braking in the target, so its upper energy limit is eV: higher voltage moves λ_{min} shorter (0.0496 nm → 0.0354 nm) and raises the whole curve. The Kα photon energy is the difference between copper's L and K energy levels, a property of the copper atom, not of the tube. The voltage only has to exceed the 8.98 kV needed to make the K vacancy; above that, more voltage makes Kα *brighter* but cannot shift it.
3. $\mu = (\mu/\rho)\rho = 30.3 \times 11.35 = 344\ \text{cm}^{-1}$. For $x = 0.025$ cm: $T = e^{-344 \times 0.025} = e^{-8.6} = 1.8 \times 10^{-4}$, about 0.02%. That is why a thin, flexible apron works so well at diagnostic energies: photoelectric absorption in high-Z lead is enormous there. Compare the 1 MeV case in Chapter 10, where lead's HVL is 0.86 cm.
4. Breast tissue consists of fat, glandular tissue, and tiny calcium deposits, all with similar low atomic numbers and densities. Contrast comes from differences in photoelectric absorption, which grows as roughly $1/E^3$. At low energies those small differences in composition produce noticeable differences in transmission (Worked Example 11.1 shows contrast collapsing from 8.6:1 to 1.2:1 between 30 and 100 keV for bone). Low kV trades a somewhat higher absorbed dose for the contrast needed to see subtle features.
5. **Right:** the geometry looks like mirror reflection. The scattered beam leaves the planes at the same angle θ that the incident beam made, and $\vec{d}$ is along the plane normal. **Wrong:** a mirror reflects at every angle, while crystal planes give a strong beam only at the discrete angles where $2d\sin\theta = n\lambda$. The beam is not a bounce off a surface. It is the constructive interference of weak wavelets scattered by electrons on many thousands of planes deep inside the crystal. Between Bragg angles, the wavelets cancel and the "reflection" is essentially zero.

6. $d_{111} = 0.4049/\sqrt{3} = 0.2338$ nm; $\sin\theta = 0.1542/(2 \times 0.2338) = 0.3298$, so $\theta = 19.26°$ and $2\theta = 38.5°$. $d_{200} = 0.4049/2 = 0.2025$ nm; $\sin\theta = 0.1542/0.4049 = 0.3808$, so $\theta = 22.39°$ and $2\theta = 44.8°$. These are the first two peaks used in Worked Example 11.4.
7. First order: $\sin\theta = \lambda/(2d_{110}) = 0.1542/0.4054 = 0.3804$, so $\theta = 22.36°$ and $2\theta = 44.7°$. Second order: $\sin\theta = 2\lambda/(2d_{110}) = 0.7608$, so $\theta = 49.54°$ and $2\theta = 99.1°$. With $d_{220} = 0.2866/\sqrt{8} = 0.1013$ nm and $n = 1$: $\sin\theta = 0.1542/0.2027 = 0.7608$, the same angle. A second-order (110) reflection and a first-order (220) reflection are the same peak, just labeled differently (compare the 99.07° peak in Figure 11.8b).
8. BCC appears when $h + k + l$ is even: **(110), (200), (211)** appear; **(100), (111), (210)** are absent. Physically, the body-center atoms form a second, identical set of atoms displaced by $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})a$. For a given (hkl), that displacement adds a path difference of $(h + k + l)/2$ wavelengths relative to the corner atoms. If $h + k + l$ is odd, that is a half-wavelength: the two sets, with equal numbers of atoms, scatter exactly out of step and cancel. If it is even, it is a whole number of wavelengths and they reinforce.
9. $a = 0.2866$ nm, $n_v = 2/a^3 = 85.0$ nm^{-3}. (110): a rectangle $a \times a\sqrt{2}$ contains 4 corners × ¼ + 1 body center = 2 atoms, so $PD = 2/(\sqrt{2}a^2) = \sqrt{2}/a^2 = 17.2$ nm^{-2}. Adjacent (110) planes are $a/\sqrt{2} = 0.2027$ nm apart, and $n_v d = 85.0 \times 0.2027 = 17.2$; matches. (100): a face has 4 corners × ¼ = 1 atom on a^2, so $PD = 12.2$ nm^{-2}. Adjacent atom-bearing planes are $a/2 = 0.1433$ nm apart (body centers fill the halfway planes), and $n_v d = 12.2$; matches. The (110) plane is densest and widest-spaced, so expect slip on {110}, as observed in BCC metals.
10. Compute $\sin^2\theta$: 0.1187, 0.2373, 0.3563, 0.4747, 0.5937, 0.7121, 0.8307. Divided by the first: 1, 2, 3, 4, 5, 6, 7. Simple cubic can never give 7, because no integers satisfy $h^2 + k^2 + l^2 = 7$. So the pattern must be BCC with N = 2, 4, 6, …, 14: (110), (200), (211), (220), (310), (222), (321). From (110): $a = \lambda\sqrt{2}/(2\sin\theta) = 0.1542 \times 1.414/(2 \times 0.3445) = 0.3165$ nm, which matches **tungsten** (BCC, a = 0.3165 nm; from Callister's R = 0.1371 nm, $a = 4R/\sqrt{3} = 0.3166$ nm).
11. $eV = 200$ keV and $m_e c^2 = 511$ keV, so the correction factor is $1 + 200/(2 \times 511) = 1.196$. Nonrelativistic: $\lambda = h/\sqrt{2m_e eV} = 6.626 \times 10^{-34}/\sqrt{2(9.109 \times 10^{-31})(3.204 \times 10^{-14})} = 2.74 \times 10^{-12}$ m $= 0.00274$ nm. Relativistic: divide by $\sqrt{1.196} = 1.094$ to get $\lambda = 0.00251$ nm. The nonrelativistic answer is 9% too long, a correction no electron microscopist can ignore.

Where to Go Deeper

Every chapter was checked against standard references. The best free places to keep exploring:

- **HyperPhysics** (Georgia State University): short, linked concept pages on waves, optics, quantum physics and radiation.
- **OpenStax *University Physics***, Volumes 2 and 3 (free online): electromagnetic waves, geometric and wave optics, photons and atoms.
- **NASA Science, "Tour of the Electromagnetic Spectrum"**: one illustrated page per band.
- **Callister & Rethwisch, *Materials Science and Engineering***: Ch. 3 (crystal planes and X-ray diffraction) and the Optical Properties chapter (band gaps, color, opacity).
- **U.S. NRC, EPA and CDC radiation pages**: doses, shielding and health effects.
- **NIST**: physical constants, atomic spectra and X-ray attenuation data.

The full per-chapter source lists (titles and URLs) are kept with the book's source files in `EM-Radiation-Textbook/source/sources/`.